PENGARUH VOLATILITAS NILAI TUKAR, GEOPOLITIK, DAN EUROIZATION TERHADAP EKSPOR MANUFAKTUR DI CEECS EU-8

SKRIPSI

DIAJUKAN UNTUK MEMENUHI SEBAGIAN PERSYARATAN DALAM MEMPEROLEH GELAR SARJANA EKONOMI

Oleh:

HEIDIR ROYYAN FIRDAUS

NIM. 042111133107

PROGRAM STUDI S1 EKONOMI PEMBANGUNAN

FAKULTAS EKONOMI DAN BISNIS

UNIVERSITAS AIRLANGGA

SURABAYA

2024

EXCHANGE RATE VOLATILITY, GEOPOLITICS AND EUROIZATION EFFECTS ON MANUFACTURING EXPORTS IN CEECS EU-8

THESIS

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF ECONOMICS

By:

HEIDIR ROYYAN FIRDAUS

NIM. 042111133107

DEVELOPMENT ECONOMICS STUDY PROGRAM

FACULTY OF ECONOMICS AND BUSINESS

UNIVERSITAS AIRLANGGA

SURABAYA

2024

Surabaya, 30 Desember 2024

Skripsi telah selesai dan siap untuk diuji

Dosen Pembimbing

Dr. Rossanto Dwi Handoyo, S.E., M.Si., Ph.D.

NIP. 197608242003121001

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DIAJUKAN OLEH:
HEIDIR ROYYAN FIRDAUS
NIM: 042111133107

TELAH DISETUJUI DAN DITERIMA DENGAN BAIK OLEH

DOSEN PEMBIMBING.

Dr. ROSSANTO DWI HANDOYO, S.E., M.Si., Ph.D. **TANGGAL 30 DESEMBER 2024**

NIP. 197608242003121001

KOORDINATOR PROGRAM STUDI,

RUMAYYA, SE., M.Reg Dev., Ph.D **TANGGAL 30 DESEMBER 2024**

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KATA PENGANTAR

Puji syukur atas ke hadirat Allah SWT yang telah memberikan rahmat dan hidayah-Nya sehingga penulis dapat menyelesaikan skripsi yang berjudul "Pengaruh Volatilitas Nilai Tukar, Geopolitik, dan Euroization terhadap Ekspor Manufaktur di CEECs EU-8". Skripsi ini disusun untuk memenuhi sebagian persyaratan dalam memperoleh gelar sarjana ekonomi. Berbagai pihak telah membantu dalam menyelesaikan penelitian ini. Dalam kesempatan ini penulis ingin mengucapkan terima kasih kepada:

1. Kedua orang tua dan kedua kakak penulis, serta keluarga besar penulis yang telah memberikan dunia dan segalanya bagi penulis atas cinta dan kasih sayang, nasihat, dukungan, dan doa tiada henti yang dipanjatkan kepada penulis.
2. Dr. Rossanto Dwi Handoyo, S.E., M.Si., Ph.D. selaku dosen pembimbing yang telah meluangkan waktu untuk membimbing, memberi nasehat, semangat, serta saran untuk penyelesaian skripsi ini.
3. Dr. Lilik Sugiharti, SE., M.Si. selaku dosen wali yang terus memberikan dorongan, semangat, dan juga pentunjuk selama masa perkuliahan di Universitas Airlangga.
4. Ibu Prof. Dian Agustia, SE., M.Si., Ak., CMA., CA selaku Dekan Fakultas Ekonomi dan Bisnis yang telah memberikan kesempatan kepada penulis untuk berkuliah di Program Studi S1 Ekonomi Pembangunan.
5. Bapak Rumayya, SE., M.Reg.Dev., Ph.D selaku Koordinator Program Studi S1 Ekonomi Pembangunan yang senantiasa memberikan semangat kepada mahasiswa Ekonomi Pembangunan untuk terus melanjutkan skripsi.
6. Seluruh dosen pengajar Program Studi Ekonomi Pembangunan Universitas Airlangga atas ilmu, pengetahuan, dan pengalaman selama masa studi.
7. Teman-teman PPKI (Tegar, Anton, Rezky, Akmal, Naufal, Ricky, Zaki, dan Epafras) atas segala pengalaman dan kenangan selama masa pengerjaan skripsi ini.
8. Temen-teman UKM ACT atas segala pengalaman dan kenangan selama perkuliahan.
9. Kakak tingkat yang menulis skripsi yang sama (Kak Sesot, Kak Rosa, Kak Ilham, Kak Nadya, dan Kak Feliks), terutama Kak Tito yang sudah bersedia memberi ilmu untuk mengerjakan skripsi ini.
10. Kak Niluh yang bersedia membantu dalam administrasi sidang skripsi.
11. Teman-teman seperjuangan Ekonomi Pembangunan Angkatan 2021.
12. Semua pihak yang tidak dapat disebutkan satu per satu.

Pengaruh Volatilitas Nilai Tukar, Geopolitik, dan Euroization terhadap Ekspor Manufaktur di CEECs EU-8

Heidir Royyan Firdaus

ABSTRAK

Efek volatilitas nilai tukar terhadap ekspor masih ambigu. Penelitian ini bertujuan untuk meneliti efek volatilitas nilai tukar terhadap ekspor manufaktur di CEECs EU-8 dengan menggunakan metode GARCH untuk membuat variabel nilai tukar. Sementara itu, penelitian ini menggunakan metode ARDL untuk melihat efek simetris dan NARDL untuk melihat efek asimetris dalam jangka pendek dan jangka panjang dengan data periode 2009M1-2022M12. Sebagian besar, volatilitas nilai tukar berpengaruh positif terhadap ekspor komoditas di CEECs EU-8. Dalam model asimetris, volatilitas nilai tukar negatif atau penurunan volatilitas ditemukan mendominasi karena *currency union* mengurangi volatilitas nilai tukar di Uni Eropa. Selain itu, geopolitik atau perang Rusia dan Ukraine dan euroization meningkatkan nilai ekspor sejalan dengan peningkatan *trade cost*. Hal ini sesuai dengan asumsi yang menyatakan perang Rusia dan Ukraina meningkatkan harga komoditas, biaya input, harga energi, dan logistik maritim, sedangkan adopsi euro meningkatkan harga ekspor dan meningkatkan perdagangan *intra-trade* dalam *Economic and Monetary Union* (EMU) Uni Eropa. Oleh karena itu, penelitian ini bisa menjadi landasan teori jika suatu negara ingin mengadopsi mata uang euro untuk mendapatkan manfaat dari *currency union*.

Kata Kunci: volatilitas nilai tukar, geopolitik, euroization, CEECs EU-8.

Title Exchange Rate Volatility, Geopolitics, and Euroization Effects on Manufacturing Exports in CEECs EU-8

Heidir Royyan Firdaus

ABSTRACT

The effect of exchange rate volatility on exports is still ambiguous. This study aims to examine the effect of exchange rate volatility on manufacturing exports in CEECs EU-8 using the GARCH method to create exchange rate variables. Meanwhile, this study uses the ARDL method to see the symmetric effect and NARDL to see the asymmetric effect in the short and long term with data from 2009M1-2022M12. Mostly, exchange rate volatility has a positive effect on commodity exports in EU-8 CEECs. In the asymmetric model, negative exchange rate volatility or a decrease in volatility is found to dominate because currency union reduces exchange rate volatility in the European Union. In addition, geopolitics or the Russia-Ukraine war and euroization increase export value in line with increasing trade costs. This aligns with the assumption that the Russia-Ukraine war increases commodity prices, input costs, energy prices, and maritime logistics, while the adoption of the euro increases export prices and increases intra-trade within the Economic and Monetary Union (EMU) European Union. Therefore, this research can be a theoretical basis if a country wants to adopt the euro currency to gain benefits from currency union.

Keywords: exchange rate volatility, geopolitics, euroization, CEECs EU-8.

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BAB 1
PENDAHULUAN

1.1 Latar Belakang/Fenomena

Sistem volatilitas nilai tukar yang digunakan sekarang mempunyai sejarah yang panjang. Dalam perkembangan mata uang sebagai nilai tukar, logam mulia seperti emas dan perak sudah digunakan sebagai alat tukar dalam perdagangan (Williams, Cribb, & Errington, 1997). Kemudian sistem standar emas dimulai pada akhir abad ke-19, setelah itu digantikan dengan sistem Bretton Woods pada 1944 hingga tahun 1971 ketika dollar AS dipatok ke emas (Eichengreen, 2008). Dalam sistem Bretton Woods, dollar AS dapat dikonversi menjadi emas dengan nilai tukar tetap $35 per ons dan mata uang negara lain dipatok dengan dollar AS, akan tetapi dollar AS tidak dapat mempertahankan nilai tukar tetap $35 per ons karena mengalami deficit neraca pembayaran (Ghizoni, 2013). Oleh karena itu, sistem nilai tukar mengambang digunakan sampai sekarang.

Para peneliti berusaha meneliti dampak dari volatilitas nilai tukar, terutama terhadap perdagangan internasional. Namun, hasil penelitian masih ambigu. Teori ekonomi menyatakan dampak peralihan ke nilai tukar mengambang mungkin mengurangi perdagangan internasional karena meningkatnya ketidakpastian (Clark, 1973). Biarpun begitu, volatilitas nilai tukar mungkin tidak selalu secara langsung menurunkan volume perdagangan, akan tetapi membuat *traders* ragu untuk berpartisipasi karena risiko yang terkait dengan fluktuasi nilai tukar (*risk-averse*) (Hansen & Hodrick, 1980). Bukti penelitian tentang volatilitas menurunkan perdagangan internasional ditemukan oleh Grauwe (1988), pertumbuhan perdagangan internasional berkurang lebih dari setengahnya di negara-negara industri sejak diberlakukan nilai tukar mengambang walaupun ada faktor lain yang seperti penurunan tingkat pertumbuhan output dan perlambatan proses integrasi perdagangan dalam dunia industri sejak awal tahun 1970-an.

Sebaliknya, walaupun dikatakan bahwa peningkatan volatilitas nilai tukar akan mengurangi volume perdagangan internasional karena mempunyai sifat *risk-*

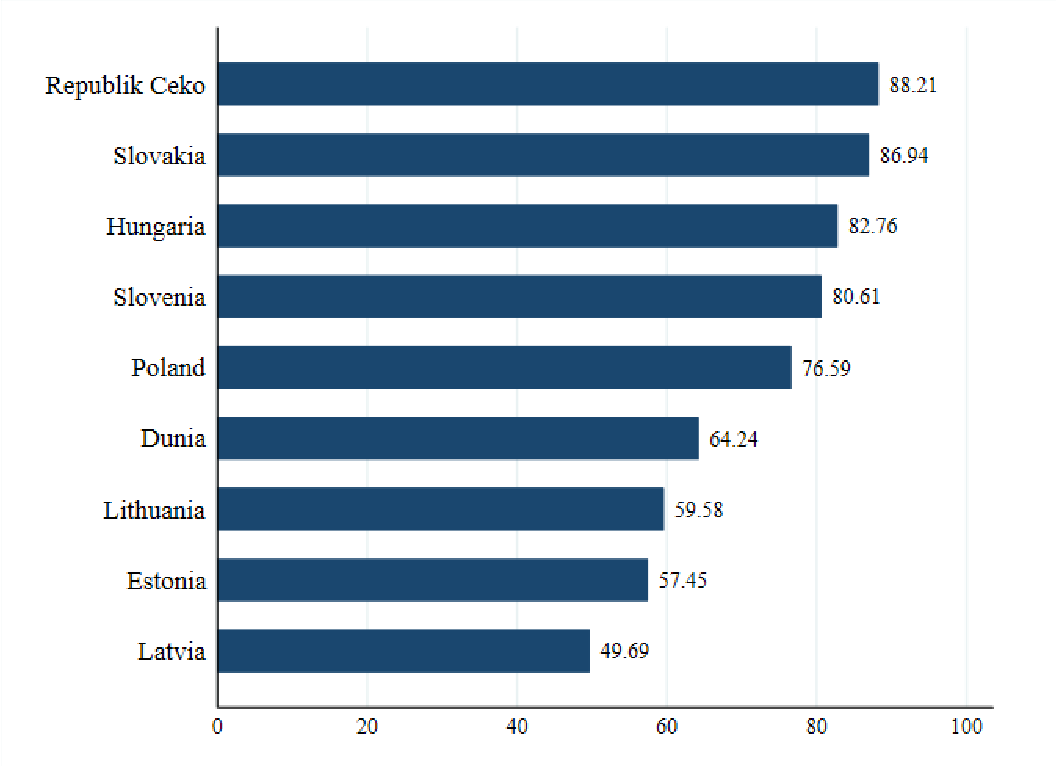
averse, *trader* mungkin dapat beradaptasi dan mendapatkan keuntungan dari volatilitas nilai tukar (*risk-taker*) (Franke, 1991). Namun, Hooper dan Kohlhagen (1978) menemukan hasil penelitian yang berbeda, penelitian ini menemukan volatilitas nilai tukar mempunyai dampak yang tidak signifikan terhadap volume perdagangan, akan tetapi volatilitas nilai tukar berpengaruh negatif terhadap harga barang sejalan dengan penurunan permintaan. Oleh karena itu, literatur menyimpulkan bahwa volatilitas nilai tukar dapat mempunyai pengaruh negatif (*risk-averse*), positif (*risk-taker*), dan tidak signifikan terhadap perdagangan internasional.

Penelitian terbaru menggunakan data bilateral yang *disaggregated* menghasilkan hasil ambigu. Hasil ambigu ini berarti volatilitas nilai tukar dapat mempengaruhi perdagangan bilateral secara positif, negatif, dan tidak signifikan walaupun hanya berbeda kode produk perdagangan baik jangka pendek maupun jangka panjang. Hal ini didukung oleh beberapa literatur yang meneliti pengaruh volatilitas terhadap perdagangan internasional bilateral, seperti negara Pakistan dengan Jepang (Bahmani-Oskooee, Iqbal, & Salam, 2016), negara-negara ASEAN dengan mitra dagang terbesarnya (Handoyo, Alfani, Ibrahim, Sarmidi, & Haryanto, 2023), negara Thailand dengan Malaysia (Aftab, Syed, & Katper, 2017), negara Thailand dengan China (Bahmani-Oskooee & Kanitpong, 2019), negara Indonesia dengan mitra dagang terbesarnya (Sugiharti, Esquivias, & Setyorani, 2020), negara Indonesia dengan negara-negara OIC (Handoyo, Sari, Ibrahim, & Sarmidi, 2022), negara Indonesia dengan Amerika Serikat (Bahmani-Oskooee, Harvey, & Hegerty, 2015; Handoyo et al., 2024), negara Amerika Serikat dengan Korea (Bahmani-Oskooee & Baek, 2016), dan negara Malaysia dengan Amerika Serikat (Bahmani-Oskooee & Aftab, 2017; Bahmani-Oskooee & Harvey, 2011).

Aftab, Syed, dan Katper (2017) menyarankan menggunakan data *disaggregated* dengan mitra negara yang spesifik agar menghindari *aggregation bias* yang menyebabkan temuan terlalu digeneralisasi, sehingga dapat menarik implikasi kebijakan untuk industri yang spesifik. Penelitian yang menggunakan kode SITC sudah banyak yang menggunakan komoditas *disaggregated* dengan kode 2 digit (Aftab et al., 2017; Bahmani-Oskooee & Aftab, 2017; Bahmani-Oskooee et al., 2016) dan 3 digit (Bahmani-Oskooee & Baek, 2016; Bahmani-Oskooee & Harvey, 2011; Bahmani-Oskooee et al., 2015). Sebaliknya, penelitian yang menggunakan kode HS sebagian besar menggunakan HS 2 digit seperti beberapa penelitian sebelumnya (Bahmani-Oskooee & Kanitpong, 2019; Handoyo et al., 2024, 2022; Sugiharti et al., 2020), kecuali Handoyo dkk. (2023) yang menggunakan komoditas *disaggregated* dengan kode HS 4 digit. Oleh karena itu, penelitian ini juga akan menggunakan kode HS 4 digit yang dapat digunakan oleh pembuat kebijakan untuk perdagangan komoditas yang lebih *disaggregated*.

Central and Eastern European Countries (CEECs) adalah regional geopolitik yang berisi negara Bulgaria, Republik Ceko, Hungaria, Polandia, Rumania, Slovakia, Slovenia, dan negara-negara di wilayah Baltik yang terdiri dari Estonia, Latvia, dan Lithuania (OECD, 2001). Akan tetapi, negara-negara yang dipilih dalam penelitian ini adalah 8 negara CEECs yang masuk Uni Eropa (EU-8) yang terdiri dari Republik Ceko, Hungaria, Polandia, Slovakia, Slovenia, Estonia, Latvia, dan Lithuania karena sebagian besar mempunyai persentasi ekspor manufaktur terhadap total ekspor barang lebih tinggi dari rata-rata dunia pada gambar 1.1.

Lall (2000) membagi produk manufaktur berdasarkan kategori teknologi dengan data SITC 3 digit, misalnya susu masuk dalam kategori produk primer, sementara jika susu tersebut diubah menjadi keju, produk tersebut masuk dalam kategori manufaktur berbasis sumber daya alam. Sementara itu, UNCTAD (2002) mengklasifikasikan produk manufaktur dengan menggunakan kode SITC satu digit, yaitu produk kimia (SITC 5), barang manufaktur (SITC 6), peralatan mesin (SITC 7), dan barang manufaktur lainnya (SITC 8). Oleh karena itu, penelitian ini mengasumsikan jika kode HS 28 sampai HS 97 adalah barang manufaktur. Di CEECs EU-8, semua komoditas unggulan adalah barang manufaktur kecuali bahan bakar mineral (HS 27) pada gambar 1.2.

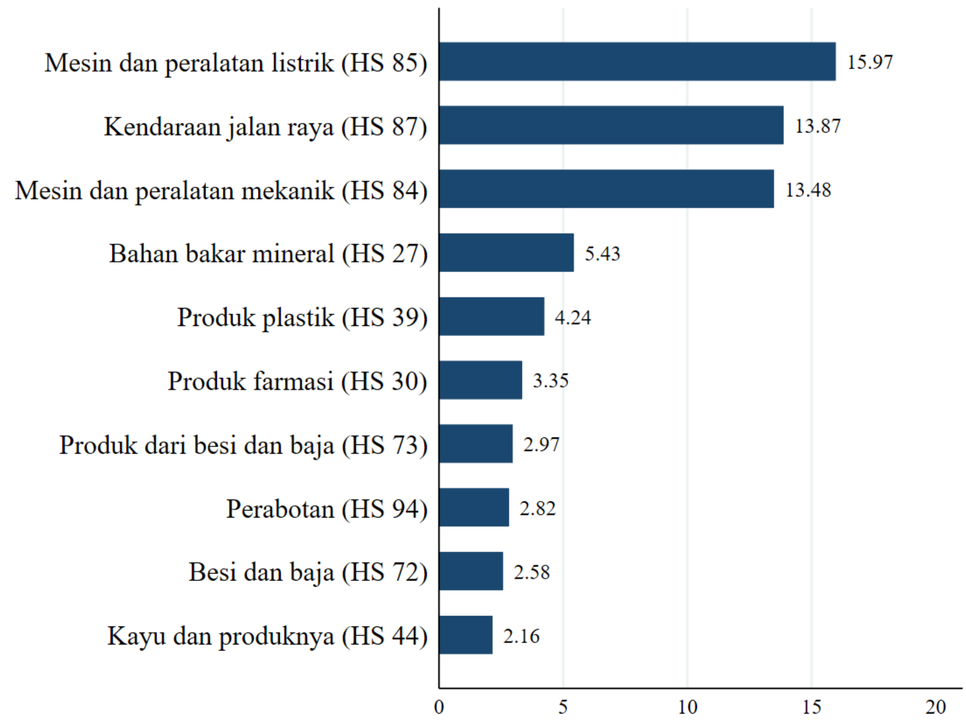


Gambar 1. 1

Persentasi Ekspor Manufaktur Dibandingkan Total Ekspor Barang Tahun 2022 (%)

Sumber: UNCTAD (2024), diolah.

Pada periode 1995 hingga 2004, pertumbuhan produktivitas tenaga kerja tahunan rata-rata di bidang manufaktur dan jasa secara signifikan lebih tinggi di CEECs EU-8 dibandingkan dengan EU-15 dengan sektor ICT memimpin pertumbuhan, akan tetapi produktivitas tenaga kerja rata-rata CEECs EU-8 masih berada di bawah dari level EU-15 (World Bank, 2008). EU-15 adalah negara-negara uni eropa sebelum CEECs bergabung seperti Italia, Spanyol, dan Inggris. Piatkowski (2003) menyatakan bahwa ekspansi sektor ICT di CEECs EU-8 sebagian besar didorong oleh FDI karena industri dalam negeri kurang memiliki daya saing. Selain itu, integrasi CEECs dalam regional uni eropa mempengaruhi arus FDI, terutama intra-regional FDI dari negara-negara uni eropa (ASEAN Investment Report, 2024).



Gambar 1. 2

Produk Unggulan CEECs EU-8 Tahun 2022 (%)

Sumber: Trade Map (2024), diolah.

Selain FDI, beberapa faktor lain yang dapat mempengaruhi industri manufaktur adalah reformasi industri jasa dan pendidikan. Arnold, Javorcik, dan Mattoo (2011) menemukan hubungan positif antara reformasi sektor jasa dan kinerja perusahaan domestik di sektor manufaktur hilir sehingga meningkatkan produktivitas manufaktur di Republik Ceko. Sementara itu, Javorcik (2004) menemukan FDI dapat meningkatkan produktivitas bisnis lokal di sektor manufaktur hulu ke perusahaan afiliasi asing di Lithuania. Peningkatan ketersediaan pendidikan tinggi, khususnya tenaga kerja terampil ICT, mendukung transisi pekerja dari kegiatan yang produktivitasnya rendah ke kegiatan yang produktivitasnya tinggi di UE-8 (World Bank, 2008)

Penelitian serupa di CEECs sudah pernah dilakukan oleh Tomanova (2013) dengan teknik estimasi *autoregressive distributive lag* (ARDL) untuk meneliti pengaruh volatilitas nilai tukar terhadap ekspor bilateral agregat dengan negara-

negara area euro secara kolektif. Penelitian ini menunjukkan tidak ada hubungan signifikan antara volatilitas nilai tukar dan ekspor CEECs dan dampak volatilitas nilai tukar ambigu. Hal ini menekankan pentingnya menggunakan mitra dagang dan produk yang spesifik. Selain itu, penelitian ini menemukan bahwa penerapan euro di negara Slovakia sebagai mata uang tunggal membawa keuntungan dalam menurunkan ketidakpastian nilai tukar dalam perdagangan internasional untuk negara dengan ekonomi terbuka kecil. Oleh karena itu, penelitian ini menggunakan variabel *dummy* euroization atau mengganti nilai tukar ke euro di Estonia, Latvia, dan Lithuania untuk meneliti dampak euroization terhadap ekspor.

Penelitian ini juga membahas pengaruh perang Rusia dan Ukraina atau geopolitik terhadap ekspor CEECs. Pada tahun 2022, pertumbuhan perdagangan barang global tumbuh di bawah proyeksi WTO, sebesar 2.7% karena adanya perang Rusia-Ukraina sejalan dengan terganggunya perdagangan barang-barang yang dipasok oleh Rusia dan Ukraina seperti gandum dan energi (World Trade Organization, 2023a). Pertumbuhan *volume* jauh lebih rendah dari value yang sebesar 12% karena terjadi peningkatan harga. Hal ini dikonfirmasi ekspor Rusia mengalami peningkatan sebesar 15.6% karena didorong oleh peningkatan harga di komoditas unggulan Russia seperti bahan bakar dan pupuk, walaupun *volume* ekspor Rusia mungkin menurun (World Trade Organization, 2023b).

1.2 Kesenjangan Penelitian

Penelitian sebelumnya yang membahas pengaruh nilai tukar terhadap ekspor di CEECs menggunakan produk dan mitra dagang yang tidak spesifik, serta metode volatilitas nilai tukar simetris. Sementara penelitian ini membahas pengaruh volatilitas nilai tukar terhadap ekspor komoditas manufaktur dan mitra dagang yang spesifik baik menggunakan pendekatan simetris maupun asimetris di CEECs EU-8.

1.3 Tujuan Penelitian

Tujuan dari penelitian ini adalah untuk membandingkan hasil efek simetris dan efek asimetris volatilitas nilai tukar terhadap tiga ekspor komoditas manufaktur unggulan dan satu mitra dagang unggulan di delapan negara CEE menggunakan pendekatan ARDL dan NARDL dengan periode dari 2009M1 sampai 2022M12 dalam jangka pendek dan panjang. Selain itu, penelitian ini juga menganalisis dampak euroization dan geopolitik terhadap ekspor.

1.4 Ringkasan Metode Penelitian

Penelitian ini menggunakan metode GARCH untuk membuat variabel nilai tukar. Sementara itu, penelitian ini menggunakan metode *autoregressive distributive lag* (ARDL) untuk melihat efek simetris dan metode *nonlinear autoregressive distributive lag* (NARDL) untuk melihat efek asimetris dalam jangka pendek dan jangka panjang dengan data periode 2009M1-2022M12. Sementara itu, metode *quantile regression* digunakan sebagai *robustness*.

1.5 Ringkasan Hasil Penelitian

Sebagian besar, volatilitas nilai tukar berpengaruh positif terhadap ekspor komoditas di CEECs EU-8. Dalam model asimetris, volatilitas nilai tukar negatif atau penurunan volatilitas ditemukan mendominasi karena *currency union* mengurangi volatilitas nilai tukar di Uni Eropa. Selain itu, geopolitik atau perang Rusia dan Ukraina dan euroization meningkatkan nilai ekspor sejalan dengan peningkatan *trade cost*. Hal ini sesuai dengan asumsi yang menyatakan perang Rusia dan Ukraina meningkatkan harga komoditas, biaya input, harga energi, dan logistik maritim, sedangkan adopsi euro meningkatkan harga ekspor dan meningkatkan perdagangan *intra-trade* dalam *Economic and Monetary Union* (EMU) Uni Eropa.

1.6 Kontribusi Riset

Kontribusi penelitian ini adalah menunjukkan perbandingan hasil pada model simetris dan asimetris untuk variabel volatilitas nilai tukar terhadap ekspor komoditas manufaktur di CEECs EU-8. Diharapkan penelitian ini dapat menjadi rekomendasi kebijakan terkait volatilitas nilai tukar dan ekspor komoditas manufaktur.

Penelitian ini menggunakan *dummy* mengganti nilai tukar menjadi euro sebagai pembanding pengaruh yang terjadi pada ekspor di CEECs EU-8 sebelum dan selama negara tersebut mengganti mata uang menjadi euro. Penelitian ini juga menggunakan *dummy* perang Russia dan Ukraina sebagai pembanding pengaruh yang terjadi pada ekspor CEECs EU-8 sebelum dan selama terjadi perang Rusia dan Ukraina, serta penambahan metode *quantile regression* untuk menunjukkan konsistensi pengaruh variabel volatilitas nilai tukar yang diuji.

1.7 Uji Ketahanan (*Robustness*)

Penelitian ini menggunakan metode *quantile regression* untuk menunjukkan konsistensi pengaruh variabel volatilitas nilai tukar yang diuji.

1.8 Sistematika Penulisan

Penelitian ini terdiri dari lima bab. Bab pertama membahas mengenai latar belakang, kesenjangan penelitian, keunikan penelitian, tujuan penelitian, dan lain-lain. Bab kedua membahas teori dan hasil penelitian sebelumnya dan hipotesis. Bab ketiga membahas pendekatan penelitian, sumber data, populasi dan sampel, teknik analisis yang digunakan untuk mencapai tujuan penelitian, dan lain-lain. Bab keempat membahas mengenai hasil dan pembahasan, serta bab kelima yang membahas kesimpulan, saran, dan keterbatasan penulis.

BAB 2
TINJAUAN PUSTAKA

2.1 Landasan Teori

Penelitian ini bertujuan untuk menyelidiki perbandingan efek simetris dan asimetris volatilitas nilai tukar terhadap ekspor komoditas manufaktur di CEECs EU-8. Selain itu, penelitian ini juga membahas dampak euroization dan geopolitik terhadap ekspor manufaktur. Oleh karena itu, perlu adanya teori dan penelitian terdahulu sebagai landasan dan acuan dalam penelitian ini.

2.1.1 Teori Perdagangan

Salah satu cara negara perekonomian terbuka berinteraksi dengan negara perekonomian terbuka lainnya adalah dengan membeli dan menjual barang dan jasa di pasar produk global (Mankiw, 2018, p. 654). Teori tentang perdagangan internasional telah mengalami perkembangan, teori merkantilis percaya bahwa kekayaan itu tetap dan menganjurkan surplus perdagangan untuk mengumpulkan emas, tetapi Adam Smith mengusulkan keunggulan absolut, di mana negara-negara harus mengkhususkan diri pada barang-barang yang mereka produksi secara efisien, David Ricardo memperluas ini dengan keunggulan komparatif yang menyatakan bahwa bahkan negara-negara yang kurang efisien pun mendapat manfaat dari perdagangan dengan berfokus pada produksi yang relatif lebih murah (Carbaugh, 2015, p. 29).

2.1.2 Teori Nilai Tukar

Dalam perdagangan internasional, harga mempengaruhi konsumen dan produsen dalam membuat keputusan (Mankiw, 2018, p. 664). Harga ini dibagi menjadi dua, yaitu nilai tukar nominal yang merupakan harga relatif satu mata uang terhadap mata uang lainnya, dan nilai tukar riil yang menyesuaikan nilai tukar nominal dengan perbedaan tingkat harga antar negara. Perubahan nilai tukar dibagi menjadi dua, yaitu depresiasi dan apresiasi. Depresiasi mata uang berarti dibutuhkan lebih banyak unit mata uang suatu negara untuk membeli satu unit mata uang asing, sebaliknya apresiasi mata uang berarti dibutuhkan lebih sedikit unit mata uang suatu negara untuk membeli satu unit mata uang asing (Carbaugh, 2015, p. 365).

Seperti harga, nilai tukar seringkali mengalami perubahan atau fluktuasi yang disebabkan oleh permintaan dan penawaran. Permintaan valuta domestik meningkat seiring dengan depresiasi nilai tukar domestik sehingga barang domestik menjadi lebih murah, sementara *supply* valuta asing meningkat jika penduduk asing ingin membeli barang domestik dengan menawarkan mata uang mereka untuk ditukar dengan mata uang domestik, hal ini akan menyebabkan mata uang domestik terapresiasi (Carbaugh, 2015, p. 369). Oleh karena itu, peningkatan permintaan mata uang asing menghasilkan dampak depresiasi mata uang domestik, sementara peningkatan *supply* menghasilkan dampak apreasiasi mata uang domestik.

Penelitian ini menggunakan nilai tukar riil. Cara menghitung nilai tukar rill adalah nilai tukar nominal dikali dengan IHK (indeks harga konsumen) negara asing dan dibagi dengan IHK negara domestik (Carbaugh, 2015, p. 373). Tambahan lagi, Catão (2019) mendefinisikan nilai tukar riil dollar-euro adalah nilai tukar nominal dollar-euro dikali dengan harga barang dari Jerman dan dibagi dengan harga barang Amerika Serikat sehingga jika ada kenaikan nilai menunjukan apresiasi dari Jerman dan depresiasi dari Amerika Serikat. Oleh karena itu, jika ada kenaikan dari nilai tukar riil, maka nilai tukar domestik mengalami depresiasi.

Ekspor dipengaruhi oleh nilai tukar riil (e) dan peningkatan nilai tukar riil menurunkan permintaan ekspor domestik (Blanchard & Johnson, 2017, p. 371). Namun, peningkatan neraca dagang dari depresiasi nilai tukar domestik mungkin butuh waktu untuk menyesuaikan, neraca dagang memburuk kemudian membaik membentuk kurva J (Blanchard & Johnson, 2017, p. 384). Hal ini disetujui oleh Stojanov, Engel, dan Varela (2024) yang menyatakan ekspor tidak langsung meningkat karena mencari pembeli baru, membangun kepercayaan, dan membangun hubungan memerlukan waktu kecuali barang homogen seperti kedelai atau gandum ketika mata uang terdepresiasi.

2.1.3 Teori Indeks Produksi Industri

Aftab dkk. (2017) menggunakan indeks produksi industri agregat negara tujuan dijadikan sebagai pengganti PDB untuk menjadi indikator ukuran aktivitas ekonomi di negara tujuan karena ketidaksediaan data PDB secara bulanan. Indeks produksi industri negara tujuan dimasukkan dalam model karena PDB negara tujuan mempengaruhi ekspor domestik. Ekspor bergantung pada kondisi ekonomi di mitra dagang (Y*) dan peningkatan ekonomi di mitra dagang meningkatkan permintaan ekspor domestik (Blanchard & Johnson, 2017, p. 371).

2.1.4 Teori Geopolitik

Dampak perang Rusia dan Ukraina adalah peningkatan indeks harga pangan FAO, harga energi seperti minyak mentah dan gas alam Eropa, harga pupuk, dan biaya transportasi laut yang berkontribusi terhadap inflasi dan biaya barang yang lebih tinggi (UNCTAD, 2022a). Lebih lanjut lagi, logistik maritim terganggu karena operasi pelabuhan tidak beroperasi di Ukraina dan banyak infrastruktur yang hancur, meningkatnya biaya asuransi dan bahan bakar yang menyebabkan jarak pengiriman yang lebih jauh, waktu transit yang lebih lama, dan biaya yang lebih tinggi (UNCTAD, 2022b). Pada tahun 2022, nilai total perdagangan meningkat sebesar 12% dengan peningkatan terbesar di komoditas bahan bakar dan tambang (59%), pertanian (19%), manufaktur (21%), dan layanan transportasi (24%) karena peningkatan biaya bahan bakar (World Trade Organization, 2023a). Oleh karena itu, nilai ekspor manufaktur selama perang Rusia dan Ukraina diasumsikan mengalami peningkatan sejalan dengan peningkatan harga komoditas, biaya input, energi, dan logistik maritim.

2.1.5 Teori Euroization

Euroization terjadi di negara-negara ekonomi berkembang, khususnya bagi negara-negara ekonomi kecil dan terbuka yang memiliki hubungan ekonomi erat dengan kawasan euro (Valle, 2018). Persatuan mata uang mengurangi kebutuhan akan mata uang nasional, memperkuat hubungan perdagangan, dan mengurangi hambatan ekonomi (Alesina & Barro, 2001). Hal ini disetujui oleh Salvatore (2004) integrasi ekonomi akan berjalan lebih lancar jika biaya transaksi dikurangi dengan memberlakukan nilai tukar yang sama dan harga tetap stabil. Sementara itu, Rose (2000) menemukan bahwa dampak positif dari penyatuan mata uang terhadap perdagangan internasional dan dampak negatif terhadap volatilitas nilai tukar, akan tetapi juga dapat mengalihkan perdagangan dari produsen non-Eropa berbiaya rendah ke produsen Eropa yang kurang efisien sehingga penyatuan mata uang meningkatkan perdagangan di dalam Economic and Monetary Union (EMU) Uni Eropa dengan mengorbankan perdagangan dengan negara non-anggota.

2.1.6 Penelitian Terdahulu

Beberapa penelitian sebelumnya yang membahas pengaruh nilai tukar terhadap ekspor komoditas manufaktur menyimpulkan sebagian besar volatilitas nilai tukar berpengaruh negatif terhadap ekspor. Hal ini disetujui oleh penelitian Aftab et al. (2017) yang menemukan nilai tukar berdampak buruk pada dua industri ekspor Malaysia dan tujuh industri impor Malaysia dengan Thailand. Hal serupa juga ditemukan pada studi kasus ekspor Indonesia ke negara-negara OKI, Handoyo et al. (2022) menemukan bahwa volatilitas nilai tukar memiliki efek negatif yang signifikan terhadap lima komoditas ekspor utama ke negara-negara OKI, sedangkan dalam jangka panjang, volatilitas nilai tukar berdampak negatif terhadap dua belas komoditas ekspor utama ke negara-negara OKI. Sebaliknya, Handoyo et al. (2024) menemukan volatilitas nilai tukar ARDL linier mempunyai efek positif pada 9 komoditas ekspor Indonesia ke Amerika Serikat dan 6 impor komoditas dari AS, sementara ditemukan dalam 12 ekspor komoditas dan 8 impor dalam model *nonlinear*.

Bahmani-Oskooee & Aftab (2017) menyatakan bahwa volatilitas dapat memiliki efek asimetris dan menemukan efek asimetris di sekitar sepertiga industri. Penelitian ini menggunakan model ARDL dan NARDL dengan data bilateral 54 industri ekspor Malaysia dan 63 industri impor dari AS. Demikian juga, Bahmani-Oskooee & Baek (2016) melakukan uji asimetri dan menemukan efek asimetri dalam jangka pendek hanya untuk 21 industri, sementara efek asimetri jangka panjang hanya intuk untuk 42 industri. Penelitian ini menggunakan model ARDL dan NARDL dengan data bilateral Korea Selatan dan Amerika Serikat dalam 73 industri, penelitian ini juga memberikan pemahaman yang lebih baik tentang pengaruh nilai tukar terhadap perdagangan dengan menggunakan asimetris dibandingkan dengan simetris.

Bahmani-Oskooee et al., (2016) menemukan dampak jangka pendek dari volatilitas nilai tukar di tujuh industri ekspor Pakistan dan di dua belas industri impor Pakistan, akan tetapi berlangsung dalam ke panjang hanya terjadi di empat industri ekspor dan tujuh industri impor. Sementara itu, penelitian Bahmani-Oskooee et al. (2015) menggunakan studi kasus 108 industri ekspor Amerika Serikat ke Indonesia dan 32 industri impor dan menemukan lebih dari separuh industri ekspor dan impor dipengaruhi oleh volatilitas nilai tukar riil dalam jangka pendek, akan tetapi hanya sepertiga industri ekspor dan impor yang mencatat dampak jangka panjang.

Hasil yang sama ditemukan oleh Bahmani-Oskooee & Harvey (2011), volatilitas nilai tukar memberikan efek jangka pendek hampir 2/3 industri dari 101 industri ekspor Amerika Serikat dan 17 industri impor dari Malaysia, akan tetapi hanya bertahan dalam jangka panjang di 38 industri Amerika Serikat dan di 10 industri impor Amerika Serikat dari Malaysia. Lebih lanjut lagi, Bahmani-Oskooee & Harvey (2011) menemukan bahwa volatilitas nilai tukar dalam jangka panjang berpengaruh positif pada komoditas perabotan (SITC 821), sementara volatilitas nilai tukar dalam jangka pendek berpengaruh positif pada komoditas perabotan dan kayu manufaktur (SITC 632) dan berpengaruh negatif pada komoditas obat-obatan dan farmasi (SITC 541). Selain itu, nilai tukar riil berpengaruh negatif pada komoditas kayu manufaktur, sedangkan pendapatan negara mitra berpengaruh positif pada komoditas perabotan dan kayu manufaktur, serta obat-obatan dan farmasi.

Model asimetris dapat menemukan pengaruh volatilitas nilai tukar yang diabaikan oleh model simetris. Bahmani-Oskooee & Kanitpong (2019) menemukan efek volatilitas nilai tukar jangka pendek yang signifikan pada 22 industri ekspor Thailand dan 32 industri impor Thailand dari China, sementara efek volatilitas nilai tukar jangka pendek yang signifikan ditemukan pada hampir semua industri dalam model asimetris. Handoyo et al. (2023) juga menyetujui hal ini dan menemukan volatilitas nilai tukar memiliki pengaruh signifikan terhadap 13 komoditas ekspor dalam jangka pendek jika menggunakan metode ARDL, sementara volatilitas nilai tukar memiliki pengaruh signifikan terhadap 19 komoditas ekspor dalam jangka pendek jika menggunakan metode NARDL.

Lebih lanjut lagi, Handoyo et al. (2023) menemukan bahwa volatilitas nilai tukar dalam jangka panjang berpengaruh negatif pada bagian dari mobil (HS 8708), sementara volatilitas nilai tukar dan volatilitas nilai tukar negatif dalam jangka pendek berpengaruh positif. Selain itu, volatilitas nilai tukar, volatilitas nilai tukar positif, dan volatilitas nilai tukar negatif berpengaruh negatif pada komoditas telepon (HS 8517). Sedangkan, nilai tukar riil dan indeks produksi industry berpengaruh positif pada komoditas telepon.

Literatur telah menerapkan berbagai pendekatan untuk volatilitas nilai tukar, Hayakawa & Kimura (2009) menggunakan deviasi standar dari *first difference* logaritma natural bulanan nilai tukar riil bilateral dalam periode 5 tahun sebelum t yang diperkenalkan oleh Rose (2000). Sementara itu, Arize, Osang, & Slottje (2000) menggunakan simpangan baku standar deviasi dari nilai tukar riil. Namun, penerapan deviasi standar dari rata-rata bergerak logaritma nilai tukar gagal menangkap efek potensial dari nilai puncak tinggi dan rendah dari nilai tukar (Serenis & Tsounis, 2013). Dua pendekatan pertama mengabaikan aspek-aspek kunci dari proses stokastik atau acak, sementara model berbasis GARCH lebih tepat dan akurat menangkap risiko karena mereka menggabungkan varians kondisional yang bervariasi dari waktu ke waktu (Sharma & Pal, 2018). Oleh karena itu, penelitian ini menggunakan metode GARCH.

2.2 Hipotesis

Hipotesis dari penelitian ini yang menggunakan model ARDL adalah sebagai berikut:

1. Variabel indeks produksi industri berpengaruh signifikan terhadap ekspor komoditas dalam jangka panjang.

2. Variabel nilai tukar berpengaruh signifikan terhadap ekspor komoditas dalam jangka panjang.
3. Variabel geopolitik berpengaruh signifikan terhadap ekspor komoditas dalam jangka panjang.
4. Variabel euroization berpengaruh signifikan terhadap ekspor komoditas dalam jangka panjang.
5. Variabel volatilitas nilai tukar, volatilitas nilai tukar negatif, dan volatilitas nilai tukar positif berpengaruh signifikan terhadap ekspor komoditas dalam jangka panjang dan jangka pendek.

BAB 3
METODE PENELITIAN

3.1 Pendekatan penelitian

Penelitian ini menganalisis efek simetris dan asimetris volatilitas nilai tukar terhadap ekspor di negara-negara CEE dengan ekspor komoditas manufaktur 4 digit unggulan dan mitra dagang terbesar di negara-negara CEE.

Penelitian ini menggunakan metode kuantitatif dengan data sekunder dengan data bulanan *time series* dengan periode waktu 2009M1-2022M12. Data dalam penelitian ini mencakup data ekspor, indeks produksi industri, nilai tukar riil, volatilitas nilai tukar, volatilitas nilai tukar positif, volatilitas nilai tukar negatif, *dummy* geopolitik, serta *dummy* euroization. Volatilitas nilai tukar diperoleh dengan menggunakan metode GARCH (*generalized autoregressive conditional heteroskedasticity*). Penelitian ini juga menggunakan metode ARDL untuk mengetahui efek simetris volatilitas nilai tukar terhadap ekspor. Sedangkan, metode *nonlinear* ARDL digunakan untuk mengetahui efek asimetris volatilitas nilai tukar terhadap ekspor. Selain itu, penelitian ini menambahkan metode *quantile regression* sebagai *robustness* untuk melihat konsistensi pengaruh dari volatilitas nilai tukar simetris terhadap ekspor manufaktur.

3.2 Sumber Data

Data penelitian ini menggunakan data bulanan dengan sifat *time series* dalam periode 2009M1-2022:M12. Data jenis ini termasuk jenis data sekunder, yang bersumber dari Trade Map IMF (International Monetary Fund).

3.3 Populasi dan Sampel

Penelitian ini menggunakan 11 komoditas ekspor ekspor dan tiga negara mitra terbesar dengan data yang digunakan dari periode 2009M1-2022M12. Rata-rata persentase komoditas yang dipilih adalah 6.2% dibandingkan total komoditas ekspor, sementara rata-rata persentase negara mitra adalah 21.8% dibandingkan total komoditas negara tujuan mitra dalam satu komoditas.

Tabel 3. 1

Daftar Komoditas, Negara Asal, dan Negara Mitra yang Digunakan

Negara		Komoditas		Persantase	
Asal	**Mitra**	**Kode**	**Nama**	**Produk**	**Mitra**
Republik Ceko	Jerman	8703	Mobil penumpang	10.56	22.93
Republik Ceko	Jerman	8517	Telepon	6.66	43.71
Republik Ceko	Jerman	8708	Bagian dari mobil	5.95	41.9
Hungaria	Jerman	8703	Mobil penumpang	8.13	24.47
Hungaria	Jerman	8708	Bagian dari mobil	5.2	38.46
Hungaria	Jerman	8507	Baterai	4.97	43.41
Polandia	Jerman	8708	Bagian dari mobil	5.08	32.36
Polandia	Jerman	8507	Baterai	3.66	46.37
Polandia	Jerman	8703	Mobil penumpang	2.16	24.47
Slovakia	Britania Raya	8703	Mobil penumpang	24.39	8.17
Slovakia	Britania Raya	8708	Bagian dari mobil	5.03	3.91
Slovakia	Britania Raya	8528	Monitor dan projektor	4.44	11.38
Slovenia	Jerman	3004	Obat	25	1.67
Slovenia	Jerman	8703	Mobil penumpang	4.76	30.39
Slovenia	Jerman	8516	Pemanas air	1.12	43.91
Estonia	Jerman	8517	Telepon	5.42	6.64
Estonia	Jerman	8703	Mobil penumpang	2.74	3.72
Estonia	Jerman	9406	Rumah Cetakan	2.68	17.91
Latvia	Lithuania	4407	Olahan Kayu	5.64	4.47
Latvia	Lithuania	8517	Telepon	3.58	29.61
Latvia	Lithuania	3004	Obat	2.62	18.31
Lithuania	Jerman	9403	Perabotan	4.65	12.25
Lithuania	Jerman	8703	Mobil penumpang	2.44	5.65
Lithuania	Jerman	3102	Pupuk	1.98	7.14

Sumber: Trade Map (2024), diolah.

3.4 Model Empiris

Penelitian ini menggunakan volatilitas nilai tukar yang dihasilkan dari metode GARCH. Bollerslev (1986) menggunakan spesifikasi yang paling umum digunakan untuk varians kondisional dengan *first-order* variabel dan biasanya ditulis GARCH(1,1). Demikian juga, Gujarati dan Porter (2009:796) memberikan contoh model paling sederhana dari GARCH adalah GARCH (1,1) sebagai berikut:

$$\sigma_t^2 = \gamma_0 + \gamma_1 u_{t-1}^2 + \rho_1 \omega_{t-1}^2 \qquad (3.1)$$

Keterangan:

σ_t^2 = *Conditional Variance*

γ_0 = Konstanta

u_{t-1}^2 = Lag satu ARCH

ω_{t-1}^2 = Lag satu GARCH

γ_1 = Koefisien ARCH

ω_1 = Koefisien GARCH

Penelitian ini mengadopsi model dalam penelitian sebelumnya yang menjadikan ekspor sebagai variabel dependen dan volatilitas nilai tukar, nilai tukar riil, dan indeks produksi industri negara mitra sebagai independen (Aftab et al., 2017). Akan tetapi, fungsi dari model tersebut tidak menggambarkan kondisi saat terjadinya perang Rusia dan Ukraina, sehingga variabel *dummy* geopolitik ditambahkan dalam variabel. Perang Rusia dan Ukraina terjadi sejak akhir Februari 2022 (World Trade Organization, 2023a).

Di samping itu, negara Estonia, Latvia, dan Lithuania mengadopsi mata uang Euro saat waktu penelitian ini dilakukan, sehingga variabel *dummy* eurozation ditambahkan. Estonia mengadopsi euro pada 1 Januari 2011, Latvia mengadopsi euro pada 1 Januari 2014, sedangkan Lithuania mengadopsi euro pada 1 Januari 2015 (European Central Bank, 2011, 2014, 2015). Akan tetapi, negara-negara tersebut diasumsikan menggunakan mata uang euro di periode 2009M1-2022M12 untuk membuat variabel bilateral nilai tukar riil. Asumsi ini didukung oleh OECD (2024) dalam data nilai tukar nominal yang menggunakan mata uang Euro di periode 2009M1-2022M12 untuk negara Estonia, Latvia, dan Lithuania.

Model persamaan 3.2 adalah ekspor negara Republik Ceko, Hungaria, Polandia, Slovakia, dan Slovenia, sedangkan model persamaan 3.3 adalah ekspor negara Estonia, Latvia, dan Lithuania.

$$lnExp_t^i = \beta_0 + \beta_1 lnVol_t + \beta_2 lnRER_t + \beta_3 lnIPI_t + \beta_4 D_{Geopolitik_t} + \varepsilon_t \qquad (3.2)$$

$$\mathrm{lnExp}_t^i = \beta_0 + \beta_1 \mathrm{lnVol}_t + \beta_2 \mathrm{lnRER}_t + \beta_3 \mathrm{lnIPI}_t + \beta_4 \mathrm{D}_{\mathrm{Geopolitik}_t} + \beta_5 \mathrm{D}_{\mathrm{Euroization}_t} + \varepsilon_t \quad (3.3)$$

Selain itu, penelitian ini juga menggunakan model dalam jangka pendek. Bahmani-Oskooee dan Aftab (2017) mengatakan efek jangka pendek dan jangka panjang yang diperoleh dengan menggunakan model ECM (*Error Correction Model*). ECM diperkenalkan oleh Pasaran, Shine, dan Smith (2001) yang memperkirakan dampak jangka pendek dan jangka panjang secara bersamaan dalam model yang sama, ECM dikembangkan menjadi model ARDL, akan tetapi harus memenuhi beberapa syarat, yaitu variabel terintegrasi pada tingkat I(0) dan I(1), serta saling terkointegrasi.

Koefisien error correction diekspektasikan negatif dan signifikan (Gujarati & Porter, 2009, p. 765). Selain itu, koefisien *error correction* diperlihatkan menunjukan *speed-of-adjustment* atau kecepatan penyesuaian ke ekuilibrium jangka panjang di model ARDL (Kripfganz dan Schneider 2023). Sementara itu, cara menghitung kecepatan penyesuaiannya adalah 1/koefisien *error correction* (Handoyo et al., 2023).

Model persamaan 3.4 adalah ekspor negara Republik Ceko, Hungaria, Polandia, Slovakia, dan Slovenia, sedangkan model persamaan 3.5 adalah ekspor negara Estonia, Latvia, dan Lithuania. Pemilihan empat lag maksimum berdasarkan AIC untuk memilih berapa jumlah optimal lag dalam model di jangka pendek diadopsi dari beberapa penelitian Bahmani-Oskooee dan Aftab (2017). Namun, penelitian ini hanya menampilkan variabel volatilitas nilai tukar untuk model simetris, serta volatilitas nilai tukar positif dan volatilitas nilai tukar negatif untuk model asimetris dalam jangka pendek karena hanya mau melihat bagaimana pengaruh volatilitas nilai tukar sebelum mencapai equilibrium atau jangka panjang.

$$\Delta lnEXP_t^i = \alpha_0 + \sum_{k=1}^{n1} \alpha_1 \Delta lnExp_{t-k} + \sum_{k=0}^{n2} \alpha_2 \Delta lnVol_{t-k} + \sum_{k=0}^{n3} \alpha_3 \Delta lnRER_{t-k} + \sum_{k=0}^{n4} \alpha_4 \Delta lnIPI_{t-k} + \sum_{k=0}^{n5} \alpha_5 \Delta D_{Geopolitik_{t-k}} + \beta_0 lnExp_{t-1} + \beta_1 lnVol_{t-1} + \beta_2 lnRER_{t-1} + \beta_3 lnIPI_{i,t-1} + \beta_4 D_{Geopolitik_{t-1}} + \varepsilon_t \quad (3.4)$$

$$\Delta lnEXP_t^i = \alpha_0 + \sum_{k=1}^{n1} \alpha_1 \Delta lnExp_{t-k} + \sum_{k=0}^{n2} \alpha_2 \Delta lnVol_{t-k} + \sum_{k=0}^{n3} \alpha_3 \Delta lnRER_{t-k} + \sum_{k=0}^{n4} \alpha_4 \Delta lnIPI_{t-k} + \sum_{k=0}^{n5} \alpha_5 \Delta D_{Geopolitik_{t-k}} + \sum_{k=0}^{n6} \alpha_6 \Delta D_{Euroization_{t-k}} + \beta_0 lnExp_{t-1} + \beta_1 lnVol_{t-1} + \beta_2 lnRER_{t-1} + \beta_3 lnIPI_{t-1} + \beta_4 D_{Geopolitik_{t-1}} + \beta_5 D_{Euroization_{t-1}} + \varepsilon_t \quad (3.5)$$

Penelitian ini kemudian membuat dua variabel baru untuk model asimetris yang dikembangkan oleh Shin, Yu, dan Greenwood-Nimmo (2014). Volatilitas nilai tukar positif dapat dihasilkan dari *first-difference* volatilitas nilai tukar dan mengganti nilai yang negatif dengan nol, sementara volatilitas nilai tukar negatif dibuat sebaliknya. Satu yang mencerminkan peningkatan volatilitas yang dilambangkan dengan Volpos dan satu yang mencerminkan penurunan volatilitas yang dilambangkan oleh Volneg, sebagaimana dirumuskan oleh model di persamaan 3.6 dan persamaan 3.7.

$$\sum_{k=1}^{n} \Delta Volpos_k = \sum_{k=1}^{n} \min(\Delta lnVolj_k, 0) \quad (3.6)$$

$$\sum_{k=1}^{n} \Delta Volneg_k = \sum_{k=1}^{n} \min(\Delta lnVolj_k, 0) \quad (3.7)$$

Kemudian, volatilitas dalam model persamaan 3.4 dan 3.5 diganti dengan volatilitas positif dan volatilitas negatif pada model persamaan 3.6 dan 3.7. Model persamaan 3.8 adalah ekspor negara Republik Ceko, Hungaria, Polandia, Slovakia, dan Slovenia sedangkan model persamaan 3.9 adalah ekspor negara Estonia, Latvia, dan Lithuania. Oleh karena itu, berikut ini adalah model asimetris yang akan digunakan dalam penelitian ini dengan metode NARDL:

$$\Delta lnEXP_t^i = \alpha_0 + \sum_{k=1}^{n1} \alpha_1 \Delta lnExp_{t-k} + \sum_{k=0}^{n2} \alpha_2 \Delta lnVolpos_{t-k} + \sum_{k=0}^{n3} \alpha_3 \Delta lnVolneg_{t-k} + \sum_{k=0}^{n4} \alpha_4 \Delta lnRER_{t-k} + \sum_{k=0}^{n5} \alpha_5 \Delta lnIPI_{t-k} + \sum_{k=0}^{n6} \alpha_6 \Delta D_{Geopolitik_{t-k}} + \beta_0 lnExp_{t-1} + \beta_1 lnVolpos_{t-1} + \beta_2 lnVolneg_{t-1} + \beta_3 lnRER_{t-1} + \beta_4 lnIPI_{t-1} + \beta_5 D_{Geopolitik_{t-1}} + \varepsilon_t \quad (3.8)$$

$$\Delta LnEXP_t^i = \alpha_0 + \sum_{k=1}^{n1} \alpha_1 \, \Delta lnExp_{t-k} + \sum_{k=0}^{n2} \alpha_2 \, \Delta lnVolpos_{t-k} + \sum_{k=0}^{n3} \alpha_3 \, \Delta LnVolneg_{t-k} + \sum_{k=0}^{n4} \alpha_4 \, \Delta LnRER_{t-k} + \sum_{k=0}^{n5} \alpha_5 \, \Delta LnIPI_{t-k} + \sum_{k=0}^{n6} \alpha_6 \, \Delta D_{Geopolitik_{t-k}} + \sum_{k=0}^{n7} \alpha_7 \, \Delta D_{Euroization_{t-k}} + \beta_0 lnExp_{t-1} + \beta_1 lnVolpos_{t-1} + \beta_2 lnVolneg_{t-1} + \beta_3 lnRER_{t-1} + \beta_4 lnIPI_{t-1} + \beta_5 D_{Geopolitik_{t-1}} + \beta_6 D_{Euroization_{t-1}} + \varepsilon_t \tag{3.9}$$

Keterangan:

$lnEXP_t^i$ = Logaritma natural ekspor dari komoditas i ke negara mitra pada periode t.

$lnVOL_t$ = Logaritma natural volatilitas nilai tukar riil bilateral pada periode t.

$lnRER_t$ = Logaritma natural nilai tukar riil bilateral pada periode t.

$lnIPI_t$ = Logaritma natural indeks produksi industri negara mitra pada periode t.

$D_{Geopolitik_t}$ = Dummy krisis perang Rusia dan Ukraina pada periode t (1 = terjadi krisis perang Rusia dan Ukraina dan 0 = tidak terjadi krisis perang Rusia dan Ukraina)

$D_{Euroization_t}$ = Dummy negara mengadopsi nilai tukar euro pada periode t (1 = mengadopsi nilai tukar euro dan 0 = mengadopsi nilai tukar euro)

$lnVolpos_t$ = Logaritma natural volatilitas nilai tukar riil positif pada periode t.

$lnVolneg_t$ = Logaritma natural volatilitas nilai tukar riil negatif pada periode t.

α_0, β_0 = Konstanta dari variabel independen.

α_1, ..., α_n, β_1, ..., β_n = Koefisien dari variabel independen.

ε_t, μ_t = *Error term* pada periode t.

Tabel 3. 2

Definisi Kegiatan Operasional

Variabel	Definisi dan Pengukuran	Sumber
Ekspor	Kegiatan negara Republik Ceko, Hungaria, Polandia, Slovakia, Slovenia, Estonia, Latvia, dan Lithuania untuk mengirimkan komoditas unggulan industri manufakturnya ke negara mitra terbesarnya. Nilai dalam satuan ribu dollar. Variabel ditransformasikan ke logaritma natural. Daftar komoditas dan negara mitra negara-negara CEE ditampilkan di tabel 3.1.	Trade Map
Nilai tukar riil	Perbandingan harga antara mata uang negara asal dan negara mitra yang disesuaikan dengan tingkat harga domestik dan luar negeri. Nilai tukar riil dapat diperoleh dari hitungan nilai tukar riil = nilai tukar nominal x IHK negara tujuan / IHK negara asal. Variabel ditransformasikan ke logaritma natural. Daftar negara asal dan negara mitra ditampilkan di tabel 3.1.	IMF
Indeks produksi industri	Indeks yang mengukur produksi industri negara mitra sebagai proksi kegiatan perekonomian suata negara. Variabel ditransformasikan ke logaritma natural. Daftar mitra dagang komoditas unggulan negara mitra ditampilkan di tabel 3.1.	IMF
Volatilitas nilai tukar	Volatilitas nilai tukar digunakan untuk mengukur ketidakpastian nilai tukar. Berdasarkan penelitian Bahmani-Oskooee dan Aftab (2017), volatilitas dapat dihasilkan dari metode GARCH. Variabel ditransformasikan ke logaritma natural.	Data diolah
Volatilitas nilai tukar positif	Volatilitas nilai tukar positif digunakan sebagai proksi peningkatan volatilitas nilai tukar. Berdasarkan Shin, Yu, & Greenwood-Nimmo (2014), volatilitas nilai tukar positif dapat dihasilkan dari *first-difference* volatilitas nilai tukar dan mengganti nilai yang negatif dengan nol.	Data diolah
Volatilitas nilai tukar negatif	Volatilitas nilai tukar negatif digunakan sebagai proksi penurunan volatilitas nilai tukar. Berdasarkan Shin, Yu, & Greenwood-Nimmo (2014), volatilitas nilai tukar negatif dapat dihasilkan dari *first-difference* volatilitas nilai tukar dan mengganti nilai yang positif dengan nol.	Data diolah
Geopolitik	Variabel geopolitik dalam penelitian ini menggunakan variabel *dummy*. Bernilai 1 untuk periode jika mengalami perang Russia dan Ukraina yakni 2022:M2 - 2022:M12 dan bernilai 0 untuk periode yang tidak mengalami perang (World Trade Organization, 2023a).	WTO
Euroization	Variabel euroization dalam penelitian ini menggunakan variabel *dummy*. Bernilai 1 untuk periode jika mengadopsi euro, yakni 2011:M1 - 2022:M12 di Estonia, 2014:M1 - 2022:M12 di Latvia, 2015:M1 - 2022:M12 di Lithuania, dan bernilai 0 untuk periode yang tidak mengadopsi euro.	European Central Bank

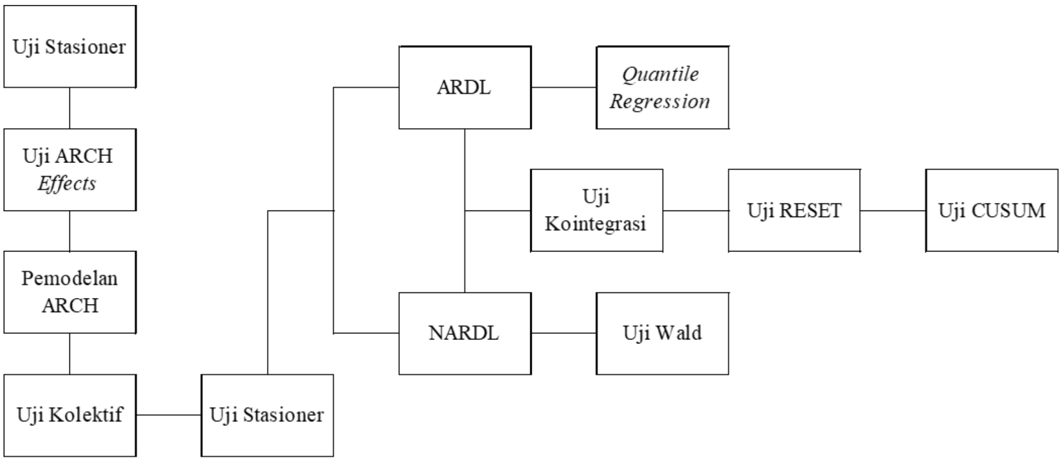
Sumber: Diolah.

3.5 Deskripsi Operasional Variabel

Penelitian ini menggunakan tujuh variabel dimana terdiri dari variabel ekspor sebagai variabel dependen, sedangkan variabel independen terdiri dari variabel volatilitas nilai tukar, volatilitas nilai tukar positif, volatilitas nilai tukar negatif, nilai tukar riil, indeks industri produksi, geopolitik, dan euroization. Setiap variabel memiliki definisi dan sumber data yang berbeda yang dirinci secara jelas dalam tabel 3.2.

3.6 Teknik Analisis

Penelitian ini menggunakan metode GARCH untuk membuat variabel volatilitas nilai tukar. Penelitian ini juga menggunakan metode ARDL untuk melihat pengaruh volatilitas nilai tukar terhadap ekspor komoditas industri manufaktur secara simetris dan metode *nonlinear* ARDL untuk melihat pengaruh volatilitas nilai tukar terhadap ekspor komoditas industri manufaktur secara asimetris. Teknik yang digunakan pada penelitian ini terdiri dari beberapa tahap yang secara ringkas dapat dilihat pada gambar 3.1.



Gambar 3. 1

***Flowchart* Teknik Analisis**

Sumber: Diolah.

3.6.1 Metode GARCH

Metode GARCH dikembangkan oleh Bollerslev (1986) untuk dapat menangkap sifat volatilitas data yang berubah-ubah setiap waktu atau keberadaan ARCH (*Autoregressive Conditional Heteroskedasticity*) yang diusulkan oleh Engle (1982). Pendekatan ARCH digunakan untuk membuat variabel volatilitas nilai tukar seperti penelitian Bahmani-Oskooee dan Aftab (2017). Sebelum melakukan metode GARCH, ada beberapa langkah. Pertama, variabel harus mempunyai keberadaan ARCH dengan uji ARCH *effects*. Kedua, sebelum menerapkan metode GARCH untuk memprediksi volatilitas nilai tukar, uji stasioneritas variabel (Sharma & Pal, 2018).

3.6.1.1 Uji Stasioneritas Nilai Tukar Riil

Uji stasioneritas variabel nilai tukar dalam penelitian ini menggunakan uji ADF (*Augmented Dickey Fuller*). Dickey dan Fuller (1979) mengembangkan prosedur untuk menguji apakah suatu variabel mempunyai *unit root* atau secara acak berfluktuasi. Hipotesis dalam uji ADF adalah:

H0: terdapat unit root atau tidak stasioner.

H1: stasioner.

H0 ditolak apabila *p-value* < 5%. Maka dari itu, variabel dikatakan stasioner apabila *p-value* < 5%. Hasil stasioner dibutuhkan dalam penelitian ini, artinya rata-rata dan varians variabel tersebut tidak berubah seiring waktu.

3.6.1.2 Uji ARCH *Effects*

Selanjutnya untuk mengetahui adanya ARCH *Effect* (kluster volatilitas) dalam model, maka perlu dilakukan uji ARCH *effects*. Engle (1982) mengembangkan uji LM (Lagrange Multiplier) untuk memeriksa keberadaan ARCH (*Autoregressive Conditional Heteroskedasticity*) atau varians berubah-ubah sepanjang waktu. Langkah yang harus dilakukan adalah melakukan regresi OLS dengan model konstan atau hanya menggunakan variabel konstan sebagai variabel independen, kemudian menguji efek ARCH dengan menggunakan uji LM (StataCorp, 2009). Hipotesis dari pengujian ini adalah:

H0: tidak terdapat ARCH *Effect* (Tidak terdapat kluster volatilitas).

H1: terdapat ARCH *Effect* (Terdapat kluster volatilitas).

Jika hasil uji LM *p-value* < 5%, maka akan menolak H0 atau terdapat efek ARCH. Hasil terdapat ARCH *Effect* dibutuhkan dalam penelitian ini, artinya model dapat dilanjutkan menggunakan model GARCH.

3.6.1.3 Estimasi Metode GARCH

Hasil *p-value* koefisien ARCH dan GARCH harus menunjukan signifikan. Namun, jika hasil salah satu koefisien ARCH dan GARCH tidak signifikan, penulis bisa mengkonfirmasi menggunakan uji koefisien secara kolektif (StataCorp, 2009).

H0: secara kolektif, variabel tidak signifikan.

H1: secara kolektif, variabel signifikan.

Model dikatakan signifikan secara kolektif jika hasil uji koefisien *p-value* < 5%. Setelah itu, maka dapat memperoleh varians bersyarat untuk menjadikannya sebagai variabel volatilitas untuk uji berikutnya.

3.6.2 Metode ARDL dan NARDL

Penelitian ini menggunakan teknik estimasi menggunakan metode ARDL *autoregressive distributed lag* (ARDL) yang diusulkan oleh Pesaran, Shin, dan Smith (2001b) yang akomodatif terhadap variabel dengan integrasi campuran. Sedangkan, teknik estimasi *nonlinear* ARDL menggunakan pendekatan asimetris yang diusulkan Shin, Yu, dan Greenwood-Nimmo (2014). Sebelum melakukan metode ARDL dan NARDL, langkah yang harus dilalui adalah melakukan uji stasioner seluruh variabel dan pemilihan lag optimum.

Bahmani-Oskooee dan Kanitpong (2019) mengatakan lag optimum yang ideal untuk jangka pendek adalah berdasarkan nilai AIC (*Akaike Information Criterion*) yang paling minimum. Pemilihan ini ditentukan otomatis dalam model di perangkat lunak STATA. Pemilihan empat lag maksimum berdasarkan AIC untuk memilih berapa jumlah optimal lag dalam model di jangka pendek diadopsi dari penelitian Bahmani-Oskooee dan Aftab (2017).

Uji diagnostik untuk regresi OLS juga dapat digunakan setelah ARDL, jika dalam bentuk *error correction* atau bentuk *first difference*, uji RESET dapat digunakan, akan tetapi uji yang berbasis residual seperti uji bgodfrey atau Breusch–Godfrey untuk *higher-order serial correlation* dan uji imtest untuk *information matrix* tidak terpengaruh (Kripfganz & Schneider, 2023). Oleh karena itu, uji diagnostik yang akan dilakukan adalah melakukan uji kointegrasi dan uji diagnostik untuk model ARDL dan *nonlinear* ARDL mencakup, uji spesifikasi model *regression specification-error test* (RESET), uji *error correction*, uji stabilitas model *cumulative sum* (CUSUM). Koefisien dalam *error correction* diekspektasikan negatif dan signifikan (Gujarati & Porter, 2009, p. 765). Sementara itu, dilakukan uji efek asimetris atau uji Wald khusus model NARDL dan uji *robustness* untuk model *quantile regression* untuk model ARDL.

3.6.2.1 Uji Stasioner Seluruh Variabel

Uji stasioneritas variabel nilai tukar dalam penelitian ini menggunakan uji *Augmented Dickey Fuller* (ADF). Dickey dan Fuller (1979) mengembangkan prosedur untuk menguji apakah suatu variabel mempunyai *unit root* atau secara acak berfluktuasi. Hipotesis dalam uji ADF adalah:

H0: terdapat unit root atau tidak stasioner

H1: stasioner

H0 ditolak apabila *p-value* < 5%. Maka dari itu, variabel dikatakan stasioner apabila *p-value* < 5%. Hasil stasioner pada tingkat *level* dan *first difference* dibutuhkan dalam penelitian ini, artinya variabel dapat melakukan estimasi ARDL dan NARDL.

3.6.2.2 Estimasi Metode ARDL dan NARDL

Pada tahap ini, dilakukan estimasi pada model ARDL maupun *nonlinear* ARDL. Model ARDL menggunakan volatilitas nilai tukar (simetris), sedangkan model *nonlinear* ARDL menggunakan volatilitas nilai tukar positif dan negatif (asimetris).

3.6.2.3 Uji Kointegrasi

Uji kointegrasi digunakan untuk mengetahui hubungan jangka panjang antara variabel independen dan dependen dalam sebuah model. Uji kointegrasi yang dilakukan dalam penelitian ini menggunakan uji F (Pasaran et al., 2001). Uji F dalam hal ini memiliki nilai kritis berupa *upper bound* dan *lower bound.* Model dikatakan terkointegrasi apabila nilai F statistik berada di atas *upper bound* atau I(1). Sebaliknya, jika nilai F ada di bawah *lower bound* I(0) maka model tidak terkointegrasi. Sementara itu, jika nilai F berada di antara *lower bound* I(0) dan *upper bound* I(1) maka model tidak konsisten. Uji kointegrasi dinyatakan jika hasil H0 diterima (ACC), H0 ditolak (REJ), dan inkosisten (INC). Hipotesis dalam uji kointegrasi adalah:

H0: model tidak terkointegrasi

H1: model terkointegrasi

H0 ditolak apabila *p-value* < 5%. Maka dari itu, variabel dikatakan terkointegrasi apabila *p-value* < 5%. Hasil terkointegrasi dibutuhkan dalam penelitian ini, artinya ada hubungan jangka panjang dalam model.

3.6.2.4 Uji RESET

Ramsey (1969) mengembangkan uji RESET untuk mengidentifikasi potensi masalah *omitted* variabel atau variabel independent tidak masuk dalam model. Hipotesis dalam uji RESET adalah:

H0: tidak ada *omitted* variabel

H1: ada *omitted* variabel

H0 diterima apabila *p-value* > 5%. Maka dari itu, model dikatakan tidak ada *omitted* variabel apabila *p-value* > 5%. Hasil ada *omitted* variabel dibutuhkan dalam penelitian ini, artinya variabel independen yang relevan mempengaruhi variabel dependen disertakan dalam model.

3.6.2.5 Uji CUSUM

Brown, Durbin, dan Evans (1975) memperkenalkan uji CUSUM, residual *recursive* diasumsikan terdistribusi normal dengan rata-rata nol dan varians konstan. Ploberger dan Krämer (1992) menemukan bahwa uji CUSUM berdasarkan residual *recursive* untuk mengidentifikasi ketidakstabilan parameter yang terjadi di awal waktu, sedangkan pengujian CUSUM berdasarkan residual OLS dalam mendeteksi ketidakstabilan di akhir data. Oleh karena itu, uji CUSUM menggunakan uji CUSUM *recursive* dan OLS. Hipotesis dalam uji CUSUM menggunakan uji CUSUM *recursive* dan OLS adalah:

H0: parameter stabil.

H1: parameter tidak stabil.

H0 diterima apabila *p-value* > 5%. Maka dari itu, model dikatakan mempunyai residual diasumsikan terdistribusi normal dengan rata-rata nol dan varians konstan apabila *p-value* > 5%. Hasil residual diasumsikan terdistribusi normal dengan rata-rata nol dan varians konstan dibutuhkan dalam penelitian ini, artinya model stabil dengan varians konstan. Hasil dari uji stabil dibagi menjadi dua, H0 diterima dinotasikan sebagai ACC dan H0 ditolak dinotasikan sebagai REJ. Model dikatakan menerima hipotesis jika nilai *test statistic* lebih rendah dari 5% *critical value*.

3.6.2.6 Uji Wald

Khusus model *nonlinier* ARDL, penulis juga menambahkan uji wald untuk mengetahui efek asimetris dalam jangka pendek secara kumulatif maupun jangka panjang (Bahmani-Oskooee & Aftab, 2017). Model dikatakan terdapat efek asimetris apabila hasil uji wald memiliki *p-value* < 5%. Ho & Saadaoui (2021) menggunakan uji wald untuk menguji efek asimetris, jika H0 ditolak menunjukkan bahwa model harus menggunakan spesifikasi yang memungkinkan efek asimetris dari volatilitas nilai tukar dalam jangka pendek atau jangka panjang. Model persamaan 3.10 adalah uji wald dalam jangka panjang, sedangkan model persamaan 3.11 adalah uji wald dalam jangka pendek. Hipotesis dalam uji Wald adalah:

$$L_{volpos} = L_{volneg} \quad (3.10)$$

$$\sum_{k=0}^{n} S_{volpos} = \sum_{k=0}^{n} S_{volnos} \quad (3.11)$$

H0: tidak ada efek asimetris.

H1: ada efek asimetris.

H0 ditolak apabila *p-value* < 5%. Maka dari itu, model dikatakan mempunyai efek simetris dalam jangka panjang dan jangka pendek apabila *p-value* < 5%. Hasil H0 ditolak menunjukkan bahwa model harus menggunakan spesifikasi yang memungkinkan efek asimetris dari volatilitas nilai tukar dalam jangka pendek atau jangka panjang.

3.6.3 Teknik Estimasi Quantile Regression

Rousseeuw dan Leroy (1987) membahas estimator *quantile regression* yang lebih kuat untuk analisis regresi karena sensitif terhadap outlier dan dapat memberikan hasil yang lebih andal jika ada outlier. Berbeda dengan regresi biasa yang berfokus pada rata-rata, regresi kuantil memberikan gambaran yang lebih komprehensif dengan memperkirakan berbagai kuantil, seperti median (q50), kuartil bawah (q25), atau kuartil atas (q75). Penelitian ini menggunakan q10, q25, q50, q75, dan q90. Metode *quantile regression* digunakan untuk melihat konsistensi pengaruh dari variabel volatilitas nilai tukar terhadap ekspor komoditas manufaktur.

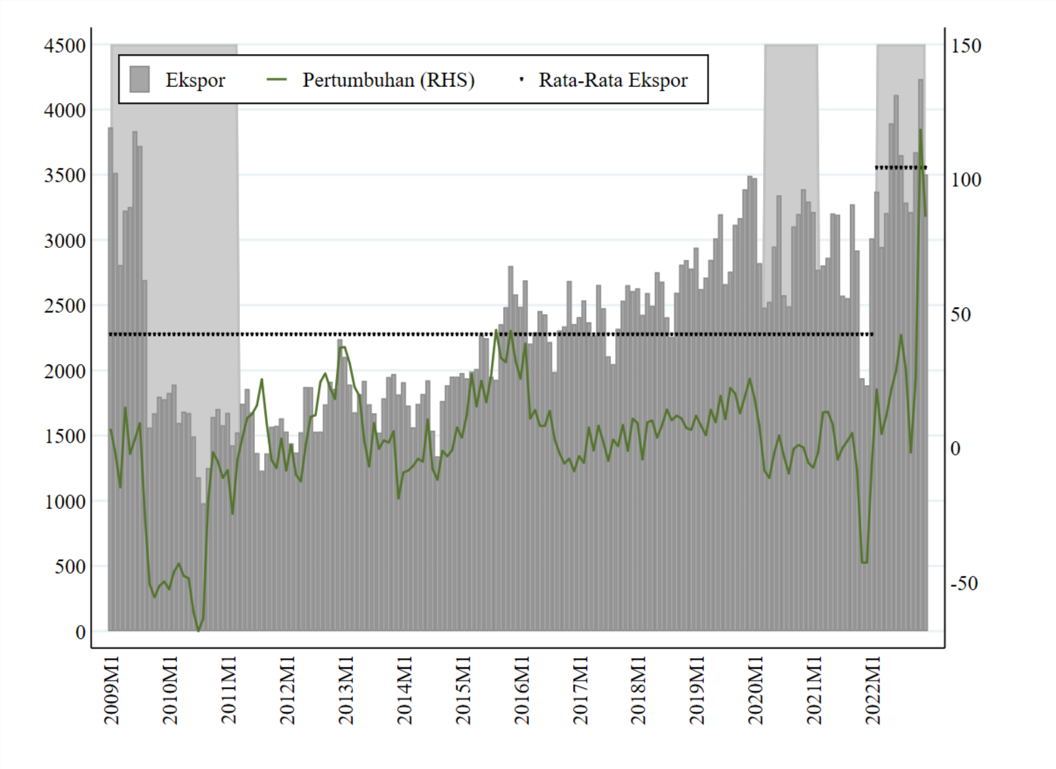
BAB 4
HASIL DAN PEMBAHASAN

4.1 Gambaran Umum

Negara-negara CEE EU-8 beserta negara mitranya memiliki tren ekspor komoditas manufaktur, indeks produksi industri, dan nilai tukar rill yang berbeda-beda. Oleh karena itu, berikut adalah tren terkait ekspor komoditas manufaktur, nilai tukar riil, indeks produksi industri negara mitra di CEECs EU-8 yang digunakan.

4.1.1 Tren Ekspor Komoditas Manufaktur CEECs EU-8

Jika dilihat dari tren akumulatif ekspor komoditas manufaktur CEECs EU-8 yang digunakan, ekspor mengalami peningkatan tiap tahun dari 2011. Aizenman & Saadaoui (2024) menyatakan krisis Keuangan Global yang bermula di AS, berkembang menjadi krisis global melalui globalisasi keuangan, krisis ini terjadi pada periode 2008M9-2009M7, sedangkan peningkatan suku bunga terjadi sampai 2011M5. Akumulatif ekspor komoditas manufaktur mempunyai tren penurunan pada periode ini. Tren penurunan juga terjadi saat periode COVID-19, Handoyo et al., (2024) membuat *dummy* COVID-19 pada periode 2020M3-2022M12. Sementara itu, periode perang Russia dan Ukraina adalah 2022:M2 - 2022:M12 (World Trade Organization, 2023a). Selama perang, rata-rata ekspor CEECs EU-8 mempunyai nilai ekspor yang lebih tinggi dibandingkan dari periode tidak mengalami perang pada gambar 4.1, hal ini dikarenakan harga komoditas naik saat periode perang. Oleh karena itu, diasumsikan variabel *dummy* geopolitik mempunyai efek positif terhadap ekspor manufaktur CEECs EU-8 yang digunakan.



Gambar 4. 1

Tren Ekspor Komoditas Manufaktur CEECs EU-8

Keterangan: Y-axis sebelah kiri adalah nilai ekspor ($ juta), Y-axis sebelah kanan adalah pertumbuhan ekspor (%). Rata-rata ekspor adalah rata-rata ekspor sebelum dan sesudah geopolitik. Nilai ekspor adalah akumulasi ekspor manufaktur yang digunakan.

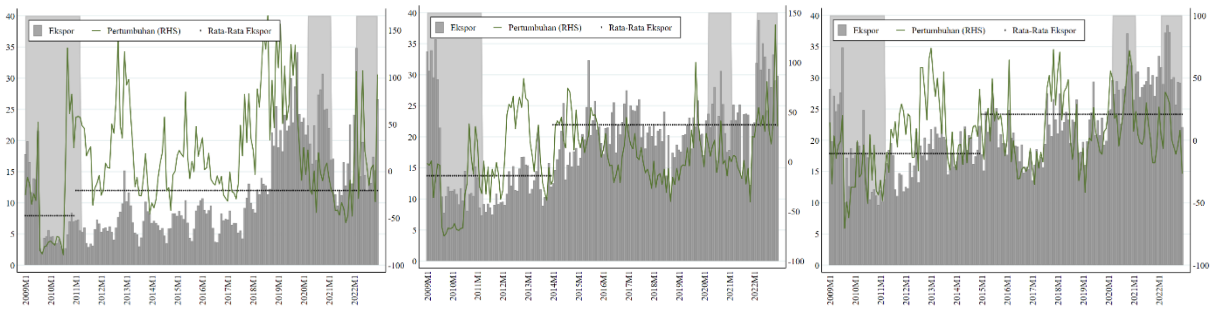
Sumber: Trade Map (2024), diolah.

Akumulatif ekspor komoditas manufaktur negara Estonia, Latvia, dan Lithuania yang dipilih nilainya lebih tinggi saat mengadopsi euro dibanding periode tidak mengadopsi euro pada gambar 4.2. Hal ini dikonfirmasi oleh Drivas, Kalyvitis, & Katsimi (2023) yang menemukan adopsi euro di Yunani meningkatkan harga ekspor sebesar 8.9% jika perusahaan tersebut mempunyai produktivitas yang tinggi. Sementara itu, Rose (2000) menemukan bahwa negara yang mengadopsi currency union meningkatkan perdagangan terutama intra-trade dalam *Economic and Monetary Union* (EMU) Uni Eropa. Oleh karena itu, diasumsikan variabel dummy euroization berpengaruh positif terhadap ekspor manufaktur CEECs EU-8 yang digunakan.

4.1.2 Tren IPI Negara Mitra CEECs EU-8

Jika dilihat dari tren rata-rata IPI negara mitra CEECs EU-8 yang digunakan, IPI mengalami pertumbuhan positif dari tahun 2010. Tren IPI mempunyai tren yang mirip dengan ekspor manufaktur. IPI mempunyai kecenderungan pertumbuhan negatif saat krisis keuangan global dan COVID-19, akan tetapi IPI mempunyai kecenderungan pertumbuhan positif saat awal perang Rusia dan Ukraina pada gambar 4.3. Oleh karena itu, diasumsikan variabel IPI berpengaruh positif terhadap ekspor manufaktur CEECs EU-8 yang digunakan. Hal ini sesuai dengan teori yang menyatakan peningkatan ekonomi di mitra dagang meningkatkan permintaan ekspor domestik (Blanchard & Johnson, 2017, p. 371).

a. Estonia b. Latvia c. Lithuania



Gambar 4. 2

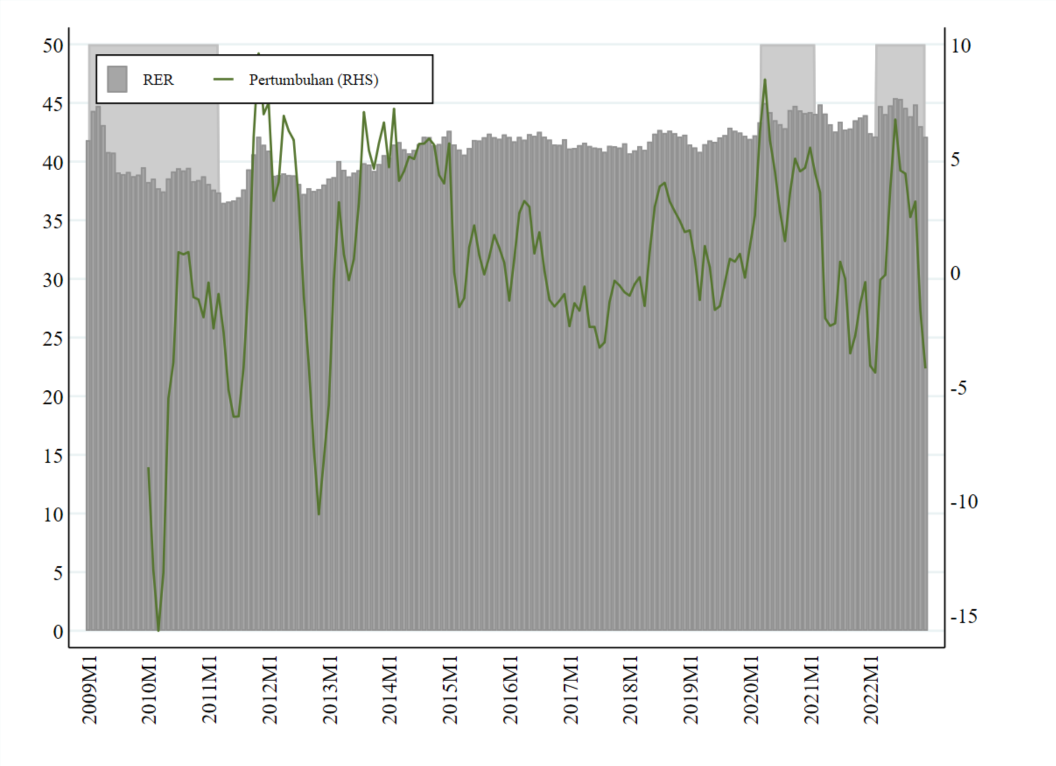
Tren Ekspor Estonia, Latvia, dan Lithunia dalam Sebelum dan Sesudah Euroization

Keterangan: Y-axis sebelah kiri adalah nilai ekspor ($ juta), Y-axis sebelah kanan adalah pertumbuhan ekspor (%). Rata-rata ekspor adalah rata-rata ekspor sebelum dan sesudah euroization. Nilai ekspor adalah akumulasi ekspor manufaktur yang digunakan.

Sumber: Trade Map (2024), diolah.

4.1.3 Tren RER CEECs EU-8

Jika dilihat dari tren rata-rata RER negara mitra CEECs EU-8 yang digunakan, RER mengalami pertumbuhan positif dari tahun 2012. Tren RER mempunyai tren yang mirip dengan ekspor manufaktur. RER mempunyai kecenderungan pertumbuhan negatif saat krisis keuangan global dan RER mempunyai kecenderungan pertumbuhan positif saat awal perang Rusia dan Ukraina dan COVID-19 pada gambar 4.4. Oleh karena itu, diasumsikan variabel RER mempunyai efek positif terhadap ekspor manufaktur CEECs EU-8 yang digunakan. Hal ini sesuai dengan teori yang menyatakan peningkatan RER atau depresiasi mata uang meningkatkan permintaan ekspor domestik (Blanchard & Johnson, 2017, p. 371).



Gambar 4. 3

Tren Indeks Produksi Industri Negara Mitra CEECs EU-8

Keterangan: Y-axis sebelah kiri adalah nilai indeks, Y-axis sebelah kanan adalah pertumbuhan IPI (%). Nilai IPI adalah rata-rata IPI yang digunakan.

Sumber: Trade Map (2024), diolah.

4.2 Deskripsi Statistik

Deskripsi statistik variabel penelitian meliputi nilai *mean*, median, standar deviasi, jumlah data, minimum dan maksimum, standard deviasi. Tabel 4.1 menunjukkan deskriptif statistik dari delapan negara CEE. Satuan nilai ekspor adalah ribu dollar, nilai variabel *dummy* geopolitik dan euroization antara 0 dan 1, sementara nilai minimum volpos adalah 0 dan nilai maksimum volneg adalah 0. Semua median volatilitas nilai tukar positif bernilai 0, sementara median volatilitas nilai tukar negatif bernilai negatif, artinya volatilitas nilai tukar negatif atau penurunan volatilitas nilai tukar mendominasi kecuali di Slovenia.

4.3 Deskripsi Hasil Penelitian

4.3.1 Metode ARCH

4.3.1.1 Uji ARCH *Effects*

Sebelum melakukan metode ARCH, dibutuhkan uji ARCH *effects* untuk mengetahui adanya ARCH *effect* (kluster volatilitas) dalam model. Langkah pertama adalah menentukan level stasioner dari variabel nilai tukar. Hasil uji stasioner menunjukan bahwa Sebagian besar variabel nilai tukar CEECs EU-8 berada di *first difference*, kecuali negara Polandia dan Slovenia yang berada di level atau I(0). Jika variabel nilai tukar mempunyai tingkat stasioner di *first difference*, langkah yang harus dilakukan adalah melakukan regresi OLS dengan model konstan atau hanya menggunakan *first difference* variabel RER.

Setelah itu, dilakukan uji ARCH *effects* dalam satu lag dan dua lag untuk menentukan jumlah ARCH. Jika *p-value* lag satu di bawah 5%, maka negara tersebut menggunakan ARCH(1), sementara jika di atas 5% maka menggunakan ARCH(2). Negara yang menggunakan ARCH(1) adalah Republik Ceko, Polandia, Slovenia, Estonia, sedangkan Hungaria, Slovakia, Latvia, Lithuania menggunakan ARCH(2) yang ditunjukan pada tabel 4.2.

4.3.1.2 Estimasi Metode ARCH

Syarat model ARCH sudah tepat adalah variabel ARCH dan GARCH signifikan. Namun, jika salah satu variabel tidak signifikan, maka uji kolektif akan dilakukan untuk memastikan apakah kedua variabel, yakni ARCH dan GARCH signifikan. Pada tabel 4.3, *p-value* uji kolektif dari semua variabel ditemukan signifkan. Oleh karena itu, variabel nilai tukar riil dapat ditangkap volatilitasnya.

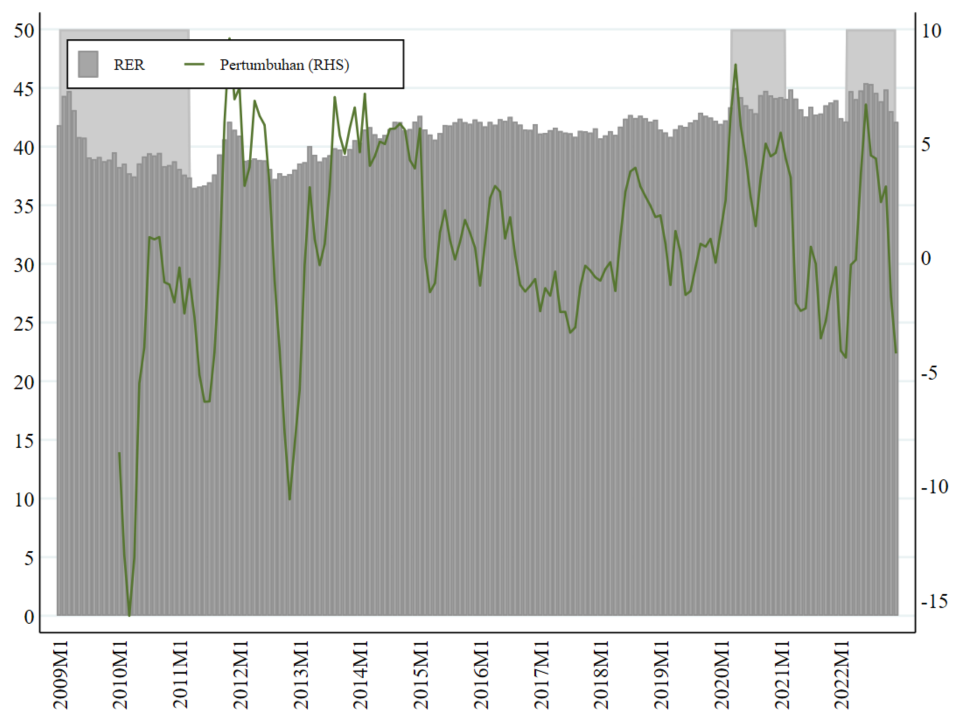
4.3.2 Metode ARDL dan NARDL

4.3.2.1 Uji Stasioner Seluruh Variabel

Sebelum melakukan metode ARDL dan NARDL dibutuhkan uji stasioner seluruh variabel. Metode ARDL dan NARDL dapat dilakukan jika variabel stasioner dalam level atau *first difference*. Hasil uji stasioner seluruh variabel pada tabel 4.4 menunjukan bahwa seluruh variabel stasioner dalam level atau *first difference*. Oleh karena itu, dapat dilanjutkan ke tahap selanjutnya, estimasi metode ARDL dan NARDL.

4.3.2.2 Estimasi Metode ARDL *Short Run*

Tabel 4.5 menunjukan hubungan jangka pendek volatilitas nilai tukar terhadap ekspor komoditas manufaktur CEECs EU-8. Volatilitas nilai tukar berpengaruh negatif signifikan di negara Republik Ceko pada komoditas 8703 dan 8708, volatilitas nilai tukar juga berpengaruh negatif signifikan di negara Hungaria pada komoditas 8507. Sebaliknya, volatilitas nilai tukar berpengaruh negatif signifikan di negara Polandia pada komoditas 8507 dan 8703, volatilitas tukar juga berpengaruh negatif signifikan di negara Estonia pada komoditas 8703 dan di negara Lithuania pada komoditas 9403.



Gambar 4. 4

Tren *Real Exchange Rate* CEECs EU-8

Sumber: IMF (2024), diolah.

Tabel 4. 1

Deskripsi Statistik Variabel

Variabel	Negara Asal	N	Mean	Median	Std. Dev	Min	Max
Exp 8703	Republik Ceko	168	365891.9	361771	125135.6	97347	711868
Exp 8517	Republik Ceko	168	178896.3	122310	152360	15992	777720
Exp 8708	Republik Ceko	168	445580	445602	105547.4	108499	671135
Vol	Republik Ceko	168	0.138354	0.113872	0.069605	0.098326	0.512958
Volpos	Republik Ceko	168	0.071567	0	0.213327	0	1.524439
Volneg	Republik Ceko	168	-0.07187	-0.02213	0.110084	-0.54449	0
RER	Republik Ceko	168	25.34069	25.29671	1.572011	20.81238	28.49134
IPI	Republik Ceko	168	105.8947	106.1575	9.051847	77.14811	126.2328
Geopolitik	Republik Ceko	168	0.065476	0	0.248104	0	1
Exp 8703	Hungaria	168	238417.9	239422.5	109082.7	59128	485147
Exp 8708	Hungaria	168	184511.3	192261	51367.5	56596	283376
Exp 8507	Hungaria	168	35909.79	6787	66616.19	337	358384
Vol	Hungaria	168	30.74073	22.80523	21.77101	17.51446	158.2843
Volpos	Hungaria	168	0.154208	0	0.30613	0	1.700424
Volneg	Hungaria	168	-0.1527	-0.04602	0.230267	-1.13217	0
RER	Hungaria	168	295.8572	297.0855	16.95381	259.2631	333.9146
IPI	Hungaria	168	105.8947	106.1575	9.051847	77.14811	126.2328
Geopolitik	Hungaria	168	0.065476	0	0.248104	0	1
Exp 8708	Polandia	168	337154.1	338929	82078.05	87058	516265
Exp 8507	Polandia	168	53540.19	10891.5	118221.3	2875	669619
Exp 8703	Polandia	168	153735	150370.5	41508.27	49696	340659
Vol	Polandia	168	0.026633	0.014854	0.039141	0.003566	0.357417
Volpos	Polandia	168	0.277409	0	0.425008	0	2.529333
Volneg	Polandia	168	-0.27806	-0.00115	0.446276	-2.05186	0
RER	Polandia	168	4.181727	4.192511	0.157206	3.824601	4.782053
IPI	Polandia	168	105.8947	106.1575	9.051847	77.14811	126.2328
Geopolitik	Polandia	168	0.065476	0	0.248104	0	1
Exp 8703	Slovakia	168	123168.5	124547	63331.9	5351	276375
Exp 8708	Slovakia	168	11132.81	10591	5399.586	895	22804
Exp 8528	Slovakia	168	66493.55	58141	32587.86	15360	197473
Vol	Slovakia	168	0.000583	0.000406	0.000442	0.000211	0.002598
Volpos	Slovakia	168	0.143655	0	0.30689	0	1.753416
Volneg	Slovakia	168	-0.14262	-0.09052	0.164566	-0.58326	0
RER	Slovakia	168	1.199972	1.185617	0.076482	1.055067	1.452569
IPI	Slovakia	168	105.8947	106.1575	9.051847	77.14811	126.2328
Geopolitik	Slovakia	168	0.065476	0	0.248104	0	1
Exp 3004	Slovenia	168	15221.63	15508	6202.103	2247	35943
Exp 8703	Slovenia	168	81926.73	72870.5	44906.13	26969	410613
Exp 8516	Slovenia	168	15615.18	15158.5	4352.522	6739	30859

Variabel	Negara Asal	N	Mean	Median	Std. Dev	Min	Max
Vol	Slovenia	168	0.000152	9.95E-05	0.000132	8.86E-06	0.000991
Volpos	Slovenia	168	0.247696	0.002907	0.416394	0	2.745556
Volneg	Slovenia	168	-0.25247	0	0.519566	-3.64731	0
RER	Slovenia	168	1.010291	1.010176	0.012328	0.985368	1.048231
IPI	Slovenia	168	105.8947	106.1575	9.051847	77.14811	126.2328
Geopolitik	Slovenia	168	0.065476	0	0.248104	0	1
Exp 8517	Estonia	168	4853.72	1786	5963.563	10	25672
Exp 9406	Estonia	168	4880.292	4714	2331.011	1074	13899
Exp 8703	Estonia	168	1670.667	1523.5	922.5367	146	4425
Vol	Estonia	168	3.29E-05	2.29E-05	3.37E-05	1.18E-05	0.000195
Volpos	Estonia	168	0.093902	0	0.183523	0	0.893499
Volneg	Estonia	168	-0.08653	-0.06637	0.087521	-0.28997	0
RER	Estonia	168	0.945409	0.948483	0.045503	0.780682	1.026997
IPI	Estonia	168	105.8947	106.1575	9.051847	77.14811	126.2328
Geopolitik	Estonia	168	0.065476	0	0.248104	0	1
Euroization	Estonia	168	0.857143	1	0.350973	0	1
Exp 4407	Latvia	168	1864.315	1819.5	895.8971	498	6242
Exp 8517	Latvia	168	10435.14	10869	5727.367	728	25033
Exp 3004	Latvia	168	6686.524	6764	2098.546	2823	16194
Vol	Latvia	168	2.16E-05	1.92E-05	7.85E-06	1.66E-05	8.08E-05
Volpos	Latvia	168	0.055277	0	0.140105	0	1.154472
Volneg	Latvia	168	-0.05302	-0.02214	0.079199	-0.47557	0
RER	Latvia	168	1.013035	1.012327	0.019457	0.963883	1.07881
IPI	Latvia	168	105.8947	106.1575	9.051847	77.14811	126.2328
Geopolitik	Latvia	168	0.065476	0	0.248104	0	1
Euroization	Latvia	168	0.642857	1	0.48059	0	1
Exp 9403	Lithuania	168	12580.71	12011	4144.236	5675	24614
Exp 8703	Lithuania	168	3821.268	2751	3237.819	81	16846
Exp 3102	Lithuania	168	5024.482	4362.5	3019.321	382	15183
Vol	Lithuania	168	1.38E-05	1.03E-05	8.51E-06	6.07E-06	4.29E-05
Volpos	Lithuania	168	0.034244	0	0.098921	0	0.953172
Volneg	Lithuania	168	-0.03638	-0.04092	0.031872	-0.10112	0
RER	Lithuania	168	0.23068	0.229458	0.013556	0.193756	0.259038
IPI	Lithuania	168	105.8947	106.1575	9.051847	77.14811	126.2328
Geopolitik	Lithuania	168	0.065476	0	0.248104	0	1
Euroization	Lithuania	168	0.571429	1	0.496351	0	1

Sumber: Diolah.

Keterangan: IPI = indeks produksi industry; RER = *real exchange rate*; Vol = Volatilitas nilai tukar; Volpos = volatilitas nilai tukar positif; Volneg = volatilitas nilai tukar negatif; Exp = komoditas ekspor HS 4 digit. Daftar komoditas ekspor HS 4 dapat dilihat di tabel 3.1.

4.3.2.3 Estimasi Metode ARDL *Long Run*

Tabel 4.6 menunjukan hubungan jangka panjang variabel volatilitas nilai tukar, IPI, RER, geopolitik, euroization terhadap ekspor komoditas manufaktur CEECs EU-8. Koefisien *error correction* seluruh komoditas menunjukan hasil negatif signifikan, hal ini menunjukan bahwa *error correction* sudah valid. Nilai koefisien *error correction* menjelaskan kecepatan penyesuaian menuju ekuilibrium jangka panjangnya. Nilai koefisien *error correction* terbesar pada negara Hungaria pada komoditas 8507, yaitu -0.112 yang berarti kecepatan penyesuaian ke ekuilibrium jangka panjang selama 8 bulan. Sebaliknya, nilai koefisien *error correction* terkecil pada negara Estonia pada komoditas 9406, yaitu -0.557 yang berarti kecepatan penyesuaian ke ekuilibirum jangka panjang selama 1 bulan.

Tabel 4. 2

Uji ARCH *Effects*

Negara	**ADF Level**		**ADF *First Difference***		**I(#)**	***p-value* ARCH**		**ARCH(#)**
	Test	***p-value***	**Test**	***p-value***		**Lag(1)**	**Lag(2)**	
Republik Ceko	-0.4709	0.8976	-13.3790	0.0000	I(1)	0.0105	0.6453	ARCH(1)
Hungaria	-2.0634	0.2595	-11.9922	0.0000	I(1)	0.3037	0.0059	ARCH(2)
Polandia	-3.1676	0.0219	-13.4255	0.0000	I(0)	0.0000	0.0000	ARCH(1)
Slovakia	-2.2903	0.1751	-13.5769	0.0000	I(1)	0.1072	0.0220	ARCH(2)
Slovenia	-3.9812	0.0015	-12.0896	0.0000	I(0)	0.0000	0.0000	ARCH(1)
Estonia	1.9136	0.9985	-11.8677	0.0000	I(1)	0.0028	0.0000	ARCH(1)
Latvia	-1.2457	0.6536	-12.7893	0.0000	I(1)	0.6102	0.0120	ARCH(2)
Lithuania	-1.2357	0.6581	-12.5709	0.0000	I(1)	0.3403	0.0017	ARCH(2)

Keterangan: *** = signifikan 1%; ** = signifikan 5%; * = signifikan 10%. -3.488 = 1% *critical value*; -2.886 = 5% *critical value*; -2.576 = 10% *critical value*.

Sumber: Diolah.

Variabel volatilitas nilai tukar berpengaruh positif terhadap ekspor di negara Republik Ceko pada komoditas 8703 dan 8708. Selain itu, pengaruh positif juga ditemukan di negara Estonia pada komoditas 9406 dan di negara Latvia pada komoditas 4407. Sebaliknya, volatilitas nilai tukar berpengaruh negatif terhadap ekspor di Estonia pada komoditas 8703 dan di negara Lithuania pada komoditas 9403.

Variabel indeks produksi industri negara mitra berpengaruh positif di negara Republik Ceko pada komoditas 8703, 8517, dan 8708. Selain itu, pengaruh positif juga ditemukan di negara Slovakia pada komoditas 8708 dan di Slovenia pada komoditas 3004. Sebaliknya, indeks produksi industri berpengaruh negatif terhadap ekspor di Slovakia pada komoditas 8528 dan di negara Slovenia pada komoditas 8516, serta di negara Lithuania pada komoditas 9403.

Variabel nilai tukar riil berpengaruh positif di Hungaria pada komoditas 8703 dan 8517, serta di Polandia pada komoditas 8507. Selain itu, pengaruh positif juga ditemukan di negara Slovenia pada komoditas 3004 dan 8703. Sebaliknya, nilai tukar riil berpengaruh negatif terhadap ekspor di Republik Ceko pada komoditas 8517. Sementara itu, pengaruh negatif juga ditemukan di negara Estonia pada komoditas 8517 dan 8703, serta di Lithuania pada komoditas 9403.

Variabel geopolitik berpengaruh positif di negara Republik Ceko pada komoditas 8708 dan di Polandia pada komoditas 8507. Selain itu, pengaruh positif juga ditemukan di negara Slovenia pada komoditas 8703 dan 8516. geopolitik berpengaruh negatif terhadap ekspor di Slovakia pada komoditas 8528 dan di Estonia pada komoditas 8517.

Tabel 4. 3

Estimasi Model ARCH

Negara	**RER**	**ARCH**				***p-value* Uji Kolektif**	**N**
	cons	**ARCH(1)**	**ARCH(2)**	**GARCH(1)**	**cons**		
Republik Ceko	-0.021	0.152	-	0.465	0.052*	0.0000	167
Hungaria	0.197	-	0.353**	0.236	13.186***	0.0000	167
Polandia	4.176***	0.965***	-	-0.072	0.005**	0.0000	168
Slovakia	0.002	-	0.364**	0.498***	0.000**	0.0000	167
Slovenia	1.012***	0.701***	-	-0.246*	0.000***	0.0003	168
Estonia	-0.001**	0.214***	-	0.702***	0.000	0.0000	167
Latvia	0.001	-	0.116	0.523	0.000	0.0001	167
Lithuania	0.000	-	0.064*	0.888***	0.000*	0.0000	167

Keterangan: *** = signifikan 1%; ** = signifikan 5%; * = signifikan 10%. Cons = konstanta.

Sumber: Diolah.

Tabel 4. 4

Uji Stasioner Seluruh Variabel

Variabel	**Negara**	**Level**		**First Difference**	
		Test	***p-value***	**Test**	***p-value***
lnExp 8703	Republik Ceko	-6.4795	0.0000	-17.1515	0.0000
lnExp 8517	Republik Ceko	-4.2834	0.0005	-17.8000	0.0000
lnExp 8708	Republik Ceko	-6.3853	0.0000	-17.6239	0.0000
lnVol	Republik Ceko	-5.5307	0.0000	-14.7927	0.0000
lnVolpos	Republik Ceko	-13.6299	0.0000	-23.0780	0.0000
lnVolneg	Republik Ceko	-9.9049	0.0000	-19.7252	0.0000
lnRER	Republik Ceko	-0.2635	0.9305	-13.1494	0.0000
lnIPI	Republik Ceko	-8.2138	0.0000	-19.1738	0.0000
Geopolitik	Republik Ceko	-0.2517	0.9321	-12.8841	0.0000
lnExp 8703	Hungaria	-4.4420	0.0002	-17.8774	0.0000
lnExp 8708	Hungaria	-4.8241	0.0000	-18.4923	0.0000
lnExp 8507	Hungaria	-2.1272	0.2337	-13.5507	0.0000
lnVol	Hungaria	-7.3958	0.0000	-19.0711	0.0000
lnVolpos	Hungaria	-14.8578	0.0000	-24.7471	0.0000
lnVolneg	Hungaria	-13.4593	0.0000	-24.9733	0.0000
lnRER	Hungaria	-2.0330	0.2723	-11.8702	0.0000
lnIPI	Hungaria	-8.2138	0.0000	-19.1738	0.0000
Geopolitik	Hungaria	-0.2517	0.9321	-12.8841	0.0000
lnExp 8708	Polandia	-6.0784	0.0000	-17.2316	0.0000
lnExp 8507	Polandia	-2.1700	0.2173	-14.1688	0.0000
lnExp 8703	Polandia	-7.0052	0.0000	-16.9251	0.0000
lnVol	Polandia	-5.6684	0.0000	-13.9220	0.0000
lnVolpos	Polandia	-13.9568	0.0000	-21.3771	0.0000
lnVolneg	Polandia	-13.6046	0.0000	-20.4724	0.0000
lnRER	Polandia	-3.1125	0.0256	-13.1150	0.0000
lnIPI	Polandia	-8.2138	0.0000	-19.1738	0.0000
Geopolitik	Polandia	-0.2517	0.9321	-12.8841	0.0000
lnExp 8703	Slovakia	-4.2469	0.0005	-15.7236	0.0000
lnExp 8708	Slovakia	-3.9340	0.0018	-13.5100	0.0000
lnExp 8528	Slovakia	-6.2647	0.0000	-13.5872	0.0000
lnVol	Slovakia	-4.4624	0.0002	-15.6140	0.0000
lnVolpos	Slovakia	-14.5458	0.0000	-23.5408	0.0000
lnVolneg	Slovakia	-12.0144	0.0000	-22.6169	0.0000
lnRER	Slovakia	-2.4146	0.1376	-13.8672	0.0000
lnIPI	Slovakia	-8.2138	0.0000	-19.1738	0.0000
Geopolitik	Slovakia	-0.2517	0.9321	-12.8841	0.0000
lnExp 3004	Slovenia	-6.5988	0.0000	-23.1445	0.0000
lnExp 8703	Slovenia	-6.3830	0.0000	-17.5819	0.0000
lnExp 8516	Slovenia	-5.9426	0.0000	-14.8428	0.0000

Variabel	Negara	Level		First Difference	
		Test	***p-value***	**Test**	***p-value***
lnVolpos	Slovenia	-13.6396	0.0000	-22.3471	0.0000
lnVolneg	Slovenia	-13.8363	0.0000	-22.1183	0.0000
lnRER	Slovenia	-3.9755	0.0015	-12.0419	0.0000
lnIPI	Slovenia	-8.2138	0.0000	-19.1738	0.0000
Geopolitik	Slovenia	-0.2517	0.9321	-12.8841	0.0000
lnExp 8517	Estonia	-4.4022	0.0003	-18.8921	0.0000
lnExp 9406	Estonia	-3.8209	0.0027	-11.4847	0.0000
lnExp 8703	Estonia	-6.1660	0.0000	-16.9520	0.0000
lnVol	Estonia	-2.0711	0.2563	-13.2343	0.0000
lnVolpos	Estonia	-12.8757	0.0000	-21.7827	0.0000
lnVolneg	Estonia	-12.2818	0.0000	-21.8629	0.0000
lnRER	Estonia	2.1617	0.9988	-11.4511	0.0000
lnIPI	Estonia	-8.2138	0.0000	-19.1738	0.0000
Geopolitik	Estonia	-0.2517	0.9321	-12.8841	0.0000
Euroization	Estonia	-2.4785	0.1208	-12.8841	0.0000
lnExp 4407	Latvia	-4.3343	0.0004	-17.4229	0.0000
lnExp 8517	Latvia	-3.6929	0.0042	-19.5286	0.0000
lnExp 3004	Latvia	-5.8601	0.0000	-18.9466	0.0000
lnVol	Latvia	-4.7467	0.0001	-15.7537	0.0000
lnVolpos	Latvia	-12.6550	0.0000	-20.7113	0.0000
lnVolneg	Latvia	-12.4435	0.0000	-27.8388	0.0000
lnRER	Latvia	-1.2889	0.6342	-12.8958	0.0000
lnIPI	Latvia	-8.2138	0.0000	-19.1738	0.0000
Geopolitik	Latvia	-0.2517	0.9321	-12.8841	0.0000
Euroization	Latvia	-1.3386	0.6113	-12.8841	0.0000
lnExp 9403	Lithuania	-3.9618	0.0016	-16.6798	0.0000
lnExp 8703	Lithuania	-3.9171	0.0019	-17.4097	0.0000
lnExp 3102	Lithuania	-8.2894	0.0000	-19.3594	0.0000
lnVol	Lithuania	-1.5248	0.5212	-13.2349	0.0000
lnVolpos	Lithuania	-12.9995	0.0000	-22.9777	0.0000
lnVolneg	Lithuania	-11.8464	0.0000	-22.6529	0.0000
lnRER	Lithuania	-1.1481	0.6955	-12.8354	0.0000
lnIPI	Lithuania	-8.2138	0.0000	-19.1738	0.0000
Geopolitik	Lithuania	-0.2517	0.9321	-12.8841	0.0000
Euroization	Lithuania	-1.1498	0.6948	-12.8841	0.0000

Keterangan: *** = signifikan 1%; ** = signifikan 5%; * = signifikan 10%. -3.488 = 1% *critical value*; -2.886 = 5% *critical value*; -2.576 = 10% *critical value*.

Sumber: Diolah.

Variabel euroization berpengaruh positif di negara Estonia pada komoditas 9406. Selain itu, hubungan positif juga ditemukan di negara Latvia pada komoditas

4407 dan 8517. Sebaliknya, euroization berpengaruh negatif terhadap ekspor di Estonia pada komoditas 8703.

Tabel 4. 5

Estimasi Metode ARDL *Short Run* Volatilitas Nilai Tukar

Negara	Variabel	D.lnVol	LD.lnVol	L3D.lnVol	L2D.lnVol
Republik Ceko	Exp 8703	-0.471***	-	-	-
Republik Ceko	Exp 8517	-	-	-	-
Republik Ceko	Exp 8708	-0.169**	-	-	-
Hungaria	Exp 8703	-	-	-	-
Hungaria	Exp 8708	-	-	-	-
Hungaria	Exp 8507	0.094	0.183	-0.182**	-0.105
Polandia	Exp 8708	-	-	-	-
Polandia	Exp 8507	-0.052	0.122**	-	-
Polandia	Exp 8703	0.050*	-	-	-
Slovakia	Exp 8703	-	-	-	-
Slovakia	Exp 8708	-	-	-	-
Slovakia	Exp 8528	-	-	-	-
Slovenia	Exp 3004	-	-	-	-
Slovenia	Exp 8703	-	-	-	-
Slovenia	Exp 8516	-	-	-	-
Estonia	Exp 8517	-	-	-	-
Estonia	Exp 9406	-	-	-	-
Estonia	Exp 8703	-0.118	0.467***	-	0.276
Latvia	Exp 4407	-	-	-	-
Latvia	Exp 8517	-	-	-	-
Latvia	Exp 3004	-	-	-	-
Lithuania	Exp 9403	0.270*	0.176	-	-
Lithuania	Exp 8703	-	-	-	-
Lithuania	Exp 3102	-	-	-	-

Keterangan: *** = signifikan 1%; ** = signifikan 5%; * = signifikan 10%. D = *first difference*; L = lag; - = tidak dilakukan estimasi.

Sumber: Diolah.

Hasil Uji Diagnostik ARDL

Tabel 4.7 menunjukan hasil uji diagnostik model ARDL berupa uji kointegrasi, uji RESET, dan uji CUSUM. Uji kointegrasi digunakan untuk mengetahui hubungan jangka panjang antara variabel independen dan dependen dalam sebuah model. Model dikatakan terkointegrasi apabila nilai F statistik berada di atas *upper bound* atau I(1), hasil ini akan menolak hipotesis dari model tidak terkointegrasi (REJ). Sebaliknya, jika nilai F ada di bawah *lower bound* atau I(0) maka model tidak terkointegrasi (ACC). Sementara itu, jika nilai F berada di antara *lower bound* I(0) dan *upper bound* atau I(1) maka model tidak konsisten (INC). Komoditas yang terkointegrasi dalam jangka panjang setidaknya pada tingkat signifikansi 10% ditemukan di negara Republik Ceko (8703), Polandia (8507), Slovakia (8528), Slovenia (3004 8703 8516), Estonia (9406 8703 4407), Latvia (4407 8517), Lithuania (9403 3102).

Tabel 4. 6

Estimasi Metode ARDL *Long Run*

Negara	Variabel	L.lnExp	lnVol	lnIPI	lnRER	Geopolitik	Euroization
Republik Ceko	Exp 8703	-0.301***	0.732**	5.395***	-0.761	0.239	-
Republik Ceko	Exp 8517	-0.160***	0.936	14.334***	-10.608**	-1.637	-
Republik Ceko	Exp 8708	-0.197***	0.787*	4.805**	3.249	1.060*	-
Hungaria	Exp 8703	-0.201***	-0.106	1.51	6.135***	0.205	-
Hungaria	Exp 8708	-0.113*	0.613	8.426	0.978	1.257	-
Hungaria	Exp 8507	-0.112***	0.628	4.05	19.089***	0.836	-
Polandia	Exp 8708	-0.210**	0.002	3.303	2.777	0.567	-
Polandia	Exp 8507	-0.208***	0.177	1.198	16.363***	3.871***	-
Polandia	Exp 8703	-0.328***	-0.001	0.385	-1.079	0.303	-
Slovakia	Exp 8703	-0.117**	0.261	5.216	2.286	1.775	-
Slovakia	Exp 8708	-0.109**	0.194	12.386*	-5.44	1.178	-
Slovakia	Exp 8528	-0.464***	0.043	-1.924*	0.424	-0.424*	-
Slovenia	Exp 3004	-0.353***	0.062	3.544**	15.143**	0.259	-
Slovenia	Exp 8703	-0.313***	-0.038	1.113	15.244**	0.528*	-
Slovenia	Exp 8516	-0.489***	0.056	-1.074**	-3.579	0.336***	-
Estonia	Exp 8517	-0.274***	0.038	1.19	-49.977***	-4.972**	-1.635
Estonia	Exp 9406	-0.557***	0.422***	-0.181	-2.309	-0.313	0.444**
Estonia	Exp 8703	-0.476***	-0.720**	0.554	-14.169***	-0.295	-0.863*
Latvia	Exp 4407	-0.281***	1.074**	1.498	8.728	-0.499	0.561***
Latvia	Exp 8517	-0.279***	0.102	-0.377	15.23	-0.325	1.066***
Latvia	Exp 3004	-0.212***	-0.193	0.195	-4.569	0.485	0.116
Lithuania	Exp 9403	-0.436***	-0.362***	-1.452***	-2.272**	0.143	-0.019
Lithuania	Exp 8703	-0.132**	-0.264	-5.825	2.891	-0.928	1.491
Lithuania	Exp 3102	-0.600***	-0.046	1.046	3.696	0.364	-0.031

Keterangan: *** = signifikan 1%; ** = signifikan 5%; * = signifikan 10%. L = lag; - = tidak dilakukan estimasi.

Sumber: Diolah.

Tabel 4. 7

Uji Diagnostik ARDL

Negara	**Variable**	**Uji Kointegrasi**			**Uji RESET**	**Uji CUSUM**		**R^2**	**N**
		10%	**5%**	**1%**		**OLS**	***Recursive***		
Republik Ceko	Exp 8703	REJ	INC	INC	0.153	ACC	ACC	0.478	164
Republik Ceko	Exp 8517	INC	INC	ACC	0.536	ACC	ACC	0.368	164
Republik Ceko	Exp 8708	INC	INC	ACC	0.126	ACC	ACC	0.523	164
Hungaria	Exp 8703	ACC	ACC	ACC	0.614	ACC	ACC	0.381	164
Hungaria	Exp 8708	ACC	ACC	ACC	0.018	ACC	ACC	0.502	164
Hungaria	Exp 8507	INC	INC	ACC	0.000	ACC	ACC	0.182	164
Polandia	Exp 8708	ACC	ACC	ACC	0.029	ACC	ACC	0.491	164
Polandia	Exp 8507	REJ	REJ	INC	0.000	ACC	ACC	0.208	164
Polandia	Exp 8703	INC	ACC	ACC	0.076	ACC	ACC	0.392	164
Slovakia	Exp 8703	ACC	ACC	ACC	0.574	ACC	ACC	0.289	164
Slovakia	Exp 8708	ACC	ACC	ACC	0.121	ACC	ACC	0.329	164
Slovakia	Exp 8528	REJ	REJ	REJ	0.386	ACC	ACC	0.271	164
Slovenia	Exp 3004	REJ	REJ	REJ	0.035	ACC	ACC	0.443	164
Slovenia	Exp 8703	REJ	INC	INC	0.011	ACC	ACC	0.396	164
Slovenia	Exp 8516	REJ	REJ	REJ	0.109	ACC	ACC	0.332	164
Estonia	Exp 8517	REJ	REJ	INC	0.004	ACC	ACC	0.388	164
Estonia	Exp 9406	REJ	REJ	REJ	0.007	ACC	ACC	0.365	164
Estonia	Exp 8703	REJ	REJ	REJ	0.056	ACC	ACC	0.296	164
Latvia	Exp 4407	REJ	REJ	INC	0.147	ACC	ACC	0.365	164
Latvia	Exp 8517	REJ	REJ	INC	0.338	ACC	ACC	0.337	164
Latvia	Exp 3004	ACC	ACC	ACC	0.181	ACC	ACC	0.328	164
Lithuania	Exp 9403	REJ	REJ	REJ	0.001	ACC	ACC	0.314	164
Lithuania	Exp 8703	ACC	ACC	ACC	0.85	ACC	ACC	0.373	164
Lithuania	Exp 3102	REJ	REJ	REJ	0.154	ACC	ACC	0.347	164

Keterangan: *** = signifikan 1%; ** = signifikan 5%; * = signifikan 10%. L = lag; - = tidak dilakukan estimasi. REJ = menolak hipotesis; INC = inkosisten; ACC = menerima hipotesis.

Sumber: Diolah.

Uji RESET dilakukan untuk identifikasi potensi masalah *omitted* variabel atau variabel independen tidak masuk dalam model. Hasil uji RESET menunjukan semua model tidak mempunyai potensi masalah *omitted* variabel pada tingkat signifikansi 5% kecuali sembilan komoditas di negara Hungaria (8708 8517), Polandia (8708 8507), Slovenia (3004 8703), Estonia (8517 9406), dan Lithuania (9403). Sementara itu, uji CUSUM menemukan tidak ada ketidakstabilan parameter di seluruh model dan residual diasumsikan terdistribusi normal dengan rata-rata nol dan varians konstan. Namun, R^2 di hampir seluruh model hanya memiliki nilai di bawah 50%, artinya variabel independen mampu menjelaskan variabel dependen sebesar kurang dari 50% di hampir seluruh model kecuali komoditas 8708 di Republik Ceko dan 8507 di negara Hungaria.

Tabel 4. 8

Estimasi Metode NARDL *Short Run* Volatilitas Nilai Tukar Positif

Negara	Variabel	D.Volpos	LD.Volpos	L2D.Volpos	L3D.Volpos
Republik Ceko	Exp 8703	-	-	-	-
Republik Ceko	Exp 8517	-	-	-	-
Republik Ceko	Exp 8708	-	-	-	-
Hungaria	Exp 8703	-	-	-	-
Hungaria	Exp 8708	-0.119*	-	-	-
Hungaria	Exp 8507	-	-	-	-
Polandia	Exp 8708	-	-	-	-
Polandia	Exp 8507	-	-	-	-
Polandia	Exp 8703	-	-	-	-
Slovakia	Exp 8703	-	-	-	-
Slovakia	Exp 8708	-	-	-	-
Slovakia	Exp 8528	-	-	-	-
Slovenia	Exp 3004	-	-	-	-
Slovenia	Exp 8703	0.1	-	-	-
Slovenia	Exp 8516	-	-	-	-
Estonia	Exp 8517	-	-	-	-
Estonia	Exp 9406	-	-	-	-
Estonia	Exp 8703	-0.475**	-	-	-
Latvia	Exp 4407	-0.182	0.181	0.138	0.298**
Latvia	Exp 8517	-	-	-	-
Latvia	Exp 3004	-	-	-	-
Lithuania	Exp 9403	0.27	0.341	0.378*	0.428***
Lithuania	Exp 8703	0.427	0.14	-0.828*	-0.564*
Lithuania	Exp 3102	2.380**	2.084**	2.147***	2.117***

Keterangan: *** = signifikan 1%; ** = signifikan 5%; * = signifikan 10%. D = *first difference*; L = lag; - = tidak dilakukan estimasi.

Sumber: Diolah.

4.3.2.4 Hasil Model NARDL Jangka Pendek

Tabel 4.8 menunjukan hubungan jangka pendek volatilitas nilai tukar positif terhadap ekspor komoditas manufaktur CEECs EU-8. Volatilitas nilai tukar positif berpengaruh positif signifikan di negara Hungaria pada komoditas 8507 dan di negara Estonia pada komoditas 9406. Sementara itu, tabel 4.9 menunjukan hubungan jangka pendek volatilitas nilai tukar negatif terhadap ekspor komoditas manufaktur CEECs EU-8. Volatilitas nilai tukar negatif berpengaruh negatif signifikan di Republik Ceko pada komoditas 8703. Sebaliknya, volatilitas nilai tukar negatif berpengaruh positif di Polandia pada komoditas 8507 dan di Estonia pada komoditas 8517, serta di Lithuania pada komoditas 9403.

Tabel 4. 9

Estimasi Metode NARDL *Short Run* Volatilitas Nilai Tukar Negatif

Negara	Variabel	D.Volneg	LD.Volneg	L2D.Volneg	L3D.Volneg
Republik Ceko	Exp 8703	-	-	-	-
Republik Ceko	Exp 8517	-	-	-	-
Republik Ceko	Exp 8708	-	-	-	-
Hungaria	Exp 8703	-	-	-	-
Hungaria	Exp 8708	-	-	-	-
Hungaria	Exp 8507	-0.409	0.031	-0.014	-0.264*
Polandia	Exp 8708	-0.05	-	-	-
Polandia	Exp 8507	-0.267***	-	-	-
Polandia	Exp 8703	-	-	-	-
Slovakia	Exp 8703	-	-	-	-
Slovakia	Exp 8708	-	-	-	-
Slovakia	Exp 8528	-	-	-	-
Slovenia	Exp 3004	-	-	-	-
Slovenia	Exp 8703	-	-	-	-
Slovenia	Exp 8516	-0.073**	-	-	-
Estonia	Exp 8517	-2.670*	-3.305***	-1.613**	-
Estonia	Exp 9406	-1.218***	-0.866***	-0.470**	-
Estonia	Exp 8703	-	-	-	-
Latvia	Exp 4407	-	-	-	-
Latvia	Exp 8517	-	-	-	-
Latvia	Exp 3004	-	-	-	-
Lithuania	Exp 9403	-	-	-	-
Lithuania	Exp 8703	-2.271	-1.866	1.815	-
Lithuania	Exp 3102	-	-	-	-

Keterangan: *** = signifikan 1%; ** = signifikan 5%; * = signifikan 10%. D = *first difference*; L = lag; - = tidak dilakukan estimasi.

Sumber: Diolah.

4.3.2.5 Hasil Model NARDL Jangka Panjang

Tabel 4.10 menunjukan hubungan jangka panjang variabel volatilitas nilai tukar positif, volatilitas nilai tukar negative, IPI, RER, geopolitik, euroization terhadap ekspor komoditas manufaktur CEECs EU-8. Koefisien *error correction* seluruh komoditas menunjukan hasil negatif signifikan, hal ini menunjukan bahwa *error correction* sudah valid. Nilai koefisien *error correction* menjelaskan kecepatan penyesuaian menuju ekuilibrium jangka panjangnya. Nilai koefisien *error correction* terkecil pada negara Hungaria pada komoditas 8507, yaitu -0.107 yang berarti kecepatan penyesuaian ke ekuilibirum jangka panjang selama 9 bulan. Sebaliknya, nilai koefisien *error correction* terbesar pada negara Lithuania pada komoditas 9406, yaitu -0.601 yang berarti kecepatan penyesuaian ke ekuilibirum jangka panjang selama 1 bulan.

Variabel volatilitas nilai tukar positif berpengaruh positif terhadap ekspor di Hungaria pada komoditas 8507 dan di Estonia pada komoditas 9406. Sementara itu, variabel volatilitas nilai tukar negatif berpengaruh positif tehadap ekspor di Polandia pada komoditas 8507 dan di Estonia pada komoditas 8517, serta di Lithuania pada komoditas 9403. Sebaliknya, variabel volatilitas nilai tukar negatif berpengaruh negatif tehadap ekspor di Republik Ceko pada komoditas 8703.

Variabel indeks produksi industri negara mitra berpengaruh positif di negara Republik Ceko pada komoditas 8703, 8517, dan 8708. Selain itu, pengaruh positif juga ditemukan di negara Slovakia pada komoditas 8708 dan 8528, serta di Slovenia pada komoditas 3004. Sebaliknya, indeks produksi industri berpengaruh negatif terhadap ekspor di Slovenia pada komoditas 8528 dan di negara Lithuania pada komoditas 9403.

Variabel nilai tukar riil berpengaruh positif di Hungaria pada komoditas 8703 dan 8517, serta di Polandia pada komoditas 8507. Selain itu, pengaruh positif juga ditemukan di negara Slovenia pada komoditas 3004 dan 8703, serta di Lithuania pada komoditas 3102. Sebaliknya, nilai tukar riil berpengaruh negatif terhadap ekspor di Republik Ceko pada komoditas 8517, serta di di Estonia pada komoditas 8517, 9406, dan 8703.

Variabel geopolitik berpengaruh positif di Polandia pada komoditas 8708 dan 8507. Selain itu, pengaruh positif juga ditemukan di negara Slovenia pada komoditas 8703 dan 8516. Sebaliknya, geopolitik berpengaruh negatif terhadap ekspor di Slovakia pada komoditas 8528 dan di Estonia pada komoditas 8517.

Variabel euroization berpengaruh positif di negara Lithuania pada komoditas 9403. Selain itu, hubungan positif juga ditemukan di negara Latvia pada komoditas 4407 dan 8517. Sebaliknya, euroization berpengaruh negatif terhadap ekspor di Estonia pada komoditas 8517.

Tabel 4. 10

Estimasi Metode NARDL *Long Run*

Negara	Variabel	L.lnExp	Volpos	Volneg	lnIPI	lnRER	Geopolitik	Euroization
Republik Ceko	Exp 8703	-0.277***	-1.019	-2.196**	4.476***	-0.984	0.369	-
Republik Ceko	Exp 8517	-0.135***	1.855	0.71	12.373***	-11.632**	-1.765	-
Republik Ceko	Exp 8708	-0.144**	0.313	-1.921	4.914*	3.475	1.126	-
Hungaria	Exp 8703	-0.196***	-0.193	-0.211	1.874	6.201***	0.166	-
Hungaria	Exp 8708	-0.114*	1.65	0.292	7.884	1.578	1.168	-
Hungaria	Exp 8507	-0.107***	3.442**	1.838	4.601	19.079***	0.856	-
Polandia	Exp 8708	-0.208**	-0.055	0.185	3.106	2.876	0.647*	-
Polandia	Exp 8507	-0.188***	-0.675	1.645**	0.811	15.890***	4.234***	-
Polandia	Exp 8703	-0.328***	0.147	0.156	0.381	-1.074	0.305	-
Slovakia	Exp 8703	-0.126***	1.06	-0.131	4.721	1.587	1.338	-
Slovakia	Exp 8708	-0.112**	0.451	-1.241	12.180*	-5.169	1.036	-
Slovakia	Exp 8528	-0.461***	0.068	0.064	-2.049*	0.446	-0.441*	-
Slovenia	Exp 3004	-0.349***	0.097	0.032	3.568**	14.554*	0.234	-
Slovenia	Exp 8703	-0.319***	-0.367	0.09	0.834	14.542**	0.755**	-
Slovenia	Exp 8516	-0.456***	0.172	0.054	-1.105**	-3.527	0.291*	-
Estonia	Exp 8517	-0.324***	-0.603	9.453*	-0.504	-51.781***	-5.482***	-2.166**
Estonia	Exp 9406	-0.502***	0.769***	1.599	-0.326	-5.039***	0.102	0.115
Estonia	Exp 8703	-0.427***	0.328	-0.775	0.953	-8.630**	-0.815	-0.278
Latvia	Exp 4407	-0.262***	1.41	2.505	1.302	2.833	0.607	0.659***
Latvia	Exp 8517	-0.272***	0.386	-2.912	0.325	16.15	-0.652	1.025***
Latvia	Exp 3004	-0.218***	-0.502	1.707	-0.029	-5.947	0.678	0.155
Lithuania	Exp 9403	-0.334***	-1.709	4.160**	-1.564**	-1.538	0.183	0.210*
Lithuania	Exp 8703	-0.126**	8.775	0.205	-4.876	-0.057	-1.775	1.434
Lithuania	Exp 3102	-0.601***	-3.765	-2.089	-0.471	5.337**	0.527	0.222

Keterangan: *** = signifikan 1%; ** = signifikan 5%; * = signifikan 10%. L = lag; - = tidak dilakukan estimasi.

Sumber: Diolah.

Tabel 4. 11

Uji Diagnostik NARDL

Negara	Variabel	Uji Kointegrasi			Uji RESET	Uji CUSUM		Uji *Wald*		R^2	N
		10%	5%	1%		OLS	*Recursive*	*Long Run*	*Short Run*		
Republik Ceko	Exp 8703	INC	INC	INC	0.112	ACC	ACC	0.319	-	0.463	164
Republik Ceko	Exp 8517	INC	INC	ACC	0.813	ACC	ACC	0.719	-	0.366	164
Republik Ceko	Exp 8708	ACC	ACC	ACC	0.787	ACC	ACC	0.216	-	0.517	164
Hungaria	Exp 8703	ACC	ACC	ACC	0.580	ACC	ACC	0.984	-	0.383	164
Hungaria	Exp 8708	ACC	ACC	ACC	0.030	ACC	ACC	0.300	0.060	0.524	164
Hungaria	Exp 8507	INC	INC	ACC	0.000	ACC	ACC	0.654	0.430	0.228	164
Polandia	Exp 8708	ACC	ACC	ACC	0.105	ACC	ACC	0.549	0.130	0.500	164
Polandia	Exp 8507	REJ	INC	INC	0.000	ACC	ACC	0.044	0.000	0.233	164
Polandia	Exp 8703	INC	INC	ACC	0.077	ACC	ACC	0.972	-	0.392	164
Slovakia	Exp 8703	ACC	ACC	ACC	0.485	ACC	ACC	0.589	-	0.286	164
Slovakia	Exp 8708	ACC	ACC	ACC	0.150	ACC	ACC	0.458	-	0.331	164
Slovakia	Exp 8528	REJ	REJ	REJ	0.463	ACC	ACC	0.994	-	0.271	164
Slovenia	Exp 3004	REJ	REJ	INC	0.043	ACC	ACC	0.872	-	0.443	164
Slovenia	Exp 8703	INC	INC	INC	0.007	ACC	ACC	0.266	0.123	0.408	164
Slovenia	Exp 8516	REJ	REJ	REJ	0.009	ACC	ACC	0.480	0.035	0.382	164
Estonia	Exp 8517	REJ	REJ	REJ	0.001	ACC	ACC	0.106	0.011	0.436	164
Estonia	Exp 9406	REJ	REJ	REJ	0.050	ACC	ACC	0.458	0.003	0.396	164
Estonia	Exp 8703	REJ	REJ	REJ	0.156	ACC	ACC	0.506	0.029	0.272	164
Latvia	Exp 4407	INC	INC	INC	0.044	ACC	ACC	0.551	0.685	0.396	164
Latvia	Exp 8517	REJ	INC	INC	0.181	ACC	ACC	0.157	-	0.356	164
Latvia	Exp 3004	ACC	ACC	ACC	0.192	ACC	ACC	0.242	-	0.335	164
Lithuania	Exp 9403	REJ	REJ	REJ	0.008	ACC	ACC	0.016	0.078	0.306	164
Lithuania	Exp 8703	ACC	ACC	ACC	0.618	ACC	ACC	0.726	0.784	0.502	164
Lithuania	Exp 3102	REJ	REJ	REJ	0.057	ACC	ACC	0.723	0.005	0.424	164

Keterangan: *** = signifikan 1%; ** = signifikan 5%; * = signifikan 10%. L = lag; - = tidak dilakukan estimasi. REJ = menolak hipotesis; INC = inkosisten; ACC = menerima hipotesis.

Sumber: Diolah.

4.3.2.6 Hasil Uji Diagnostik Metode NARDL

Tabel 4.11 menunjukan hasil uji diagnostik model NARDL berupa uji kointegrasi, uji RESET, uji CUSUM, dan uji Wald. Uji kointegrasi digunakan untuk mengetahui hubungan jangka panjang antara variabel independen dan dependen dalam sebuah model. Model dikatakan terkointegrasi apabila nilai F statistik berada di atas *upper bound* atau I(1), hasil ini akan menolak hipotesis dari model tidak terkointegrasi (REJ). Sebaliknya, jika nilai F ada di bawah *lower bound* atau I(0) maka model tidak terkointegrasi (ACC). Sementara itu, jika nilai F berada di antara *lower bound* I(0) dan *upper bound* atau I(1) maka model tidak konsisten (INC). Komoditas yang terkointegrasi dalam jangka panjang setidaknya pada tingkat signifikansi 10% ditemukan di Polandia (8507), Slovakia (8528), Slovenia (3004, 8516), Estonia (9406, 8703, 4407), Latvia (8517), dan Lithuania (9403, 3102).

Uji RESET dilakukan untuk identifikasi potensi masalah *omitted* variabel atau variabel independen tidak masuk dalam model. Hasil uji RESET menunjukan semua model tidak mempunyai potensi masalah *omitted* variabel pada tingkat signifikansi 5% kecuali komoditas di negara Hungaria (8708, 8517), Polandia (8507), Slovenia (3004, 8703, 8516), Estonia (8517, 9406), Latvia (4407), dan Lithuania (9403). Sementara itu, uji CUSUM menemukan tidak ada ketidakstabilan parameter di seluruh model dan residual diasumsikan terdistribusi normal dengan rata-rata nol dan varians konstan. Namun, R^2 di hampir seluruh model hanya memiliki nilai di bawah 50%, artinya variabel independen mampu menjelaskan variabel dependen sebesar kurang dari 50% di hampir seluruh model kecuali di Republik Ceko pada komoditas 8708, di Hungaria pada komoditas 8708, dan di Polandia pada komoditas 8708, serta di Lithuania pada komoditas 8703.

Sementara itu, uji Wald dilakukan untuk mengetahui efek asimetris dalam jangka pendek secara kumulatif dan jangka panjang. Dalam jangka panjang, hanya dua model yang mempunyai efek jangka panjang, yaitu di negara Polandia pada komoditas 8507 dan di Lithuania pada komoditas 9403. Sementara itu, hanya ditemukan enam model yang mempunyai efek jangka pendek, yaitu di Polandia pada komoditas 8507, di Slovenia pada komoditas 8516, dan di Estonia pada komoditas 8517, 9406, dan 8703, serta di Lithuania pada komoditas 3102.

4.4 Pembahasan

Sebagian besar, volatilitas nilai tukar di jangka panjang berpengaruh positif terhadap ekspor manufaktur. Pengaruh positif ini ditemukan pada komoditas di Republik Ceko (8703, 8708), dan di Estonia (9406), serta di Latvia (4407). Sebaliknya, volatilitas nilai tukar di jangka panjang berpengaruh negatif pada komoditas di Estonia (8703) dan di Lithuania (9403). Sedangkan pada model NARDL, volatilitas nilai tukar dibagi menjadi volatilitas nilai tukar positif dan volatilitas nilai tukar negatif. Volatilitas nilai tukar positif di jangka panjang berpengaruh positif di Hungaria pada komoditas 8507 dan di Estonia pada komoditas 9406. Sementara itu, volatilitas nilai tukar negatif di jangka panjang berpengaruh positif di Polandia pada komoditas 8507 dan di Estonia pada komoditas 8517, serta di Lithuania pada komoditas 9403. Sebaliknya, volatilitas nilai tukar negatif di jangka panjang berpengaruh negatif di Republik Ceko pada komoditas 8703. Volatilitas nilai tukar berpengaruh positif pada komoditas 9403 dan berpengaruh negatif pada komoditas 8708 sesuai dengan penelitian Handoyo et al. (2023).

Hasil tersebut berbeda dari kebanyakan penelitian sebelumnya yang menyatakan volatilitas nilai tukar menurunkan ekspor. Hal ini dikarenakan kebanyakan volatilitas nilai tukar mempunyai efek asimetris negatif atau penurunan nilai tukar. Rose (2000) menemukan bahwa dampak positif dari penyatuan mata uang terhadap perdagangan internasional menurunkan volatilitas nilai tukar.

Sebagian besar, indeks produksi industri negara mitra berpengaruh positif terhadap ekspor manufaktur pada model ARDL. Pengaruh positif ini ditemukan pada komoditas di Republik Ceko (8703, 8517, 8708), dan di Slovakia (8708), serta di Slovenia (3004). Sebaliknya, indeks produksi industri negara mitra berpengaruh negatif pada komoditas di negara Slovakia (8528) dan di Slovenia (8516), serta di Lithuania (9406). Sedangkan pada model NARDL, indeks produksi industri negara mitra berpengaruh positif pada komoditas di Republik Ceko (8703, 8517, 8708), dan di Slovakia (8708, 8528), serta di Slovenia (3004). Sebaliknya indeks produksi industri negara mitra berpengaruh negatif pada komoditas di Slovenia (8528) dan Lithuania (9403). Dalam model NARDL, indeks produksi industri negara mitra berpengaruh positif pada komoditas 8517 sesuai dalam penelitian Handoyo et al. (2023), sementara indeks produksi industri negara mitra berpengaruh positif pada komoditas obat-obatan dan farmasi (SITC 541) sesuai dalam penelitian Bahmani-Oskooee & Kanitpong (2019). Hal ini sesuai dengan teori yang menyatakan peningkatan pendapatan negara mitra meningkatkan ekspor ke negara mitra.

Sebagian besar, nilai tukar riil berpengaruh positif terhadap ekspor manufaktur pada model ARDL. Pengaruh positif ini ditemukan pada komoditas di Hungaria (8703, 8517) dan di Polandia (8507), serta Slovenia (3004, 8703). Sebaliknya, nilai tukar riil berpengaruh negatif pada komoditas di Republik Ceko (8517) dan di Estonia (8517, 8703), serta di Lithuania (9403). Dalam model NARDL, nilai tukar riil berpengaruh positif pada komoditas di Hungaria (8703, 8507), di Polandia (8507), dan di Slovenia (3004, 8703), serta di Lithuania (3102). Sebaliknya, nilai tukar riil berpengaruh negatif pada komoditas di Republik Ceko (8517) dan Estonia (8517, 9406, 8703). Hal ini sesuai dengan teori yang menyatakan peningkatan nilai tukar riil atau depresiasi mata uang terhadap negara mitra meningkatkan ekspor ke negara mitra.

Sebagian besar, geopolitik berpengaruh positif terhadap ekspor manufaktur pada model ARDL. Pengaruh positif ini ditemukan pada komoditas di Republik Ceko (8708) dan di Polandia (8507), serta di Slovenia (8703, 8516). Sebaliknya, geopolitik berpengaruh negatif pada komoditas Slovakia (8528) dan Estonia (8517). Dalam model NARDL, geopolitik berpengaruh positif pada komoditas di Polandia (8708, 8507) dan Slovenia (8703, 8516). Sebaliknya, geopolitik berpengaruh negatif pada komoditas di Slovakia (8528) dan Estonia (8517). Hal ini sesuai dengan teori yang perang Rusia dan Ukraina meningkatkan nilai ekspor ke negara tersebut sejalan dengan peningkatan biaya perdagangan.

Sebagian besar, euroization berpengaruh positif terhadap ekspor manufaktur pada model ARDL. Pengaruh positif ini ditemukan pada komoditas di Estonia (9406) dan Latvia (4407, 8517). Sebaliknya, geopolitik berpengaruh negatif pada komoditas di Estonia (8703). Dalam model NARDL, euroization berpengaruh positif pada komoditas di Latvia (4407, 8517) dan Lithuania (9403). Sebaliknya, euroization berpengaruh negatif pada komoditas di Estonia (8517). Hal ini sesuai dengan teori yang euroization meningkatkan nilai ekspor ke negara tersebut sejalan dengan peningkatan biaya perdagangan dan peningkatan perdagangan *intra-trade* dalam *Economic and Monetary Union* (EMU) Uni Eropa.

Sebagian besar, volatilitas nilai tukar di jangka pendek berpengaruh positif terhadap ekspor manufaktur. Pengaruh positif ini ditemukan pada komoditas di Polandia (8507, 8703), di Estonia (8703), dan di Lithuania (9403). Sebaliknya, volatilitas nilai tukar di jangka panjang berpengaruh negatif pada komoditas di Republik Ceko (8703, 8708) dan Hungaria (8507). Sedangkan pada model NARDL, volatilitas nilai tukar dibagi menjadi volatilitas nilai tukar positif dan volatilitas nilai tukar negatif. Volatilitas nilai tukar positif di jangka pendek berpengaruh positif pada komoditas di Latvia (4407) dan di Lithuania (9403, 3102). Sebaliknya, volatilitas nilai tukar positif di jangka pendek berpengaruh negatif pada komoditas di Hungaria (8708) dan di Estonia (8703), serta di Lithuania (8703). Sementara itu, volatilitas nilai tukar negatif di jangka pendek berpengaruh negatif pada komoditas di Hungaria (8507), di Polandia (8507), di Slovenia (8516), dan di Estonia (8517, 9406).

Volatilitas nilai tukar negatif berpengaruh negatif pada komoditas 8517 sesuai dengan penelitian Handoyo et al. (2023). Dalam model ARDL, efek volatilitas nilai tukar jangka pendek yang bertahan sampai jangka panjang ditemukan pada komoditas di Republik Ceko (8703, 8708), di Estonia (8703), dan di Lithuania (9403). Sedengkan dalam model NARDL, efek volatilitas nilai tukar negatif jangka pendek yang bertahan sampai jangka panjang ditemukan pada komoditas di Polandia (8507) dan di Estonia (8517). Efek volatilitas nilai tukar jangka pendek yang bertahan sampai jangka panjang dalam model NARDL lebih sedikit dari model ARDL karena sedikitnya efek asimetris yang ditemukan dalam model NARDL.

4.5 Uji Ketahanan (*Robustness Check*)

Tabel 4.11 menunjukan pengaruh variabel volatilitas nilai tukar terhadap ekspor komoditas manufaktur dalam hasil metode *quantile regression.* Metode *quantile regression* digunakan untuk melihat konsistensi pengaruh dari variabel volatilitas nilai tukar terhadap ekspor komoditas manufaktur. Volatilitas nilai tukar berpengaruh positif dalam komoditas di Republik Ceko (8703), Estonia (8517 8703 9406), Hungaria (8507, 8703, 8708), Latvia (3004, 4407), Polandia (8703), Slovakia (8528), dan Slovenia (8516). Sebaliknya, volatilitas nilai tukar berpengaruh negatif dalam komoditas di Lithuania (8703, 9403), Polandia (8507, 8708), dan Slovakia (8703, 8708). Sementara itu, volatilitas nilai tukar berpengaruh negatif di *quantile* 0.25 dan berpengaruh positif di *quantile* 0.75 di Republik Ceko pada komoditas 8517. Sebagian besar volatilitas nilai tukar berpengaruh positif terhadap ekspor manufaktur.

Tabel 4. 12

Hasil Metode *Quantile Regression*

Negara	**Variabel**	***Quantile***				
		0.1	**0.25**	**0.5**	**0.75**	**0.9**
Republik Ceko	Exp 8517	-0.457	-0.610*	0.462	0.528*	0.387
Republik Ceko	Exp 8703	0.088	0.131	0.144	0.226**	0.183*
Republik Ceko	Exp 8708	0.189	0.006	0.08	0.066	0.033
Estonia	Exp 8517	1.005***	0.747***	0.556**	0.292	0.029
Estonia	Exp 8703	0.283	0.262**	0.097	0.093	0.019
Estonia	Exp 9406	0.496***	0.509***	0.432***	0.307***	0.305***
Hungaria	Exp 8507	0.085	0.151	1.527***	1.294***	1.760***
Hungaria	Exp 8703	-0.104	-0.197	-0.211	-0.028	0.178***
Hungaria	Exp 8708	0.047	-0.093	-0.12	0.073*	0.04
Latvia	Exp 3004	0.075	0.488***	0.348***	0.345***	0.285*
Latvia	Exp 4407	0.713**	0.506*	0.301*	0.405***	0.611**
Latvia	Exp 8517	0.494	0.829	0.543**	0.454***	0.292*
Lithuania	Exp 3102	-0.02	0.013	-0.049	0.12	0.072
Lithuania	Exp 8703	-1.079***	-1.170***	-0.764***	-0.690***	-0.899***
Lithuania	Exp 9403	-0.318***	-0.356***	-0.291***	-0.259**	0.231***
Polandia	Exp 8507	-0.197***	-0.131***	-0.151**	0.157	0.308
Polandia	Exp 8703	-0.05	0.01	0.012	0.061**	0.097**
Polandia	Exp 8708	-0.099**	-0.070*	-0.048	-0.032	-0.001
Slovakia	Exp 8528	0.083	0.046	0.075	0.093	0.272**
Slovakia	Exp 8703	-0.680***	-0.075	0.132	0.044	0.047
Slovakia	Exp 8708	-0.567***	-0.555***	-0.233***	-0.206**	0.003
Slovenia	Exp 3004	-0.342	-0.072	-0.008	0.024	0.027
Slovenia	Exp 8516	0.122	0.127**	0.091**	0.066	0.023
Slovenia	Exp 8703	0.071	-0.05	-0.091	-0.063	-0.088

Keterangan: *** = signifikan 1%; ** = signifikan 5%; * = signifikan 10%.

Sumber: Diolah.

BAB 5
SIMPULAN DAN SARAN

5.1 Kesimpulan

Sebagian besar, volatilitas nilai tukar berpengaruh positif terhadap ekspor komoditas di CEECs EU-8. Hal ini tidak sesuai dengan literatur yang menyebutkan volatilitas nilai tukar menurunkan ekspor. Dalam model asimetris, ditemukan jika volatilitas nilai tukar negatif atau penurunan volatilitas mendominasi karena *currency union* mengurangi volatilitas nilai tukar di Uni Eropa. Namun, volatilitas nilai tukar negatif yang dominan juga ditemukan di negara-negara asal yang tidak bergabung dengan *Economic and Monetary Union* (EMU) Uni Eropa seperti Republik Ceko, Hungaria, dan Polandia, serta negara mitra seperti Britania Raya. Sementara itu, efek volatilitas nilai tukar jangka pendek yang bertahan sampai jangka panjang dalam model NARDL lebih sedikit dari model ARDL. Hal ini dapat dijelaskan dengan sedikitnya efek asimetris yang ditemukan dalam model NARDL.

Nilai tukar riil dan indeks produksi industri sebagian besar berpengaruh positif terhadap ekspor manufaktur. Hal ini sesuai dengan teori yang menyatakan peningkatan pendapatan negara mitra dan depresiasi mata uang terhadap negara mitra meningkatkan ekspor ke negara mitra. Sedangkan perang Rusia dan Ukraina dan euroization meningkatkan nilai ekspor sejalan dengan peningkatan *trade cost*. Hal ini sesuai dengan asumsi yang menyatakan perang Rusia dan Ukraina meningkatkan harga komoditas, biaya input, harga energi, dan logistik maritim, sementara adopsi euro meningkatkan harga ekspor dan meningkatkan perdagangan *intra-trade* dalam *Economic and Monetary Union* (EMU) Uni Eropa.

5.2 Saran

Kebijakan *currency union* dapat meningkatkan ekspor dan mengurangi volatilitas nilai tukar dalam jangka panjang di CEECs EU-8. Penelitian ini dapat menjadi landasan kebijakan untuk mengadopsi kebijakan *currency union*. Sementara itu, *currency union* mungkin hanya dapat meningkatkan perdagangan *intra-trade*, akan tetapi mengurangi perdagangan yang bukan anggota *currency* *union* karena mengalihkan perdagangan dari non-Uni Eropa yang mempunyai biaya rendah ke Uni Eropa yang mempunyai mata uang sama. Hal ini dapat menurunkan efisiensi dalam perdagangan dan meningkatkan harga perdagangan. Pemerintah harus membuat kebijakan bahwa *currency union* tidak hanya bermanfaat dalam Uni Eropa, akan tetapi juga non-Uni Eropa. Sementara itu, pemerintah dapat menurunkan dampak dari geopolitik dengan mengganti impor yang berasal dari Rusia dan Ukraina seperti gas, minyak, dan gandum dan melakukan subsidi dari biaya logistik yang terganggu.

Depresiasi nilai tukar riil terhadap mata uang mitra berpengaruh positif terhadap ekspor manufaktur, terutama CEECs EU-8 adalah negara-negara dengan persentase ekspor manufaktur yang tinggi, sebagian besar di atas rata-rata dunia. Oleh karena itu, kebijakan *dumping* mata uang bisa dipertimbangkan untuk pembuat kebijakan.

5.3 Keterbatasan Penelitian

Sebagian besar, model ARDL dan NARDL memiliki nilai R^2 di bawah 50%. Oleh karena itu variabel lain yang mungkin mempengaruhi ekspor manufaktur dapat ditambahkan dalam model. Variabel lain seperti liberalisasi FDI dengan *proxy* OECD FDI *regulatory restrictiveness index*, dan integrasi *supply chain* global dengan *proxy* persentasi ekspor dan impor oleh perusahaan-perusahaan afiliasi asing di CEECs terhadap total ekspor dan impor dapat ditambahkan dalam model.

Penelitian ini menggunakan metode GARCH yang biasa. Metode GARCH menggunakan pendekatan ARIMA dapat digunakan untuk mengontrol efek *pattern seasonal* (StataCorp, 2009). Sementara itu, Nelson (1991) menyatakan metode EGARCH dapat digunakan untuk merespons secara berbeda terhadap peningkatan yang tidak diantisipasi dibandingkan dengan penurunan yang tidak diantisipasi. Oleh karena itu, metode GARCH dengan pendekatan yang berbeda dapat digunakan dan disesuaikan dengan kondisi negara tersebut.

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LAMPIRAN

Lampiran 1 Uji Stasioner Nilai Tukar Riil

Republik Ceko

A. *Level*

```
. dfuller rer_czech

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -0.471            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.8976
```

B. *First Difference*

```
. dfuller D.rer_czech

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.379            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Hungaria

A. *Level*

```
. dfuller rer_hungary

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -2.063            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.2595
```

B. *First Difference*

```
.
. dfuller D.rer_hungary

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -11.992            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Polandia

A. *Level*

```
. dfuller rer_poland

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -3.168            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0219
```

B. *First Difference*

```
. dfuller D.rer_poland

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.426            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Slovakia

A. *Level*

B. *First Difference*

```
. dfuller rer_slovakia

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -2.290            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.1751
```

```
. dfuller D.rer_slovakia

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.577            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Slovenia

A. *Level*

```
. dfuller rer_slovenia

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -3.981            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0015
```

B. *First Difference*

```
. dfuller D.rer_slovenia

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.090            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Estonia

A. *Level*

```
. dfuller rer_estonia

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)              1.914            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.9985
```

B. *First Difference*

```
. dfuller D.rer_estonia

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -11.868            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Latvia

A. *Level*

```
. dfuller rer_latvia

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.246            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.6536
```

B. *First Difference*

```
. dfuller D.rer_latvia

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.789            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Lithuania

A. *Level*

B. *First Difference*

```
. dfuller rer_lithuania

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.236            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.6581
```

```
. dfuller D.rer_lithuania

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.571            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Lampiran 2 Uji ARCH *Effects*

Republik Ceko

. estat archlm, lags(1 2)
LM test for autoregressive conditional heteroskedasticity (ARCH)

lags(*p*)	chi2	df	Prob > chi2
1	6.541	1	0.0105
2	0.876	2	0.6453

H0: no ARCH effects *vs.* H1: ARCH(*p*) disturbance

Hungaria

. estat archlm, lags(1 2)
LM test for autoregressive conditional heteroskedasticity (ARCH)

lags(*p*)	chi2	df	Prob > chi2
1	1.058	1	0.3037
2	10.255	2	0.0059

H0: no ARCH effects *vs.* H1: ARCH(*p*) disturbance

Poland

. estat archlm, lags(1 2)
LM test for autoregressive conditional heteroskedasticity (ARCH)

lags(*p*)	chi2	df	Prob > chi2
1	65.348	1	0.0000
2	108.917	2	0.0000

H0: no ARCH effects *vs.* H1: ARCH(*p*) disturbance

Slovakia

. estat archlm, lags(1 2)
LM test for autoregressive conditional heteroskedasticity (ARCH)

lags(*p*)	chi2	df	Prob > chi2
1	2.595	1	0.1072
2	7.633	2	0.0220

H0: no ARCH effects *vs.* H1: ARCH(*p*) disturbance

Slovenia

. estat archlm, lags(1 2)
LM test for autoregressive conditional heteroskedasticity (ARCH)

lags(*p*)	chi2	df	Prob > chi2
1	49.168	1	0.0000
2	50.576	2	0.0000

H0: no ARCH effects *vs.* H1: ARCH(*p*) disturbance

Estonia

. estat archlm, lags(1 2)
LM test for autoregressive conditional heteroskedasticity (ARCH)

lags(*p*)	chi2	df	Prob > chi2
1	8.928	1	0.0028
2	26.520	2	0.0000

H0: no ARCH effects *vs.* H1: ARCH(*p*) disturbance

Latvia

. estat archlm, lags(1 2)
LM test for autoregressive conditional heteroskedasticity (ARCH)

lags(*p*)	chi2	df	Prob > chi2
1	0.260	1	0.6102
2	8.841	2	0.0120

H0: no ARCH effects *vs.* H1: ARCH(*p*) disturbance

Lithuania

. estat archlm, lags(1 2)
LM test for autoregressive conditional heteroskedasticity (ARCH)

lags(*p*)	chi2	df	Prob > chi2
1	0.909	1	0.3403
2	12.739	2	0.0017

H0: no ARCH effects *vs.* H1: ARCH(*p*) disturbance

Lampiran 3 Estimasi Metode GARCH

Republik Ceko

Hungaria

. arch arch_czech, arch(1) garch(1)

ARCH family regression

Sample: 589 - 755 Number of obs = 167
Distribution: Gaussian Wald chi2(.) = .
Log likelihood = -69.33659 Prob > chi2 = .

arch_czech	Coef.	OPG Std. Err.	z	P>|z|	[95% Conf.	Interval]
arch_czech						
_cons	-.0210391	.0276291	-0.76	0.446	-.075191	.0331129
ARCH						
arch L1.	.1516555	.0986711	1.54	0.124	-.0417364	.3450474
garch L1.	.4645809	.2858174	1.63	0.104	-.0956109	1.024773
_cons	.0522405	.0288097	1.81	0.070	-.0042255	.1087065

. arch arch_hungary, arch(2) garch(1)

ARCH family regression

Sample: 589 - 755 Number of obs = 167
Distribution: Gaussian Wald chi2(.) = .
Log likelihood = -513.3165 Prob > chi2 = .

arch_hungary	Coef.	OPG Std. Err.	z	P>|z|	[95% Conf.	Interval]
arch_hungary						
_cons	.196674	.3968739	0.50	0.620	-.5811845	.9745325
ARCH						
arch L2.	.3531888	.1401496	2.52	0.012	.0785005	.627877
garch L1.	.2355862	.1723681	1.37	0.172	-.1022491	.5734215
_cons	13.18628	3.95514	3.33	0.001	5.43435	20.93821

Poland

. arch arch_poland, arch(1) garch(1)

ARCH family regression

Sample: 588 - 755 Number of obs = 168
Distribution: Gaussian Wald chi2(.) = .
Log likelihood = 106.5288 Prob > chi2 = .

arch_poland	Coef.	OPG Std. Err.	z	P>|z|	[95% Conf.	Interval]
arch_poland						
_cons	4.17585	.0087789	475.67	0.000	4.158644	4.193056
ARCH						
arch L1.	.9648889	.1820746	5.30	0.000	.6080293	1.321749
garch L1.	-.0719063	.1194872	-0.60	0.547	-.306097	.1622844
_cons	.0048867	.0024468	2.00	0.046	.0000911	.0096824

Slovakia

.
. arch arch_slovakia, arch(2) garch(1)

ARCH family regression

Sample: 589 - 755 Number of obs = 167
Distribution: Gaussian Wald chi2(.) = .
Log likelihood = 401.6603 Prob > chi2 = .

arch_slovakia	Coef.	OPG Std. Err.	z	P>|z|	[95% Conf.	Interval]
arch_slovakia						
_cons	.0020621	.001666	1.24	0.216	-.0012033	.0053275
ARCH						
arch L2.	.3641928	.1581224	2.30	0.021	.0542785	.674107
garch L1.	.4983838	.1656729	3.01	0.003	.1736709	.8230966
_cons	.0000943	.000046	2.05	0.040	4.25e-06	.0001844

Slovenia

. arch arch_slovenia, arch(1) garch(1)

ARCH family regression

Sample: 588 - 755 Number of obs = 168
Distribution: Gaussian Wald chi2(.) = .
Log likelihood = 521.5859 Prob > chi2 = .

arch_slovenia	Coef.	OPG Std. Err.	z	P>|z|	[95% Conf.	Interval]
arch_slovenia						
_cons	1.011886	.0009457	1070.01	0.000	1.010032	1.013739
ARCH						
arch L1.	.7006587	.2580638	2.72	0.007	.194863	1.206454
garch L1.	-.2455679	.1313278	-1.87	0.061	-.5029658	.0118299
_cons	.000081	.0000181	4.48	0.000	.0000456	.0001165

Estonia

. arch arch_estonia, arch(1) garch(1)

ARCH family regression

Sample: 589 - 755 Number of obs = 167
Distribution: Gaussian Wald chi2(.) = .
Log likelihood = 644.6149 Prob > chi2 = .

arch_estonia	Coef.	OPG Std. Err.	z	P>|z|	[95% Conf.	Interval]
arch_estonia						
_cons	-.0008511	.0003619	-2.35	0.019	-.0015604	-.0001419
ARCH						
arch L1.	.2141951	.0800037	2.68	0.007	.0573907	.3709995
garch L1.	.7017485	.1250289	5.61	0.000	.4566965	.9468006
_cons	2.74e-06	2.00e-06	1.37	0.170	-1.18e-06	6.65e-06

Slovenia

Estonia

```
. arch arch_slovenia, arch(1) garch(1)

ARCH family regression

Sample: 588 - 755                               Number of obs   =        168
Distribution: Gaussian                          Wald chi2(.)    =          .
Log likelihood =  521.5859                      Prob > chi2     =          .

------------------------------------------------------------------------------
              |                 OPG
arch_slovenia |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
--------------+---------------------------------------------------------------
arch_slovenia |
        _cons |   1.011886   .0009457  1070.01   0.000     1.010032    1.013739
--------------+---------------------------------------------------------------
ARCH          |
         arch |
          L1. |   .7006587   .2580638     2.72   0.007      .194863    1.206454
              |
        garch |
          L1. |  -.2455679   .1313278    -1.87   0.061    -.5029658    .0118299
              |
        _cons |    .000081   .0000181     4.48   0.000     .0000456    .0001165
------------------------------------------------------------------------------
```

```
. arch arch_estonia, arch(1) garch(1)

ARCH family regression

Sample: 589 - 755                               Number of obs   =        167
Distribution: Gaussian                          Wald chi2(.)    =          .
Log likelihood =  644.6149                      Prob > chi2     =          .

------------------------------------------------------------------------------
             |                 OPG
arch_estonia |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
arch_estonia |
       _cons |  -.0008511   .0003619    -2.35   0.019    -.0015604   -.0001419
-------------+----------------------------------------------------------------
ARCH         |
        arch |
         L1. |   .2141951   .0800037     2.68   0.007     .0573907    .3709995
             |
       garch |
         L1. |   .7017485   .1250289     5.61   0.000     .4566965    .9468006
             |
       _cons |   2.74e-06   2.00e-06     1.37   0.170    -1.18e-06    6.65e-06
------------------------------------------------------------------------------
```

Latvia

```
. arch arch_latvia, arch(1) garch(1)

ARCH family regression

Sample: 589 - 755                               Number of obs   =        167
Distribution: Gaussian                          Wald chi2(.)    =          .
Log likelihood =  667.2998                      Prob > chi2     =          .

------------------------------------------------------------------------------
             |                 OPG
 arch_latvia |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
arch_latvia  |
       _cons |   .0004171   .0003447     1.21   0.226    -.0002586    .0010928
-------------+----------------------------------------------------------------
ARCH         |
        arch |
         L1. |   .2785786   .0636351     4.38   0.000     .1538562    .4033011
             |
       garch |
         L1. |   .4901136   .1329094     3.69   0.000      .229616    .7506111
             |
       _cons |   5.59e-06   1.98e-06     2.83   0.005     1.72e-06    9.46e-06
------------------------------------------------------------------------------
```

Lithuania

```
.
. arch arch_lithuania, arch(1) garch(1)

ARCH family regression

Sample: 589 - 755                               Number of obs   =        167
Distribution: Gaussian                          Wald chi2(.)    =          .
Log likelihood =  698.4087                      Prob > chi2     =          .

------------------------------------------------------------------------------
               |                 OPG
arch_lithuania |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
---------------+--------------------------------------------------------------
arch_lithuania |
         _cons |  -.0003329   .0002314    -1.44   0.150    -.0007864    .0001206
---------------+--------------------------------------------------------------
ARCH           |
          arch |
           L1. |   .0757062   .0420577     1.80   0.072    -.0067254    .1581377
               |
         garch |
           L1. |   .8731989   .0610011    14.31   0.000      .753639    .9927588
               |
         _cons |   5.23e-07   3.56e-07     1.47   0.142    -1.75e-07    1.22e-06
------------------------------------------------------------------------------
```

Lampiran 4 Uji Kolektif

Republik Ceko

```
. test [ARCH]L1.arch [ARCH]L1.garch

 ( 1)  [ARCH]L.arch = 0
 ( 2)  [ARCH]L.garch = 0

           chi2(  2) =   25.58
         Prob > chi2 =    0.0000
```

Hungaria

```
. test [ARCH]L2.arch [ARCH]L1.garch

 ( 1)  [ARCH]L2.arch = 0
 ( 2)  [ARCH]L.garch = 0

           chi2(  2) =   25.09
         Prob > chi2 =    0.0000
```

Poland

```
. test [ARCH]L1.arch [ARCH]L1.garch

 ( 1)  [ARCH]L.arch = 0
 ( 2)  [ARCH]L.garch = 0

           chi2(  2) =   28.23
         Prob > chi2 =    0.0000
```

Slovakia

```
. test [ARCH]L2.arch [ARCH]L1.garch

 ( 1)  [ARCH]L2.arch = 0
 ( 2)  [ARCH]L.garch = 0

           chi2(  2) =   73.13
         Prob > chi2 =    0.0000
```

Slovenia

```
. test [ARCH]L1.arch [ARCH]L1.garch

 ( 1)  [ARCH]L.arch = 0
 ( 2)  [ARCH]L.garch = 0

           chi2(  2) =   16.36
         Prob > chi2 =    0.0003
```

Estonia

```
. test [ARCH]L1.arch [ARCH]L1.garch

 ( 1)  [ARCH]L.arch = 0
 ( 2)  [ARCH]L.garch = 0

           chi2(  2) =  165.30
         Prob > chi2 =    0.0000
```

Latvia

```
. test [ARCH]L1.arch [ARCH]L1.garch

 ( 1)  [ARCH]L.arch = 0
 ( 2)  [ARCH]L.garch = 0

           chi2(  2) =   71.86
         Prob > chi2 =    0.0000
```

Lithuania

```
. test [ARCH]L1.arch [ARCH]L1.garch

 ( 1)  [ARCH]L.arch = 0
 ( 2)  [ARCH]L.garch = 0

           chi2(  2) = 1325.55
         Prob > chi2 =    0.0000
```

Lampiran 5 Uji Stasioner Seluruh Variabel

Republik Ceko

A. *Level*

```
. dfuller lnvol

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -5.531            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller D.lnvol

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -14.793            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.630            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller D.vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -23.078            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -9.905            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller D.vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -19.725            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller lnrer

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -0.263            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.9305
```

B. *First Difference*

```
. dfuller D.lnrer

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.149            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller lnipi

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -8.214            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller D.lnipi

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -19.174            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -0.252            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.9321
```

```
. dfuller D.geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.884            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller lncze_8703

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -6.479            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller D.lncze_8703

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -17.151            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller lncze_8517

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -4.283            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0005
```

```
. dfuller D.lncze_8517

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -17.800            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
.
. dfuller lncze_8708

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -6.385            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller D.lncze_8708

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -17.624            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Hungaria

A. *Level*

B. *First Difference*

```
. dfuller lnvol

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -7.396            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller D.lnvol

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -19.071            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -14.858            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller D.vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -24.747            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.459            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller D.vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -24.973            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnrer

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -2.033            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.2723
```

A. *Level*

```
. dfuller D.lnrer

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -11.870            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnipi

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -8.214            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller D.lnipi

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -19.174            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -0.252            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.9321
```

A. *Level*

```
. dfuller D.geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.884            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnhun_8703

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -4.442            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0002
```

```
.
. dfuller D.lnhun_8703

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -17.877            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller lnhun_8708

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -4.824            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller lnhun_8507

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -2.127            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.2337
```

A. *Level*

B. *First Difference*

```
. dfuller lnhun_8507

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -2.127            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.2337
```

```
. dfuller D.lnhun_8507

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.551            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Polandia

A. *Level*

B. *First Difference*

```
. dfuller lnvol

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -5.668            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller D.lnvol

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.922            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.957            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller D.vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -21.377            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.605            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller D.vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -20.472            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnrer

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -3.113            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0256
```

A. *Level*

```
. dfuller D.lnrer

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.115            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnipi

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -8.214            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller D.lnipi

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -19.174            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -0.252            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.9321
```

A. *Level*

```
. dfuller D.geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.884            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnpol_8708

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -6.078            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller D.lnpol_8708

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -17.232            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnpol_8507

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -2.170            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.2173
```

```
. dfuller D.lnpol_8507

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -14.169            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller lnpol_8703

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -7.005            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller D.lnpol_8703

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -16.925            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Slovakia

A. *Level*

B. *First Difference*

```
. dfuller lnvol

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -4.462            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0002
```

```
. dfuller D.lnvol

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -15.614            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -14.546            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller D.vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -23.541            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.014            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller D.vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -22.617            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller lnrer

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -2.415            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.1376
```

A. *Level*

```
. dfuller D.lnrer

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.867            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnipi

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -8.214            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller D.lnipi

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -19.174            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -0.252            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.9321
```

A. *Level*

```
. dfuller D.geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.884            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnsvk_8703

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -4.247            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0005
```

A. *Level*

```
. dfuller D.lnsvk_8703

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -15.724            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnsvk_8708

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -3.934            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0018
```

A. *Level*

```
. dfuller D.lnsvk_8708

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.510            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnsvk_8528

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ────────── Interpolated Dickey-Fuller ──────────
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
──────────────────────────────────────────────────────────────────────────────
 Z(t)             -6.265            -3.488            -2.886            -2.576
──────────────────────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller D.lnsvk_8528

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ────────── Interpolated Dickey-Fuller ──────────
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
──────────────────────────────────────────────────────────────────────────────
 Z(t)            -13.587            -3.488            -2.886            -2.576
──────────────────────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000
```

Slovenia

A. *Level*

B. *First Difference*

```
. dfuller lnvol

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ────────── Interpolated Dickey-Fuller ──────────
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
──────────────────────────────────────────────────────────────────────────────
 Z(t)             -8.549            -3.488            -2.886            -2.576
──────────────────────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller D.lnvol

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ────────── Interpolated Dickey-Fuller ──────────
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
──────────────────────────────────────────────────────────────────────────────
 Z(t)            -17.474            -3.488            -2.886            -2.576
──────────────────────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ────────── Interpolated Dickey-Fuller ──────────
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
──────────────────────────────────────────────────────────────────────────────
 Z(t)            -13.640            -3.488            -2.886            -2.576
──────────────────────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller D.vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ────────── Interpolated Dickey-Fuller ──────────
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
──────────────────────────────────────────────────────────────────────────────
 Z(t)            -22.347            -3.488            -2.886            -2.576
──────────────────────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ────────── Interpolated Dickey-Fuller ──────────
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
──────────────────────────────────────────────────────────────────────────────
 Z(t)            -13.836            -3.488            -2.886            -2.576
──────────────────────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller D.vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ────────── Interpolated Dickey-Fuller ──────────
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
──────────────────────────────────────────────────────────────────────────────
 Z(t)            -22.118            -3.488            -2.886            -2.576
──────────────────────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller lnrer

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ────────── Interpolated Dickey-Fuller ──────────
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
──────────────────────────────────────────────────────────────────────────────
 Z(t)             -3.975            -3.488            -2.886            -2.576
──────────────────────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0015
```

```
. dfuller D.lnrer

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ────────── Interpolated Dickey-Fuller ──────────
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
──────────────────────────────────────────────────────────────────────────────
 Z(t)            -12.042            -3.488            -2.886            -2.576
──────────────────────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

. dfuller lnipi

Dickey-Fuller test for unit root Number of obs = 167

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-8.214	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

. dfuller D.lnipi

Dickey-Fuller test for unit root Number of obs = 166

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-19.174	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

A. *Level* B. *First Difference*

. dfuller geopolitik

Dickey-Fuller test for unit root Number of obs = 167

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-0.252	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.9321

. dfuller D.geopolitik

Dickey-Fuller test for unit root Number of obs = 166

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-12.884	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

A. *Level* B. *First Difference*

. dfuller lnsvn_8703

Dickey-Fuller test for unit root Number of obs = 167

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-6.383	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

. dfuller D.lnsvn_8703

Dickey-Fuller test for unit root Number of obs = 166

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-17.582	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

A. *Level* B. *First Difference*

. dfuller lnsvn_3004

Dickey-Fuller test for unit root Number of obs = 167

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-6.599	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

. dfuller D.lnsvn_3004

Dickey-Fuller test for unit root Number of obs = 166

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-23.145	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

A. *Level* B. *First Difference*

. dfuller lnsvn_8703

Dickey-Fuller test for unit root Number of obs = 167

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-6.383	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

. dfuller D.lnsvn_8703

Dickey-Fuller test for unit root Number of obs = 166

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-17.582	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

Estonia

A. *Level* B. *First Difference*

. dfuller lnvol

Dickey-Fuller test for unit root Number of obs = 167

		Interpolated Dickey-Fuller		
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-2.071	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.2563

A. *Level*

. dfuller D.lnvol

Dickey-Fuller test for unit root Number of obs = 166

		Interpolated Dickey-Fuller		
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-13.234	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

B. *First Difference*

. dfuller vol_pos

Dickey-Fuller test for unit root Number of obs = 167

		Interpolated Dickey-Fuller		
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-12.876	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

A. *Level*

. dfuller D.vol_pos

Dickey-Fuller test for unit root Number of obs = 166

		Interpolated Dickey-Fuller		
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-21.783	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

B. *First Difference*

. dfuller vol_neg

Dickey-Fuller test for unit root Number of obs = 167

		Interpolated Dickey-Fuller		
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-12.282	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

A. *Level*

. dfuller D.vol_neg

Dickey-Fuller test for unit root Number of obs = 166

		Interpolated Dickey-Fuller		
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-21.863	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

B. *First Difference*

. dfuller lnrer

Dickey-Fuller test for unit root Number of obs = 167

		Interpolated Dickey-Fuller		
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	2.162	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.9988

A. *Level*

. dfuller D.lnrer

Dickey-Fuller test for unit root Number of obs = 166

		Interpolated Dickey-Fuller		
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-11.451	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

B. *First Difference*

. dfuller lnipi

Dickey-Fuller test for unit root Number of obs = 167

		Interpolated Dickey-Fuller		
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-8.214	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

A. *Level*

. dfuller D.lnipi

Dickey-Fuller test for unit root Number of obs = 166

		Interpolated Dickey-Fuller		
	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-19.174	-3.488	-2.886	-2.576

MacKinnon approximate p-value for Z(t) = 0.0000

B. *First Difference*

```
. dfuller geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -0.252            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.9321
```

```
. dfuller D.geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.884            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller lnest_8517

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -4.402            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0003
```

B. *First Difference*

```
. dfuller D.lnest_8517

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -18.892            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller lnest_9406

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -3.821            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0027
```

B. *First Difference*

```
. dfuller D.lnest_9406

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -11.485            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller lnest_8703

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -6.166            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller D.lnest_8703

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -16.952            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller euro

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -2.478            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.1208
```

B. *First Difference*

```
. dfuller D.euro

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.884            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Latvia

A. *Level*

B. *First Difference*

```
. dfuller lnvol

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -4.747            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0001
```

A. *Level*

```
. dfuller D.lnvol

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -15.754            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.655            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller D.vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -20.711            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.444            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller D.vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -27.839            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnrer

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.289            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.6342
```

A. *Level*

```
. dfuller D.lnrer

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.896            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnipi

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -8.214            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
.
. dfuller D.lnipi

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -19.174            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -0.252            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.9321
```

```
. dfuller D.geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.884            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller lnlva_4407

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -4.334            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0004
```

```
. dfuller D.lnlva_4407

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -17.423            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller lnlva_8517

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -3.693            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0042
```

```
. dfuller D.lnlva_8517

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -19.529            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller lnlva_3004

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -5.860            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

```
. dfuller D.lnlva_3004

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -18.947            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

B. *First Difference*

```
. dfuller euro

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.339            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.6113
```

```
. dfuller D.euro

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ----------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.884            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Lithuania

A. *Level*

B. *First Difference*

```
. dfuller lnvol

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.525            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.5212
```

A. *Level*

```
. dfuller D.lnvol

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.235            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -13.000            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller D.vol_pos

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -22.978            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -11.846            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller D.vol_neg

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -22.653            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
.
. dfuller lnrer

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.148            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.6955
```

A. *Level*

```
. dfuller D.lnrer

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.835            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnipi

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -8.214            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller D.lnipi

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -19.174            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -0.252            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.9321
```

A. *Level*

```
. dfuller D.geopolitik

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.884            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnltu_9403

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -3.962            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0016
```

A. *Level*

```
. dfuller D.lnltu_9403

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -16.680            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnltu_8703

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -3.917            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0019
```

A. *Level*

```
. dfuller D.lnltu_8703

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -17.410            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller lnltu_3102

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -8.289            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

A. *Level*

```
. dfuller D.lnltu_3102

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -19.359            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

B. *First Difference*

```
. dfuller euro

Dickey-Fuller test for unit root                   Number of obs   =       167

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.150            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.6948
```

```
. dfuller D.euro

Dickey-Fuller test for unit root                   Number of obs   =       166

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -12.884            -3.488            -2.886            -2.576
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

Lampiran 6 Estimasi Metode ARDL

Republik Ceko

A. Exp 8703

B. Exp 8517

```
. ardl lncze_8703 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,1,3,4,1) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.4778
                                                Adj R-squared     =     0.4170
Log likelihood = -2.1469655                     Root MSE          =     0.2598

------------------------------------------------------------------------------
D.lncze_8703 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ADJ          |
  lncze_8703 |
         L1. |  -.3006535   .0788193    -3.81   0.000    -.4564276   -.1448794
-------------+----------------------------------------------------------------
LR           |
       lnvol |   .7322875   .3402051     2.15   0.033     .0599246     1.40465
       lnrer |  -.7612056   1.682027    -0.45   0.652    -4.085473    2.563061
       lnipi |   5.395411   1.760614     3.06   0.003     1.915829    8.874993
  geopolitik |   .2385004   .4274569     0.56   0.578    -.6063021    1.083303
-------------+----------------------------------------------------------------
SR           |
  lncze_8703 |
         LD. |  -.3035996   .0881477    -3.44   0.001    -.4778099   -.1293893
        L2D. |  -.3302683   .0815051    -4.05   0.000    -.4913506   -.1691861
        L3D. |  -.2954674   .0729968    -4.05   0.000    -.4397342   -.1512005
             |
       lnvol |
         D1. |   -.470865   .1110184    -4.24   0.000    -.6902757   -.2514542
             |
       lnrer |
         D1. |  -3.371098   1.607458    -2.10   0.038    -6.547991   -.1942055
         LD. |   5.953738   1.711089     3.48   0.001     2.572035    9.335442
        L2D. |  -2.957452   1.546734    -1.91   0.058    -6.014333    .0994283
             |
       lnipi |
         D1. |  -1.200425    .448031    -2.68   0.008    -2.085889   -.3149605
         LD. |  -.3659469     .40325    -0.91   0.366    -1.162908    .4310146
        L2D. |   .1642483   .3527037     0.47   0.642    -.5328161    .8613126
        L3D. |   .6970675   .2952814     2.36   0.020     .1134894    1.280645
             |
  geopolitik |
         D1. |   .7790501    .393475     1.98   0.050     .0014075    1.556693
             |
       _cons |  -2.555145   2.639442    -0.97   0.335    -7.771595    2.661305
------------------------------------------------------------------------------
```

```
. ardl lncze_8517 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(3,0,0,3,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3684
                                                Adj R-squared     =     0.3271
Log likelihood = -77.866816                     Root MSE          =     0.4028

------------------------------------------------------------------------------
D.lncze_8517 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ADJ          |
  lncze_8517 |
         L1. |  -.1596325   .0477939    -3.34   0.001    -.2540536   -.0652114
-------------+----------------------------------------------------------------
LR           |
       lnvol |   .9363272   .8029108     1.17   0.245    -.6498957     2.52255
       lnrer |  -10.60781   4.856134    -2.18   0.030    -20.20154   -1.014073
       lnipi |   14.33384   4.341908     3.30   0.001     5.756004    22.91167
  geopolitik |  -1.637015   1.291514    -1.27   0.207    -4.188519    .9144878
-------------+----------------------------------------------------------------
SR           |
  lncze_8517 |
         LD. |  -.4409692   .0770196    -5.73   0.000    -.5931284     -.28881
        L2D. |  -.2479377   .0648404    -3.82   0.000    -.3760358   -.1198396
             |
       lnipi |
         D1. |  -1.140906   .6124935    -1.86   0.064    -2.350942    .0691299
         LD. |  -1.760542   .5161546    -3.41   0.001    -2.780252   -.7408323
        L2D. |  -.6175496   .4370428    -1.41   0.160    -1.480967    .2458679
             |
       _cons |  -2.983748     3.40151    -0.88   0.382    -9.703738    3.736242
------------------------------------------------------------------------------
```

C. Exp 8708

```
. ardl lncze_8708 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,1,4,4,0) regression

Sample:       592 -       755                  Number of obs     =        164
                                               R-squared         =     0.5227
                                               Adj R-squared     =     0.4672
Log likelihood =   62.02242                    Root MSE          =     0.1757
```

D.lncze_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lncze_8708						
L1.	-.1970303	.0652992	-3.02	0.003	-.326084	-.0679766
LR						
lnvol	.7865423	.4180084	1.88	0.062	-.0395868	1.612671
lnrer	3.24889	2.058235	1.58	0.117	-.8188934	7.316674
lnipi	4.805395	2.051013	2.34	0.020	.7518845	8.858905
geopolitik	1.059588	.5499014	1.93	0.056	-.0272069	2.146383
SR						
lncze_8708						
LD.	-.3658035	.0826562	-4.43	0.000	-.5291608	-.2024463
L2D.	-.361699	.0825779	-4.38	0.000	-.5249014	-.1984965
L3D.	-.5134216	.0740559	-6.93	0.000	-.6597816	-.3670615
lnvol						
D1.	-.1690614	.0696403	-2.43	0.016	-.3066946	-.0314282
lnrer						
D1.	-2.760908	1.073397	-2.57	0.011	-4.882312	-.6395049
LD.	1.353829	1.060994	1.28	0.204	-.743061	3.450719
L2D.	-1.788256	1.016896	-1.76	0.081	-3.797993	.2214815
L3D.	-2.059077	1.011385	-2.04	0.044	-4.057923	-.0602297
lnipi						
D1.	-1.011517	.300964	-3.36	0.001	-1.606326	-.4167085
LD.	-.44858	.275054	-1.63	0.105	-.9921817	.0950217
L2D.	.181867	.2414729	0.75	0.453	-.2953669	.659101
L3D.	.4791819	.201808	2.37	0.019	.0803395	.8780243
_cons	-3.628689	1.902158	-1.91	0.058	-7.38801	.1306318

Hungaria

A. Exp 8703

B. Exp 8708

```
. ardl lnhun_8703 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,0,2,2,3) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3810
                                                Adj R-squared     =     0.3182
Log likelihood = -20.773035                     Root MSE          =     0.2891
```

D.lnhun_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnhun_8703						
L1.	-.2008135	.0693563	-2.90	0.004	-.33787	-.063757
LR						
lnvol	-.1063454	.3076159	-0.35	0.730	-.7142321	.5015412
lnrer	6.135048	2.284066	2.69	0.008	1.621454	10.64864
lnipi	1.510047	2.103653	0.72	0.474	-2.647028	5.667122
geopolitik	.204586	.5948988	0.34	0.731	-.971007	1.380179
SR						
lnhun_8703						
LD.	-.3288743	.0914473	-3.60	0.000	-.5095854	-.1481632
L2D.	-.2470199	.0960682	-2.57	0.011	-.4368624	-.0571773
L3D.	-.2603337	.0864857	-3.01	0.003	-.43124	-.0894273
lnrer						
D1.	-.4313085	1.392859	-0.31	0.757	-3.183768	2.321151
LD.	-3.645814	1.409676	-2.59	0.011	-6.431506	-.8601223
lnipi						
D1.	.1351211	.3850453	0.35	0.726	-.6257755	.8960178
LD.	-.6207534	.321331	-1.93	0.055	-1.255743	.0142359
geopolitik						
D1.	.4635096	.3478616	1.33	0.185	-.2239075	1.150927
LD.	.1024615	.3484356	0.29	0.769	-.5860901	.791013
L2D.	.7448976	.3495593	2.13	0.035	.0541256	1.43567
_cons	-5.8914	3.237836	-1.82	0.071	-12.28976	.5069618

```
. ardl lnhun_8708 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,0,0,3,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.5020
                                                Adj R-squared     =     0.4659
Log likelihood =  68.749756                     Root MSE          =     0.1653
```

D.lnhun_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnhun_8708						
L1.	-.1128364	.06356	-1.78	0.078	-.2384115	.0127387
LR						
lnvol	.6126251	.4943562	1.24	0.217	-.3640715	1.589322
lnrer	.9781482	2.743341	0.36	0.722	-4.441854	6.39815
lnipi	8.425987	5.58939	1.51	0.134	-2.616937	19.46891
geopolitik	1.256939	1.005281	1.25	0.213	-.7291891	3.243066
SR						
lnhun_8708						
LD.	-.4371875	.0874086	-5.00	0.000	-.6098801	-.2644949
L2D.	-.3547475	.0824037	-4.30	0.000	-.5175519	-.1919431
L3D.	-.3971377	.0739967	-5.37	0.000	-.5433326	-.2509428
lnipi						
D1.	-.3628816	.2673634	-1.36	0.177	-.8911098	.1653466
LD.	-.9416836	.2271494	-4.15	0.000	-1.390461	-.4929059
L2D.	-.4980074	.1924294	-2.59	0.011	-.8781889	-.1178259
_cons	-3.932304	1.732892	-2.27	0.025	-7.355968	-.5086397

C. Exp 8507

```
. ardl lnhun_8507 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(1,4,0,0,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.1818
                                                Adj R-squared     =     0.1340
Log likelihood = -85.202187                     Root MSE          =     0.4198
```

D.lnhun_8507	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnhun_8507						
L1.	-.1117687	.0351431	-3.18	0.002	-.1811934	-.042344
LR						
lnvol	.627669	1.131418	0.55	0.580	-1.607433	2.862771
lnrer	19.08869	6.072498	3.14	0.002	7.092541	31.08483
lnipi	4.049686	4.480455	0.90	0.367	-4.801399	12.90077
geopolitik	.8356463	1.385758	0.60	0.547	-1.901902	3.573195
SR						
lnvol						
D1.	.0942721	.1254198	0.75	0.453	-.1534931	.3420373
LD.	.1828724	.1116091	1.64	0.103	-.03761	.4033547
L2D.	-.1052986	.1035171	-1.02	0.311	-.3097953	.0991982
L3D.	-.1823417	.0892642	-2.04	0.043	-.358682	-.0060013
_cons	-13.44731	4.7621	-2.82	0.005	-22.85479	-4.039842

Polandia

A. Exp 8708

B. Exp 8507

. ardl lnpol_8708 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,0,2,3,4) regression

Sample: 592 - 755	Number of obs	=	164
	R-squared	=	0.4910
	Adj R-squared	=	0.4317
Log likelihood = 59.356605	Root MSE	=	0.1786

D.lnpol_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnpol_8708						
L1.	-.210193	.1006825	-2.09	0.039	-.4091764	-.0112095
LR						
lnvol	.0024788	.0869685	0.03	0.977	-.1694011	.1743586
lnrer	2.777386	2.214923	1.25	0.212	-1.600068	7.15484
lnipi	3.303027	2.068562	1.60	0.112	-.785166	7.391219
geopolitik	.5667438	.3749357	1.51	0.133	-.1742588	1.307746
SR						
lnpol_8708						
LD.	-.2976271	.1080345	-2.75	0.007	-.5111407	-.0841136
L2D.	-.302771	.10528	-2.88	0.005	-.5108406	-.0947013
L3D.	-.4936972	.094262	-5.24	0.000	-.6799914	-.307403
lnrer						
D1.	.5731874	.9549072	0.60	0.549	-1.314039	2.460414
LD.	-2.435651	.9537015	-2.55	0.012	-4.320494	-.550807
lnipi						
D1.	-.4406858	.2690791	-1.64	0.104	-.9724792	.0911075
LD.	-.9577433	.2399288	-3.99	0.000	-1.431926	-.483561
L2D.	-.5039081	.2027046	-2.49	0.014	-.9045224	-.1032938
geopolitik						
D1.	-.0552067	.2361783	-0.23	0.816	-.5219767	.4115632
LD.	-.2549649	.2366813	-1.08	0.283	-.7227289	.2127991
L2D.	.491034	.2315439	2.12	0.036	.0334233	.9486447
L3D.	.3359899	.2072949	1.62	0.107	-.0736964	.7456762
_cons	-1.411745	1.468678	-0.96	0.338	-4.31436	1.49087

. ardl lnpol_8507 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(1,2,1,0,0) regression

Sample: 592 - 755	Number of obs	=	164
	R-squared	=	0.2081
	Adj R-squared	=	0.1672
Log likelihood = -79.829335	Root MSE	=	0.4050

D.lnpol_8507	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnpol_8507						
L1.	-.2075977	.0456738	-4.55	0.000	-.2978211	-.1173742
LR						
lnvol	.1766655	.2167205	0.82	0.416	-.2514414	.6047725
lnrer	16.36289	4.652694	3.52	0.001	7.17202	25.55376
lnipi	1.198215	2.087896	0.57	0.567	-2.926187	5.322618
geopolitik	3.870995	.6556775	5.90	0.000	2.575778	5.166212
SR						
lnvol						
D1.	-.0516772	.0509077	-1.02	0.312	-.1522396	.0488851
LD.	.1215182	.0494308	2.46	0.015	.0238733	.2191631
lnrer						
D1.	-3.858588	2.136408	-1.81	0.073	-8.078821	.3616444
_cons	-3.897997	2.2531	-1.73	0.086	-8.348743	.552748

C. Exp 8703

. ardl lnpol_8703 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,1,2,4,1) regression

Sample: 592 - 755	Number of obs	=	164
	R-squared	=	0.3920
	Adj R-squared	=	0.3258
Log likelihood = 27.008672	Root MSE	=	0.2168

D.lnpol_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnpol_8703						
L1.	-.3279755	.0932407	-3.52	0.001	-.5122409	-.1437101
LR						
lnvol	-.0007712	.071987	-0.01	0.991	-.1430343	.1414919
lnrer	-1.078635	1.597522	-0.68	0.501	-4.235712	2.078443
lnipi	.3847951	.9107456	0.42	0.673	-1.415051	2.184641
geopolitik	.3027989	.2285405	1.32	0.187	-.1488504	.7544481
SR						
lnpol_8703						
LD.	-.1392043	.0954804	-1.46	0.147	-.3278958	.0494871
L2D.	-.2062436	.0882048	-2.34	0.021	-.3805568	-.0319305
L3D.	-.1247166	.0785114	-1.59	0.114	-.2798735	.0304403
lnvol						
D1.	.0499806	.0268579	1.86	0.065	-.0030968	.103058
lnrer						
D1.	-1.959158	1.122974	-1.74	0.083	-4.178417	.260101
LD.	2.223937	1.119424	1.99	0.049	.011694	4.43618
lnipi						
D1.	-.3056672	.3397037	-0.90	0.370	-.977001	.3656666
LD.	.3213254	.3299537	0.97	0.332	-.3307402	.9733909
L2D.	.2237802	.2952479	0.76	0.450	-.3596985	.8072588
L3D.	.7869485	.2584589	3.04	0.003	.2761734	1.297724
geopolitik						
D1.	.361464	.2548162	1.42	0.158	-.1421123	.8650402
_cons	3.813294	1.964384	1.94	0.054	-.0687872	7.695376

Slovakia

A. Exp 8703

. ardl lnsvk_8703 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,0,0,1,2) regression

Sample: 592 - 755	Number of obs	=	164
	R-squared	=	0.2889
	Adj R-squared	=	0.2375
Log likelihood = -69.511675	Root MSE	=	0.3840

D.lnsvk_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
ADJ						
lnsvk_8703						
L1.	-.1172968	.0492857	-2.38	0.019	-.2146703	-.0199233
LR						
lnvol	.2606681	.4634784	0.56	0.575	-.6550233	1.17636
lnrer	2.286396	4.675866	0.49	0.626	-6.951684	11.52448
lnipi	5.215544	4.571583	1.14	0.256	-3.816505	14.24759
geopolitik	1.77496	1.305394	1.36	0.176	-.8040986	4.354019
SR						
lnsvk_8703						
LD.	-.250762	.0845117	-2.97	0.003	-.4177312	-.0837928
L2D.	-.3018465	.0838781	-3.60	0.000	-.467564	-.136129
L3D.	-.1172558	.0844208	-1.39	0.167	-.2840455	.0495339
lnipi						
D1.	.6919279	.4273522	1.62	0.107	-.1523893	1.536245
geopolitik						
D1.	.5398441	.4367896	1.24	0.218	-.3231184	1.402807
LD.	-.7899185	.4358057	-1.81	0.072	-1.650937	.0711001
_cons	-1.327794	2.187033	-0.61	0.545	-5.648702	2.993113

B. Exp 8708

. ardl lnsvk_8708 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,0,1,3,4) regression

Sample: 592 - 755	Number of obs	=	164
	R-squared	=	0.3291
	Adj R-squared	=	0.2561
Log likelihood = -33.266192	Root MSE	=	0.3131

D.lnsvk_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
ADJ						
lnsvk_8708						
L1.	-.1086782	.0498035	-2.18	0.031	-.2071016	-.0102549
LR						
lnvol	.1942138	.4593593	0.42	0.673	-.7135874	1.102015
lnrer	-5.440414	4.962635	-1.10	0.275	-15.24774	4.366911
lnipi	12.38574	7.238033	1.71	0.089	-1.918305	26.68978
geopolitik	1.177545	1.184559	0.99	0.322	-1.163419	3.51851
SR						
lnsvk_8708						
LD.	-.1216504	.0847472	-1.44	0.153	-.2891306	.0458298
L2D.	-.2174318	.0970948	-2.24	0.027	-.4093138	-.0255498
L3D.	-.3261087	.1025496	-3.18	0.002	-.5287706	-.1234467
lnrer						
D1.	1.94215	1.416009	1.37	0.172	-.8562137	4.740514
lnipi						
D1.	-1.109884	.4758761	-2.33	0.021	-2.050326	-.1694417
LD.	-1.539957	.412425	-3.73	0.000	-2.355005	-.7249092
L2D.	-.8827724	.3424073	-2.58	0.011	-1.559449	-.2060957
geopolitik						
D1.	.73532	.4284613	1.72	0.088	-.1114195	1.58206
LD.	-.5086124	.4330998	-1.17	0.242	-1.364519	.3472939
L2D.	.8479929	.3850135	2.20	0.029	.0871165	1.608869
L3D.	.6310842	.3823914	1.65	0.101	-.1246103	1.386779
_cons	-5.026788	2.093472	-2.40	0.018	-9.163977	-.8895992

C. Exp 8528

. ardl lnsvk_8528 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(2,0,0,3,0) regression

Sample: 592 - 755	Number of obs	=	164
	R-squared	=	0.2712
	Adj R-squared	=	0.2286
Log likelihood = -55.054834	Root MSE	=	0.3493

D.lnsvk_8528	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
ADJ						
lnsvk_8528						
L1.	-.4640652	.0676118	-6.86	0.000	-.5976316	-.3304988
LR						
lnvol	.043001	.1006112	0.43	0.670	-.1557553	.2417573
lnrer	.4238678	1.109937	0.38	0.703	-1.768799	2.616535
lnipi	-1.924203	1.103311	-1.74	0.083	-4.10378	.2553745
geopolitik	-.4241239	.2424573	-1.75	0.082	-.9030954	.0548477
SR						
lnsvk_8528						
LD.	.1495533	.075785	1.97	0.050	-.000159	.2992656
lnipi						
D1.	.9627718	.4999739	1.93	0.056	-.0249206	1.950464
LD.	1.207007	.4395733	2.75	0.007	.3386356	2.075379
L2D.	.678311	.3648297	1.86	0.065	-.0424057	1.399028
_cons	9.396963	2.379344	3.95	0.000	4.696597	14.09733

Slovenia

A. Exp 3004

```
. ardl lnsvn_3004 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(2,0,1,3,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.4426
                                                Adj R-squared     =     0.4062
Log likelihood = -68.378274                     Root MSE          =     0.3801
```

D.lnsvn_3004	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnsvn_3004						
L1.	-.3525234	.0710982	-4.96	0.000	-.4929844	-.2120624
LR						
lnvol	.0622723	.1269444	0.49	0.624	-.1885178	.3130625
lnrer	15.14307	7.5122	2.02	0.046	.3020361	29.98409
lnipi	3.544332	1.437546	2.47	0.015	.7043311	6.384333
geopolitik	.2588342	.3397159	0.76	0.447	-.4123053	.9299737
SR						
lnsvn_3004						
LD.	-.3341845	.0718815	-4.65	0.000	-.4761929	-.1921761
lnrer						
D1.	-10.21136	4.435308	-2.30	0.023	-18.97371	-1.449005
lnipi						
D1.	-1.918036	.5294139	-3.62	0.000	-2.963941	-.8721312
LD.	-1.800632	.4743729	-3.80	0.000	-2.737799	-.8634658
L2D.	-1.631997	.4019702	-4.06	0.000	-2.426126	-.8378689
_cons	-2.33237	2.318044	-1.01	0.316	-6.911876	2.247135

B. Exp 8703

```
. ardl lnsvn_8703 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(3,0,2,4,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3960
                                                Adj R-squared     =     0.3436
Log likelihood = -25.900525                     Root MSE          =     0.2963
```

D.lnsvn_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnsvn_8703						
L1.	-.3127018	.0842663	-3.71	0.000	-.479204	-.1461996
LR						
lnvol	-.037733	.1123355	-0.34	0.737	-.2596974	.1842314
lnrer	15.24379	7.244851	2.10	0.037	.9286453	29.55893
lnipi	1.11269	1.503896	0.74	0.461	-1.858866	4.084247
geopolitik	.5278346	.301891	1.75	0.082	-.0686735	1.124343
SR						
lnsvn_8703						
LD.	-.1858187	.0866464	-2.14	0.034	-.3570239	-.0146135
L2D.	-.183895	.073847	-2.49	0.014	-.3298098	-.0379803
lnrer						
D1.	-9.958372	3.494769	-2.85	0.005	-16.86371	-3.053038
LD.	-8.277674	3.7238	-2.22	0.028	-15.63555	-.9197977
lnipi						
D1.	-.8948624	.4762416	-1.88	0.062	-1.835871	.0461459
LD.	-.2914814	.4677709	-0.62	0.534	-1.215752	.6327896
L2D.	-.3210991	.4218886	-0.76	0.448	-1.154711	.5125129
L3D.	.7726817	.3520759	2.19	0.030	.077013	1.46835
_cons	1.70796	2.505459	0.68	0.496	-3.24259	6.658509

C. Exp 8516

```
. ardl lnsvn_8516 lnvol lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(1,0,0,4,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3321
                                                Adj R-squared     =     0.2931
Log likelihood =  41.988854                     Root MSE          =     0.1933
```

D.lnsvn_8516	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnsvn_8516						
L1.	-.4885043	.0656298	-7.44	0.000	-.6181551	-.3588534
LR						
lnvol	.0560772	.0463917	1.21	0.229	-.0355691	.1477234
lnrer	-3.578531	2.752024	-1.30	0.195	-9.015123	1.85806
lnipi	-1.07366	.5222999	-2.06	0.042	-2.105457	-.0418625
geopolitik	.3359224	.1240402	2.71	0.008	.0908826	.5809623
SR						
lnipi						
D1.	.6509483	.2968482	2.19	0.030	.0645282	1.237368
LD.	.9755666	.2893371	3.37	0.001	.4039846	1.547148
L2D.	1.131822	.2638291	4.29	0.000	.610631	1.653013
L3D.	.6339534	.223551	2.84	0.005	.1923311	1.075576
_cons	7.387805	1.596369	4.63	0.000	4.234198	10.54141

Estonia

A. Exp 8517

B. Exp 9406

```
. ardl lnest_8517 lnvol lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(3,0,4,0,1,4) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3882
                                                Adj R-squared     =     0.3170
Log likelihood = -185.11224                     Root MSE          =     0.7929
```

D.lnest_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnest_8517						
L1.	-.2737603	.0625317	-4.38	0.000	-.3973445	-.1501761
LR						
lnvol	.0383233	.7738005	0.05	0.961	-1.490974	1.567621
lnrer	-49.97677	14.28376	-3.50	0.001	-78.20641	-21.74713
lnipi	1.190417	3.35511	0.35	0.723	-5.44044	7.821274
geopolitik	-4.971982	2.03126	-2.45	0.016	-8.986454	-.9575097
euro	-1.635074	1.280045	-1.28	0.204	-4.164885	.8947363
SR						
lnest_8517						
LD.	-.222504	.0846918	-2.63	0.010	-.3898842	-.0551238
L2D.	-.1120213	.0763675	-1.47	0.145	-.2629499	.0389074
lnrer						
D1.	25.78322	10.38049	2.48	0.014	5.267789	46.29865
LD.	5.739466	10.80312	0.53	0.596	-15.61123	27.09016
L2D.	2.25298	10.91137	0.21	0.837	-19.31166	23.81762
L3D.	36.93961	11.39268	3.24	0.001	14.42373	59.45548
geopolitik						
D1.	2.252052	.9586829	2.35	0.020	.3573634	4.146741
euro						
D1.	-.3130593	.8207864	-0.38	0.703	-1.935217	1.309098
LD.	1.409858	.8124067	1.74	0.085	-.1957388	3.015454
L2D.	-2.244806	.8151047	-2.75	0.007	-3.855735	-.6338774
L3D.	1.355874	.8416002	1.61	0.109	-.3074186	3.019167
_cons	.4336175	4.333605	0.10	0.920	-8.131083	8.998318

```
. ardl lnest_9406 lnvol lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(3,0,1,0,0,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3655
                                                Adj R-squared     =     0.3284
Log likelihood =  3.2883961                     Root MSE          =     0.2447
```

D.lnest_9406	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnest_9406						
L1.	-.5572508	.0639189	-8.72	0.000	-.6835219	-.4309797
LR						
lnvol	.4224402	.1132992	3.73	0.000	.198619	.6462614
lnrer	-2.309037	1.948853	-1.18	0.238	-6.158972	1.540899
lnipi	-.1809617	.4992028	-0.36	0.717	-1.167131	.8052074
geopolitik	-.3125474	.2797376	-1.12	0.266	-.8651658	.240071
euro	.4440726	.1830237	2.43	0.016	.0825116	.8056337
SR						
lnest_9406						
LD.	.3612436	.0767978	4.70	0.000	.2095305	.5129568
L2D.	.4378717	.0741376	5.91	0.000	.2914137	.5843297
lnrer						
D1.	4.746716	3.133646	1.51	0.132	-1.443764	10.9372
_cons	7.347112	1.602673	4.58	0.000	4.181051	10.51317

C. Exp 8703

```
. ardl lnest_8703 lnvol lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(1,3,0,0,0,1) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.2964
                                                Adj R-squared     =     0.2504
Log likelihood = -107.41998                     Root MSE          =     0.4823
```

D.lnest_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnest_8703						
L1.	-.4755682	.0677304	-7.02	0.000	-.6093758	-.3417606
LR						
lnvol	-.7195927	.3333284	-2.16	0.032	-1.378113	-.0610723
lnrer	-14.16905	4.505537	-3.14	0.002	-23.07014	-5.267952
lnipi	.5543828	1.12826	0.49	0.624	-1.674597	2.783363
geopolitik	-.2950173	.6909452	-0.43	0.670	-1.660042	1.070007
euro	-.8633407	.440993	-1.96	0.052	-1.734562	.0078807
SR						
lnvol						
D1.	-.1183789	.1925296	-0.61	0.540	-.4987385	.2619808
LD.	.4669518	.1772606	2.63	0.009	.1167575	.8171462
L2D.	.2755831	.1717814	1.60	0.111	-.0637866	.6149529
euro						
D1.	.8420731	.5023222	1.68	0.096	-.1503097	1.834456
_cons	-1.437015	2.657408	-0.54	0.589	-6.686965	3.812934

Latvia

A. Exp 4407

```
. ardl lnlva_4407 lnvol lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(4,0,0,4,0,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3649
                                                Adj R-squared     =     0.3098
Log likelihood =  8.9257726                     Root MSE          =     0.2396
```

D.lnlva_4407	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnlva_4407						
L1.	-.2813063	.0658138	-4.27	0.000	-.4113481	-.1512645
LR						
lnvol	1.073947	.5038001	2.13	0.035	.0784854	2.069408
lnrer	8.727704	6.516621	1.34	0.182	-4.148522	21.60393
lnipi	1.49842	1.518605	0.99	0.325	-1.5022	4.49904
geopolitik	-.4994569	.5470415	-0.91	0.363	-1.580359	.5814453
euro	.5613823	.1849935	3.03	0.003	.1958527	.9269118
SR						
lnlva_4407						
LD.	-.27812	.0841597	-3.30	0.001	-.4444116	-.1118284
L2D.	-.1168239	.0822484	-1.42	0.158	-.2793389	.0456912
L3D.	-.161782	.0739901	-2.19	0.030	-.3079794	-.0155847
lnipi						
D1.	-.3678597	.4140519	-0.89	0.376	-1.185987	.4502678
LD.	-.2183326	.3833569	-0.57	0.570	-.9758095	.5391444
L2D.	.6070548	.3330034	1.82	0.070	-.0509284	1.265038
L3D.	.7099246	.2784374	2.55	0.012	.1597587	1.260091
_cons	3.2402	2.303099	1.41	0.162	-1.310505	7.790905

B. Exp 8517

```
. ardl lnlva_8517 lnvol lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(2,0,1,4,0,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3374
                                                Adj R-squared     =     0.2847
Log likelihood = -52.318727                     Root MSE          =     0.3469
```

D.lnlva_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnlva_8517						
L1.	-.2792775	.0645124	-4.33	0.000	-.406741	-.151814
LR						
lnvol	.1021787	.6355203	0.16	0.872	-1.153482	1.357839
lnrer	15.22957	9.833593	1.55	0.124	-4.199637	34.65877
lnipi	-.3766441	1.937474	-0.19	0.846	-4.204704	3.451415
geopolitik	-.3249694	.7982633	-0.41	0.685	-1.902177	1.252238
euro	1.065929	.2673077	3.99	0.000	.5377827	1.594075
SR						
lnlva_8517						
LD.	-.2896784	.0755819	-3.83	0.000	-.439013	-.1403438
lnrer						
D1.	-12.37842	6.417266	-1.93	0.056	-25.05765	.3008031
lnipi						
D1.	.7101665	.5771053	1.23	0.220	-.4300775	1.85041
LD.	.1890957	.547453	0.35	0.730	-.8925613	1.270753
L2D.	.3850377	.4832609	0.80	0.427	-.5697886	1.339864
L3D.	.9754196	.3918086	2.49	0.014	.2012847	1.749554
_cons	3.069412	3.164394	0.97	0.334	-3.182794	9.321617

C. Exp 3004

```
. ardl lnlva_3004 lnvol lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(4,0,0,0,2,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3277
                                                Adj R-squared     =     0.2790
Log likelihood =  21.266501                     Root MSE          =     0.2208
```

D.lnlva_3004	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnlva_3004						
L1.	-.2122575	.0766259	-2.77	0.006	-.3636467	-.0608682
LR						
lnvol	-.1925237	.5601046	-0.34	0.732	-1.299119	.9140715
lnrer	-4.569231	7.775867	-0.59	0.558	-19.93196	10.7935
lnipi	.1949241	1.055608	0.18	0.854	-1.890634	2.280482
geopolitik	.4853137	.6536614	0.74	0.459	-.8061213	1.776749
euro	.1159443	.2219282	0.52	0.602	-.3225179	.5544064
SR						
lnlva_3004						
LD.	-.3196131	.0925516	-3.45	0.001	-.5024668	-.1367594
L2D.	-.150541	.0870211	-1.73	0.086	-.3224681	.0213861
L3D.	-.1907708	.077802	-2.45	0.015	-.3444838	-.0370578
geopolitik						
D1.	.7387154	.2478215	2.98	0.003	.2490959	1.228335
LD.	.4738489	.2620626	1.81	0.073	-.0439065	.9916044
_cons	1.211979	1.88139	0.64	0.520	-2.505073	4.929031

Lithuania

A. Exp 9403

. ardl lnltu_9403 lnvol lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(1,2,3,1,0,0) regression

Sample: 592 - 755		Number of obs =	164
		R-squared =	0.3139
		Adj R-squared =	0.2593
Log likelihood = 71.499851		Root MSE =	0.1631

D.lnltu_9403	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnltu_9403						
L1.	-.4363465	.0593187	-7.36	0.000	-.5535483	-.3191448
LR						
lnvol	-.3617651	.0990617	-3.65	0.000	-.5574911	-.1660391
lnrer	-2.272225	.9659499	-2.35	0.020	-4.180747	-.363702
lnipi	-1.452271	.5142944	-2.82	0.005	-2.468413	-.4361284
geopolitik	.1432105	.1622258	0.88	0.379	-.1773151	.463736
euro	-.0192241	.110561	-0.17	0.862	-.2376704	.1992221
SR						
lnvol						
D1.	.2704459	.1486612	1.82	0.071	-.0232786	.5641705
LD.	.1764947	.1170955	1.51	0.134	-.0548625	.407852
lnrer						
D1.	1.543454	.9112009	1.69	0.092	-.2568958	3.343804
LD.	-1.035737	.8626614	-1.20	0.232	-2.740182	.6687087
L2D.	2.591948	.8596534	3.02	0.003	.8934456	4.29045
lnipi						
D1.	.4489872	.1819794	2.47	0.015	.0894324	.8085419
_cons	3.803686	1.285122	2.96	0.004	1.264542	6.342829

B. Exp 8703

. ardl lnltu_8703 lnvol lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(4,0,0,3,0,0) regression

Sample: 592 - 755		Number of obs =	164
		R-squared =	0.3729
		Adj R-squared =	0.3231
Log likelihood = -69.419562		Root MSE =	0.3851

D.lnltu_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnltu_8703						
L1.	-.1323544	.0522486	-2.53	0.012	-.2355871	-.0291217
LR						
lnvol	-.2642365	.7974277	-0.33	0.741	-1.839793	1.31132
lnrer	2.890526	8.206494	0.35	0.725	-13.32386	19.10491
lnipi	-5.824979	5.826617	-1.00	0.319	-17.3372	5.687244
geopolitik	-.9276084	1.282568	-0.72	0.471	-3.461706	1.606489
euro	1.490932	.9462598	1.58	0.117	-.3786875	3.360551
SR						
lnltu_8703						
LD.	-.3423786	.0765844	-4.47	0.000	-.4936941	-.1910632
L2D.	-.2766836	.075227	-3.68	0.000	-.4253171	-.1280501
L3D.	-.1476272	.0645339	-2.29	0.024	-.2751332	-.0201212
lnipi						
D1.	1.135654	.6376388	1.78	0.077	-.1241921	2.3955
LD.	2.171765	.551441	3.94	0.000	1.082228	3.261301
L2D.	1.043976	.4415648	2.36	0.019	.171533	1.91642
_cons	4.720909	3.178049	1.49	0.140	-1.558277	11.00009

C. Exp 3102

. ardl lnltu_3102 lnvol lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(1,0,0,2,0,0) regression

Sample: 592 - 755		Number of obs =	164
		R-squared =	0.3468
		Adj R-squared =	0.3130
Log likelihood = -161.13338		Root MSE =	0.6648

D.lnltu_3102	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnltu_3102						
L1.	-.6003947	.0732635	-8.20	0.000	-.7451185	-.4556709
LR						
lnvol	-.0457817	.2909917	-0.16	0.875	-.620603	.5290396
lnrer	3.69573	2.67669	1.38	0.169	-1.591769	8.983228
lnipi	1.046261	1.730157	0.60	0.546	-2.371469	4.463992
geopolitik	.3636677	.4773504	0.76	0.447	-.579284	1.306619
euro	-.0309872	.3125433	-0.10	0.921	-.6483812	.5864068
SR						
lnipi						
D1.	.1700608	.8975422	0.19	0.850	-1.602932	1.943054
LD.	-1.427676	.7240285	-1.97	0.050	-2.857913	.0025602
_cons	5.004733	5.184017	0.97	0.336	-5.235707	15.24517

Lampiran 7 Uji Kointegrasi ARDL

Republik Ceko

A. Exp 8703

B. Exp 8517

. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship F = 4.661
Case 3 t = -3.814

Finite sample (4 variables, 164 observations, 12 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.440	3.570	2.875	4.098	3.826	5.228	0.002	0.023
t	-2.532	-3.610	-2.841	-3.953	-3.442	-4.600	0.003	0.067

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

	10%	5%	1%
decision	.r	.	.

. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship F = 3.635
Case 3 t = -3.340

Finite sample (4 variables, 164 observations, 5 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.468	3.566	2.906	4.090	3.860	5.207	0.015	0.092
t	-2.552	-3.642	-2.859	-3.980	-3.455	-4.620	0.014	0.169

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

	10%	5%	1%
decision	.	.	.a

C. Exp 8708

. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship F = 4.781
Case 3 t = -3.017

Finite sample (4 variables, 164 observations, 12 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.440	3.570	2.875	4.098	3.826	5.228	0.002	0.019
t	-2.532	-3.610	-2.841	-3.953	-3.442	-4.600	0.032	0.256

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

	10%	5%	1%
decision	.	.	.a

Hungaria

A. Exp 8703

B. Exp 8708

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     1.937
Case 3                                                      t =    -2.895

Finite sample (4 variables, 164 observations, 10 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%             | 5%              | 1%              | p-value
   |    I(0)    I(1) |    I(0)    I(1) |    I(0)    I(1) |    I(0)    I(1)
---+-----------------+-----------------+-----------------+----------------
 F |   2.448   3.569 |   2.883   4.096 |   3.836   5.222 |   0.213   0.555
 t |  -2.538  -3.619 |  -2.846  -3.961 |  -3.446  -4.606 |   0.044   0.303

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |    10%      5%      1%
-------------+-------------------------
    decision |     .a      .a      .a
```

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     5.849
Case 3                                                      t =    -1.775

Finite sample (4 variables, 164 observations, 6 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%             | 5%              | 1%              | p-value
   |    I(0)    I(1) |    I(0)    I(1) |    I(0)    I(1) |    I(0)    I(1)
---+-----------------+-----------------+-----------------+----------------
 F |   2.464   3.567 |   2.901   4.091 |   3.855   5.210 |   0.000   0.004
 t |  -2.549  -3.638 |  -2.856  -3.977 |  -3.453  -4.617 |   0.372   0.747

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |    10%      5%      1%
-------------+-------------------------
    decision |     .a      .a      .a
```

C. Exp 8507

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     3.434
Case 3                                                      t =    -3.180

Finite sample (4 variables, 164 observations, 4 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%             | 5%              | 1%              | p-value
   |    I(0)    I(1) |    I(0)    I(1) |    I(0)    I(1) |    I(0)    I(1)
---+-----------------+-----------------+-----------------+----------------
 F |   2.472   3.566 |   2.910   4.089 |   3.865   5.204 |   0.021   0.118
 t |  -2.555  -3.647 |  -2.861  -3.984 |  -3.457  -4.622 |   0.022   0.218

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |    10%      5%      1%
-------------+-------------------------
    decision |      .       .      .a
```

Polandia

A. Exp 8708

B. Exp 8507

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     3.132
Case 3                                                      t =    -2.088

Finite sample (4 variables, 164 observations, 12 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%               | 5%                | 1%                | p-value
   |    I(0)      I(1) |    I(0)      I(1) |    I(0)      I(1) |    I(0)      I(1)
---+-------------------+-------------------+-------------------+------------------
 F |   2.440     3.570 |   2.875     4.098 |   3.826     5.228 |   0.033     0.171
 t |  -2.532    -3.610 |  -2.841    -3.953 |  -3.442    -4.600 |   0.228     0.617

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |       10%         5%         1%
-------------+---------------------------------
    decision |        .a         .a         .a
```

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     5.469
Case 3                                                      t =    -4.545

Finite sample (4 variables, 164 observations, 3 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%               | 5%                | 1%                | p-value
   |    I(0)      I(1) |    I(0)      I(1) |    I(0)      I(1) |    I(0)      I(1)
---+-------------------+-------------------+-------------------+------------------
 F |   2.476     3.565 |   2.915     4.088 |   3.870     5.201 |   0.001     0.007
 t |  -2.558    -3.652 |  -2.864    -3.988 |  -3.459    -4.625 |   0.000     0.013

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |       10%         5%         1%
-------------+---------------------------------
    decision |        .r         .r          .
```

C. Exp 8703

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     2.757
Case 3                                                      t =    -3.518

Finite sample (4 variables, 164 observations, 11 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%               | 5%                | 1%                | p-value
   |    I(0)      I(1) |    I(0)      I(1) |    I(0)      I(1) |    I(0)      I(1)
---+-------------------+-------------------+-------------------+------------------
 F |   2.444     3.569 |   2.879     4.097 |   3.831     5.225 |   0.061     0.260
 t |  -2.535    -3.614 |  -2.843    -3.957 |  -3.444    -4.603 |   0.008     0.119

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |       10%         5%         1%
-------------+---------------------------------
    decision |         .         .a         .a
```

Slovakia

A. Exp 8703

B. Exp 8708

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     1.731
Case 3                                                      t =    -2.380

Finite sample (4 variables, 164 observations, 6 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%             | 5%              | 1%              | p-value
   |  I(0)     I(1)  |  I(0)     I(1)  |  I(0)     I(1)  |  I(0)     I(1)
---+-----------------+-----------------+-----------------+-----------------
 F |  2.464    3.567 |  2.901    4.091 |  3.855    5.210 |  0.287    0.644
 t | -2.549   -3.638 | -2.856   -3.977 | -3.453   -4.617 |  0.141    0.521

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |    10%      5%      1%
-------------+-------------------------
    decision |     .a      .a      .a
```

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     2.838
Case 3                                                      t =    -2.182

Finite sample (4 variables, 164 observations, 11 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%             | 5%              | 1%              | p-value
   |  I(0)     I(1)  |  I(0)     I(1)  |  I(0)     I(1)  |  I(0)     I(1)
---+-----------------+-----------------+-----------------+-----------------
 F |  2.444    3.569 |  2.879    4.097 |  3.831    5.225 |  0.053    0.238
 t | -2.535   -3.614 | -2.843   -3.957 | -3.444   -4.603 |  0.196    0.584

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |    10%      5%      1%
-------------+-------------------------
    decision |     .a      .a      .a
```

C. Exp 8528

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =    10.006
Case 3                                                      t =    -6.864

Finite sample (4 variables, 164 observations, 4 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%             | 5%              | 1%              | p-value
   |  I(0)     I(1)  |  I(0)     I(1)  |  I(0)     I(1)  |  I(0)     I(1)
---+-----------------+-----------------+-----------------+-----------------
 F |  2.472    3.566 |  2.910    4.089 |  3.865    5.204 |  0.000    0.000
 t | -2.555   -3.647 | -2.861   -3.984 | -3.457   -4.622 |  0.000    0.000

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |    10%      5%      1%
-------------+-------------------------
    decision |     .r      .r      .r
```

Slovenia

A. Exp 3004

B. Exp 8703

. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship F = 5.267
Case 3 t = -4.958

Finite sample (4 variables, 164 observations, 5 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.468	3.566	2.906	4.090	3.860	5.207	0.001	0.009
t	-2.552	-3.642	-2.859	-3.980	-3.455	-4.620	0.000	0.004

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

	10%	5%	1%
decision	.r	.r	.r

. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship F = 3.987
Case 3 t = -3.711

Finite sample (4 variables, 164 observations, 8 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.456	3.568	2.892	4.093	3.846	5.216	0.008	0.058
t	-2.543	-3.628	-2.851	-3.969	-3.450	-4.611	0.004	0.085

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

	10%	5%	1%
decision	.r	.	.

C. Exp 8516

. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship F = 11.296
Case 3 t = -7.443

Finite sample (4 variables, 164 observations, 4 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.472	3.566	2.910	4.089	3.865	5.204	0.000	0.000
t	-2.555	-3.647	-2.861	-3.984	-3.457	-4.622	0.000	0.000

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

	10%	5%	1%
decision	.r	.r	.r

Estonia

A. Exp 8517

B. Exp 9406

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                     F =     4.115
Case 3                                                        t =    -4.378

Finite sample (5 variables, 164 observations, 11 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%              | 5%               | 1%               | p-value
   |    I(0)     I(1) |    I(0)     I(1) |    I(0)     I(1) |    I(0)     I(1)
---+------------------+------------------+------------------+------------------
 F |   2.263    3.412 |   2.644    3.884 |   3.475    4.888 |   0.003    0.035
 t |  -2.531   -3.826 |  -2.840   -4.174 |  -3.444   -4.831 |   0.000    0.032

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |      10%        5%        1%
-------------+------------------------------
    decision |       .r        .r         .
```

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                     F =    13.320
Case 3                                                        t =    -8.718

Finite sample (5 variables, 164 observations, 3 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%              | 5%               | 1%               | p-value
   |    I(0)     I(1) |    I(0)     I(1) |    I(0)     I(1) |    I(0)     I(1)
---+------------------+------------------+------------------+------------------
 F |   2.293    3.407 |   2.678    3.875 |   3.512    4.867 |   0.000    0.000
 t |  -2.555   -3.866 |  -2.861   -4.208 |  -3.459   -4.856 |   0.000    0.000

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |      10%        5%        1%
-------------+------------------------------
    decision |       .r        .r        .r
```

C. Exp 8703

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                     F =     8.326
Case 3                                                        t =    -7.021

Finite sample (5 variables, 164 observations, 4 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%              | 5%               | 1%               | p-value
   |    I(0)     I(1) |    I(0)     I(1) |    I(0)     I(1) |    I(0)     I(1)
---+------------------+------------------+------------------+------------------
 F |   2.290    3.408 |   2.674    3.876 |   3.507    4.869 |   0.000    0.000
 t |  -2.552   -3.861 |  -2.859   -4.204 |  -3.458   -4.853 |   0.000    0.000

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |      10%        5%        1%
-------------+------------------------------
    decision |       .r        .r        .r
```

Latvia

A. Exp 4407

B. Exp 8517

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                    F =     6.611
Case 3                                                       t =    -4.274

Finite sample (5 variables, 164 observations, 7 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   |  10%              |  5%               |  1%               |  p-value
   |    I(0)     I(1)  |    I(0)     I(1)  |    I(0)     I(1)  |    I(0)     I(1)
---+-------------------+-------------------+-------------------+------------------
 F |   2.278    3.410  |   2.661    3.880  |   3.493    4.877  |   0.000    0.000
 t |  -2.543   -3.846  |  -2.851   -4.191  |  -3.452   -4.843  |   0.001    0.042

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |      10%        5%        1%
-------------+------------------------------
    decision |       .r        .r         .
```

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                    F =     4.511
Case 3                                                       t =    -4.329

Finite sample (5 variables, 164 observations, 6 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   |  10%              |  5%               |  1%               |  p-value
   |    I(0)     I(1)  |    I(0)     I(1)  |    I(0)     I(1)  |    I(0)     I(1)
---+-------------------+-------------------+-------------------+------------------
 F |   2.282    3.409  |   2.665    3.878  |   3.498    4.875  |   0.001    0.018
 t |  -2.546   -3.851  |  -2.854   -4.195  |  -3.454   -4.846  |   0.001    0.037

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |      10%        5%        1%
-------------+------------------------------
    decision |       .r        .r         .
```

C. Exp 3004

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                    F =     1.754
Case 3                                                       t =    -2.770

Finite sample (5 variables, 164 observations, 5 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   |  10%              |  5%               |  1%               |  p-value
   |    I(0)     I(1)  |    I(0)     I(1)  |    I(0)     I(1)  |    I(0)     I(1)
---+-------------------+-------------------+-------------------+------------------
 F |   2.286    3.409  |   2.669    3.877  |   3.503    4.872  |   0.241    0.641
 t |  -2.549   -3.856  |  -2.856   -4.200  |  -3.456   -4.849  |   0.061    0.444

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |      10%        5%        1%
-------------+------------------------------
    decision |       .a        .a        .a
```

Lithuania

A. Exp 9403

B. Exp 8703

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                               F =     9.709
Case 3                                                  t =    -7.356

Finite sample (5 variables, 164 observations, 6 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

  | 10%             | 5%              | 1%              | p-value
  |   I(0)     I(1) |   I(0)     I(1) |   I(0)     I(1) |   I(0)     I(1)
--+-----------------+-----------------+-----------------+----------------
F |  2.282    3.409 |  2.665    3.878 |  3.498    4.875 |  0.000    0.000
t | -2.546   -3.851 | -2.854   -4.195 | -3.454   -4.846 |  0.000    0.000

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |     10%       5%       1%
-------------+---------------------------
    decision |      .r       .r       .r
```

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                               F =     2.363
Case 3                                                  t =    -2.533

Finite sample (5 variables, 164 observations, 6 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

  | 10%             | 5%              | 1%              | p-value
  |   I(0)     I(1) |   I(0)     I(1) |   I(0)     I(1) |   I(0)     I(1)
--+-----------------+-----------------+-----------------+----------------
F |  2.282    3.409 |  2.665    3.878 |  3.498    4.875 |  0.087    0.371
t | -2.546   -3.851 | -2.854   -4.195 | -3.454   -4.846 |  0.103    0.537

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |     10%       5%       1%
-------------+---------------------------
    decision |      .a       .a       .a
```

C. Exp 3102

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                               F =    11.533
Case 3                                                  t =    -8.195

Finite sample (5 variables, 164 observations, 2 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

  | 10%             | 5%              | 1%              | p-value
  |   I(0)     I(1) |   I(0)     I(1) |   I(0)     I(1) |   I(0)     I(1)
--+-----------------+-----------------+-----------------+----------------
F |  2.297    3.407 |  2.682    3.874 |  3.517    4.864 |  0.000    0.000
t | -2.558   -3.871 | -2.864   -4.213 | -3.461   -4.859 |  0.000    0.000

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |     10%       5%       1%
-------------+---------------------------
    decision |      .r       .r       .r
```

Lampiran 8 Uji RESET ARDL

Republik Ceko

A. Exp 8703

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lncze_8703
       Ho:  model has no omitted variables
                F(3, 143) =      1.78
                 Prob > F =      0.1529
```

B. Exp 8703

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lncze_8517
       Ho:  model has no omitted variables
                F(3, 150) =      0.73
                 Prob > F =      0.5363
```

C. Exp 8708

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lncze_8708
       Ho:  model has no omitted variables
                F(3, 143) =      1.94
                 Prob > F =      0.1259
```

Hungaria

A. Exp 8703

B. Exp 8708

C. Exp 8507

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnhun_8703
       Ho:  model has no omitted variables
                 F(3, 145) =      0.60
                  Prob > F =      0.6142
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnhun_8708
       Ho:  model has no omitted variables
                 F(3, 149) =      3.46
                  Prob > F =      0.0180
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnhun_8507
       Ho:  model has no omitted variables
                 F(3, 151) =     13.21
                  Prob > F =      0.0000
```

Polandia

A. Exp 8708 B. Exp 8507 C. Exp 8703

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnpol_8708
       Ho:  model has no omitted variables
                 F(3, 143) =      3.10
                  Prob > F =      0.0289
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnpol_8507
       Ho:  model has no omitted variables
                 F(3, 152) =     61.59
                  Prob > F =      0.0000
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnpol_8703
       Ho:  model has no omitted variables
                 F(3, 144) =      2.34
                  Prob > F =      0.0756
```

Slovakia

A. Exp 8703 B. Exp 8708 C. Exp 8528

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnsvk_8703
       Ho:  model has no omitted variables
                 F(3, 149) =      0.67
                  Prob > F =      0.5741
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnsvk_8708
       Ho:  model has no omitted variables
                 F(3, 144) =      1.97
                  Prob > F =      0.1212
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnsvk_8528
       Ho:  model has no omitted variables
                 F(3, 151) =      1.02
                  Prob > F =      0.3859
```

Slovenia

A. Exp 3004 B. Exp 8703 C. Exp 8516

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnsvn_3004
       Ho:  model has no omitted variables
                 F(3, 150) =      2.93
                  Prob > F =      0.0354
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnsvn_8703
       Ho:  model has no omitted variables
                 F(3, 147) =      3.82
                  Prob > F =      0.0114
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnsvn_8516
       Ho:  model has no omitted variables
                 F(3, 151) =      2.05
                  Prob > F =      0.1091
```

Estonia

A. Exp 8517 B. Exp 9406 C. Exp 8703

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnest_8517
       Ho:  model has no omitted variables
                 F(3, 143) =      4.68
                  Prob > F =      0.0038
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnest_9406
       Ho:  model has no omitted variables
                 F(3, 151) =      4.22
                  Prob > F =      0.0067
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnest_8703
       Ho:  model has no omitted variables
                 F(3, 150) =      2.57
                  Prob > F =      0.0563
```

Latvia

A. Exp 4407 B. Exp 8517 C. Exp 3004

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnlva_4407
       Ho:  model has no omitted variables
                 F(3, 147) =      1.81
                  Prob > F =      0.1471
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnlva_8517
       Ho:  model has no omitted variables
                 F(3, 148) =      1.13
                  Prob > F =      0.3378
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnlva_3004
       Ho:  model has no omitted variables
                 F(3, 149) =      1.65
                  Prob > F =      0.1810
```

Lithuania

A. Exp 9403 B. Exp 8703 C. Exp 3102

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnltu_9403
       Ho:  model has no omitted variables
                F(3, 148) =      5.47
                 Prob > F =      0.0014
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnltu_8703
       Ho:  model has no omitted variables
                F(3, 148) =      0.27
                 Prob > F =      0.8501
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnltu_3102
       Ho:  model has no omitted variables
                F(3, 152) =      1.78
                 Prob > F =      0.1538
```

Lampiran 9 Uji CUSUM ARDL

Republik Ceko

A. Exp 8703

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

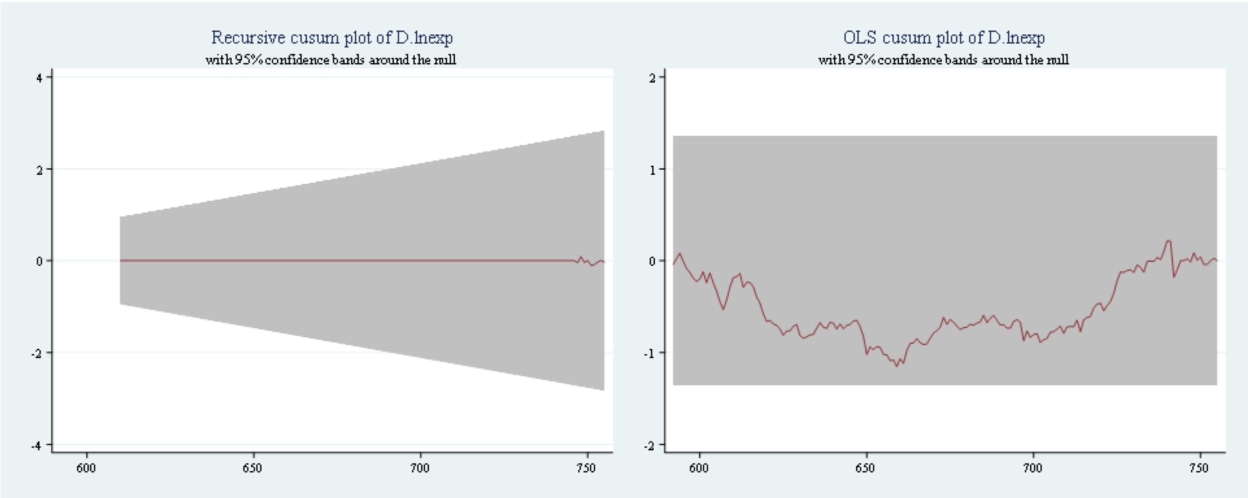
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0352	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	1.1536	1.6276	1.3581	1.224



B. Exp 8517

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

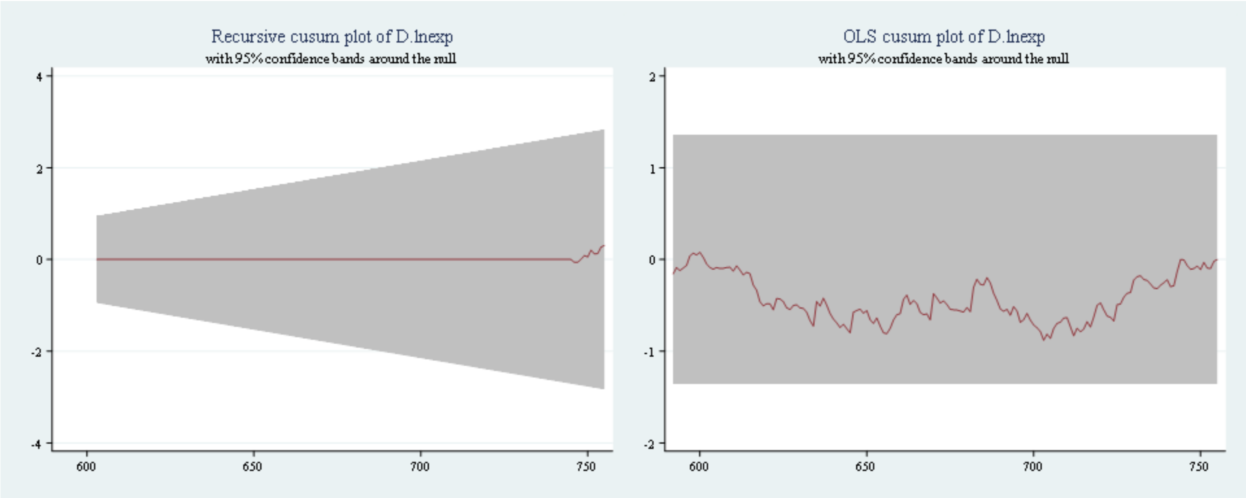
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.1018	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.8815	1.6276	1.3581	1.224



C. Exp 8708

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

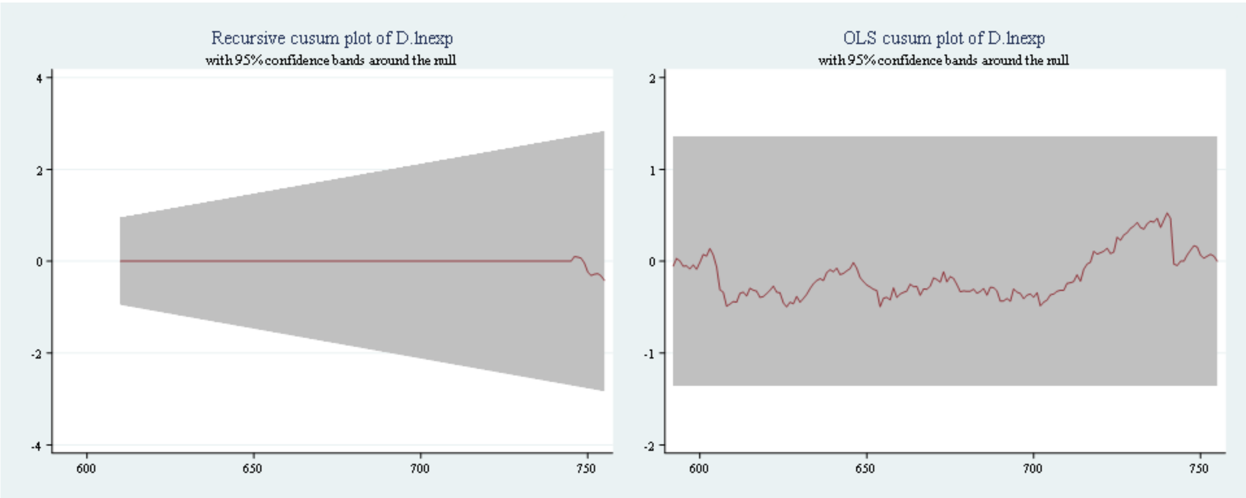
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.1394	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.5259	1.6276	1.3581	1.224

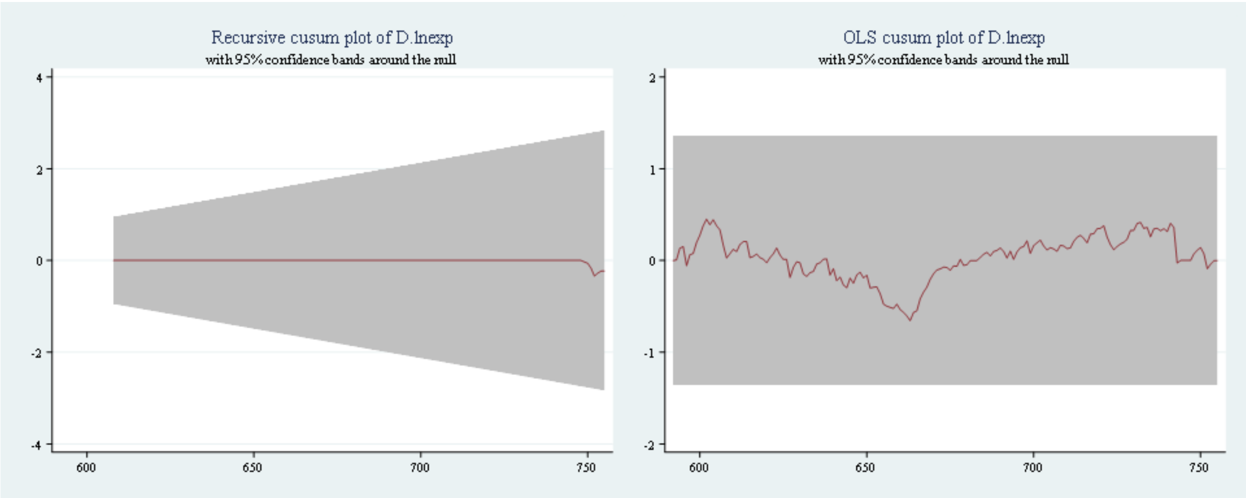


Hungaria

A. Exp 8703

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =       164
Ho: No structural break

                                1% Critical       5% Critical      10% Critical
 Statistic    Test Statistic      Value             Value             Value

 recursive       0.1147          1.1430            0.9479            0.850
```

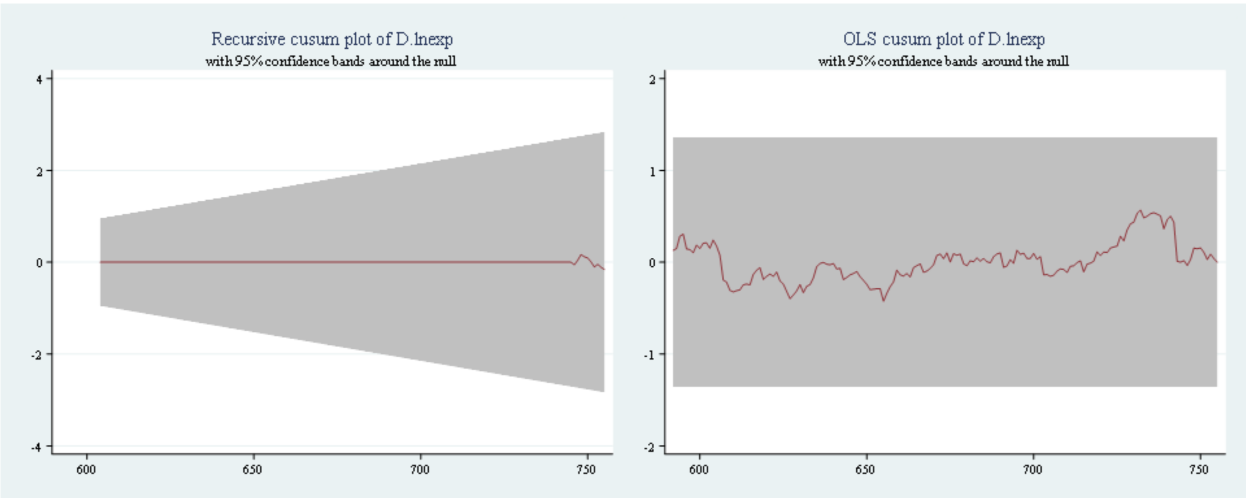
```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =       164
Ho: No structural break

                                1% Critical       5% Critical      10% Critical
 Statistic    Test Statistic      Value             Value             Value

 ols             0.6569          1.6276            1.3581            1.224
```



B. Exp 8708

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =       164
Ho: No structural break

                                1% Critical       5% Critical      10% Critical
 Statistic    Test Statistic      Value             Value             Value

 recursive       0.0569          1.1430            0.9479            0.850
```

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =       164
Ho: No structural break

                                1% Critical       5% Critical      10% Critical
 Statistic    Test Statistic      Value             Value             Value

 ols             0.5680          1.6276            1.3581            1.224
```



C. Exp 8507

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

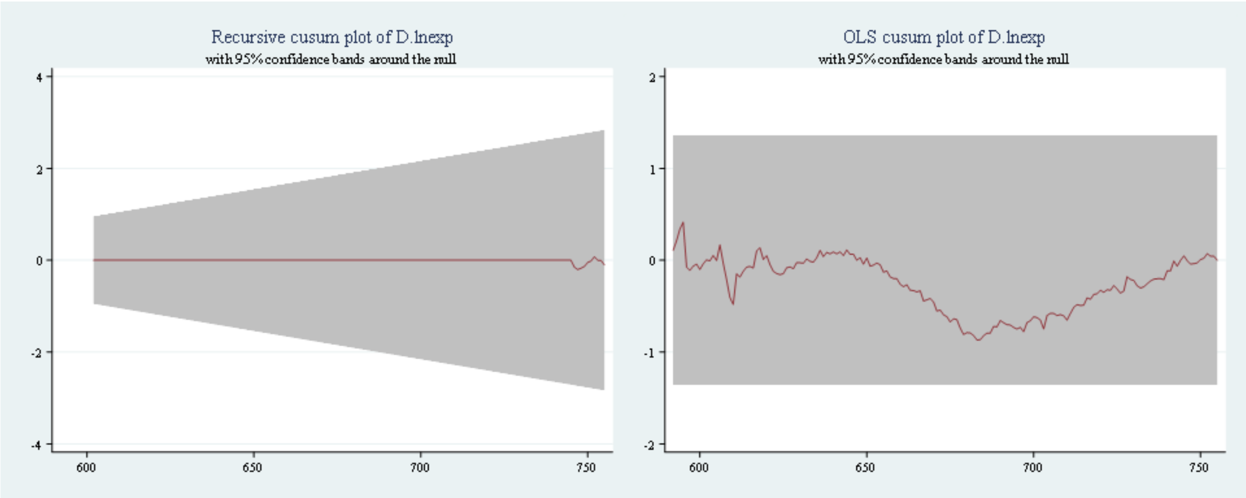
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0703	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.8684	1.6276	1.3581	1.224



Polandia

A. Exp 8708

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

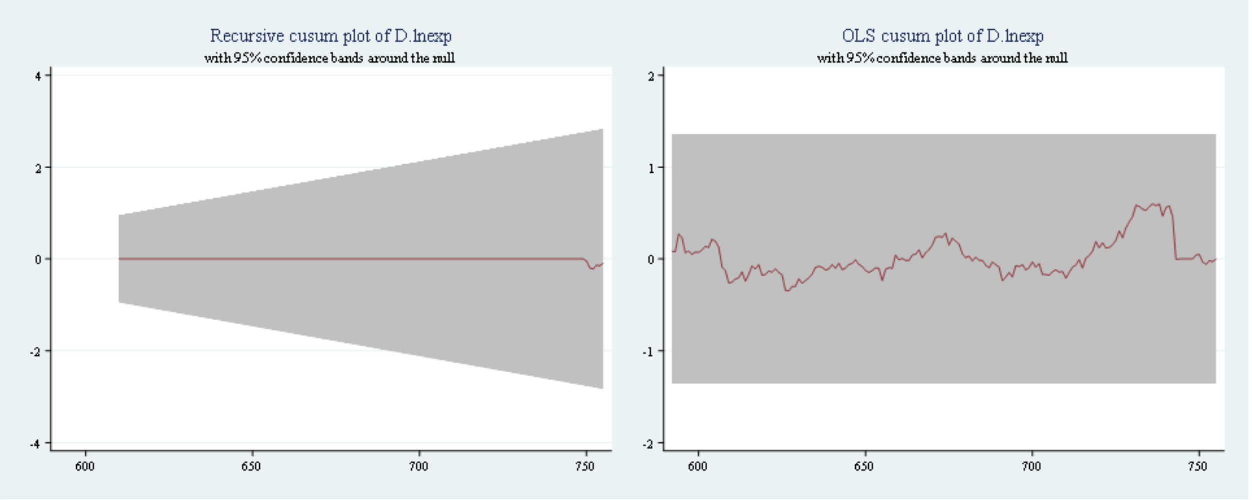
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0743	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.6018	1.6276	1.3581	1.224



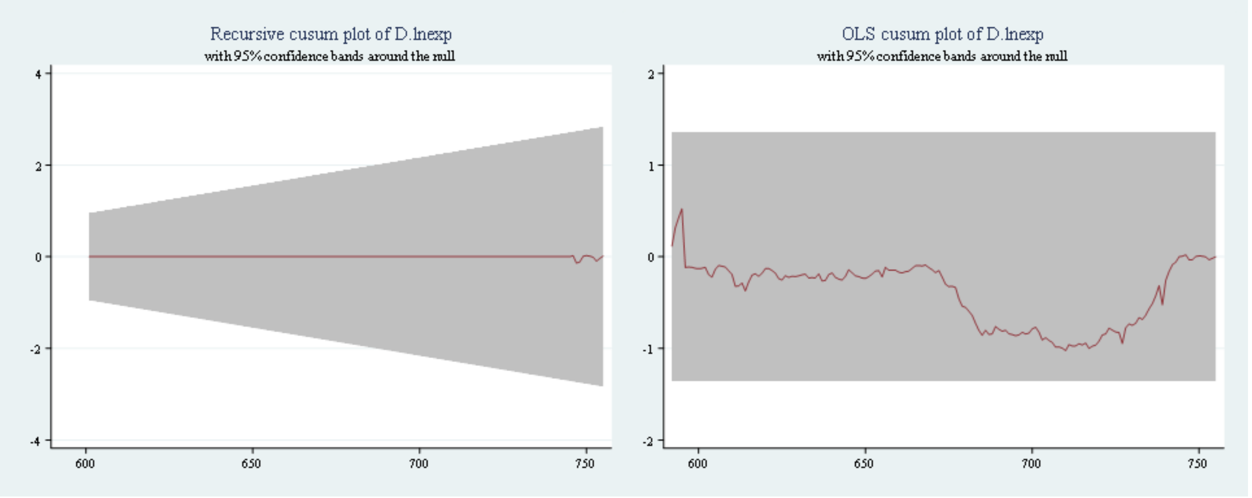
B. Exp 8507

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                              Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0485	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                              Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	1.0253	1.6276	1.3581	1.224



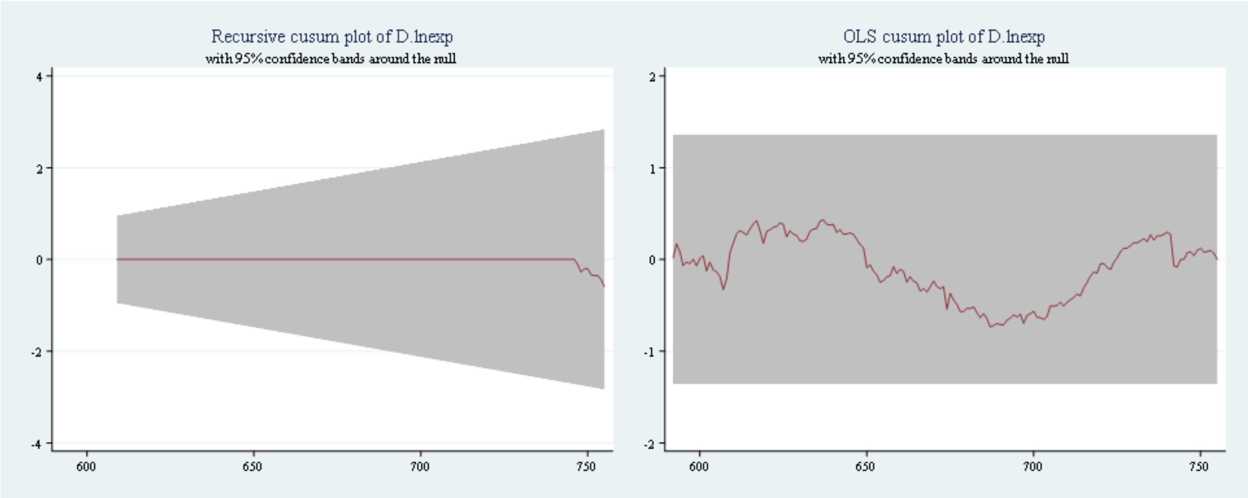
C. Exp 8703

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                              Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.1978	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                              Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.7373	1.6276	1.3581	1.224



Slovakia

A. Exp 8703

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168   Number of obs = 164
Ho: No structural break

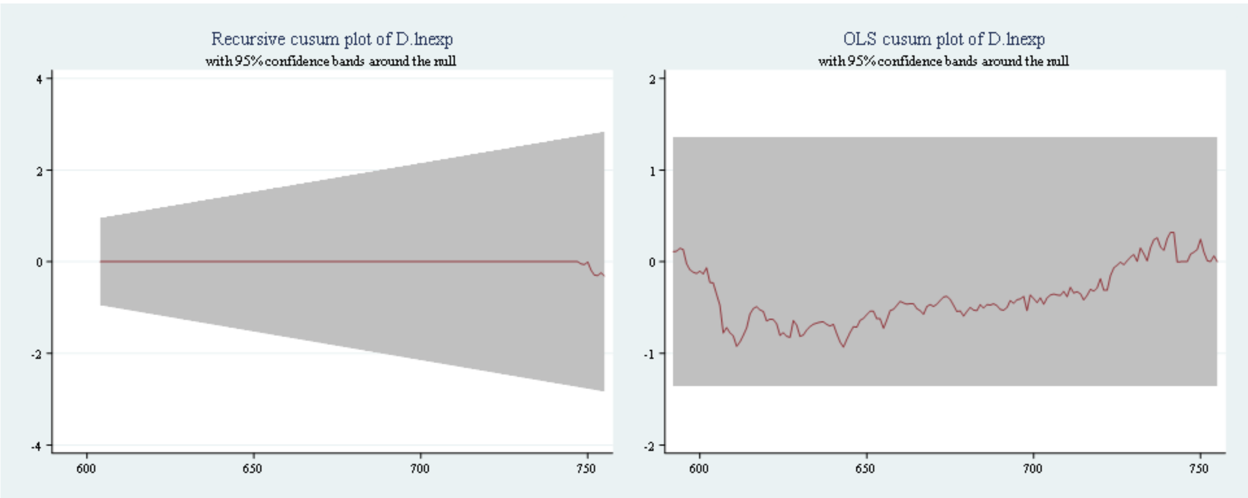
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.1035	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168   Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.9327	1.6276	1.3581	1.224



B. Exp 8708

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

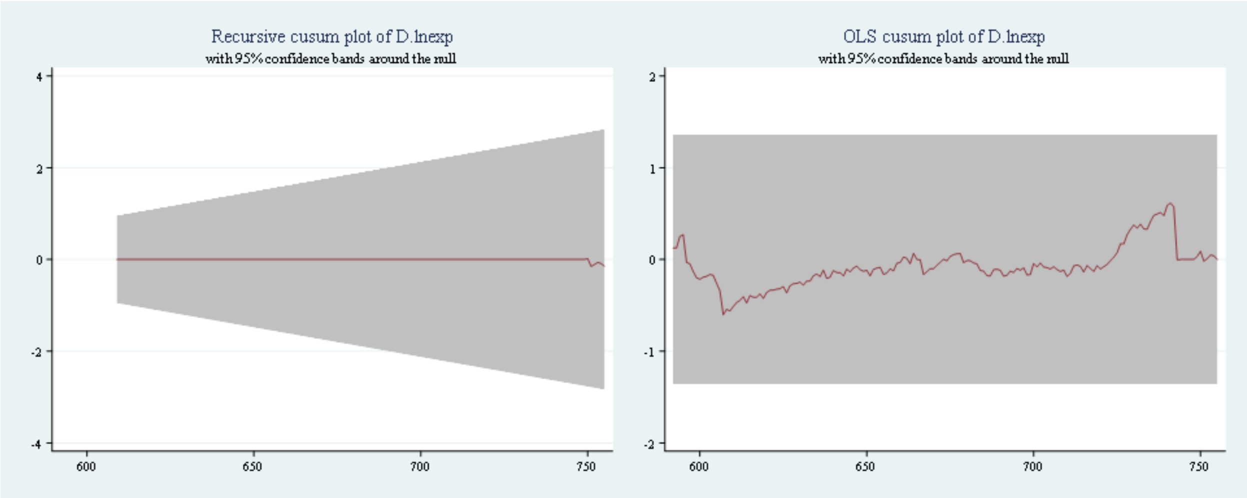
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0519	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.6155	1.6276	1.3581	1.224



C. Exp 8528

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

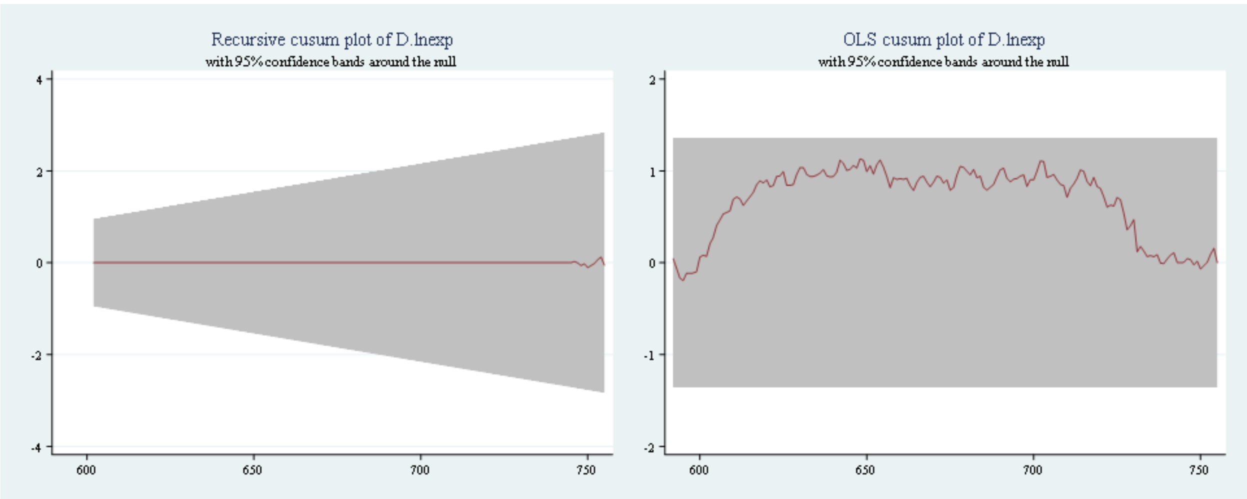
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0410	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	1.1308	1.6276	1.3581	1.224



Slovenia

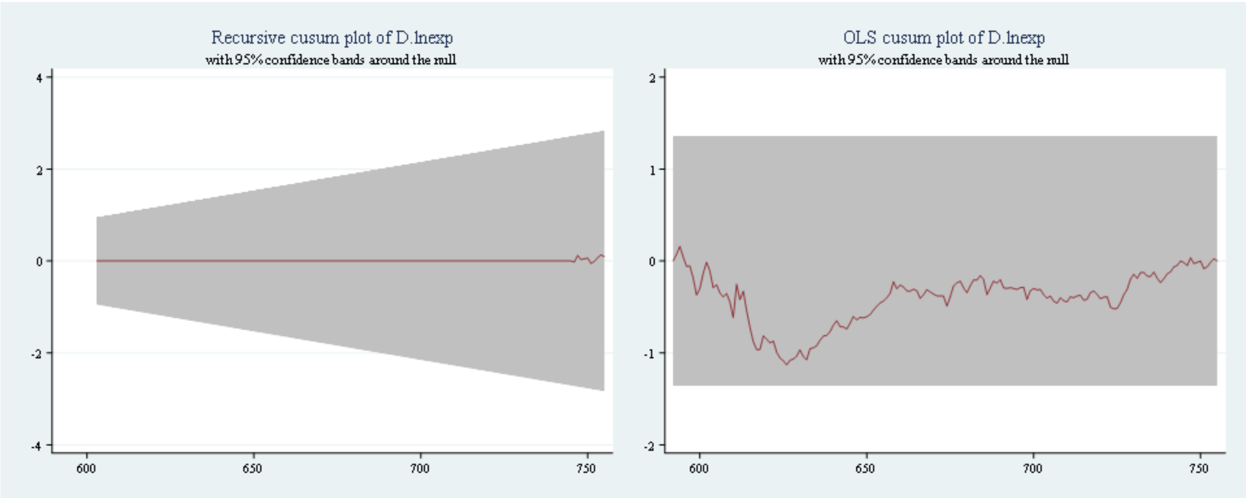
A. Exp 3004

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0446	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	1.1304	1.6276	1.3581	1.224



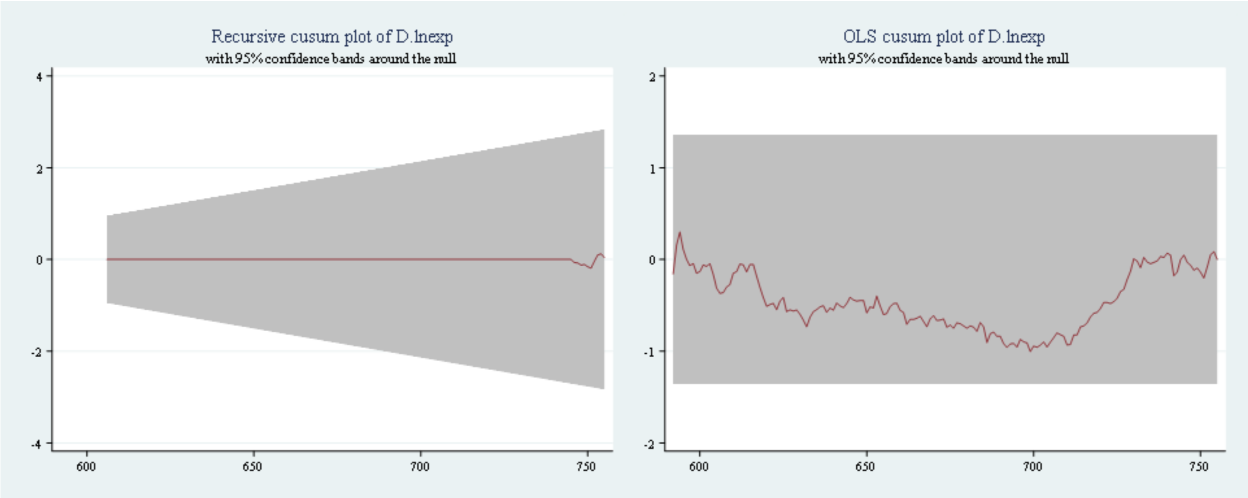
B. Exp 8703

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0641	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

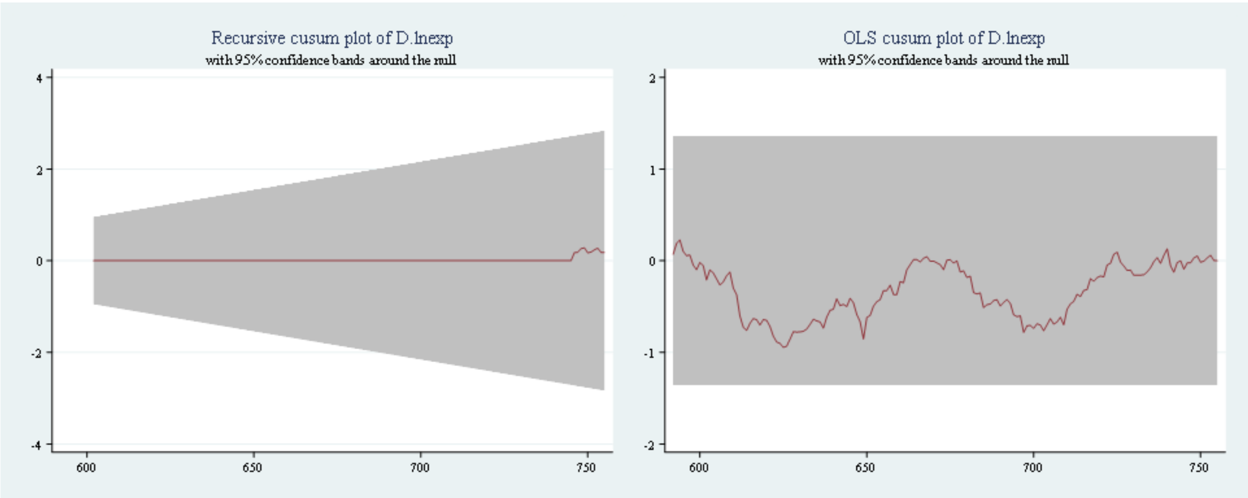
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	1.0042	1.6276	1.3581	1.224



C. Exp 8516

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                Number of obs =        164
Ho: No structural break

                                1% Critical      5% Critical      10% Critical
 Statistic    Test Statistic       Value            Value            Value

 recursive       0.0959           1.1430           0.9479           0.850
```

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                Number of obs =        164
Ho: No structural break

                                1% Critical      5% Critical      10% Critical
 Statistic    Test Statistic       Value            Value            Value

 ols             0.9461           1.6276           1.3581           1.224
```



Estonia

A. Exp 8517

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

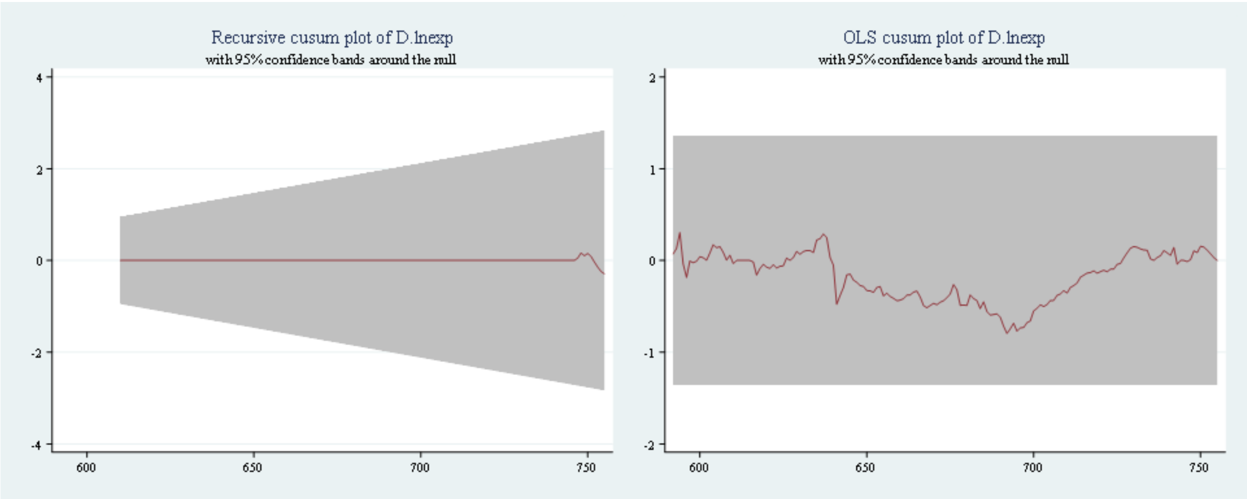
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0971	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.7961	1.6276	1.3581	1.224



B. Exp 9406

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

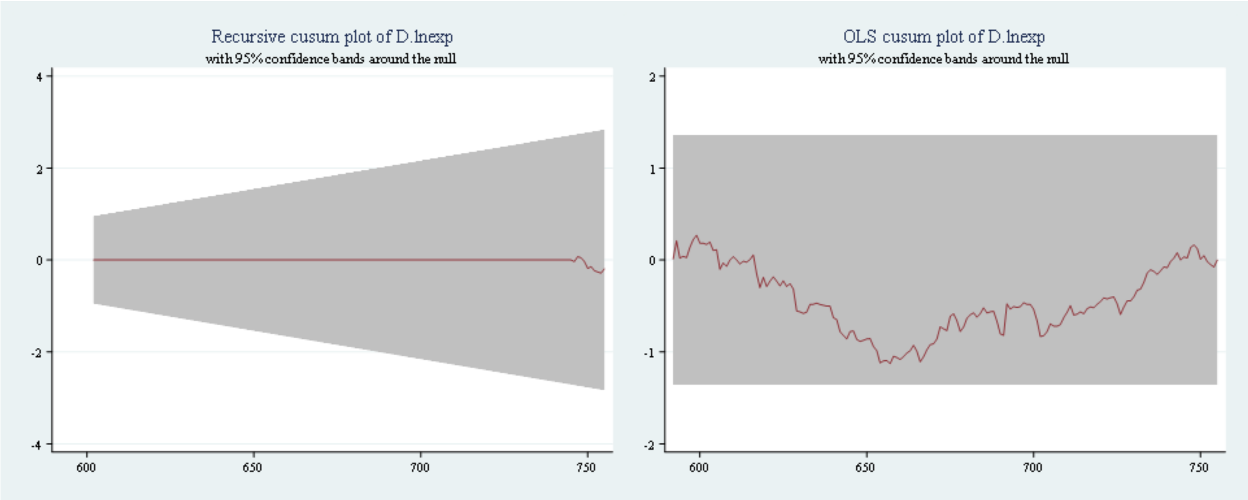
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0954	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	1.1285	1.6276	1.3581	1.224



C. Exp 8703

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

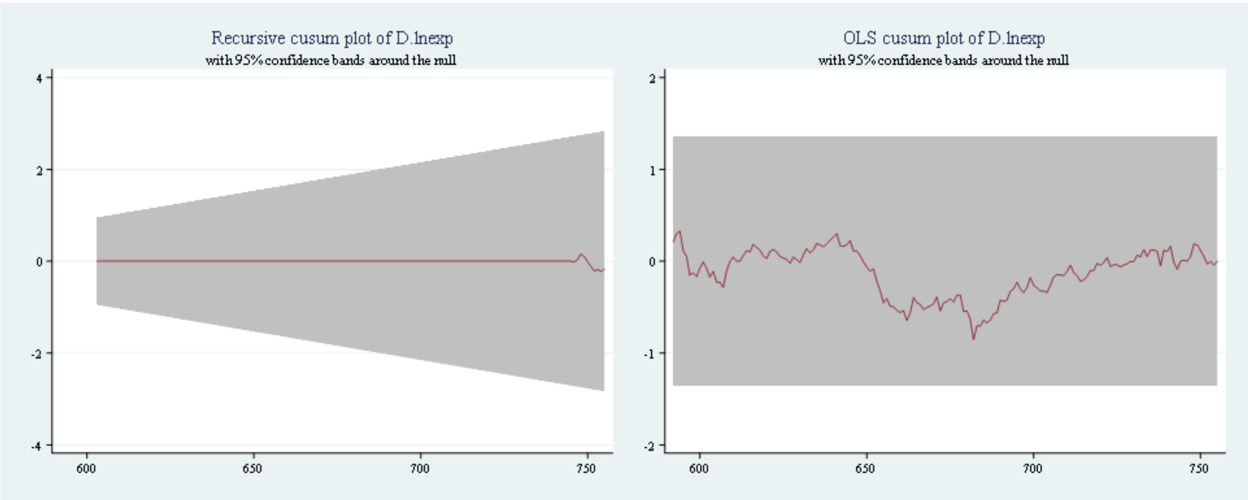
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0746	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.8561	1.6276	1.3581	1.224



Latvia

A. Exp 4407

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

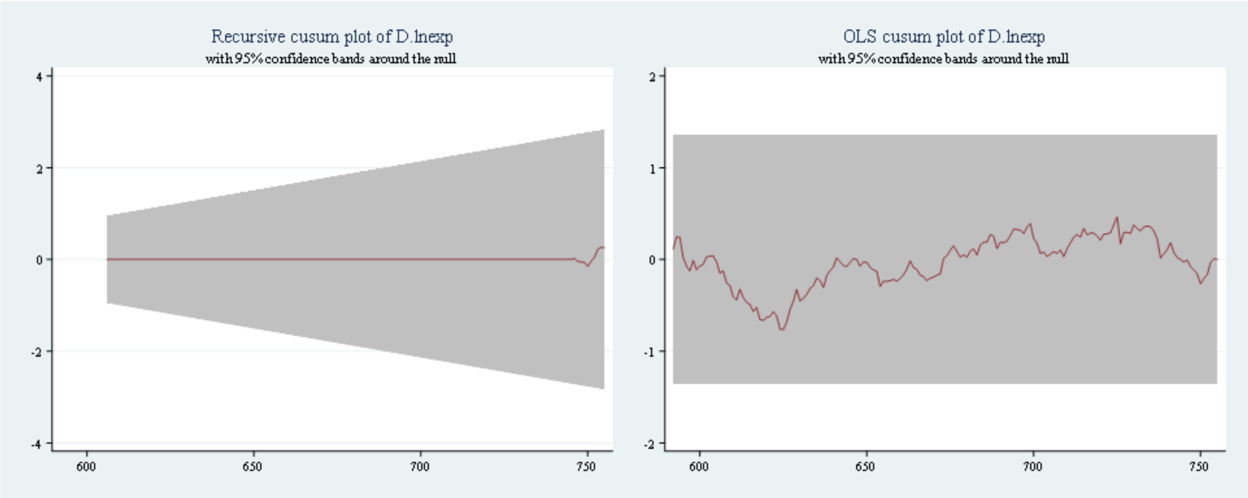
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0892	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.7662	1.6276	1.3581	1.224



B. Exp 8517

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

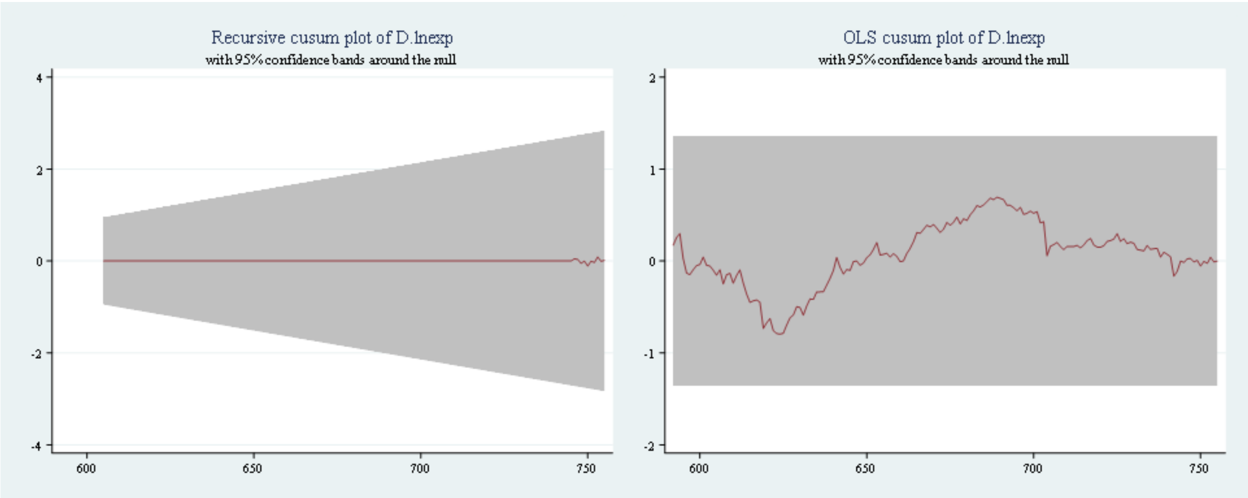
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0375	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.7966	1.6276	1.3581	1.224



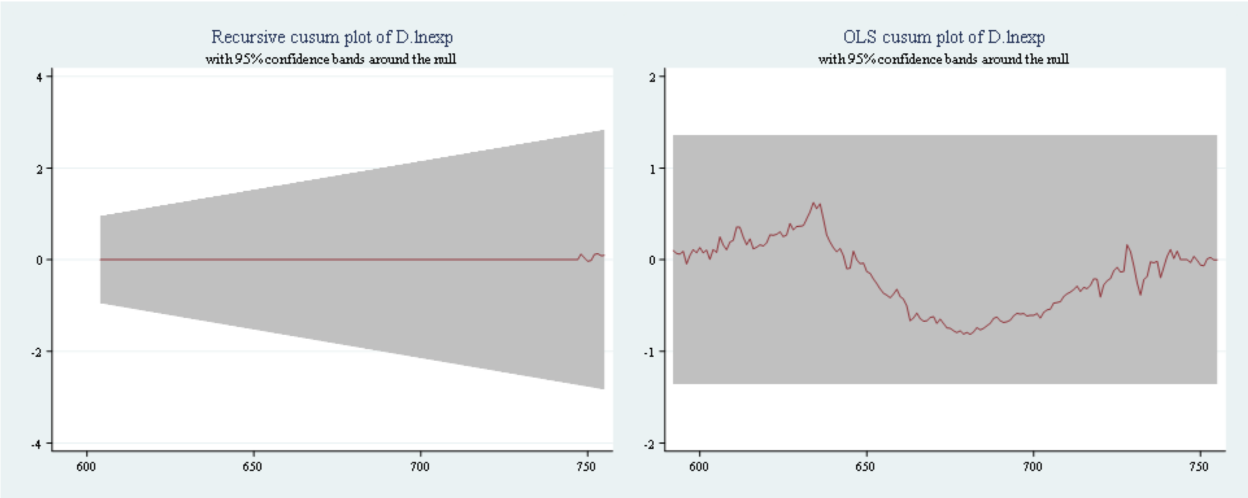
C. Exp 3004

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0447	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.8150	1.6276	1.3581	1.224



Lithuania

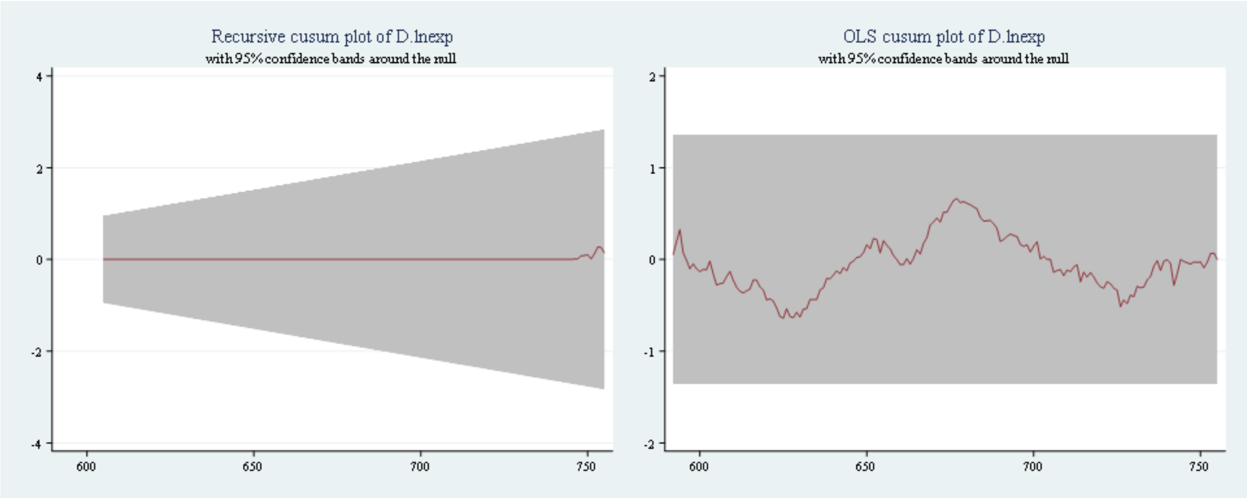
A. Exp 9403

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0910	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

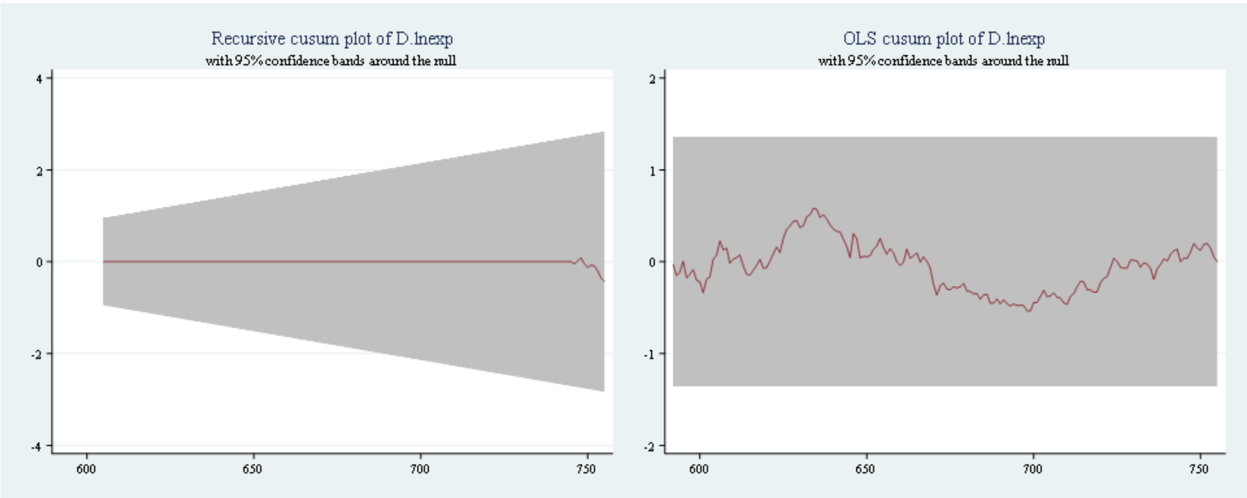
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.6644	1.6276	1.3581	1.224



B. Exp 8703

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =        164
Ho: No structural break

                              1% Critical      5% Critical     10% Critical
Statistic   Test Statistic         Value            Value            Value
---------------------------------------------------------------------------
recursive       0.1430           1.1430           0.9479            0.850
---------------------------------------------------------------------------
```

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =        164
Ho: No structural break

                              1% Critical      5% Critical     10% Critical
Statistic   Test Statistic         Value            Value            Value
---------------------------------------------------------------------------
ols             0.5824           1.6276           1.3581            1.224
---------------------------------------------------------------------------
```



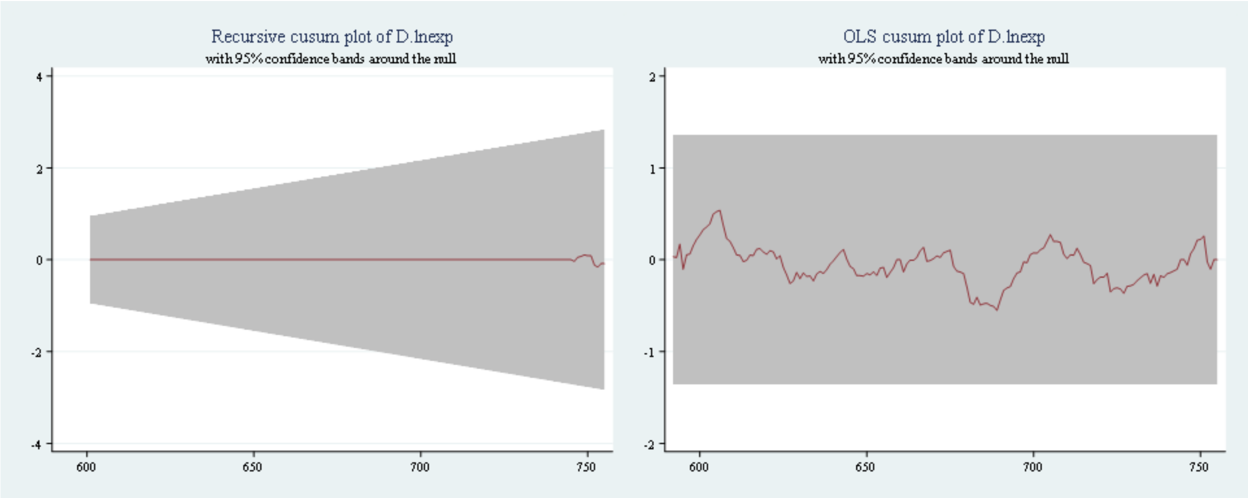
C. Exp 3102

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0542	1.1430	0.9479	0.850

```
.
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.5521	1.6276	1.3581	1.224



Lampiran 10 Estimasi Metode *Quantile Regression*

Republik Ceko

A. Exp 8703

```
. qreg lncze_8703 lnvol lnrer lnipi geopolitik, quantile(.1)

.1 Quantile regression                              Number of obs =        168
  Raw sum of deviations 12.52083 (about 12.215904)
  Min sum of deviations 11.24818                    Pseudo R2     =     0.1016
```

lncze_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.1154387	.2456432	-0.47	0.639	-.6004919	.3696145
lnrer	1.367962	1.538291	0.89	0.375	-1.669586	4.405509
lnipi	.7130912	.8820908	0.81	0.420	-1.028707	2.454889
geopolitik	1.047779	.4075008	2.57	0.011	.243118	1.85244
_cons	4.236135	6.396718	0.66	0.509	-8.394982	16.86725

```
. qreg lncze_8703 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression                             Number of obs =        168
  Raw sum of deviations 21.86246 (about 12.478475)
  Min sum of deviations 19.76733                    Pseudo R2     =     0.0958
```

lncze_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0058251	.1336751	-0.04	0.965	-.2697832	.258133
lnrer	-.3419566	.837113	-0.41	0.683	-1.994941	1.311027
lnipi	1.046439	.4800195	2.18	0.031	.0985805	1.994297
geopolitik	.547676	.2217553	2.47	0.015	.1097925	.9855595
_cons	8.676951	3.48099	2.49	0.014	1.803303	15.5506

```
. qreg lncze_8703 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression                                   Number of obs =        168
  Raw sum of deviations 25.16603 (about 12.79714)
  Min sum of deviations 23.21905                    Pseudo R2     =     0.0774
```

lncze_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.1992922	.1584193	1.26	0.210	-.1135265	.5121108
lnrer	-.6213484	.9920688	-0.63	0.532	-2.580312	1.337615
lnipi	.2869068	.5688747	0.50	0.615	-.8364071	1.410221
geopolitik	.2815519	.2628039	1.07	0.286	-.2373871	.8004909
_cons	13.8823	4.125347	3.37	0.001	5.736293	22.02832

```
. qreg lncze_8703 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression                             Number of obs =        168
  Raw sum of deviations 17.92574 (about 13.040268)
  Min sum of deviations 15.35722                    Pseudo R2     =     0.1433
```

lncze_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0649407	.0960767	0.68	0.500	-.1247746	.2546561
lnrer	-2.451789	.6016606	-4.08	0.000	-3.639843	-1.263736
lnipi	-.2821958	.3450058	-0.82	0.415	-.9634527	.3990611
geopolitik	-.1105086	.1593828	-0.69	0.489	-.4252299	.2042127
_cons	22.40278	2.501902	8.95	0.000	17.46246	27.3431

```
. qreg lncze_8703 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression                              Number of obs =        168
  Raw sum of deviations 8.958678 (about 13.198598)
  Min sum of deviations 7.073444                    Pseudo R2     =     0.2104
```

lncze_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0715273	.0657751	1.09	0.278	-.0583539	.2014085
lnrer	-1.728825	.4119034	-4.20	0.000	-2.54218	-.9154707
lnipi	-.576866	.2361947	-2.44	0.016	-1.043262	-.1104701
geopolitik	-.0290131	.1091152	-0.27	0.791	-.2444747	.1864485
_cons	21.54232	1.712829	12.58	0.000	18.16013	24.92452

B. Exp 8517

```
. qreg lncze_8517 lnvol lnrer lnipi geopolitik, quantile(.1)

.1 Quantile regression                              Number of obs =        168
  Raw sum of deviations 27.63098 (about 10.396505)
  Min sum of deviations 21.25539                    Pseudo R2     =     0.2307
```

lncze_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.1247136	.2889592	0.43	0.667	-.4458723	.6952995
lnrer	2.12189	1.809548	1.17	0.243	-1.451288	5.695069
lnipi	4.354359	1.037636	4.20	0.000	2.305417	6.4033
geopolitik	2.053117	.4793582	4.28	0.000	1.106565	2.999669
_cons	-16.33448	7.524694	-2.17	0.031	-31.19292	-1.476033

```
. qreg lncze_8517 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression                             Number of obs =        168
  Raw sum of deviations  50.6594 (about 11.161622)
  Min sum of deviations 41.28515                    Pseudo R2     =     0.1850
```

lncze_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0283606	.2885763	0.10	0.922	-.5414692	.5981904
lnrer	-.7468673	1.80715	-0.41	0.680	-4.315311	2.821576
lnipi	4.365246	1.036261	4.21	0.000	2.31902	6.411472
geopolitik	1.262061	.478723	2.64	0.009	.3167625	2.207359
_cons	-6.832816	7.514723	-0.91	0.365	-21.67157	8.005941

```
. qreg lncze_8517 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression                                   Number of obs =        168
  Raw sum of deviations 63.24829 (about 11.700856)
  Min sum of deviations 53.49607                    Pseudo R2     =     0.1542
```

lncze_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.5974209	.3252771	1.84	0.068	-.0448792	1.239721
lnrer	-5.534367	2.036982	-2.72	0.007	-9.556641	-1.512092
lnipi	3.248652	1.168051	2.78	0.006	.9421888	5.555115
geopolitik	-.2068843	.5396064	-0.38	0.702	-1.272404	.8586358
_cons	15.7251	8.470437	1.86	0.065	-1.000829	32.45104

```
. qreg lncze_8517 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression                             Number of obs =        168
  Raw sum of deviations 47.49243 (about 12.454592)
  Min sum of deviations 40.35544                    Pseudo R2     =     0.1503
```

lncze_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.291667	.2658899	1.10	0.274	-.2333658	.8166998
lnrer	-7.035991	1.665082	-4.23	0.000	-10.3239	-3.748079
lnipi	.7814845	.9547955	0.82	0.414	-1.103878	2.666847
geopolitik	-.6354681	.4410883	-1.44	0.152	-1.506452	.2355158
_cons	31.99792	6.923955	4.62	0.000	18.32571	45.67014

```
. qreg lncze_8517 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression                              Number of obs =        168
  Raw sum of deviations 23.83392 (about 12.897726)
  Min sum of deviations 21.53877                    Pseudo R2     =     0.0963
```

lncze_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.2307228	.3671609	0.63	0.531	-.4942821	.9557278
lnrer	-5.362166	2.299271	-2.33	0.021	-9.902363	-.8219691
lnipi	.3856076	1.318454	0.29	0.770	-2.217844	2.989059
geopolitik	-.7266879	.6090881	-1.19	0.235	-1.929408	.4760323
_cons	28.74061	9.56112	3.01	0.003	9.860981	47.62023

C. Exp 8708

```
. qreg lncze_8708 lnvol lnrer lnipi geopolitik, quantile(.1)

.1 Quantile regression                              Number of obs =        168
  Raw sum of deviations 9.451193 (about 12.626252)
  Min sum of deviations  8.23699                    Pseudo R2     =     0.1285
```

lncze_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0162229	.1634887	0.10	0.921	-.3066058	.3390517
lnrer	2.268569	1.023815	2.22	0.028	.2469197	4.290219
lnipi	-.6541345	.5870785	-1.11	0.267	-1.813394	.5051251
geopolitik	.7896889	.2712135	2.91	0.004	.254144	1.325234
_cons	8.348771	4.257357	1.96	0.052	-.0579105	16.75545

```
. qreg lncze_8708 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression                             Number of obs =        168
  Raw sum of deviations 14.81247 (about 12.831393)
  Min sum of deviations 14.11448                    Pseudo R2     =     0.0471
```

lncze_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0297925	.1091376	-0.27	0.785	-.2452984	.1857134
lnrer	1.09972	.6834524	1.61	0.110	-.2498419	2.449282
lnipi	-.1194243	.391907	-0.30	0.761	-.8932936	.654445
geopolitik	.5021361	.1810499	2.77	0.006	.1446305	.8596416
_cons	9.725435	2.842019	3.42	0.001	4.113515	15.33736

```
. qreg lncze_8708 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression                                   Number of obs =        168
  Raw sum of deviations  17.5078 (about 13.001716)
  Min sum of deviations 16.96364                    Pseudo R2     =     0.0311
```

lncze_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.22951	.1107171	2.07	0.040	.0108853	.4481348
lnrer	.0576161	.6933436	0.08	0.934	-1.311477	1.42671
lnipi	.5917557	.3975789	1.49	0.139	-.1933134	1.376825
geopolitik	.1473445	.1836701	0.80	0.424	-.215335	.510024
_cons	10.53758	2.88315	3.65	0.000	4.844439	16.23072

```
. qreg lncze_8708 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression                             Number of obs =        168
  Raw sum of deviations 11.98503 (about 13.178328)
  Min sum of deviations 11.46934                    Pseudo R2     =     0.0430
```

lncze_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0928318	.0579406	1.60	0.111	-.0215791	.2072428
lnrer	-.554198	.3628413	-1.53	0.129	-1.270673	.1622774
lnipi	.299528	.2080614	1.44	0.152	-.1113152	.7103712
geopolitik	.0057728	.0961184	0.06	0.952	-.1840251	.1955706
_cons	13.75933	1.508813	9.12	0.000	10.77999	16.73867

```
. qreg lncze_8708 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression                              Number of obs =        168
  Raw sum of deviations  5.79051 (about 13.264703)
  Min sum of deviations  5.39588                    Pseudo R2     =     0.0682
```

lncze_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0534467	.0643704	0.83	0.408	-.0736607	.1805541
lnrer	-.956628	.4031068	-2.37	0.019	-1.752613	-.1606435
lnipi	.0927714	.2311505	0.40	0.689	-.3636641	.5492069
geopolitik	-.0429814	.106785	-0.40	0.688	-.2538416	.1678788
_cons	16.03253	1.67625	9.56	0.000	12.72257	19.3425

Hungaria

A. Exp 8703

```
. qreg lnhun_8703 lnvol lnrer lnipi geopolitik, quantile(.1)

.1 Quantile regression                              Number of obs =        168
  Raw sum of deviations  16.1274 (about 11.567244)
  Min sum of deviations 12.89186                    Pseudo R2     =     0.2006
```

lnhun_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.044984	.1733153	-0.26	0.796	-.3872166	.2972487
lnrer	5.518929	1.355744	4.07	0.000	2.841844	8.196013
lnipi	.6977778	.8383526	0.83	0.406	-.9576539	2.353209
geopolitik	.6422103	.3225628	1.99	0.048	.0052699	1.279151
_cons	-22.8415	8.657287	-2.64	0.009	-39.9364	-5.746611

```
. qreg lnhun_8703 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression                             Number of obs =        168
  Raw sum of deviations 29.80272 (about 11.857614)
  Min sum of deviations 22.96491                    Pseudo R2     =     0.2294
```

lnhun_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.1823881	.133669	-1.36	0.174	-.4463342	.081558
lnrer	5.661786	1.045614	5.41	0.000	3.597091	7.726481
lnipi	.9776402	.6465774	1.51	0.132	-.2991075	2.254388
geopolitik	.345092	.2487758	1.39	0.167	-.1461468	.8363307
_cons	-24.15594	6.676912	-3.62	0.000	-37.34033	-10.97155

```
. qreg lnhun_8703 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression                                   Number of obs =        168
  Raw sum of deviations 37.21212 (about 12.383042)
  Min sum of deviations 26.51024                    Pseudo R2     =     0.2876
```

lnhun_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.1771819	.0978836	-1.81	0.072	-.3704654	.0161015
lnrer	5.493815	.765686	7.18	0.000	3.981873	7.005758
lnipi	.6813055	.4734781	1.44	0.152	-.2536359	1.616247
geopolitik	.2240947	.1821744	1.23	0.220	-.1356314	.5838208
_cons	-21.54284	4.889393	-4.41	0.000	-31.19756	-11.88813

```
. qreg lnhun_8703 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression                             Number of obs =        168
  Raw sum of deviations 24.24424 (about 12.704819)
  Min sum of deviations  19.3223                    Pseudo R2     =     0.2030
```

lnhun_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.1762899	.0984378	-1.79	0.075	-.3706676	.0180879
lnrer	4.964509	.7700212	6.45	0.000	3.444006	6.485012
lnipi	-.1142027	.4761589	-0.24	0.811	-1.054438	.8260323
geopolitik	.2507605	.1832059	1.37	0.173	-.1110024	.6125233
_cons	-14.5673	4.917077	-2.96	0.004	-24.27668	-4.857915

. qreg lnhun_8703 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression Number of obs = 168
Raw sum of deviations 11.56375 (about 12.822436)
Min sum of deviations 9.125518 Pseudo R2 = 0.2109

lnhun_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0595983	.0576276	1.03	0.303	-.0541945	.1733912
lnrer	2.228384	.4507866	4.94	0.000	1.33825	3.118518
lnipi	.3519969	.2787534	1.26	0.208	-.1984364	.9024302
geopolitik	.1470922	.1072526	1.37	0.172	-.0646913	.3588758
_cons	-1.762947	2.87856	-0.61	0.541	-7.447022	3.921128

B. Exp 8708

. qreg lnhun_8708 lnvol lnrer lnipi geopolitik, quantile(.1)

.1 Quantile regression Number of obs = 168
Raw sum of deviations 11.38425 (about 11.597845)
Min sum of deviations 8.271374 Pseudo R2 = 0.2734

lnhun_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.019077	.1113935	-0.17	0.864	-.2390375	.2008834
lnrer	4.401977	.8713661	5.05	0.000	2.681356	6.122598
lnipi	1.043362	.5388276	1.94	0.055	-.0206205	2.107344
geopolitik	-.064	.2073182	-0.31	0.758	-.4733756	.3453755
_cons	-18.06529	5.564228	-3.25	0.001	-29.05256	-7.078032

. qreg lnhun_8708 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression Number of obs = 168
Raw sum of deviations 18.56733 (about 11.895006)
Min sum of deviations 13.9187 Pseudo R2 = 0.2504

lnhun_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0556549	.0758559	-0.73	0.464	-.2054417	.094132
lnrer	4.021387	.5933759	6.78	0.000	2.849692	5.193081
lnipi	1.136638	.3669265	3.10	0.002	.4120957	1.86118
geopolitik	-.0389371	.1411779	-0.28	0.783	-.3177104	.2398361
_cons	-16.03526	3.789083	-4.23	0.000	-23.51727	-8.55324

. qreg lnhun_8708 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression Number of obs = 168
Raw sum of deviations 21.57858 (about 12.155495)
Min sum of deviations 15.1226 Pseudo R2 = 0.2992

lnhun_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.037955	.0529037	0.72	0.474	-.06651	.14242
lnrer	3.244921	.4138346	7.84	0.000	2.427753	4.062089
lnipi	.3541952	.2559033	1.38	0.168	-.1511178	.8595082
geopolitik	-.0959848	.0984608	-0.97	0.331	-.2904079	.0984384
_cons	-8.086056	2.642598	-3.06	0.003	-13.30419	-2.867918

. qreg lnhun_8708 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression Number of obs = 168
Raw sum of deviations 13.82963 (about 12.335894)
Min sum of deviations 10.59234 Pseudo R2 = 0.2341

lnhun_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0102139	.0490818	-0.21	0.835	-.1071321	.0867042
lnrer	3.011971	.3839382	7.84	0.000	2.253837	3.770105
lnipi	.2187955	.2374163	0.92	0.358	-.2500125	.6876035
geopolitik	-.0561148	.0913478	-0.61	0.540	-.2364924	.1242628
_cons	-5.872211	2.45169	-2.40	0.018	-10.71338	-1.031043

. qreg lnhun_8708 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression Number of obs = 168
Raw sum of deviations 6.331562 (about 12.413359)
Min sum of deviations 5.248991 Pseudo R2 = 0.1710

lnhun_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0601893	.0452158	1.33	0.185	-.0290951	.1494736
lnrer	2.862837	.3536969	8.09	0.000	2.164419	3.561256
lnipi	-.0408613	.2187159	-0.19	0.852	-.4727432	.3910206
geopolitik	-.0781354	.0841527	-0.93	0.355	-.2443053	.0880346
_cons	-3.94425	2.25858	-1.75	0.083	-8.404099	.5155979

C. Exp 8507

. qreg lnhun_8507 lnvol lnrer lnipi geopolitik, quantile(.1)

.1 Quantile regression Number of obs = 168
Raw sum of deviations 29.37132 (about 7.9707403)
Min sum of deviations 24.72994 Pseudo R2 = 0.1580

lnhun_8507	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0180804	.2850055	0.06	0.949	-.5446984	.5808592
lnrer	2.850566	2.22943	1.28	0.203	-1.551721	7.252854
lnipi	-.2651529	1.378615	-0.19	0.848	-2.9874	2.457094
geopolitik	3.166889	.530433	5.97	0.000	2.119483	4.214296
_cons	-7.017932	14.23633	-0.49	0.623	-35.12934	21.09348

. qreg lnhun_8507 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression Number of obs = 168
Raw sum of deviations 61.08004 (about 8.1803207)
Min sum of deviations 49.66933 Pseudo R2 = 0.1868

lnhun_8507	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0624677	.2455157	0.25	0.799	-.4223336	.5472691
lnrer	5.95849	1.920525	3.10	0.002	2.166174	9.750805
lnipi	-1.344736	1.187597	-1.13	0.259	-3.689794	1.000322
geopolitik	2.58592	.4569373	5.66	0.000	1.683641	3.4882
_cons	-19.51251	12.26377	-1.59	0.114	-43.72886	4.703847

. qreg lnhun_8507 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression — Number of obs = 168
Raw sum of deviations 100.6255 (about 8.7660828)
Min sum of deviations 64.83465 — Pseudo R2 = 0.3557

lnhun_8507	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.253373	.316568	0.80	0.425	-.37173	.8784761
lnrer	15.56917	2.476325	6.29	0.000	10.67936	20.45898
lnipi	-2.468109	1.531288	-1.61	0.109	-5.491828	.5556104
geopolitik	.5559962	.5891751	0.94	0.347	-.6074035	1.719396
_cons	-68.54788	15.81292	-4.33	0.000	-99.77246	-37.32331

. qreg lnhun_8507 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression — Number of obs = 168
Raw sum of deviations 87.04545 (about 10.426113)
Min sum of deviations 48.04734 — Pseudo R2 = 0.4480

lnhun_8507	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.5798857	.2717735	2.13	0.034	.0432351	1.116536
lnrer	15.29791	2.125924	7.20	0.000	11.10001	19.49581
lnipi	-3.585557	1.31461	-2.73	0.007	-6.181419	-.9896959
geopolitik	1.096173	.5058066	2.17	0.032	.097395	2.094951
_cons	-62.2381	13.57538	-4.58	0.000	-89.04439	-35.43182

. qreg lnhun_8507 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression — Number of obs = 168
Raw sum of deviations 47.37192 (about 11.560886)
Min sum of deviations 24.71071 — Pseudo R2 = 0.4784

lnhun_8507	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	1.415458	.2203656	6.42	0.000	.9803192	1.850598
lnrer	17.28385	1.72379	10.03	0.000	13.88001	20.68769
lnipi	-5.247672	1.065942	-4.92	0.000	-7.352507	-3.142836
geopolitik	.7590857	.4101296	1.85	0.066	-.0507663	1.568938
_cons	-68.01671	11.0075	-6.18	0.000	-89.75239	-46.28103

Polandia

A. Exp 8708

. qreg lnpol_8708 lnvol lnrer lnipi geopolitik, quantile(.1)

.1 Quantile regression — Number of obs = 168
Raw sum of deviations 9.457293 (about 12.331846)
Min sum of deviations 7.462722 — Pseudo R2 = 0.2109

lnpol_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0380406	.0344072	-1.11	0.271	-.105982	.0299007
lnrer	3.972107	.808956	4.91	0.000	2.374723	5.569492
lnipi	.608958	.3471442	1.75	0.081	-.0765214	1.294437
geopolitik	.270351	.123853	2.18	0.030	.0257879	.5149141
_cons	3.724551	2.055953	1.81	0.072	-.335185	7.784287

. qreg lnpol_8708 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression — Number of obs = 168
Raw sum of deviations 15.35194 (about 12.526121)
Min sum of deviations 12.49997 — Pseudo R2 = 0.1858

lnpol_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0551847	.0360804	-1.53	0.128	-.1264299	.0160606
lnrer	4.07057	.848295	4.80	0.000	2.395506	5.745634
lnipi	.7090158	.3640256	1.95	0.053	-.0097981	1.42783
geopolitik	.1266738	.1298758	0.98	0.331	-.1297822	.3831298
_cons	3.201895	2.155933	1.49	0.139	-1.055263	7.459053

. qreg lnpol_8708 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression — Number of obs = 168
Raw sum of deviations 17.78199 (about 12.732921)
Min sum of deviations 13.98003 — Pseudo R2 = 0.2138

lnpol_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0342379	.0218423	-1.57	0.119	-.0773684	.0088925
lnrer	3.340503	.5135407	6.50	0.000	2.326453	4.354553
lnipi	.3003515	.2203738	1.36	0.175	-.134804	.7355069
geopolitik	.2348302	.0786242	2.99	0.003	.0795769	.3900835
_cons	6.407191	1.305159	4.91	0.000	3.829993	8.98439

. qreg lnpol_8708 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression — Number of obs = 168
Raw sum of deviations 12.49005 (about 12.899119)
Min sum of deviations 10.08596 — Pseudo R2 = 0.1925

lnpol_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0220413	.0210104	-1.05	0.296	-.0635289	.0194464
lnrer	3.558248	.4939804	7.20	0.000	2.582822	4.533673
lnipi	.1253121	.2119799	0.59	0.555	-.2932686	.5438928
geopolitik	.3103998	.0756295	4.10	0.000	.1610599	.4597397
_cons	7.059731	1.255446	5.62	0.000	4.580696	9.538765

. qreg lnpol_8708 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression — Number of obs = 168
Raw sum of deviations 6.067881 (about 12.991439)
Min sum of deviations 5.255597 — Pseudo R2 = 0.1339

lnpol_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0336835	.0267527	-1.26	0.210	-.08651	.019143
lnrer	4.001937	.6289888	6.36	0.000	2.759921	5.243954
lnipi	.3383133	.2699155	1.25	0.212	-.1946686	.8712952
geopolitik	.2390034	.0962996	2.48	0.014	.0488479	.4291589
_cons	5.475786	1.598569	3.43	0.001	2.319214	8.632359

B. Exp 8507

.1 Quantile regression — Number of obs = 168
Raw sum of deviations 19.89747 (about 8.8724871)
Min sum of deviations 13.28922 — Pseudo R2 = 0.3321

lnpol_8507	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
lnvol	-.0979778	.046049	-2.13	0.035	-.1889072	-.0070484
lnrer	6.154129	1.082668	5.68	0.000	4.016265	8.291992
lnipi	.3909124	.4646012	0.84	0.401	-.5265007	1.308325
geopolitik	2.761479	.1657589	16.66	0.000	2.434168	3.088791
_cons	-2.082198	2.75159	-0.76	0.450	-7.515556	3.351159

. qreg lnpol_8507 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression — Number of obs = 168
Raw sum of deviations 39.44363 (about 9.0707331)
Min sum of deviations 27.75802 — Pseudo R2 = 0.2963

lnpol_8507	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
lnvol	-.0940014	.0463462	-2.03	0.044	-.1855177	-.0024852
lnrer	4.026598	1.089656	3.70	0.000	1.874937	6.178259
lnipi	.4266171	.4675998	0.91	0.363	-.496717	1.349951
geopolitik	2.994614	.1668287	17.95	0.000	2.66519	3.324038
_cons	.9633704	2.769349	0.35	0.728	-4.505054	6.431795

. qreg lnpol_8507 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression — Number of obs = 168
Raw sum of deviations 64.69402 (about 9.2921972)
Min sum of deviations 41.36429 — Pseudo R2 = 0.3606

lnpol_8507	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
lnvol	.025305	.0746361	0.34	0.735	-.1220732	.1726833
lnrer	10.1764	1.754787	5.80	0.000	6.711357	13.64145
lnipi	-.4031468	.7530252	-0.54	0.593	-1.890089	1.083795
geopolitik	3.371648	.2686619	12.55	0.000	2.841141	3.902154
_cons	-3.166396	4.459774	-0.71	0.479	-11.97278	5.639984

. qreg lnpol_8507 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression — Number of obs = 168
Raw sum of deviations 73.75972 (about 9.8877134)
Min sum of deviations 38.30868 — Pseudo R2 = 0.4806

lnpol_8507	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
lnvol	.1453418	.1074686	1.35	0.178	-.0668683	.3575519
lnrer	16.93049	2.52672	6.70	0.000	11.94117	21.91981
lnipi	-2.788053	1.084282	-2.57	0.011	-4.929102	-.6470043
geopolitik	3.383403	.3868464	8.75	0.000	2.619526	4.147279
_cons	-.7285312	6.421632	-0.11	0.910	-13.40884	11.95178

. qreg lnpol_8507 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression — Number of obs = 168
Raw sum of deviations 50.27853 (about 12.045605)
Min sum of deviations 23.31064 — Pseudo R2 = 0.5364

lnpol_8507	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
lnvol	.2723794	.1905157	1.43	0.155	-.1038176	.6485763
lnrer	20.44505	4.47926	4.56	0.000	11.6002	29.28991
lnipi	-5.109164	1.922167	-2.66	0.009	-8.904723	-1.313605
geopolitik	3.025184	.6857846	4.41	0.000	1.671017	4.379351
_cons	6.161553	11.38399	0.54	0.589	-16.31756	28.64066

C. Exp 8703

.1 Quantile regression — Number of obs = 168
Raw sum of deviations 8.859938 (about 11.511435)
Min sum of deviations 8.088056 — Pseudo R2 = 0.0871

lnpol_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
lnvol	-.0135765	.0410675	-0.33	0.741	-.0946693	.0675163
lnrer	1.434755	.9655472	1.49	0.139	-.4718382	3.341348
lnipi	-.6443392	.4143416	-1.56	0.122	-1.462508	.1738299
geopolitik	.4693533	.1478274	3.18	0.002	.1774496	.761257
_cons	12.46706	2.453928	5.08	0.000	7.621476	17.31265

.25 Quantile regression — Number of obs = 168
Raw sum of deviations 14.24769 (about 11.756922)
Min sum of deviations 13.40584 — Pseudo R2 = 0.0591

lnpol_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
lnvol	.0127287	.0321173	0.40	0.692	-.0506908	.0761482
lnrer	-.0004444	.7551165	-0.00	1.000	-1.491516	1.490627
lnipi	-.0797566	.3240403	-0.25	0.806	-.7196144	.5601013
geopolitik	.3620264	.11561	3.13	0.002	.13374	.5903127
_cons	12.18082	1.919121	6.35	0.000	8.391276	15.97036

. qreg lnpol_8703 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression — Number of obs = 168
Raw sum of deviations 17.10483 (about 11.916135)
Min sum of deviations 15.88568 — Pseudo R2 = 0.0713

lnpol_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
lnvol	.0328155	.0259178	1.27	0.207	-.0183624	.0839934
lnrer	-.6330104	.6093593	-1.04	0.300	-1.836266	.5702455
lnipi	-.1594416	.261492	-0.61	0.543	-.6757902	.356907
geopolitik	.2404024	.0932943	2.58	0.011	.0561813	.4246236
_cons	13.69318	1.54868	8.84	0.000	10.63511	16.75124

. qreg lnpol_8703 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression — Number of obs = 168
Raw sum of deviations 13.54068 (about 12.064756)
Min sum of deviations 12.27161 — Pseudo R2 = 0.0937

lnpol_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
lnvol	.0887787	.0265151	3.35	0.001	.0364214	.141136
lnrer	-.6553425	.623402	-1.05	0.295	-1.886328	.5756426
lnipi	.0440929	.2675181	0.16	0.869	-.484155	.5723408
geopolitik	.2022988	.0954442	2.12	0.036	.0138322	.3907653
_cons	13.16677	1.58437	8.31	0.000	10.03823	16.2953

. qreg lnpol_8703 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression Number of obs = 168
Raw sum of deviations 7.609524 (about 12.201035)
Min sum of deviations 6.40645 Pseudo R2 = 0.1581

lnpol_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.1128262	.0330728	3.41	0.001	.0475198	.1781326
lnrer	-2.057283	.7775833	-2.65	0.009	-3.592718	-.5218476
lnipi	.164267	.3336813	0.49	0.623	-.4946284	.8231623
geopolitik	.1134323	.1190497	0.95	0.342	-.1216462	.3485108
_cons	14.84881	1.97622	7.51	0.000	10.94652	18.7511

Slovakia

A. Exp 8703

. qreg lnsvk_8703 lnvol lnrer lnipi geopolitik, quantile(.1)

.1 Quantile regression Number of obs = 168
Raw sum of deviations 25.76025 (about 10.479651)
Min sum of deviations 22.76614 Pseudo R2 = 0.1162

lnsvk_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.4058016	.2265284	-1.79	0.075	-.8531102	.0415071
lnrer	4.909736	2.387266	2.06	0.041	.1957817	9.62369
lnipi	2.088856	1.638775	1.27	0.204	-1.147109	5.324821
geopolitik	.9757144	.5622706	1.74	0.085	-.134559	2.085988
_cons	-3.233525	7.396359	-0.44	0.663	-17.83856	11.37151

. qreg lnsvk_8703 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression Number of obs = 168
Raw sum of deviations 41.48083 (about 11.191341)
Min sum of deviations 37.1287 Pseudo R2 = 0.1049

lnsvk_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.2364012	.1687243	1.40	0.163	-.096766	.5695683
lnrer	2.562892	1.778098	1.44	0.151	-.9481839	6.073968
lnipi	1.509999	1.220602	1.24	0.218	-.9002318	3.920231
geopolitik	.6288029	.4187938	1.50	0.135	-.1981577	1.455763
_cons	5.501262	5.509001	1.00	0.319	-5.376948	16.37947

. qreg lnsvk_8703 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression Number of obs = 168
Raw sum of deviations 43.18541 (about 11.732077)
Min sum of deviations 41.53753 Pseudo R2 = 0.0382

lnsvk_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.1006059	.1048626	0.96	0.339	-.1064584	.3076703
lnrer	.9134309	1.105093	0.83	0.410	-1.268713	3.095575
lnipi	1.091803	.7586079	1.44	0.152	-.406163	2.589769
geopolitik	.2968712	.2602816	1.14	0.256	-.2170873	.8108296
_cons	7.185483	3.423861	2.10	0.037	.4246442	13.94632

. qreg lnsvk_8703 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression Number of obs = 168
Raw sum of deviations 29.11416 (about 12.044413)
Min sum of deviations 28.67284 Pseudo R2 = 0.0152

lnsvk_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0928893	.0766173	1.21	0.227	-.0584011	.2441798
lnrer	-.539615	.8074303	-0.67	0.505	-2.133987	1.054757
lnipi	.5830549	.5542727	1.05	0.294	-.5114256	1.677536
geopolitik	.2233211	.1901733	1.17	0.242	-.1521999	.5988421
_cons	10.10713	2.501625	4.04	0.000	5.167363	15.0469

. qreg lnsvk_8703 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression Number of obs = 168
Raw sum of deviations 14.07751 (about 12.272015)
Min sum of deviations 13.7675 Pseudo R2 = 0.0220

lnsvk_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0004416	.0740442	-0.01	0.995	-.1466511	.1457679
lnrer	-.3791551	.7803136	-0.49	0.628	-1.919982	1.161671
lnipi	-.1810756	.5356581	-0.34	0.736	-1.238799	.876648
geopolitik	.2193305	.1837866	1.19	0.234	-.143579	.58224
_cons	13.14603	2.417611	5.44	0.000	8.372153	17.9199

B. Exp 8708

. qreg lnsvk_8708 lnvol lnrer lnipi geopolitik, quantile(.1)

.1 Quantile regression Number of obs = 168
Raw sum of deviations 22.1576 (about 8.1565104)
Min sum of deviations 19.0654 Pseudo R2 = 0.1396

lnsvk_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.4941708	.2253552	-2.19	0.030	-.9391627	-.0491789
lnrer	.6191583	2.374902	0.26	0.795	-4.070381	5.308698
lnipi	2.364851	1.630287	1.45	0.149	-.8543539	5.584056
geopolitik	.6437888	.5593585	1.15	0.251	-.4607342	1.748312
_cons	-6.508421	7.358052	-0.88	0.378	-21.03781	8.020968

. qreg lnsvk_8708 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression Number of obs = 168
Raw sum of deviations 35.38946 (about 8.8953552)
Min sum of deviations 31.74096 Pseudo R2 = 0.1031

lnsvk_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.301366	.1297248	-2.32	0.021	-.5575238	-.0452082
lnrer	.6090128	1.367103	0.45	0.657	-2.090502	3.308527
lnipi	1.78642	.9384682	1.90	0.059	-.0667023	3.639543
geopolitik	.7428491	.3219925	2.31	0.022	.1070348	1.378663
_cons	-1.930493	4.235633	-0.46	0.649	-10.29428	6.433292

. qreg lnsvk_8708 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression Number of obs = 168
Raw sum of deviations 38.97015 (about 9.2642603)
Min sum of deviations 34.74975 Pseudo R2 = 0.1083

lnsvk_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.1680366	.0830258	-2.02	0.045	-.3319814	-.0040918
lnrer	-1.188224	.8749658	-1.36	0.176	-2.915953	.5395046
lnipi	-.3424118	.6006335	-0.57	0.569	-1.528437	.8436139
geopolitik	.3271753	.2060799	1.59	0.114	-.0797552	.7341057
_cons	9.810535	2.710867	3.62	0.000	4.45759	15.16348

. qreg lnsvk_8708 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression Number of obs = 168
Raw sum of deviations 27.59906 (about 9.6356735)
Min sum of deviations 24.01486 Pseudo R2 = 0.1299

lnsvk_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.1931528	.0723015	-2.67	0.008	-.3359212	-.0503845
lnrer	-2.433983	.7619481	-3.19	0.002	-3.938544	-.9294212
lnipi	-.1295072	.5230508	-0.25	0.805	-1.162336	.9033218
geopolitik	.1721282	.179461	0.96	0.339	-.1822398	.5264962
_cons	9.096173	2.36071	3.85	0.000	4.434657	13.75769

. qreg lnsvk_8708 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression Number of obs = 168
Raw sum of deviations 12.87654 (about 9.8627701)
Min sum of deviations 11.16321 Pseudo R2 = 0.1331

lnsvk_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.1805268	.0414005	-4.36	0.000	-.2622773	-.0987763
lnrer	-2.498843	.4362987	-5.73	0.000	-3.360369	-1.637317
lnipi	-.7149081	.2995038	-2.39	0.018	-1.306316	-.1235004
geopolitik	.1506984	.102761	1.47	0.144	-.0522161	.3536129
_cons	12.14658	1.351765	8.99	0.000	9.477347	14.8158

C. Exp 8528

. qreg lnsvk_8528 lnvol lnrer lnipi geopolitik, quantile(.1)

.1 Quantile regression Number of obs = 168
Raw sum of deviations 13.54832 (about 10.3708)
Min sum of deviations 12.8198 Pseudo R2 = 0.0538

lnsvk_8528	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.05561	.1075367	0.52	0.606	-.1567346	.2679546
lnrer	1.805711	1.133274	1.59	0.113	-.4320792	4.043501
lnipi	-.1569667	.7779528	-0.20	0.840	-1.693131	1.379198
geopolitik	-.0566518	.2669189	-0.21	0.832	-.5837165	.4704129
_cons	11.26204	3.511171	3.21	0.002	4.328794	18.19528

. qreg lnsvk_8528 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression Number of obs = 168
Raw sum of deviations 24.70941 (about 10.682927)
Min sum of deviations 24.02911 Pseudo R2 = 0.0275

lnsvk_8528	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0093789	.078008	0.12	0.904	-.1446576	.1634154
lnrer	1.253784	.8220857	1.53	0.129	-.3695268	2.877094
lnipi	-.3910886	.5643331	-0.69	0.489	-1.505435	.7232576
geopolitik	-.2710499	.1936251	-1.40	0.163	-.6533868	.111287
_cons	12.34593	2.547031	4.85	0.000	7.316502	17.37536

. qreg lnsvk_8528 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression Number of obs = 168
Raw sum of deviations 31.48608 (about 10.964709)
Min sum of deviations 30.23438 Pseudo R2 = 0.0398

lnsvk_8528	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0716207	.0833556	0.86	0.391	-.0929754	.2362167
lnrer	-.8476549	.8784417	-0.96	0.336	-2.582248	.8869378
lnipi	-.683287	.6030196	-1.13	0.259	-1.874024	.5074504
geopolitik	-.402904	.2068986	-1.95	0.053	-.811451	.0056431
_cons	14.91726	2.721637	5.48	0.000	9.543046	20.29147

. qreg lnsvk_8528 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression Number of obs = 168
Raw sum of deviations 24.82664 (about 11.343998)
Min sum of deviations 23.48406 Pseudo R2 = 0.0541

lnsvk_8528	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.1037094	.0748024	1.39	0.168	-.0439972	.251416
lnrer	-1.305354	.7883033	-1.66	0.100	-2.861957	.2512489
lnipi	.0441558	.5411427	0.08	0.935	-1.024398	1.11271
geopolitik	-.5039323	.1856684	-2.71	0.007	-.8705577	-.137307
_cons	12.16125	2.442365	4.98	0.000	7.338501	16.98401

```
. qreg lnsvk_8528 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression                          Number of obs =        168
  Raw sum of deviations 13.76338 (about 11.557278)
  Min sum of deviations 12.47273                Pseudo R2     =     0.0938
```

lnsvk_8528	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.2005106	.1008525	1.99	0.048	.0013648	.3996563
lnrer	.135914	1.062832	0.13	0.898	-1.96278	2.234608
lnipi	-1.284567	.729597	-1.76	0.080	-2.725247	.1561137
geopolitik	-.37183	.2503279	-1.49	0.139	-.8661336	.1224735
_cons	19.09533	3.292925	5.80	0.000	12.59304	25.59762

Slovenia

A. Exp 3004

```
. qreg lnsvn_3004 lnvol lnrer lnipi geopolitik, quantile(.1)

.1 Quantile regression                          Number of obs =        168
  Raw sum of deviations 21.50257 (about 8.677269)
  Min sum of deviations 17.53547                Pseudo R2     =     0.1845
```

lnsvn_3004	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0631474	.1530077	-0.41	0.680	-.3652801	.2389854
lnrer	30.00384	8.536395	3.51	0.001	13.14766	46.86002
lnipi	.4699417	1.151358	0.41	0.684	-1.803559	2.743443
geopolitik	.7068439	.41133	1.72	0.088	-.1053785	1.519066
_cons	5.647302	5.483999	1.03	0.305	-5.181538	16.47614

```
. qreg lnsvn_3004 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression                         Number of obs =        168
  Raw sum of deviations 32.28579 (about 9.3832846)
  Min sum of deviations 29.04843                Pseudo R2     =     0.1003
```

lnsvn_3004	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0305707	.1371008	-0.22	0.824	-.3012933	.240152
lnrer	19.10469	7.648942	2.50	0.013	4.000904	34.20848
lnipi	.8376821	1.031662	0.81	0.418	-1.199463	2.874827
geopolitik	.2076725	.3685677	0.56	0.574	-.5201103	.9354553
_cons	4.964792	4.913876	1.01	0.314	-4.738269	14.66785

```
. qreg lnsvn_3004 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression                               Number of obs =        168
  Raw sum of deviations 31.07592 (about 9.6484661)
  Min sum of deviations 29.10483                Pseudo R2     =     0.0634
```

lnsvn_3004	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0348504	.0463467	0.75	0.453	-.056667	.1263677
lnrer	11.13784	2.585712	4.31	0.000	6.032031	16.24365
lnipi	.1687788	.3487516	0.48	0.629	-.5198747	.8574323
geopolitik	.1824348	.1245937	1.46	0.145	-.063591	.4284605
_cons	9.012805	1.661128	5.43	0.000	5.732701	12.29291

```
. qreg lnsvn_3004 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression                         Number of obs =        168
  Raw sum of deviations 21.26773 (about 9.8444271)
  Min sum of deviations  20.2638                Pseudo R2     =     0.0472
```

lnsvn_3004	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.076805	.0543783	1.41	0.160	-.0305717	.1841818
lnrer	7.276602	3.033799	2.40	0.018	1.285987	13.26722
lnipi	.1158513	.409188	0.28	0.777	-.6921414	.9238439
geopolitik	.1837591	.146185	1.26	0.211	-.1049014	.4724195
_cons	9.912624	1.94899	5.09	0.000	6.0641	13.76115

```
. qreg lnsvn_3004 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression                          Number of obs =        168
  Raw sum of deviations 10.74339 (about 10.026722)
  Min sum of deviations 10.05764                Pseudo R2     =     0.0638
```

lnsvn_3004	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0330605	.0649574	-0.51	0.611	-.161327	.0952059
lnrer	5.535	3.624013	1.53	0.129	-1.621066	12.69107
lnipi	-.0722577	.488794	-0.15	0.883	-1.037442	.8929269
geopolitik	.4580083	.1746247	2.62	0.010	.11319	.8028266
_cons	9.978466	2.328159	4.29	0.000	5.381226	14.57571

B. Exp 8703

```
. qreg lnsvn_8703 lnvol lnrer lnipi geopolitik, quantile(.1)

.1 Quantile regression                          Number of obs =        168
  Raw sum of deviations 13.42781 (about 10.540566)
  Min sum of deviations 11.52743                Pseudo R2     =     0.1415
```

lnsvn_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.1213938	.0779889	1.56	0.122	-.032605	.2753925
lnrer	18.40084	4.351049	4.23	0.000	9.809149	26.99252
lnipi	-1.090509	.5868539	-1.86	0.065	-2.249325	.0683073
geopolitik	.0490827	.2096572	0.23	0.815	-.3649117	.463077
_cons	16.63646	2.795225	5.95	0.000	11.11694	22.15598

```
. qreg lnsvn_8703 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression                         Number of obs =        168
  Raw sum of deviations 23.68958 (about 10.89992)
  Min sum of deviations 20.82318                Pseudo R2     =     0.1210
```

lnsvn_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0233666	.0819727	0.29	0.776	-.1384987	.1852319
lnrer	13.34444	4.573308	2.92	0.004	4.313878	22.37501
lnipi	-.4866673	.6168314	-0.79	0.431	-1.704678	.7313432
geopolitik	.6680638	.2203669	3.03	0.003	.2329219	1.103206
_cons	13.26642	2.93801	4.52	0.000	7.464948	19.06788

. qreg lnsvn_8703 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression Number of obs = 168
Raw sum of deviations 29.77498 (about 11.196103)
Min sum of deviations 25.62496 Pseudo R2 = 0.1394

lnsvn_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0001344	.0562642	-0.00	0.998	-.1112351	.1109663
lnrer	10.04505	3.139015	3.20	0.002	3.84667	16.24342
lnipi	-.2722347	.4233792	-0.64	0.521	-1.10825	.5637802
geopolitik	.6147991	.1512549	4.06	0.000	.3161276	.9134707
_cons	12.34094	2.016584	6.12	0.000	8.358944	16.32294

. qreg lnsvn_8703 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression Number of obs = 168
Raw sum of deviations 24.50669 (about 11.474684)
Min sum of deviations 20.7106 Pseudo R2 = 0.1549

lnsvn_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0608197	.0559297	-1.09	0.278	-.1712599	.0496206
lnrer	13.31343	3.120356	4.27	0.000	7.1519	19.47496
lnipi	-1.001981	.4208624	-2.38	0.018	-1.833026	-.1709352
geopolitik	.521913	.1503558	3.47	0.001	.2250169	.8188092
_cons	15.42701	2.004597	7.70	0.000	11.46868	19.38533

. qreg lnsvn_8703 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression Number of obs = 168
Raw sum of deviations 13.62611 (about 11.77807)
Min sum of deviations 11.94395 Pseudo R2 = 0.1235

lnsvn_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0761436	.1167987	-0.65	0.515	-.3067773	.1544901
lnrer	7.820885	6.516276	1.20	0.232	-5.046313	20.68808
lnipi	-1.627878	.878892	-1.85	0.066	-3.363359	.1076042
geopolitik	.3854668	.3139897	1.23	0.221	-.2345449	1.005479
_cons	18.43296	4.186223	4.40	0.000	10.16674	26.69918

C. Exp 8516

. qreg lnsvn_8516 lnvol lnrer lnipi geopolitik, quantile(.1)

.1 Quantile regression Number of obs = 168
Raw sum of deviations 8.797883 (about 9.2408695)
Min sum of deviations 7.734358 Pseudo R2 = 0.1209

lnsvn_8516	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.1025244	.0904233	1.13	0.259	-.0760276	.2810764
lnrer	-4.846222	5.044771	-0.96	0.338	-14.80775	5.115306
lnipi	-.9354292	.6804207	-1.37	0.171	-2.279005	.4081462
geopolitik	.2709973	.2430845	1.11	0.267	-.2090034	.7509981
_cons	14.62104	3.24089	4.51	0.000	8.221496	21.02058

. qreg lnsvn_8516 lnvol lnrer lnipi geopolitik, quantile(.25)

.25 Quantile regression Number of obs = 168
Raw sum of deviations 15.19222 (about 9.4318829)
Min sum of deviations 13.09707 Pseudo R2 = 0.1379

lnsvn_8516	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0738204	.0425192	1.74	0.084	-.0101391	.1577799
lnrer	-5.652832	2.372174	-2.38	0.018	-10.33699	-.9686786
lnipi	-.842054	.3199504	-2.63	0.009	-1.473836	-.2102721
geopolitik	.2434422	.1143043	2.13	0.035	.0177342	.4691503
_cons	14.11696	1.523946	9.26	0.000	11.10774	17.12618

. qreg lnsvn_8516 lnvol lnrer lnipi geopolitik, quantile(.5)

Median regression Number of obs = 168
Raw sum of deviations 18.34119 (about 9.6262836)
Min sum of deviations 15.29247 Pseudo R2 = 0.1662

lnsvn_8516	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0423831	.0323369	1.31	0.192	-.0214702	.1062364
lnrer	-5.664867	1.804098	-3.14	0.002	-9.227283	-2.102451
lnipi	-.5886264	.2433303	-2.42	0.017	-1.069112	-.1081404
geopolitik	.3006824	.0869313	3.46	0.001	.1290258	.472339
_cons	12.80244	1.158999	11.05	0.000	10.51385	15.09103

. qreg lnsvn_8516 lnvol lnrer lnipi geopolitik, quantile(.75)

.75 Quantile regression Number of obs = 168
Raw sum of deviations 14.36754 (about 9.791214)
Min sum of deviations 11.87414 Pseudo R2 = 0.1735

lnsvn_8516	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0462842	.0418585	1.11	0.270	-.0363706	.1289391
lnrer	-6.53558	2.335314	-2.80	0.006	-11.14695	-1.924211
lnipi	-.5835476	.3149788	-1.85	0.066	-1.205512	.0384173
geopolitik	.309691	.1125282	2.75	0.007	.0874901	.5318918
_cons	12.95046	1.500266	8.63	0.000	9.987995	15.91292

. qreg lnsvn_8516 lnvol lnrer lnipi geopolitik, quantile(.9)

.9 Quantile regression Number of obs = 168
Raw sum of deviations 7.951922 (about 9.9671659)
Min sum of deviations 6.060416 Pseudo R2 = 0.2379

lnsvn_8516	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0279478	.0447925	0.62	0.534	-.0605006	.1163962
lnrer	-5.886073	2.499003	-2.36	0.020	-10.82066	-.9514808
lnipi	-1.624032	.3370565	-4.82	0.000	-2.289592	-.9584719
geopolitik	.2353685	.1204156	1.95	0.052	-.0024071	.473144
_cons	17.77921	1.605423	11.07	0.000	14.6091	20.94932

Estonia

A. Exp 8517

```
. qreg lnest_8517 lnvol lnrer lnipi geopolitik euro, quantile(.1)

.1 Quantile regression                              Number of obs =        168
  Raw sum of deviations 41.59134 (about 5.8051348)
  Min sum of deviations 36.23975                    Pseudo R2     =     0.1287
```

lnest_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0710919	.4648935	0.15	0.879	-.8469406	.9891244
lnrer	-9.795783	7.845566	-1.25	0.214	-25.28854	5.69698
lnipi	-.5539929	1.948341	-0.28	0.777	-4.401414	3.293428
geopolitik	.2123576	1.172694	0.18	0.857	-2.10338	2.528095
euro	.592476	.7402478	0.80	0.425	-.8693031	2.054255
_cons	8.249465	8.837985	0.93	0.352	-9.203043	25.70197

```
. qreg lnest_8517 lnvol lnrer lnipi geopolitik euro, quantile(.25)

.25 Quantile regression                             Number of obs =        168
  Raw sum of deviations 77.67162 (about 6.5250297)
  Min sum of deviations 63.19751                    Pseudo R2     =     0.1864
```

lnest_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.2166369	.3986157	0.54	0.588	-.5705158	1.00379
lnrer	-28.3053	6.727058	-4.21	0.000	-41.58933	-15.02127
lnipi	-.0634197	1.670575	-0.04	0.970	-3.362331	3.235492
geopolitik	-4.277837	1.005508	-4.25	0.000	-6.26343	-2.292244
euro	-.291918	.634714	-0.46	0.646	-1.545298	.9614618
_cons	8.379832	7.577993	1.11	0.270	-6.58455	23.34422

```
. qreg lnest_8517 lnvol lnrer lnipi geopolitik euro, quantile(.5)

Median regression                                   Number of obs =        168
  Raw sum of deviations 100.9764 (about 7.4871736)
  Min sum of deviations 77.69541                    Pseudo R2     =     0.2306
```

lnest_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0457378	.4241056	0.11	0.914	-.7917501	.8832257
lnrer	-39.33317	7.157227	-5.50	0.000	-53.46666	-25.19968
lnipi	-3.19932	1.777402	-1.80	0.074	-6.709184	.3105429
geopolitik	-5.800571	1.069807	-5.42	0.000	-7.913135	-3.688007
euro	-.7409673	.6753014	-1.10	0.274	-2.074496	.5925611
_cons	21.75377	8.062575	2.70	0.008	5.832474	37.67506

```
. qreg lnest_8517 lnvol lnrer lnipi geopolitik euro, quantile(.75)

.75 Quantile regression                             Number of obs =        168
  Raw sum of deviations  78.5207 (about 8.9618788)
  Min sum of deviations 58.67703                    Pseudo R2     =     0.2527
```

lnest_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.542552	.3904579	1.39	0.167	-.2284914	1.313595
lnrer	-47.67172	6.589387	-7.23	0.000	-60.68388	-34.65955
lnipi	-1.623758	1.636386	-0.99	0.323	-4.855156	1.60764
geopolitik	-6.877941	.9849304	-6.98	0.000	-8.822899	-4.932983
euro	-2.555017	.6217244	-4.11	0.000	-3.782746	-1.327288
_cons	21.68698	7.422907	2.92	0.004	7.028852	36.34512

```
. qreg lnest_8517 lnvol lnrer lnipi geopolitik euro, quantile(.9)

.9 Quantile regression                              Number of obs =        168
  Raw sum of deviations 36.82261 (about 9.5730371)
  Min sum of deviations 29.62312                    Pseudo R2     =     0.1955
```

lnest_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.1524922	.3468908	0.44	0.661	-.5325184	.8375029
lnrer	-48.70008	5.854146	-8.32	0.000	-60.26035	-37.1398
lnipi	-1.996436	1.453799	-1.37	0.172	-4.867275	.8744042
geopolitik	-5.725518	.8750322	-6.54	0.000	-7.453458	-3.997578
euro	-3.491044	.5523526	-6.32	0.000	-4.581784	-2.400305
_cons	20.44585	6.594662	3.10	0.002	7.423267	33.46843

B. Exp 9406

```
. qreg lnest_9406 lnvol lnrer lnipi geopolitik euro, quantile(.1)

.1 Quantile regression                              Number of obs =        168
  Raw sum of deviations 15.47594 (about 7.5786567)
  Min sum of deviations 12.12739                    Pseudo R2     =     0.2164
```

lnest_9406	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.8265179	.1929997	4.28	0.000	.4453984	1.207637
lnrer	3.301555	3.257072	1.01	0.312	-3.130237	9.733347
lnipi	-.4699338	.8088504	-0.58	0.562	-2.067183	1.127316
geopolitik	.0199286	.4868418	0.04	0.967	-.9414456	.9813029
euro	.995588	.3073125	3.24	0.001	.3887331	1.602443
_cons	18.04419	3.669073	4.92	0.000	10.79882	25.28957

```
. qreg lnest_9406 lnvol lnrer lnipi geopolitik euro, quantile(.25)

.25 Quantile regression                             Number of obs =        168
  Raw sum of deviations 28.66067 (about 8.0106916)
  Min sum of deviations 22.13991                    Pseudo R2     =     0.2275
```

lnest_9406	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.6037048	.1548357	3.90	0.000	.2979482	.9094614
lnrer	-.1542142	2.613016	-0.06	0.953	-5.314177	5.005749
lnipi	.5874161	.6489076	0.91	0.367	-.6939919	1.868824
geopolitik	-.3142739	.3905733	-0.80	0.422	-1.085545	.4569973
euro	.597573	.2465443	2.42	0.016	.1107182	1.084428
_cons	11.22307	2.943547	3.81	0.000	5.410404	17.03574

```
. qreg lnest_9406 lnvol lnrer lnipi geopolitik euro, quantile(.5)

Median regression                                   Number of obs =        168
  Raw sum of deviations 33.97974 (about 8.458292)
  Min sum of deviations 27.42615                    Pseudo R2     =     0.1929
```

lnest_9406	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.3718245	.1463342	2.54	0.012	.0828561	.6607929
lnrer	-1.502149	2.469543	-0.61	0.544	-6.378794	3.374497
lnipi	.0365109	.613278	0.06	0.953	-1.174539	1.247561
geopolitik	-.1336597	.3691281	-0.36	0.718	-.8625827	.5952633
euro	.3702666	.2330073	1.59	0.114	-.0898565	.8303898
_cons	11.76229	2.781926	4.23	0.000	6.26878	17.25581

```
. qreg lnest_9406 lnvol lnrer lnipi geopolitik euro, quantile(.75)

.75 Quantile regression                             Number of obs =        168
  Raw sum of deviations 24.95604 (about 8.7337551)
  Min sum of deviations 20.33602                    Pseudo R2     =     0.1851
```

lnest_9406	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.2411342	.1272203	1.90	0.060	-.0100897	.4923581
lnrer	-3.413566	2.146976	-1.59	0.114	-7.653233	.8261005
lnipi	-.1616271	.5331727	-0.30	0.762	-1.214492	.8912375
geopolitik	-.368436	.3209132	-1.15	0.253	-1.002148	.2652764
euro	.2184981	.2025723	1.08	0.282	-.1815246	.6185207
_cons	11.63019	2.418556	4.81	0.000	6.854225	16.40615

```
. qreg lnest_9406 lnvol lnrer lnipi geopolitik euro, quantile(.9)

.9 Quantile regression                              Number of obs =        168
  Raw sum of deviations 13.34503 (about 9.0010996)
  Min sum of deviations 10.04574                    Pseudo R2     =     0.2472
```

lnest_9406	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.1716256	.1356668	1.27	0.208	-.0962777	.4395289
lnrer	-3.373792	2.289519	-1.47	0.143	-7.894942	1.147358
lnipi	-.2777181	.5685716	-0.49	0.626	-1.400485	.8450492
geopolitik	-.1665867	.3422196	-0.49	0.627	-.8423731	.5091997
euro	.0547613	.2160216	0.25	0.800	-.37182	.4813426
_cons	11.72844	2.579131	4.55	0.000	6.63539	16.82149

C. Exp 8703

```
. qreg lnest_8703 lnvol lnrer lnipi geopolitik euro, quantile(.1)

.1 Quantile regression                              Number of obs =        168
  Raw sum of deviations 22.09534 (about 6.4118185)
  Min sum of deviations  17.9019                    Pseudo R2     =     0.1898
```

lnest_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.8622366	.3807606	-2.26	0.025	-1.614131	-.1103427
lnrer	-13.51279	6.425735	-2.10	0.037	-26.20179	-.8237882
lnipi	-.371479	1.595745	-0.23	0.816	-3.522623	2.779665
geopolitik	.161797	.9604689	0.17	0.866	-1.734856	2.05845
euro	-.382198	.6062834	-0.63	0.529	-1.579435	.8150395
_cons	-1.333749	7.238554	-0.18	0.854	-15.62784	12.96034

```
. qreg lnest_8703 lnvol lnrer lnipi geopolitik euro, quantile(.25)

.25 Quantile regression                             Number of obs =        168
  Raw sum of deviations  36.5519 (about 6.8167357)
  Min sum of deviations  31.6892                    Pseudo R2     =     0.1330
```

lnest_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.4680685	.2205908	-2.12	0.035	-.9036725	-.0324644
lnrer	-14.33244	3.722701	-3.85	0.000	-21.68372	-6.981166
lnipi	-.3678397	.9244831	-0.40	0.691	-2.193431	1.457752
geopolitik	-.8781356	.5564404	-1.58	0.116	-1.976947	.2206761
euro	-.2046382	.3512457	-0.58	0.561	-.8982487	.4889722
_cons	3.021911	4.193602	0.72	0.472	-5.25926	11.30308

```
. qreg lnest_8703 lnvol lnrer lnipi geopolitik euro, quantile(.5)

Median regression                                   Number of obs =        168
  Raw sum of deviations  41.6636 (about 7.3284373)
  Min sum of deviations 37.14697                    Pseudo R2     =     0.1084
```

lnest_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.4099559	.1967267	-2.08	0.039	-.7984351	-.0214767
lnrer	-9.601172	3.319969	-2.89	0.004	-16.15717	-3.045177
lnipi	.5469974	.82447	0.66	0.508	-1.081096	2.175091
geopolitik	-.1688344	.4962432	-0.34	0.734	-1.148774	.8111047
euro	-.3814483	.313247	-1.22	0.225	-1.000022	.2371255
_cons	.2175896	3.739926	0.06	0.954	-7.167702	7.602881

```
. qreg lnest_8703 lnvol lnrer lnipi geopolitik euro, quantile(.75)

.75 Quantile regression                             Number of obs =        168
  Raw sum of deviations 30.61458 (about 7.700748)
  Min sum of deviations 27.07942                    Pseudo R2     =     0.1155
```

lnest_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.4310469	.1696591	-2.54	0.012	-.7660754	-.0960185
lnrer	-13.18563	2.863175	-4.61	0.000	-18.83958	-7.531668
lnipi	1.068827	.7110313	1.50	0.135	-.3352575	2.472912
geopolitik	-.5905215	.4279652	-1.38	0.170	-1.435631	.254588
euro	-1.090621	.2701474	-4.04	0.000	-1.624086	-.5571571
_cons	-1.70464	3.22535	-0.53	0.598	-8.07379	4.664511

```
.9 Quantile regression                              Number of obs =        168
  Raw sum of deviations 15.41763 (about 8.0439844)
  Min sum of deviations 13.90766                    Pseudo R2     =     0.0979
```

lnest_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.5353833	.1554958	-3.44	0.001	-.8424432	-.2283233
lnrer	-8.817221	2.624155	-3.36	0.001	-13.99918	-3.635262
lnipi	-.0886171	.6516737	-0.14	0.892	-1.375488	1.198253
geopolitik	.1701232	.3922382	0.43	0.665	-.6044358	.9446822
euro	-.9117573	.2475952	-3.68	0.000	-1.400688	-.4228271
_cons	2.985979	2.956095	1.01	0.314	-2.851469	8.823427

Latvia

A. Exp 4407

. qreg lnlva_4407 lnvol lnrer lnipi geopolitik euro, quantile(.1)

```
.1 Quantile regression                          Number of obs =        168
  Raw sum of deviations 15.79575 (about 6.6618547)
  Min sum of deviations 9.626202                Pseudo R2     =     0.3906
```

lnlva_4407	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.3074765	.2936517	1.05	0.297	-.2724021	.887355
lnrer	-1.861074	3.759413	-0.50	0.621	-9.284847	5.562699
lnipi	-.5998878	.5734803	-1.05	0.297	-1.732348	.5325728
geopolitik	.1393238	.3579953	0.39	0.698	-.5676152	.8462628
euro	.9141337	.1193618	7.66	0.000	.678428	1.149839
_cons	12.554	4.333758	2.90	0.004	3.996055	21.11194

. qreg lnlva_4407 lnvol lnrer lnipi geopolitik euro, quantile(.25)

```
.25 Quantile regression                         Number of obs =        168
  Raw sum of deviations 28.80804 (about 7.1685801)
  Min sum of deviations 17.66645                Pseudo R2     =     0.3868
```

lnlva_4407	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.3319009	.2084479	1.59	0.113	-.0797243	.7435262
lnrer	.838178	2.66861	0.31	0.754	-4.431568	6.107924
lnipi	-.6128353	.4070835	-1.51	0.134	-1.416709	.1910389
geopolitik	-.0550126	.254122	-0.22	0.829	-.5568314	.4468062
euro	.7534867	.0847287	8.89	0.000	.5861716	.9208017
_cons	13.12877	3.076307	4.27	0.000	7.053941	19.2036

. qreg lnlva_4407 lnvol lnrer lnipi geopolitik euro, quantile(.5)

```
Median regression                               Number of obs =        168
  Raw sum of deviations 32.08593 (about 7.4977617)
  Min sum of deviations 22.75339                Pseudo R2     =     0.2909
```

lnlva_4407	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.2175357	.2234473	0.97	0.332	-.2237092	.6587805
lnrer	.2149752	2.860637	0.08	0.940	-5.433969	5.86392
lnipi	-.3723191	.4363763	-0.85	0.395	-1.234038	.4893999
geopolitik	.0282422	.2724081	0.10	0.918	-.5096863	.5661707
euro	.6962581	.0908256	7.67	0.000	.5169034	.8756127
_cons	11.03525	3.297671	3.35	0.001	4.523288	17.54721

. qreg lnlva_4407 lnvol lnrer lnipi geopolitik euro, quantile(.75)

```
.75 Quantile regression                         Number of obs =        168
  Raw sum of deviations 23.03266 (about 7.7222347)
  Min sum of deviations 17.93727                Pseudo R2     =     0.2212
```

lnlva_4407	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.1650699	.2519538	0.66	0.513	-.3324673	.6626071
lnrer	-5.509189	3.225586	-1.71	0.090	-11.8788	.8604265
lnipi	-.6952529	.4920475	-1.41	0.160	-1.666907	.276401
geopolitik	.5329691	.3071609	1.74	0.085	-.0735863	1.139525
euro	.4991839	.1024127	4.87	0.000	.2969478	.7014199
_cons	12.40681	3.718375	3.34	0.001	5.064078	19.74955

. qreg lnlva_4407 lnvol lnrer lnipi geopolitik euro, quantile(.9)

```
.9 Quantile regression                          Number of obs =        168
  Raw sum of deviations 12.95538 (about 7.9237103)
  Min sum of deviations 9.675636                Pseudo R2     =     0.2532
```

lnlva_4407	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.3990383	.3346547	1.19	0.235	-.2618095	1.059886
lnrer	-10.1429	4.284345	-2.37	0.019	-18.60326	-1.682534
lnipi	-1.375272	.6535561	-2.10	0.037	-2.66586	-.0846847
geopolitik	.4920609	.4079827	1.21	0.230	-.313589	1.297711
euro	.5706851	.1360285	4.20	0.000	.3020675	.8393027
_cons	18.32551	4.938887	3.71	0.000	8.572611	28.07841

B. Exp 8517

. qreg lnlva_8517 lnvol lnrer lnipi geopolitik euro, quantile(.1)

```
.1 Quantile regression                          Number of obs =        168
  Raw sum of deviations 26.03114 (about 7.7213488)
  Min sum of deviations 13.01579                Pseudo R2     =     0.5000
```

lnlva_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.007921	.3714212	0.02	0.983	-.7255304	.7413724
lnrer	-1.212803	4.755042	-0.26	0.799	-10.60266	8.177053
lnipi	-.760969	.7253586	-1.05	0.296	-2.193346	.6714081
geopolitik	.5971705	.4528054	1.32	0.189	-.2969916	1.491332
euro	1.545937	.1509732	10.24	0.000	1.247808	1.844066
_cons	11.16827	5.481494	2.04	0.043	.3438728	21.99266

. qreg lnlva_8517 lnvol lnrer lnipi geopolitik euro, quantile(.25)

```
.25 Quantile regression                         Number of obs =        168
  Raw sum of deviations 47.12879 (about 8.5857859)
  Min sum of deviations 22.87027                Pseudo R2     =     0.5147
```

lnlva_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.2238498	.2386921	0.94	0.350	-.2474993	.6951989
lnrer	5.276091	3.055805	1.73	0.086	-.7582557	11.31044
lnipi	-.4955014	.4661483	-1.06	0.289	-1.416012	.425009
geopolitik	.0602779	.2909933	0.21	0.836	-.5143511	.6349069
euro	1.37456	.0970222	14.17	0.000	1.182968	1.566151
_cons	12.49946	3.522656	3.55	0.001	5.543212	19.4557

. qreg lnlva_8517 lnvol lnrer lnipi geopolitik euro, quantile(.5)

```
Median regression                               Number of obs =        168
  Raw sum of deviations 47.65675 (about 9.2926579)
  Min sum of deviations 29.95873                Pseudo R2     =     0.3714
```

lnlva_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.2293402	.3112372	-0.74	0.462	-.8439452	.3852647
lnrer	.1061202	3.984548	0.03	0.979	-7.76223	7.97447
lnipi	-.4659445	.6078235	-0.77	0.444	-1.666223	.7343341
geopolitik	.557764	.3794341	1.47	0.144	-.1915105	1.307038
euro	1.162402	.1265099	9.19	0.000	.9125806	1.412223
_cons	7.936922	4.593288	1.73	0.086	-1.133516	17.00736

. qreg lnlva_8517 lnvol lnrer lnipi geopolitik euro, quantile(.75)

```
.75 Quantile regression                         Number of obs =        168
  Raw sum of deviations 30.93355 (about 9.576025)
  Min sum of deviations 23.64995                Pseudo R2     =     0.2355
```

lnlva_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0477086	.3254596	0.15	0.884	-.5949816	.6903988
lnrer	-7.333321	4.166628	-1.76	0.080	-15.56123	.8945839
lnipi	-.8778462	.6355988	-1.38	0.169	-2.132973	.3772808
geopolitik	.5750856	.3967729	1.45	0.149	-.208428	1.358599
euro	.8730501	.1322909	6.60	0.000	.6118131	1.134287
_cons	13.45596	4.803185	2.80	0.006	3.971032	22.94088

```
. qreg lnlva_8517 lnvol lnrer lnipi geopolitik euro, quantile(.9)

.9 Quantile regression                          Number of obs =        168
  Raw sum of deviations 15.04918 (about 9.8142738)
  Min sum of deviations 12.34083                Pseudo R2     =     0.1800
```

lnlva_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.221059	.4084262	0.54	0.589	-.5854666	1.027585
lnrer	-12.93998	5.22879	-2.47	0.014	-23.26535	-2.614602
lnipi	-1.36915	.7976265	-1.72	0.088	-2.944236	.2059354
geopolitik	.5508655	.4979187	1.11	0.270	-.4323824	1.534113
euro	.7626009	.1660147	4.59	0.000	.4347691	1.090433
_cons	18.03206	6.027619	2.99	0.003	6.129226	29.93489

C. Exp 3004

```
. qreg lnlva_3004 lnvol lnrer lnipi geopolitik euro, quantile(.1)

.1 Quantile regression                          Number of obs =        168
  Raw sum of deviations 9.274529 (about 8.2862692)
  Min sum of deviations 8.192415                Pseudo R2     =     0.1167
```

lnlva_3004	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.158522	.1958786	-0.81	0.420	-.5453266	.2282827
lnrer	-5.345739	2.507695	-2.13	0.035	-10.29772	-.3937542
lnipi	-.0815818	.3825367	-0.21	0.831	-.8369831	.6738194
geopolitik	1.037412	.2387987	4.34	0.000	.5658526	1.508972
euro	.0380777	.0796196	0.48	0.633	-.1191484	.1953038
_cons	6.991418	2.890808	2.42	0.017	1.282893	12.69994

```
. qreg lnlva_3004 lnvol lnrer lnipi geopolitik euro, quantile(.25)

.25 Quantile regression                         Number of obs =        168
  Raw sum of deviations 17.90744 (about 8.5223799)
  Min sum of deviations 16.22466                Pseudo R2     =     0.0940
```

lnlva_3004	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.2525852	.2918827	-0.87	0.388	-.8289704	.3238001
lnrer	1.385827	3.736766	0.37	0.711	-5.993224	8.764878
lnipi	.0771942	.5700255	0.14	0.892	-1.048444	1.202833
geopolitik	.7137832	.3558387	2.01	0.047	.0111029	1.416463
euro	-.1951246	.1186428	-1.64	0.102	-.4294103	.0391612
_cons	5.554014	4.307651	1.29	0.199	-2.952372	14.0604

```
. qreg lnlva_3004 lnvol lnrer lnipi geopolitik euro, quantile(.5)

Median regression                               Number of obs =        168
  Raw sum of deviations 20.75215 (about 8.8186302)
  Min sum of deviations 18.65585                Pseudo R2     =     0.1010
```

lnlva_3004	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.0192729	.1969439	-0.10	0.922	-.4081811	.3696353
lnrer	5.011971	2.521333	1.99	0.049	.0330555	9.990886
lnipi	-.3653915	.384617	-0.95	0.344	-1.124901	.3941179
geopolitik	.1838217	.2400974	0.77	0.445	-.2903024	.6579457
euro	-.0575503	.0800526	-0.72	0.473	-.2156315	.1005308
_cons	10.27456	2.906529	3.53	0.001	4.534994	16.01413

```
. qreg lnlva_3004 lnvol lnrer lnipi geopolitik euro, quantile(.75)

.75 Quantile regression                         Number of obs =        168
  Raw sum of deviations 15.25758 (about 8.9815559)
  Min sum of deviations 13.20705                Pseudo R2     =     0.1344
```

lnlva_3004	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.1606749	.1250736	1.28	0.201	-.0863099	.4076596
lnrer	5.633136	1.601228	3.52	0.001	2.471166	8.795107
lnipi	-.3074949	.2442596	-1.26	0.210	-.7898381	.1748484
geopolitik	.0397925	.1524792	0.26	0.794	-.2613105	.3408955
euro	-.0982605	.0508392	-1.93	0.055	-.1986534	.0021324
_cons	12.09951	1.845856	6.55	0.000	8.454469	15.74455

```
. qreg lnlva_3004 lnvol lnrer lnipi geopolitik euro, quantile(.9)

.9 Quantile regression                          Number of obs =        168
  Raw sum of deviations 8.472285 (about 9.1023092)
  Min sum of deviations 6.943057                Pseudo R2     =     0.1805
```

lnlva_3004	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.02977	.1489051	0.20	0.842	-.2642751	.3238152
lnrer	4.644728	1.906326	2.44	0.016	.8802769	8.409179
lnipi	-.1578529	.2908007	-0.54	0.588	-.7321017	.4163958
geopolitik	.2182064	.1815325	1.20	0.231	-.1402687	.5766814
euro	-.0717508	.0605261	-1.19	0.238	-.1912726	.047771
_cons	10.11576	2.197565	4.60	0.000	5.776199	14.45533

Lithuania

A. Exp 9403

```
. qreg lnltu_9403 lnvol lnrer lnipi geopolitik euro, quantile(.1)

.1 Quantile regression                              Number of obs =        168
  Raw sum of deviations 9.528187 (about 8.922925)
  Min sum of deviations 6.112175                    Pseudo R2     =     0.3585

------------------------------------------------------------------------------
  lnltu_9403 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       lnvol |  -.2204796     .086409    -2.55   0.012    -.3911128   -.0498464
       lnrer |  -1.714213    .7984717    -2.15   0.033    -3.290967   -.1374579
       lnipi |  -.9831003    .3687014    -2.67   0.008    -1.711181   -.2550197
  geopolitik |   .3087399    .1515988     2.04   0.043     .0093754    .6081044
        euro |   .1460121    .0979046     1.49   0.138    -.0473216    .3393458
       _cons |   8.555557    2.198574     3.89   0.000     4.213999    12.89711
------------------------------------------------------------------------------
```

```
. qreg lnltu_9403 lnvol lnrer lnipi geopolitik euro, quantile(.25)

.25 Quantile regression                             Number of obs =        168
  Raw sum of deviations 17.66681 (about 9.1560955)
  Min sum of deviations 11.54738                    Pseudo R2     =     0.3464

------------------------------------------------------------------------------
  lnltu_9403 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       lnvol |  -.1866728    .0774651    -2.41   0.017    -.3396443   -.0337014
       lnrer |  -2.687059    .7158242    -3.75   0.000    -4.100608   -1.273509
       lnipi |  -.3663602    .3305382    -1.11   0.269    -1.019079    .2863588
  geopolitik |   .1806752    .1359072     1.33   0.186     -.087703    .4490533
        euro |   .0547063    .0877707     0.62   0.534     -.118616    .2280286
       _cons |   4.843595    1.971005     2.46   0.015     .9514195    8.735771
------------------------------------------------------------------------------
```

```
. qreg lnltu_9403 lnvol lnrer lnipi geopolitik euro, quantile(.5)

Median regression                                   Number of obs =        168
  Raw sum of deviations 21.85458 (about 9.3918285)
  Min sum of deviations 14.49743                    Pseudo R2     =     0.3366

------------------------------------------------------------------------------
  lnltu_9403 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       lnvol |  -.2430931     .066118    -3.68   0.000    -.3736573   -.1125289
       lnrer |  -2.474533    .6109702    -4.05   0.000    -3.681025    -1.26804
       lnipi |   -.608924    .2821209    -2.16   0.032    -1.166033   -.0518153
  geopolitik |   .1601585    .1159995     1.38   0.169    -.0689075    .3892246
        euro |  -.0503971    .0749141    -0.67   0.502    -.1983311    .0975369
       _cons |   5.848842    1.682292     3.48   0.001     2.526793    9.170892
------------------------------------------------------------------------------
```

```
. qreg lnltu_9403 lnvol lnrer lnipi geopolitik euro, quantile(.75)

.75 Quantile regression                             Number of obs =        168
  Raw sum of deviations 17.43982 (about 9.6108599)
  Min sum of deviations 11.50896                    Pseudo R2     =     0.3401

------------------------------------------------------------------------------
  lnltu_9403 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       lnvol |  -.1489143    .0834108    -1.79   0.076    -.3136268    .0157983
       lnrer |  -2.569907    .7707661    -3.33   0.001    -4.091951   -1.047862
       lnipi |  -.6229409    .3559081    -1.75   0.082    -1.325758    .0798765
  geopolitik |   .1026558    .1463386     0.70   0.484    -.1863213    .3916329
        euro |   .0355504    .0945075     0.38   0.707     -.151075    .2221757
       _cons |   6.940523    2.122287     3.27   0.001      2.74961    11.13144
------------------------------------------------------------------------------
```

```
. qreg lnltu_3102 lnvol lnrer lnipi geopolitik euro, quantile(.9)

.9 Quantile regression                              Number of obs =        168
  Raw sum of deviations 16.42602 (about 9.1246738)
  Min sum of deviations 15.78679                    Pseudo R2     =     0.0389

------------------------------------------------------------------------------
  lnltu_3102 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       lnvol |   .0833952     .223941     0.37   0.710    -.3588246    .5256151
       lnrer |   4.587193    2.069351     2.22   0.028     .5008125    8.673573
       lnipi |   .5168334    .9555413     0.54   0.589    -1.370089    2.403756
  geopolitik |    .646124    .3928894     1.64   0.102     -.129721    1.421969
        euro |   .1793237    .2537334     0.71   0.481    -.3217277     .680375
       _cons |   14.19084    5.697911     2.49   0.014      2.93909     25.4426
------------------------------------------------------------------------------
```

B. Exp 8703

```
. qreg lnltu_8703 lnvol lnrer lnipi geopolitik euro, quantile(.1)

.1 Quantile regression                              Number of obs =        168
  Raw sum of deviations 30.92967 (about 6.6398759)
  Min sum of deviations 19.63864                    Pseudo R2     =     0.3651

------------------------------------------------------------------------------
  lnltu_8703 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       lnvol |  -.4918726    .3179856    -1.55   0.124    -1.119804    .1360586
       lnrer |  -.7829048    2.938381    -0.27   0.790    -6.585371    5.019562
       lnipi |   .9773328    1.356823     0.72   0.472    -1.702008    3.656674
  geopolitik |    .299617    .5578844     0.54   0.592    -.8020461     1.40128
        euro |   1.037463    .3602894     2.88   0.005     .3259935    1.748932
       _cons |  -4.850387    8.090764    -0.60   0.550    -20.82735    11.12657
------------------------------------------------------------------------------
```

```
. qreg lnltu_8703 lnvol lnrer lnipi geopolitik euro, quantile(.25)

.25 Quantile regression                             Number of obs =        168
  Raw sum of deviations 50.62988 (about 7.3440728)
  Min sum of deviations 36.27315                    Pseudo R2     =     0.2836

------------------------------------------------------------------------------
  lnltu_8703 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       lnvol |  -.5217588    .2292928    -2.28   0.024    -.9745469   -.0689708
       lnrer |  -.8633105    2.118805    -0.41   0.684    -5.047347    3.320726
       lnipi |   1.709063    .9783769     1.75   0.083    -.2229534    3.641079
  geopolitik |   .6777169    .4022787     1.68   0.094    -.1166693    1.472103
        euro |   .6132697    .2597971     2.36   0.019     .1002443    1.126295
       _cons |  -8.126668     5.83408    -1.39   0.166    -19.64732    3.393981
------------------------------------------------------------------------------
```

```
. qreg lnltu_8703 lnvol lnrer lnipi geopolitik euro, quantile(.5)

Median regression                                   Number of obs =        168
  Raw sum of deviations 58.96741 (about 7.9175363)
  Min sum of deviations 46.27982                    Pseudo R2     =     0.2152

------------------------------------------------------------------------------
  lnltu_8703 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       lnvol |   .2571635    .2579473     1.00   0.320    -.2522092    .7665361
       lnrer |  -6.096216     2.38359    -2.56   0.011    -10.80313   -1.389303
       lnipi |   .3457702    1.100644     0.31   0.754    -1.827689    2.519229
  geopolitik |   -.382156    .4525513    -0.84   0.400    -1.275816    .5115041
        euro |   .6007669    .2922638     2.06   0.041      .023629    1.177905
       _cons |  -.0491073    6.563161    -0.01   0.994    -13.00949    12.91127
------------------------------------------------------------------------------
```

```
. qreg lnltu_8703 lnvol lnrer lnipi geopolitik euro, quantile(.75)

.75 Quantile regression                             Number of obs =        168
  Raw sum of deviations 46.80002 (about 8.5397367)
  Min sum of deviations 32.93484                    Pseudo R2     =     0.2963

------------------------------------------------------------------------------
  lnltu_8703 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       lnvol |   .4470336    .1755793     2.55   0.012     .1003143    .7937529
       lnrer |  -7.400905     1.62246    -4.56   0.000     -10.6048   -4.197008
       lnipi |  -.7169529    .7491853    -0.96   0.340    -2.196381    .7624752
  geopolitik |  -.6490828    .3080421    -2.11   0.037    -1.257379   -.0407872
        euro |   .8587705    .1989378     4.32   0.000     .4659248    1.251616
       _cons |   5.535052    4.467406     1.24   0.217    -3.286805    14.35691
------------------------------------------------------------------------------
```

```
. qreg lnltu_8703 lnvol lnrer lnipi geopolitik euro, quantile(.9)

.9 Quantile regression                          Number of obs =        168
  Raw sum of deviations 24.06599 (about 9.099967)
  Min sum of deviations 15.96312                Pseudo R2     =     0.3367
```

lnltu_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.4815855	.1790994	2.69	0.008	.1279152	.8352559
lnrer	-7.990932	1.654987	-4.83	0.000	-11.25906	-4.722803
lnipi	-.1071824	.764205	-0.14	0.889	-1.61627	1.401905
geopolitik	-.7950428	.3142178	-2.53	0.012	-1.415534	-.174552
euro	.6754552	.2029262	3.33	0.001	.2747337	1.076177
_cons	2.575741	4.556969	0.57	0.573	-6.422977	11.57446

C. Exp 3102

```
. qreg lnltu_3102 lnvol lnrer lnipi geopolitik euro, quantile(.1)

.1 Quantile regression                          Number of obs =        168
  Raw sum of deviations  25.1915 (about 7.3821244)
  Min sum of deviations 23.27864                Pseudo R2     =     0.0759
```

lnltu_3102	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.4445257	.3457338	-1.29	0.200	-1.127252	.2382003
lnrer	3.392284	3.194791	1.06	0.290	-2.916519	9.701088
lnipi	1.786108	1.475223	1.21	0.228	-1.127038	4.699255
geopolitik	-.2182739	.6065667	-0.36	0.719	-1.416071	.979523
euro	-.2829436	.3917291	-0.72	0.471	-1.056497	.49061
_cons	-.9449188	8.796783	-0.11	0.915	-18.31606	16.42623

```
. qreg lnltu_3102 lnvol lnrer lnipi geopolitik euro, quantile(.25)

.25 Quantile regression                         Number of obs =        168
  Raw sum of deviations 42.03688 (about 7.9211726)
  Min sum of deviations 41.27672                Pseudo R2     =     0.0181
```

lnltu_3102	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.5625072	.3793288	-1.48	0.140	-1.311574	.1865593
lnrer	1.515316	3.505229	0.43	0.666	-5.406514	8.437146
lnipi	-.2176863	1.618571	-0.13	0.893	-3.413903	2.978531
geopolitik	.0634713	.6655068	0.10	0.924	-1.250716	1.377658
euro	-.3655763	.4297934	-0.85	0.396	-1.214296	.4831436
_cons	4.947902	9.651567	0.51	0.609	-14.1112	24.007

```
. qreg lnltu_3102 lnvol lnrer lnipi geopolitik euro, quantile(.5)

Median regression                               Number of obs =        168
  Raw sum of deviations 47.80051 (about 8.3772411)
  Min sum of deviations 44.97408                Pseudo R2     =     0.0591
```

lnltu_3102	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.1201679	.2024695	-0.59	0.554	-.5199875	.2796517
lnrer	4.173413	1.870941	2.23	0.027	.4788366	7.867989
lnipi	.1220933	.8639236	0.14	0.888	-1.58391	1.828097
geopolitik	.6504879	.355219	1.83	0.069	-.0509687	1.351945
euro	.0125038	.2294053	0.05	0.957	-.4405065	.4655141
_cons	12.61544	5.151593	2.45	0.015	2.442506	22.78837

```
. qreg lnltu_3102 lnvol lnrer lnipi geopolitik euro, quantile(.75)

.75 Quantile regression                         Number of obs =        168
  Raw sum of deviations 34.09541 (about 8.85495)
  Min sum of deviations 31.70389                Pseudo R2     =     0.0701
```

lnltu_3102	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	-.1285144	.1469333	-0.87	0.383	-.4186659	.161637
lnrer	6.325076	1.357753	4.66	0.000	3.643901	9.006252
lnipi	-.3309707	.6269544	-0.53	0.598	-1.569027	.907086
geopolitik	.7072898	.2577845	2.74	0.007	.1982387	1.216341
euro	.1956134	.1664808	1.17	0.242	-.1331388	.5243656
_cons	18.01432	3.738541	4.82	0.000	10.63177	25.39688

```
. qreg lnltu_3102 lnvol lnrer lnipi geopolitik euro, quantile(.9)

.9 Quantile regression                          Number of obs =        168
  Raw sum of deviations 16.42602 (about 9.1246738)
  Min sum of deviations 15.78679                Pseudo R2     =     0.0389
```

lnltu_3102	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
lnvol	.0833952	.223941	0.37	0.710	-.3588246	.5256151
lnrer	4.587193	2.069351	2.22	0.028	.5008125	8.673573
lnipi	.5168334	.9555413	0.54	0.589	-1.370089	2.403756
geopolitik	.646124	.3928894	1.64	0.102	-.129721	1.421969
euro	.1793237	.2537334	0.71	0.481	-.3217277	.680375
_cons	14.19084	5.697911	2.49	0.014	2.93909	25.4426

Lampiran 11 Estimasi Metode NARDL

Republik Ceko

A. Exp 8703

B. Exp 8517

. ardl lncze_8703 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,0,0,3,4,1) regression

Sample:	592 - 755	Number of obs =	164
		R-squared =	0.4628
		Adj R-squared =	0.4003
Log likelihood = -4.4566781		Root MSE =	0.2635

D.lncze_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lncze_8703						
L1.	-.2770332	.0792557	-3.50	0.001	-.4336699	-.1203966
LR						
vol_pos	-1.019131	.6637415	-1.54	0.127	-2.330914	.2926518
vol_neg	-2.195787	.9988949	-2.20	0.030	-4.169948	-.2216252
lnrer	-.9837683	1.84771	-0.53	0.595	-4.635481	2.667944
lnipi	4.476204	1.733113	2.58	0.011	1.050972	7.901435
geopolitik	.369264	.4670913	0.79	0.430	-.55387	1.292398
SR						
lncze_8703						
LD.	-.3109874	.0893831	-3.48	0.001	-.4876394	-.1343354
L2D.	-.3277534	.0826522	-3.97	0.000	-.4911028	-.1644041
L3D.	-.2896512	.0740967	-3.91	0.000	-.4360918	-.1432106
lnrer						
D1.	-3.184771	1.639415	-1.94	0.054	-6.424822	.0552799
LD.	6.130799	1.810354	3.39	0.001	2.552913	9.708684
L2D.	-2.469898	1.594879	-1.55	0.124	-5.62193	.6821351
lnipi						
D1.	-.9271274	.4347473	-2.13	0.035	-1.786338	-.0679164
LD.	-.2088294	.4020937	-0.52	0.604	-1.003506	.5858467
L2D.	.2755155	.3593288	0.77	0.444	-.4346425	.9856735
L3D.	.7224389	.2999152	2.41	0.017	.1297029	1.315175
geopolitik						
D1.	.7716212	.431566	1.79	0.076	-.0813023	1.624545
_cons	-1.411439	2.6607	-0.53	0.597	-6.669901	3.847023

. ardl lncze_8517 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(3,0,0,0,2,0) regression

Sample:	592 - 755	Number of obs =	164
		R-squared =	0.3664
		Adj R-squared =	0.3250
Log likelihood = -78.119203		Root MSE =	0.4034

D.lncze_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lncze_8517						
L1.	-.134732	.046235	-2.91	0.004	-.2260735	-.0433905
LR						
vol_pos	1.855363	1.363741	1.36	0.176	-.8388313	4.549558
vol_neg	.7100076	2.654993	0.27	0.790	-4.53517	5.955185
lnrer	-11.63216	5.857786	-1.99	0.049	-23.20475	-.0595773
lnipi	12.37287	4.770744	2.59	0.010	2.947839	21.79791
geopolitik	-1.765166	1.546349	-1.14	0.255	-4.820118	1.289786
SR						
lncze_8517						
LD.	-.4434676	.0769635	-5.76	0.000	-.5955159	-.2914192
L2D.	-.2599928	.0649542	-4.00	0.000	-.3883156	-.13167
lnipi						
D1.	-.4336912	.5703862	-0.76	0.448	-1.560541	.6931582
LD.	-1.229067	.4385766	-2.80	0.006	-2.095514	-.3626189
_cons	-1.107567	3.380809	-0.33	0.744	-7.78666	5.571526

C. Exp 8708

. ardl lncze_8708 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,0,0,1,4,3) regression

Sample:	592 - 755	Number of obs =	164
		R-squared =	0.5167
		Adj R-squared =	0.4604
Log likelihood = 60.983603		Root MSE =	0.1768

D.lncze_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lncze_8708						
L1.	-.1435118	.0694811	-2.07	0.041	-.2808305	-.0061932
LR						
vol_pos	.3128017	.6892311	0.45	0.651	-1.049357	1.67496
vol_neg	-1.920814	1.428659	-1.34	0.181	-4.744339	.9027101
lnrer	3.475012	2.888819	1.20	0.231	-2.234293	9.184317
lnipi	4.914379	2.859531	1.72	0.088	-.7370433	10.5658
geopolitik	1.125533	.7805875	1.44	0.151	-.4171773	2.668244
SR						
lncze_8708						
LD.	-.4074616	.0873901	-4.66	0.000	-.5801746	-.2347486
L2D.	-.4104599	.0888969	-4.62	0.000	-.586151	-.2347689
L3D.	-.5793727	.083887	-6.91	0.000	-.7451624	-.413583
lnrer						
D1.	-2.151462	1.09599	-1.96	0.052	-4.317518	.014593
lnipi						
D1.	-.7943488	.2846395	-2.79	0.006	-1.356895	-.2318027
LD.	-.2253468	.2670151	-0.84	0.400	-.7530609	.3023674
L2D.	.4437401	.2412474	1.84	0.068	-.0330481	.9205283
L3D.	.5408116	.2038857	2.65	0.009	.1378629	.9437603
geopolitik						
D1.	-.1594063	.2629749	-0.61	0.545	-.6791357	.360323
LD.	.4684386	.2238694	2.09	0.038	.0259952	.910882
L2D.	.4672794	.1986999	2.35	0.020	.0745797	.8599792
_cons	-3.079044	1.838692	-1.67	0.096	-6.712936	.5548472

Hungaria

A. Exp 8703

```
. ardl lnhun_8703 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,0,0,2,2,3) regression

Sample:       592 -       755                 Number of obs     =        164
                                              R-squared         =     0.3826
                                              Adj R-squared     =     0.3154
Log likelihood =  -20.55291                   Root MSE          =     0.2897

------------------------------------------------------------------------------
D.lnhun_8703 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ADJ          |
  lnhun_8703 |
         L1. |  -.1961124   .0689244    -2.85   0.005    -.3323232   -.0599017
-------------+----------------------------------------------------------------
LR           |
     vol_pos |  -.1928415   .4586098    -0.42   0.675    -1.099161    .7134784
     vol_neg |  -.2107287   .6207585    -0.34   0.735    -1.437492    1.016035
       lnrer |   6.200846   2.372535     2.61   0.010     1.512164    10.88953
       lnipi |   1.873716   2.112729     0.89   0.377    -2.301529    6.048962
  geopolitik |   .1664008   .6093497     0.27   0.785    -1.037816    1.370618
-------------+----------------------------------------------------------------
SR           |
  lnhun_8703 |
         LD. |  -.3308771   .0896473    -3.69   0.000    -.5080411    -.153713
        L2D. |  -.2526646   .0956429    -2.64   0.009    -.4416773   -.0636519
        L3D. |  -.2625022   .0868834    -3.02   0.003     -.434204   -.0908004
             |
       lnrer |
         D1. |  -.3030611   1.393086    -0.22   0.828    -3.056125    2.450002
         LD. |  -3.566285   1.409336    -2.53   0.012    -6.351463   -.7811078
             |
       lnipi |
         D1. |   .1040976   .3746085     0.28   0.781    -.6362162    .8444114
         LD. |  -.6421921   .3185226    -2.02   0.046    -1.271667   -.0127171
             |
  geopolitik |
         D1. |   .4751515   .3454951     1.38   0.171    -.2076274     1.15793
         LD. |   .1305132   .3537139     0.37   0.713     -.568508    .8295343
        L2D. |   .7268743   .3514599     2.07   0.040     .0323075    1.421441
             |
       _cons |  -6.228321   3.247071    -1.92   0.057    -12.64529    .1886487
------------------------------------------------------------------------------
```

B. Exp 8708

```
. ardl lnhun_8708 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,1,0,2,3,0) regression

Sample:       592 -       755                 Number of obs     =        164
                                              R-squared         =     0.5244
                                              Adj R-squared     =     0.4762
Log likelihood =  72.529372                   Root MSE          =     0.1637

------------------------------------------------------------------------------
D.lnhun_8708 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ADJ          |
  lnhun_8708 |
         L1. |  -.1135054   .0672964    -1.69   0.094    -.2464914    .0194806
-------------+----------------------------------------------------------------
LR           |
     vol_pos |   1.650189   1.273792     1.30   0.197    -.8669796    4.167358
     vol_neg |   .2923712     .81722     0.36   0.721    -1.322556    1.907298
       lnrer |   1.578487   2.706945     0.58   0.561    -3.770767    6.927742
       lnipi |   7.883727   5.426196     1.45   0.148    -2.839101    18.60655
  geopolitik |   1.168291   1.008548     1.16   0.249    -.8247232    3.161306
-------------+----------------------------------------------------------------
SR           |
  lnhun_8708 |
         LD. |   -.425615   .0904866    -4.70   0.000    -.6044275   -.2468024
        L2D. |   -.351355   .0829754    -4.23   0.000    -.5153246   -.1873855
        L3D. |  -.3794121    .075021    -5.06   0.000    -.5276628   -.2311615
             |
     vol_pos |
         D1. |  -.1188411   .0625951    -1.90   0.060    -.2425367    .0048544
             |
       lnrer |
         D1. |   .8888374   .7809682     1.14   0.257    -.6544514    2.432126
         LD. |  -1.487535   .7763817    -1.92   0.057     -3.02176    .0466902
             |
       lnipi |
         D1. |  -.3123324   .2551687    -1.22   0.223     -.816577    .1919122
         LD. |  -.9199109   .2222417    -4.14   0.000    -1.359088   -.4807342
        L2D. |  -.4734401   .1946816    -2.43   0.016    -.8581547   -.0887254
             |
       _cons |  -3.850941   1.819529    -2.12   0.036    -7.446552   -.2553292
------------------------------------------------------------------------------
```

C. Exp 8507

```
. ardl lnpol_8507 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(1,0,1,1,0,0) regression

Sample:       592 -       755                 Number of obs     =        164
                                              R-squared         =     0.2326
                                              Adj R-squared     =     0.1930
Log likelihood = -77.250703                   Root MSE          =     0.3986

------------------------------------------------------------------------------
D.lnpol_8507 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ADJ          |
  lnpol_8507 |
         L1. |  -.1877122   .0450924    -4.16   0.000    -.2767871   -.0986373
-------------+----------------------------------------------------------------
LR           |
     vol_pos |  -.6749623   .5268249    -1.28   0.202    -1.715645    .3657208
     vol_neg |   1.645144   .7937303     2.07   0.040     .0772191    3.213068
       lnrer |   15.89008   5.009756     3.17   0.002     5.993873    25.78629
       lnipi |   .8110692   2.224467     0.36   0.716    -3.583113    5.205252
  geopolitik |   4.233918   .7659241     5.53   0.000     2.720921    5.746915
-------------+----------------------------------------------------------------
SR           |
     vol_neg |
         D1. |  -.2670313   .0712715    -3.75   0.000      -.40782   -.1262425
             |
       lnrer |
         D1. |  -3.665532    2.09921    -1.75   0.083    -7.812284    .4812193
             |
       _cons |   -3.08141   2.210411    -1.39   0.165    -7.447827    1.285008
------------------------------------------------------------------------------
```

Polandia

A. Exp 8708

```
. ardl lnpol_8708 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,0,1,2,3,4) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.5005
                                                Adj R-squared     =     0.4346
Log likelihood =  60.905498                     Root MSE          =     0.1781
```

D.lnpol_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
ADJ						
lnpol_8708						
L1.	-.2078756	.0964034	-2.16	0.033	-.3984241	-.017327
LR						
vol_pos	-.055334	.2126741	-0.26	0.795	-.4757003	.3650324
vol_neg	.1850559	.2645028	0.70	0.485	-.3377536	.7078655
lnrer	2.876022	2.231549	1.29	0.200	-1.534802	7.286846
lnipi	3.105662	1.929388	1.61	0.110	-.7079185	6.919242
geopolitik	.6474311	.3906334	1.66	0.100	-.1246851	1.419547
SR						
lnpol_8708						
LD.	-.2995119	.1034219	-2.90	0.004	-.503933	-.0950909
L2D.	-.314548	.1046148	-3.01	0.003	-.5213269	-.107769
L3D.	-.490187	.0955947	-5.13	0.000	-.6791371	-.3012369
vol_neg						
D1.	-.0497477	.0326933	-1.52	0.130	-.1143685	.0148732
lnrer						
D1.	.6146283	.952434	0.65	0.520	-1.267929	2.497185
LD.	-2.477773	.9555142	-2.59	0.010	-4.366418	-.589127
lnipi						
D1.	-.3977071	.2800069	-1.42	0.158	-.9511617	.1557475
LD.	-.8927654	.2464571	-3.62	0.000	-1.379906	-.4056245
L2D.	-.4784796	.2021959	-2.37	0.019	-.878135	-.0788243
geopolitik						
D1.	-.0480504	.2356087	-0.20	0.839	-.5137487	.417648
LD.	-.2729605	.2324906	-1.17	0.242	-.7324957	.1865746
L2D.	.5034336	.2358381	2.13	0.034	.0372819	.9695854
L3D.	.2923992	.2080136	1.41	0.162	-.1187553	.7035537
_cons	-1.223244	1.431991	-0.85	0.394	-4.053681	1.607194

B. Exp 8703

```
. ardl lnpol_8703 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,0,0,2,4,1) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3920
                                                Adj R-squared     =     0.3258
Log likelihood =  27.009279                     Root MSE          =     0.2168
```

D.lnpol_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
ADJ						
lnpol_8703						
L1.	-.3280918	.0903299	-3.63	0.000	-.5066048	-.1495789
LR						
vol_pos	.1471037	.1624103	0.91	0.367	-.173857	.4680644
vol_neg	.1560949	.1482181	1.05	0.294	-.1368187	.4490086
lnrer	-1.074271	1.58265	-0.68	0.498	-4.201957	2.053415
lnipi	.3805161	.9098656	0.42	0.676	-1.417591	2.178623
geopolitik	.3054983	.2330266	1.31	0.192	-.1550166	.7660133
SR						
lnpol_8703						
LD.	-.1395979	.095826	-1.46	0.147	-.3289724	.0497767
L2D.	-.2064486	.0882372	-2.34	0.021	-.3808259	-.0320713
L3D.	-.1248768	.0785569	-1.59	0.114	-.2801235	.0303699
lnrer						
D1.	-1.962861	1.125042	-1.74	0.083	-4.186207	.2604848
LD.	2.219208	1.1267	1.97	0.051	-.0074136	4.445829
lnipi						
D1.	-.3038115	.3414408	-0.89	0.375	-.9785783	.3709552
LD.	.321577	.326132	0.99	0.326	-.3229359	.9660899
L2D.	.2239379	.2921883	0.77	0.445	-.3534944	.8013702
L3D.	.7872227	.2579364	3.05	0.003	.2774802	1.296965
geopolitik						
D1.	.3600871	.2574721	1.40	0.164	-.1487378	.8689121
_cons	3.820949	1.977099	1.93	0.055	-.0862599	7.728159

C. Exp 8507

```
. ardl lnpol_8507 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(1,0,1,1,0,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.2326
                                                Adj R-squared     =     0.1930
Log likelihood = -77.250703                     Root MSE          =     0.3986
```

D.lnpol_8507	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
ADJ						
lnpol_8507						
L1.	-.1877122	.0450924	-4.16	0.000	-.2767871	-.0986373
LR						
vol_pos	-.6749623	.5268249	-1.28	0.202	-1.715645	.3657208
vol_neg	1.645144	.7937303	2.07	0.040	.0772191	3.213068
lnrer	15.89008	5.009756	3.17	0.002	5.993873	25.78629
lnipi	.8110692	2.224467	0.36	0.716	-3.583113	5.205252
geopolitik	4.233918	.7659241	5.53	0.000	2.720921	5.746915
SR						
vol_neg						
D1.	-.2670313	.0712715	-3.75	0.000	-.40782	-.1262425
lnrer						
D1.	-3.665532	2.09921	-1.75	0.083	-7.812284	.4812193
_cons	-3.08141	2.210411	-1.39	0.165	-7.447827	1.285008

Slovakia

A. Exp 8703

```
. ardl lnsvk_8703 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(3,0,0,0,1,2) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.2864
                                                Adj R-squared     =     0.2348
Log likelihood = -69.801447                     Root MSE          =     0.3847
```

D.lnsvk_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnsvk_8703						
L1.	-.1264508	.048713	-2.60	0.010	-.2226927	-.0302089
LR						
vol_pos	1.059707	.9849645	1.08	0.284	-.8862815	3.005695
vol_neg	-.1307399	1.622122	-0.08	0.936	-3.335556	3.074076
lnrer	1.586812	4.347539	0.36	0.716	-7.002594	10.17622
lnipi	4.720785	4.059563	1.16	0.247	-3.29967	12.74124
geopolitik	1.337554	1.137246	1.18	0.241	-.9092966	3.584404
SR						
lnsvk_8703						
LD.	-.2154179	.0793737	-2.71	0.007	-.3722359	-.0585998
L2D.	-.2640262	.0792119	-3.33	0.001	-.4205247	-.1075278
lnipi						
D1.	.8423061	.406261	2.07	0.040	.0396587	1.644953
geopolitik						
D1.	.6486199	.4342116	1.49	0.137	-.2092493	1.506489
LD.	-.6296067	.4229463	-1.49	0.139	-1.465219	.2060058
_cons	-1.395715	2.228317	-0.63	0.532	-5.798189	3.006758

B. Exp 8708

```
. ardl lnsvk_8708 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(4,0,0,1,3,4) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3312
                                                Adj R-squared     =     0.2533
Log likelihood = -33.004132                     Root MSE          =     0.3136
```

D.lnsvk_8708	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnsvk_8708						
L1.	-.1115658	.0474743	-2.35	0.020	-.2053915	-.0177401
LR						
vol_pos	.4506158	.8699204	0.52	0.605	-1.268648	2.169879
vol_neg	-1.241089	1.694339	-0.73	0.465	-4.589689	2.107511
lnrer	-5.169254	4.603797	-1.12	0.263	-14.26795	3.92944
lnipi	12.17988	6.448747	1.89	0.061	-.5650705	24.92484
geopolitik	1.036319	1.159874	0.89	0.373	-1.255992	3.32863
SR						
lnsvk_8708						
LD.	-.1113545	.0825659	-1.35	0.180	-.2745333	.0518243
L2D.	-.2162744	.0972273	-2.22	0.028	-.4084293	-.0241196
L3D.	-.3275823	.1027654	-3.19	0.002	-.5306822	-.1244824
lnrer						
D1.	1.956636	1.400652	1.40	0.165	-.8115366	4.724809
lnipi						
D1.	-1.124371	.4700734	-2.39	0.018	-2.053398	-.1953431
LD.	-1.545555	.4113944	-3.76	0.000	-2.358613	-.7324977
L2D.	-.8755042	.340913	-2.57	0.011	-1.549266	-.2017424
geopolitik						
D1.	.7364144	.428964	1.72	0.088	-.1113666	1.584195
LD.	-.5076254	.429919	-1.18	0.240	-1.357294	.3420431
L2D.	.8636254	.3852521	2.24	0.026	.1022342	1.625017
L3D.	.6580293	.3852664	1.71	0.090	-.1033903	1.419449
_cons	-5.251379	2.116052	-2.48	0.014	-9.433429	-1.069329

C. Exp 8528

```
. ardl lnsvk_8528 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(2,0,0,0,3,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.2713
                                                Adj R-squared     =     0.2237
Log likelihood = -55.041156                     Root MSE          =     0.3504
```

D.lnsvk_8528	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnsvk_8528						
L1.	-.4610679	.067644	-6.82	0.000	-.5947047	-.3274311
LR						
vol_pos	.068462	.2192662	0.31	0.755	-.3647182	.5016423
vol_neg	.0644636	.410269	0.16	0.875	-.7460598	.8749871
lnrer	.4455906	1.114971	0.40	0.690	-1.757135	2.648316
lnipi	-2.049161	1.09613	-1.87	0.063	-4.214666	.1163433
geopolitik	-.4412888	.2449088	-1.80	0.074	-.9251282	.0425507
SR						
lnsvk_8528						
LD.	.1468088	.0758488	1.94	0.055	-.0030375	.296655
lnipi						
D1.	1.006213	.4958345	2.03	0.044	.0266474	1.985779
LD.	1.248722	.4378717	2.85	0.005	.3836675	2.113778
L2D.	.6954535	.3644781	1.91	0.058	-.0246059	1.415513
_cons	9.4514	2.416618	3.91	0.000	4.677153	14.22565

Slovenia

A. Exp 3004

```
. ardl lnsvn_3004 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(2,0,0,1,3,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.4428
                                                Adj R-squared     =     0.4025
Log likelihood = -68.350592                     Root MSE          =     0.3813

------------------------------------------------------------------------------
D.lnsvn_3004 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ADJ          |
  lnsvn_3004 |
         L1. |  -.3491545   .0727402    -4.80   0.000    -.4928669   -.2054422
-------------+----------------------------------------------------------------
LR           |
     vol_pos |   .0967683    .264202     0.37   0.715    -.4252139    .6187505
     vol_neg |   .0323496   .1972348     0.16   0.870     -.357326    .4220251
       lnrer |   14.55399   7.446974     1.95   0.052    -.1589526    29.26693
       lnipi |   3.568117   1.512105     2.36   0.020      .580661    6.555573
  geopolitik |   .2336327   .3986551     0.59   0.559    -.5539878    1.021253
-------------+----------------------------------------------------------------
SR           |
  lnsvn_3004 |
         LD. |  -.3350926   .0728457    -4.60   0.000    -.4790134   -.1911718
             |
       lnrer |
         D1. |  -9.559918   4.630305    -2.06   0.041    -18.70798   -.4118537
             |
       lnipi |
         D1. |   -1.90781   .5322321    -3.58   0.000    -2.959337   -.8562818
         LD. |  -1.785229   .4828736    -3.70   0.000    -2.739239   -.8312182
        L2D. |  -1.640293   .4028584    -4.07   0.000    -2.436218   -.8443678
             |
       _cons |  -2.548377   2.372206    -1.07   0.284    -7.235131    2.138376
------------------------------------------------------------------------------
```

B. Exp 8703

```
.
. gen lnsvn_8703 = ln(svn_8703)

.
. ardl lnsvn_8703 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(3,1,0,2,4,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.4077
                                                Adj R-squared     =     0.3477
Log likelihood = -24.294865                     Root MSE          =     0.2954

------------------------------------------------------------------------------
D.lnsvn_8703 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ADJ          |
  lnsvn_8703 |
         L1. |  -.3188288   .0848837    -3.76   0.000    -.4865695   -.1510882
-------------+----------------------------------------------------------------
LR           |
     vol_pos |  -.3668014   .3281546    -1.12   0.265    -1.015275    .2816723
     vol_neg |   .0903718   .1759523     0.51   0.608    -.2573314    .4380751
       lnrer |   14.54243   7.044793     2.06   0.041     .6210592    28.46381
       lnipi |   .8337795   1.469293     0.57   0.571    -2.069724    3.737283
  geopolitik |   .7554769   .3646162     2.07   0.040     .0349508    1.476003
-------------+----------------------------------------------------------------
SR           |
  lnsvn_8703 |
         LD. |  -.1878291   .0864137    -2.17   0.031    -.3585932   -.0170649
        L2D. |  -.1943218   .0739466    -2.63   0.009    -.3404493   -.0481942
             |
     vol_pos |
         D1. |   .0995624   .0641116     1.55   0.123      -.02713    .2262548
             |
       lnrer |
         D1. |  -10.47022   3.600313    -2.91   0.004    -17.58488   -3.355558
         LD. |  -9.204581   3.921438    -2.35   0.020    -16.95382   -1.455339
             |
       lnipi |
         D1. |  -.8780637   .4747534    -1.85   0.066    -1.816235    .0601071
         LD. |  -.3655616   .4718486    -0.77   0.440    -1.297992     .566869
        L2D. |  -.4338352   .4268055    -1.02   0.311    -1.277255    .4095848
        L3D. |   .7374811   .3519405     2.10   0.038     .0420036    1.432959
             |
       _cons |   2.299447   2.552004     0.90   0.369    -2.743626     7.34252
------------------------------------------------------------------------------
```

C. Exp 8516

```
. ardl lnsvn_8516 vol_pos vol_neg lnrer lnipi geopolitik, ec aic maxlags(4)

ARDL(1,0,1,4,4,0) regression

Sample:       592 -       755                  Number of obs     =        164
                                               R-squared         =     0.3818
                                               Adj R-squared     =     0.3191
Log likelihood =  48.329758                    Root MSE          =     0.1897

------------------------------------------------------------------------------
D.lnsvn_8516 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ADJ          |
  lnsvn_8516 |
         L1. |  -.4564815   .0645057    -7.08   0.000    -.5839526   -.3290103
-------------+----------------------------------------------------------------
LR           |
     vol_pos |   .1716106   .1083779     1.58   0.115    -.0425574    .3857785
     vol_neg |   .0535321   .1133164     0.47   0.637    -.1703949    .2774591
       lnrer |  -3.526675   3.310829    -1.07   0.289    -10.06928    3.015929
       lnipi |  -1.104618    .563373    -1.96   0.052    -2.217912    .0086763
  geopolitik |   .2910257   .1572643     1.85   0.066    -.0197477    .6017991
-------------+----------------------------------------------------------------
SR           |
     vol_neg |
         D1. |  -.0726831   .0341876    -2.13   0.035     -.140242   -.0051242
             |
       lnrer |
         D1. |  -.2369173   2.560922    -0.09   0.926    -5.297613    4.823778
         LD. |  -.6894602   2.437936    -0.28   0.778    -5.507119    4.128199
        L2D. |  -2.087065   2.489778    -0.84   0.403    -7.007171    2.833041
        L3D. |   6.412522   2.523153     2.54   0.012     1.426464    11.39858
             |
       lnipi |
         D1. |   .6707791   .3049445     2.20   0.029     .0681713    1.273387
         LD. |   .8062458   .2991594     2.70   0.008     .2150701    1.397421
        L2D. |   1.021496   .2656833     3.84   0.000     .4964729    1.546518
        L3D. |   .6198926   .2254338     2.75   0.007     .1744079    1.065377
             |
       _cons |   6.725499   1.549232     4.34   0.000     3.664026    9.786972
------------------------------------------------------------------------------
```

Estonia

A. Exp 8517

B. Exp 9406

```
. ardl lnest_8517 vol_pos vol_neg lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(2,0,3,4,0,4,4) regression

Sample:       592 -       755                  Number of obs     =        164
                                               R-squared         =     0.4360
                                               Adj R-squared     =     0.3434
Log likelihood =  -178.4444                    Root MSE          =     0.7774
```

D.lnest_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnest_8517						
L1.	-.3243766	.0630749	-5.14	0.000	-.449079	-.1996741
LR						
vol_pos	-.6032128	1.38166	-0.44	0.663	-3.33483	2.128404
vol_neg	9.453082	5.62116	1.68	0.095	-1.660253	20.56642
lnrer	-51.78103	11.20628	-4.62	0.000	-73.93645	-29.62561
lnipi	-.5038829	2.829116	-0.18	0.859	-6.097197	5.089431
geopolitik	-5.482497	2.037555	-2.69	0.008	-9.510852	-1.454142
euro	-2.166005	1.026426	-2.11	0.037	-4.195305	-.1367043
SR						
lnest_8517						
LD.	-.1668027	.0808683	-2.06	0.041	-.3266837	-.0069217
vol_neg						
D1.	-2.670007	1.386744	-1.93	0.056	-5.411674	.0716607
LD.	-3.304775	1.093004	-3.02	0.003	-5.465702	-1.143847
L2D.	-1.612608	.7595379	-2.12	0.036	-3.114255	-.1109605
lnrer						
D1.	27.03787	11.08936	2.44	0.016	5.113602	48.96214
LD.	15.17374	11.22799	1.35	0.179	-7.024597	37.37208
L2D.	2.461526	10.75137	0.23	0.819	-18.7945	23.71755
L3D.	38.56804	11.98062	3.22	0.002	14.88171	62.25437
geopolitik						
D1.	2.472795	.9903941	2.50	0.014	.5147323	4.430857
LD.	-.3500875	.9975836	-0.35	0.726	-2.322364	1.622189
L2D.	.5237976	1.007551	0.52	0.604	-1.468185	2.515781
L3D.	2.189599	.9821325	2.23	0.027	.2478703	4.131328
euro						
D1.	-.3898657	.816464	-0.48	0.634	-2.004059	1.224327
LD.	1.582928	.8089199	1.96	0.052	-.0163501	3.182206
L2D.	-1.835552	.815295	-2.25	0.026	-3.447434	-.2236698
L3D.	1.491396	.8316196	1.79	0.075	-.1527603	3.135553
_cons	3.34019	4.408532	0.76	0.450	-5.375715	12.05609

```
. ardl lnest_9406 vol_pos vol_neg lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(3,0,3,1,0,0,0) regression

Sample:       592 -       755                  Number of obs     =        164
                                               R-squared         =     0.3961
                                               Adj R-squared     =     0.3437
Log likelihood =  7.3397987                    Root MSE          =     0.2419
```

D.lnest_9406	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnest_9406						
L1.	-.5020764	.0571612	-8.78	0.000	-.6150215	-.3891313
LR						
vol_pos	.7689269	.2561834	3.00	0.003	.2627328	1.275121
vol_neg	1.598852	1.024619	1.56	0.121	-.4256991	3.623403
lnrer	-5.039369	1.850966	-2.72	0.007	-8.696702	-1.382036
lnipi	-.3260406	.5555633	-0.59	0.558	-1.423781	.7716998
geopolitik	.1023795	.3158538	0.32	0.746	-.5217177	.7264767
euro	.1146777	.1808731	0.63	0.527	-.2427103	.4720658
SR						
lnest_9406						
LD.	.3037165	.0745575	4.07	0.000	.1563979	.4510352
L2D.	.3548894	.0706921	5.02	0.000	.2152085	.4945702
vol_neg						
D1.	-1.2178	.3963376	-3.07	0.003	-2.000926	-.4346746
LD.	-.8658731	.3223957	-2.69	0.008	-1.502896	-.2288496
L2D.	-.4698419	.227333	-2.07	0.040	-.9190303	-.0206535
lnrer						
D1.	5.267212	3.104472	1.70	0.092	-.866932	11.40136
_cons	4.801954	1.435254	3.35	0.001	1.966027	7.637881

C. Exp 8703

```
. ardl lnest_8703 vol_pos vol_neg lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(1,1,0,0,0,0,0) regression

Sample:       592 -       755                  Number of obs     =        164
                                               R-squared         =     0.2716
                                               Adj R-squared     =     0.2340
Log likelihood = -110.25965                    Root MSE          =     0.4875
```

D.lnest_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnest_8703						
L1.	-.4269611	.0657036	-6.50	0.000	-.5567512	-.297171
LR						
vol_pos	.328211	.7546158	0.43	0.664	-1.162447	1.818869
vol_neg	-.7754706	1.26957	-0.61	0.542	-3.283363	1.732422
lnrer	-8.630175	4.255853	-2.03	0.044	-17.03713	-.2232185
lnipi	.9533788	1.235788	0.77	0.442	-1.487781	3.394539
geopolitik	-.8146055	.7275428	-1.12	0.265	-2.251784	.6225732
euro	-.2783754	.4133496	-0.67	0.502	-1.094901	.5381501
SR						
vol_pos						
D1.	-.4752422	.2162952	-2.20	0.029	-.902509	-.0479754
_cons	1.06769	2.431496	0.44	0.661	-3.735455	5.870835

Latvia

A. Exp 4407

```
. ardl lnlva_4407 vol_pos vol_neg lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(4,4,0,0,4,0,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3958
                                                Adj R-squared     =     0.3208
Log likelihood =  13.018203                     Root MSE          =     0.2377
```

D.lnlva_4407	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnlva_4407						
L1.	-.2622852	.0657133	-3.99	0.000	-.3921649	-.1324054
LR						
vol_pos	1.410431	1.826665	0.77	0.441	-2.199898	5.020759
vol_neg	2.504955	1.675186	1.50	0.137	-.8059823	5.815892
lnrer	2.832611	6.544256	0.43	0.666	-10.10185	15.76707
lnipi	1.30155	1.601956	0.81	0.418	-1.864652	4.467752
geopolitik	.6066464	.5083443	1.19	0.235	-.3980755	1.611368
euro	.6586115	.1948658	3.38	0.001	.2734671	1.043756
SR						
lnlva_4407						
LD.	-.2633053	.0847518	-3.11	0.002	-.4308138	-.0957969
L2D.	-.1081984	.0833236	-1.30	0.196	-.2728842	.0564874
L3D.	-.1459844	.075276	-1.94	0.054	-.2947644	.0027956
vol_pos						
D1.	-.18234	.4464465	-0.41	0.684	-1.064723	.7000434
LD.	.1811984	.3195801	0.57	0.572	-.4504387	.8128355
L2D.	.1381826	.2380853	0.58	0.563	-.3323834	.6087485
L3D.	.2984733	.1510082	1.98	0.050	.0000117	.5969349
lnipi						
D1.	-.3899172	.4134601	-0.94	0.347	-1.207104	.42727
LD.	-.2237295	.3839985	-0.58	0.561	-.982687	.5352279
L2D.	.6817635	.3406096	2.00	0.047	.0085625	1.354965
L3D.	.668896	.2904924	2.30	0.023	.0947495	1.243042
_cons	.2239441	1.97643	0.11	0.910	-3.68239	4.130278

B. Exp 8517

```
. ardl lnlva_8517 vol_pos vol_neg lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(2,0,0,2,4,0,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3563
                                                Adj R-squared     =     0.2958
Log likelihood = -49.942484                     Root MSE          =     0.3442
```

D.lnlva_8517	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnlva_8517						
L1.	-.2715519	.063874	-4.25	0.000	-.3977677	-.1453361
LR						
vol_pos	.3860697	.8698752	0.44	0.658	-1.332815	2.104955
vol_neg	-2.912259	1.807549	-1.61	0.109	-6.484	.6594825
lnrer	16.14952	10.3817	1.56	0.122	-4.364854	36.66389
lnipi	.3254292	2.055126	0.16	0.874	-3.735526	4.386384
geopolitik	-.6522576	.7157406	-0.91	0.364	-2.06657	.7620554
euro	1.025431	.2787372	3.68	0.000	.4746423	1.576219
SR						
lnlva_8517						
LD.	-.2801917	.0749616	-3.74	0.000	-.4283169	-.1320665
lnrer						
D1.	-12.388	6.454826	-1.92	0.057	-25.14282	.3668229
LD.	9.023594	6.350742	1.42	0.157	-3.525555	21.57274
lnipi						
D1.	.6455391	.5757899	1.12	0.264	-.4922294	1.783308
LD.	.0071202	.5440617	0.01	0.990	-1.067953	1.082193
L2D.	.2565647	.4759291	0.54	0.591	-.6838775	1.197007
L3D.	.8443305	.3944575	2.14	0.034	.0648772	1.623784
_cons	1.753066	2.760229	0.64	0.526	-3.701183	7.207316

C. Exp 3004

```
. ardl lnlva_3004 vol_pos vol_neg lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(4,0,0,0,0,2,0) regression

Sample:       592 -       755                   Number of obs     =        164
                                                R-squared         =     0.3353
                                                Adj R-squared     =     0.2825
Log likelihood =  22.200091                     Root MSE          =     0.2202
```

D.lnlva_3004	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnlva_3004						
L1.	-.2180957	.0765694	-2.85	0.005	-.3693815	-.06681
LR						
vol_pos	-.5017284	.8382768	-0.60	0.550	-2.157995	1.154538
vol_neg	1.706968	1.362959	1.25	0.212	-.985964	4.3999
lnrer	-5.947235	7.779555	-0.76	0.446	-21.31807	9.423602
lnipi	-.0290852	1.042152	-0.03	0.978	-2.088168	2.029997
geopolitik	.6778362	.5199671	1.30	0.194	-.3495143	1.705187
euro	.1547051	.2223417	0.70	0.488	-.2845974	.5940076
SR						
lnlva_3004						
LD.	-.3158654	.0928901	-3.40	0.001	-.4993976	-.1323333
L2D.	-.1530934	.087208	-1.76	0.081	-.3253989	.019212
L3D.	-.204395	.0783158	-2.61	0.010	-.3591313	-.0496588
geopolitik						
D1.	.7069772	.248297	2.85	0.005	.2163923	1.197562
LD.	.499348	.3072661	1.63	0.106	-.1077481	1.106444
_cons	1.948204	1.313652	1.48	0.140	-.647309	4.543716

Lithuania

A. Exp 9403

```
. ardl lnltu_9403 vol_pos vol_neg lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(2,4,0,0,1,0,0) regression

Sample:       592 -       755                 Number of obs     =        164
                                              R-squared         =     0.3061
                                              Adj R-squared     =     0.2460
Log likelihood =  70.579548                   Root MSE          =     0.1645
```

D.lnltu_9403	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnltu_9403						
L1.	-.3335124	.0630963	-5.29	0.000	-.4581847	-.20884
LR						
vol_pos	-1.709166	1.193263	-1.43	0.154	-4.06694	.6486083
vol_neg	4.16003	1.717226	2.42	0.017	.7669539	7.553106
lnrer	-1.53817	1.229294	-1.25	0.213	-3.967138	.8907979
lnipi	-1.563802	.6744613	-2.32	0.022	-2.896474	-.2311305
geopolitik	.1827682	.2150026	0.85	0.397	-.2420566	.607593
euro	.2098706	.1248032	1.68	0.095	-.0367287	.4564699
SR						
lnltu_9403						
LD.	-.1161192	.0771582	-1.50	0.134	-.2685766	.0363381
vol_pos						
D1.	.269524	.3007897	0.90	0.372	-.3248079	.8638559
LD.	.3409798	.2470227	1.38	0.170	-.1471136	.8290732
L2D.	.3775506	.1989812	1.90	0.060	-.0156175	.7707187
L3D.	.4284922	.1361703	3.15	0.002	.1594326	.6975518
lnipi						
D1.	.3643284	.1780903	2.05	0.043	.0124389	.716218
_cons	4.828722	1.34443	3.59	0.000	2.172256	7.485189

B. Exp 8703

```
. ardl lnltu_8703 vol_pos vol_neg lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(4,4,3,0,3,0,4) regression

Sample:       592 -       755                 Number of obs     =        164
                                              R-squared         =     0.5020
                                              Adj R-squared     =     0.4160
Log likelihood = -50.517171                   Root MSE          =     0.3576
```

D.lnltu_8703	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
ADJ						
lnltu_8703						
L1.	-.125692	.0519906	-2.42	0.017	-.2284866	-.0228974
LR						
vol_pos	8.774721	8.881702	0.99	0.325	-8.785983	26.33542
vol_neg	.2047284	19.81861	0.01	0.992	-38.98018	39.38963
lnrer	-.0566504	8.170634	-0.01	0.994	-16.21145	16.09815
lnipi	-4.875585	5.200459	-0.94	0.350	-15.15782	5.406648
geopolitik	-1.774825	1.430967	-1.24	0.217	-4.604102	1.054452
euro	1.433725	.9897449	1.45	0.150	-.5231762	3.390627
SR						
lnltu_8703						
LD.	-.3412849	.076809	-4.44	0.000	-.4931499	-.1894198
L2D.	-.323292	.0739316	-4.37	0.000	-.4694679	-.1771161
L3D.	-.1678044	.0643598	-2.61	0.010	-.2950552	-.0405536
vol_pos						
D1.	.4273554	.7438262	0.57	0.567	-1.043321	1.898032
LD.	.1403286	.5952533	0.24	0.814	-1.036593	1.31725
L2D.	-.8278378	.4605709	-1.80	0.074	-1.738468	.0827926
L3D.	-.5637994	.304446	-1.85	0.066	-1.165743	.0381445
vol_neg						
D1.	-2.270934	2.173718	-1.04	0.298	-6.56876	2.026892
LD.	-1.865764	1.712747	-1.09	0.278	-5.252168	1.52064
L2D.	1.814914	1.171392	1.55	0.124	-.5011351	4.130963
lnipi						
D1.	.9561656	.582848	1.64	0.103	-.1962285	2.10856
LD.	2.313856	.5210938	4.44	0.000	1.283561	3.344152
L2D.	1.05845	.4199212	2.52	0.013	.2281911	1.888709
euro						
D1.	-.0146127	.3787624	-0.04	0.969	-.7634933	.7342679
LD.	.2054996	.3802044	0.54	0.590	-.5462321	.9572313
L2D.	.8030874	.3808205	2.11	0.037	.0501376	1.556037
L3D.	-.866394	.4154178	-2.09	0.039	-1.687749	-.0450392
_cons	3.743617	3.297361	1.14	0.258	-2.775852	10.26309

C. Exp 3102

```
. ardl lnltu_3102 vol_pos vol_neg lnrer lnipi geopolitik euro, ec aic maxlags(4)

ARDL(1,4,0,0,2,0,2) regression

Sample:       592 -      755                 Number of obs     =        164
                                             R-squared         =     0.4236
                                             Adj R-squared     =     0.3652
Log likelihood = -150.86485                  Root MSE          =     0.6391
```

D.lnltu_3102	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
ADJ						
lnltu_3102						
L1.	-.6011684	.0711222	-8.45	0.000	-.7417145	-.4606223
LR						
vol_pos	-3.765343	2.394664	-1.57	0.118	-8.497493	.9668069
vol_neg	-2.089249	3.57698	-0.58	0.560	-9.1578	4.979302
lnrer	5.336909	2.705557	1.97	0.050	-.009603	10.68342
lnipi	-.4709058	1.602367	-0.29	0.769	-3.637379	2.695567
geopolitik	.5270579	.4651761	1.13	0.259	-.392187	1.446303
euro	.2215946	.2807219	0.79	0.431	-.3331463	.7763355
SR						
vol_pos						
D1.	2.380418	1.157886	2.06	0.042	.0922936	4.668543
LD.	2.084486	.95344	2.19	0.030	.2003713	3.9686
L2D.	2.147256	.7671083	2.80	0.006	.6313565	3.663156
L3D.	2.117385	.5259411	4.03	0.000	1.078061	3.156709
lnipi						
D1.	.6553753	.8398519	0.78	0.436	-1.004275	2.315025
LD.	-1.150759	.6863167	-1.68	0.096	-2.507005	.2054871
euro						
D1.	-.1735381	.6581274	-0.26	0.792	-1.474079	1.127002
LD.	-1.268114	.6612407	-1.92	0.057	-2.574807	.0385786
_cons	10.96796	5.476164	2.00	0.047	.1463861	21.78953

Lampiran 12 Uji Kointegrasi NARDL

Republik Ceko

A. Exp 8703

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                  F =      5.112
Case 3                                                     t =     -3.495

Finite sample (5 variables, 164 observations, 11 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values
```

	10% I(0)	10% I(1)	5% I(0)	5% I(1)	1% I(0)	1% I(1)	p-value I(0)	p-value I(1)
F	2.263	3.412	2.644	3.884	3.475	4.888	0.000	0.007
t	-2.531	-3.826	-2.840	-4.174	-3.444	-4.831	0.009	0.174

```
do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:
```

	10%	5%	1%
decision	.	.	.

B. Exp 8517

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                  F =      2.942
Case 3                                                     t =     -2.914

Finite sample (5 variables, 164 observations, 4 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values
```

	10% I(0)	10% I(1)	5% I(0)	5% I(1)	1% I(0)	1% I(1)	p-value I(0)	p-value I(1)
F	2.290	3.408	2.674	3.876	3.507	4.869	0.030	0.188
t	-2.552	-3.861	-2.859	-4.204	-3.458	-4.853	0.044	0.388

```
do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:
```

	10%	5%	1%
decision	.	.	.a

C. Exp 8708

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                     F =     2.131
Case 3                                                        t =    -2.065

Finite sample (5 variables, 164 observations, 11 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values
```

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.263	3.412	2.644	3.884	3.475	4.888	0.126	0.467
t	-2.531	-3.826	-2.840	-4.174	-3.444	-4.831	0.236	0.696

```
do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:
```

	10%	5%	1%
decision	.a	.a	.a

Hungaria

A. Exp 8703

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                     F =     1.674
Case 3                                                        t =    -2.845

Finite sample (5 variables, 164 observations, 10 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values
```

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.267	3.412	2.649	3.883	3.479	4.885	0.266	0.677
t	-2.534	-3.831	-2.843	-4.178	-3.446	-4.834	0.050	0.398

```
do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:
```

	10%	5%	1%
decision	.a	.a	.a

B. Exp 8708

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                     F =     5.409
Case 3                                                        t =    -1.687

Finite sample (5 variables, 164 observations, 9 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values
```

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.271	3.411	2.653	3.882	3.484	4.883	0.000	0.004
t	-2.537	-3.836	-2.846	-4.182	-3.448	-4.837	0.405	0.816

```
do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:
```

	10%	5%	1%
decision	.a	.a	.a

C. Exp 8507

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     4.950
Case 3                                                      t =    -3.252

Finite sample (5 variables, 164 observations, 4 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values
```

	10% I(0)	10% I(1)	5% I(0)	5% I(1)	1% I(0)	1% I(1)	p-value I(0)	p-value I(1)
F	2.290	3.408	2.674	3.876	3.507	4.869	0.001	0.009
t	-2.552	-3.861	-2.859	-4.204	-3.458	-4.853	0.018	0.262

```
do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:
```

	10%	5%	1%
decision	.	.	.a

Polandia

A. Exp 8708

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     2.699
Case 3                                                      t =    -2.156

Finite sample (5 variables, 164 observations, 13 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values
```

	10% I(0)	10% I(1)	5% I(0)	5% I(1)	1% I(0)	1% I(1)	p-value I(0)	p-value I(1)
F	2.256	3.413	2.636	3.886	3.465	4.893	0.044	0.254
t	-2.525	-3.815	-2.835	-4.165	-3.440	-4.825	0.201	0.657

```
do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:
```

	10%	5%	1%
decision	.a	.a	.a

B. Exp 8507

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     6.573
Case 3                                                      t =    -4.163

Finite sample (5 variables, 164 observations, 2 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values
```

	10% I(0)	10% I(1)	5% I(0)	5% I(1)	1% I(0)	1% I(1)	p-value I(0)	p-value I(1)
F	2.297	3.407	2.682	3.874	3.517	4.864	0.000	0.000
t	-2.558	-3.871	-2.864	-4.213	-3.461	-4.859	0.001	0.056

```
do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:
```

	10%	5%	1%
decision	.r	.	.

C. Exp 8703

. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship F = 3.050
Case 3 t = -3.632

Finite sample (5 variables, 164 observations, 10 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.267	3.412	2.649	3.883	3.479	4.885	0.023	0.164
t	-2.534	-3.831	-2.843	-4.178	-3.446	-4.834	0.006	0.141

do not reject H0 if
either F or t are closer to zero than critical values for I(0) variables
(if either p-value > desired level for I(0) variables)
reject H0 if
both F and t are more extreme than critical values for I(1) variables
(if both p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

	10%	5%	1%
decision	.	.	.a

Slovakia

A. Exp 8703

. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship F = 1.762
Case 3 t = -2.596

Finite sample (5 variables, 164 observations, 5 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.286	3.409	2.669	3.877	3.503	4.872	0.238	0.637
t	-2.549	-3.856	-2.856	-4.200	-3.456	-4.849	0.090	0.515

do not reject H0 if
either F or t are closer to zero than critical values for I(0) variables
(if either p-value > desired level for I(0) variables)
reject H0 if
both F and t are more extreme than critical values for I(1) variables
(if both p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

	10%	5%	1%
decision	.a	.a	.a

B. Exp 8708

. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship F = 2.434
Case 3 t = -2.350

Finite sample (5 variables, 164 observations, 11 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.263	3.412	2.644	3.884	3.475	4.888	0.074	0.344
t	-2.531	-3.826	-2.840	-4.174	-3.444	-4.831	0.143	0.592

do not reject H0 if
either F or t are closer to zero than critical values for I(0) variables
(if either p-value > desired level for I(0) variables)
reject H0 if
both F and t are more extreme than critical values for I(1) variables
(if both p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

	10%	5%	1%
decision	.a	.a	.a

C. Exp 8528

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     8.290
Case 3                                                      t =    -6.816

Finite sample (5 variables, 164 observations, 4 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values
```

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.290	3.408	2.674	3.876	3.507	4.869	0.000	0.000
t	-2.552	-3.861	-2.859	-4.204	-3.458	-4.853	0.000	0.000

```
do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:
```

	10%	5%	1%
decision	.r	.r	.r

Slovenia

A. Exp 3004

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     4.371
Case 3                                                      t =    -4.800

Finite sample (5 variables, 164 observations, 5 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values
```

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.286	3.409	2.669	3.877	3.503	4.872	0.002	0.023
t	-2.549	-3.856	-2.856	-4.200	-3.456	-4.849	0.000	0.011

```
do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:
```

	10%	5%	1%
decision	.r	.r	.

B. Exp 8703

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     3.528
Case 3                                                      t =    -3.756

Finite sample (5 variables, 164 observations, 9 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values
```

	10%		5%		1%		p-value	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
F	2.271	3.411	2.653	3.882	3.484	4.883	0.009	0.085
t	-2.537	-3.836	-2.846	-4.182	-3.448	-4.837	0.004	0.115

```
do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:
```

	10%	5%	1%
decision	.	.	.

C. Exp 8516

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     8.883
Case 3                                                      t =    -7.077

Finite sample (5 variables, 164 observations, 9 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%               | 5%                | 1%                | p-value
   |     I(0)     I(1) |     I(0)     I(1) |     I(0)     I(1) |     I(0)     I(1)
---+-------------------+-------------------+-------------------+------------------
 F |    2.271    3.411 |    2.653    3.882 |    3.484    4.883 |    0.000    0.000
 t |   -2.537   -3.836 |   -2.846   -4.182 |   -3.448   -4.837 |    0.000    0.000

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |       10%         5%         1%
-------------+---------------------------------
    decision |        .r         .r         .r
```

Estonia

A. Exp 8517

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     4.998
Case 3                                                      t =    -5.143

Finite sample (6 variables, 164 observations, 16 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%               | 5%                | 1%                | p-value
   |     I(0)     I(1) |     I(0)     I(1) |     I(0)     I(1) |     I(0)     I(1)
---+-------------------+-------------------+-------------------+------------------
 F |    2.114    3.299 |    2.455    3.733 |    3.195    4.653 |    0.000    0.005
 t |   -2.511   -3.989 |   -2.824   -4.346 |   -3.433   -5.020 |    0.000    0.007

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |       10%         5%         1%
-------------+---------------------------------
    decision |        .r         .r         .r
```

B. Exp 9406

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =    12.591
Case 3                                                      t =    -8.784

Finite sample (6 variables, 164 observations, 6 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%               | 5%                | 1%                | p-value
   |     I(0)     I(1) |     I(0)     I(1) |     I(0)     I(1) |     I(0)     I(1)
---+-------------------+-------------------+-------------------+------------------
 F |    2.150    3.293 |    2.494    3.722 |    3.240    4.629 |    0.000    0.000
 t |   -2.542   -4.044 |   -2.851   -4.393 |   -3.454   -5.054 |    0.000    0.000

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |       10%         5%         1%
-------------+---------------------------------
    decision |        .r         .r         .r
```

C. Exp 8703

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     6.233
Case 3                                                      t =    -6.498

Finite sample (6 variables, 164 observations, 1 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%               | 5%                | 1%                | p-value
   |    I(0)     I(1)  |    I(0)     I(1)  |    I(0)     I(1)  |    I(0)     I(1)
---+-------------------+-------------------+-------------------+------------------
 F |   2.168    3.289  |   2.514    3.716  |   3.262    4.617  |   0.000    0.000
 t |  -2.558   -4.071  |  -2.865   -4.417  |  -3.464   -5.072  |   0.000    0.000

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |      10%        5%        1%
-------------+------------------------------
    decision |       .r        .r        .r
```

Latvia

A. Exp 4407

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     4.571
Case 3                                                      t =    -3.991

Finite sample (6 variables, 164 observations, 11 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%               | 5%                | 1%                | p-value
   |    I(0)     I(1)  |    I(0)     I(1)  |    I(0)     I(1)  |    I(0)     I(1)
---+-------------------+-------------------+-------------------+------------------
 F |   2.132    3.296  |   2.475    3.727  |   3.218    4.641  |   0.000    0.011
 t |  -2.527   -4.016  |  -2.838   -4.370  |  -3.443   -5.037  |   0.002    0.105

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |      10%        5%        1%
-------------+------------------------------
    decision |        .         .         .
```

B. Exp 8517

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                   F =     4.277
Case 3                                                      t =    -4.251

Finite sample (6 variables, 164 observations, 7 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%               | 5%                | 1%                | p-value
   |    I(0)     I(1)  |    I(0)     I(1)  |    I(0)     I(1)  |    I(0)     I(1)
---+-------------------+-------------------+-------------------+------------------
 F |   2.146    3.293  |   2.490    3.723  |   3.235    4.632  |   0.001    0.019
 t |  -2.539   -4.038  |  -2.849   -4.389  |  -3.452   -5.051  |   0.001    0.066

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |      10%        5%        1%
-------------+------------------------------
    decision |       .r         .         .
```

C. Exp 3004

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                         F =     1.757
Case 3                                                            t =    -2.848

Finite sample (6 variables, 164 observations, 5 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%              | 5%               | 1%               | p-value
   |     I(0)    I(1) |     I(0)    I(1) |     I(0)    I(1) |     I(0)    I(1)
---+------------------+------------------+------------------+-----------------
 F |    2.153   3.292 |    2.498   3.720 |    3.244   4.627 |    0.208   0.645
 t |   -2.545  -4.049 |   -2.854  -4.398 |   -3.456  -5.058 |    0.051   0.484

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |      10%        5%        1%
-------------+------------------------------
    decision |       .a        .a        .a
```

Lithuania

A. Exp 9403

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                         F =     5.156
Case 3                                                            t =    -5.286

Finite sample (6 variables, 164 observations, 6 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%              | 5%               | 1%               | p-value
   |     I(0)    I(1) |     I(0)    I(1) |     I(0)    I(1) |     I(0)    I(1)
---+------------------+------------------+------------------+-----------------
 F |    2.150   3.293 |    2.494   3.722 |    3.240   4.629 |    0.000   0.004
 t |   -2.542  -4.044 |   -2.851  -4.393 |   -3.454  -5.054 |    0.000   0.005

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |      10%        5%        1%
-------------+------------------------------
    decision |       .r        .r        .r
```

B. Exp 8703

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                         F =     2.672
Case 3                                                            t =    -2.418

Finite sample (6 variables, 164 observations, 17 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%              | 5%               | 1%               | p-value
   |     I(0)    I(1) |     I(0)    I(1) |     I(0)    I(1) |     I(0)    I(1)
---+------------------+------------------+------------------+-----------------
 F |    2.111   3.300 |    2.451   3.734 |    3.191   4.655 |    0.031   0.245
 t |   -2.508  -3.984 |   -2.822  -4.341 |   -3.431  -5.016 |    0.120   0.611

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |      10%        5%        1%
-------------+------------------------------
    decision |       .a        .a        .a
```

C. Exp 3102

```
. estat ectest

Pesaran, Shin, and Smith (2001) bounds test

H0: no level relationship                                F =    10.544
Case 3                                                   t =    -8.453

Finite sample (6 variables, 164 observations, 8 short-run coefficients)

Kripfganz and Schneider (2020) critical values and approximate p-values

   | 10%             | 5%              | 1%              | p-value
   |   I(0)     I(1) |   I(0)     I(1) |   I(0)     I(1) |   I(0)     I(1)
---+-----------------+-----------------+-----------------+-----------------
 F |  2.143    3.294 |  2.487    3.724 |  3.231    4.634 |  0.000    0.000
 t | -2.536   -4.033 | -2.846   -4.384 | -3.450   -5.048 |  0.000    0.000

do not reject H0 if
    either F or t are closer to zero than critical values for I(0) variables
      (if either p-value  > desired level for I(0) variables)
reject H0 if
    both F and t are more extreme than critical values for I(1) variables
      (if both   p-values < desired level for I(1) variables)

decision: no rejection (.a), inconclusive (.), or rejection (.r) at levels:

             |     10%       5%       1%
-------------+---------------------------
    decision |      .r       .r       .r
```

Lampiran 13 Uji RESET NARDL

Republik Ceko

A. Exp 8703

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lncze_8703
       Ho:  model has no omitted variables
                F(3, 143) =      2.03
                 Prob > F =      0.1121
```

B. Exp 8517

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lncze_8517
       Ho:  model has no omitted variables
                F(3, 150) =      0.32
                 Prob > F =      0.8125
```

C. Exp 8708

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lncze_8708
       Ho:  model has no omitted variables
                F(3, 143) =      0.35
                 Prob > F =      0.7867
```

Hungaria

A. Exp 8703

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnhun_8703
       Ho:  model has no omitted variables
                F(3, 144) =      0.66
                 Prob > F =      0.5802
```

B. Exp 8708

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnhun_8708
       Ho:  model has no omitted variables
                F(3, 145) =      3.06
                 Prob > F =      0.0304
```

C. Exp 8507

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnhun_8507
       Ho:  model has no omitted variables
                F(3, 150) =     18.13
                 Prob > F =      0.0000
```

Polandia

A. Exp 8708

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnpol_8708
       Ho:  model has no omitted variables
                F(3, 141) =      2.09
                 Prob > F =      0.1047
```

B. Exp 8507

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnpol_8507
       Ho:  model has no omitted variables
                F(3, 152) =     60.00
                 Prob > F =      0.0000
```

C. Exp 8703

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnpol_8703
       Ho:  model has no omitted variables
                F(3, 144) =      2.33
                 Prob > F =      0.0773
```

Slovakia

A. Exp 8703

B. Exp 8708

C. Exp 8528

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnsvk_8703
       Ho:  model has no omitted variables
                 F(3, 149) =      0.82
                  Prob > F =      0.4846
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnsvk_8708
       Ho:  model has no omitted variables
                 F(3, 143) =      1.80
                  Prob > F =      0.1499
```

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnsvk_8528
       Ho:  model has no omitted variables
                 F(3, 150) =      0.86
                  Prob > F =      0.4634
```

Slovenia

A. Exp 3004

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnsvn_3004
       Ho:  model has no omitted variables
                 F(3, 149) =      2.79
                  Prob > F =      0.0426
```

B. Exp 8703

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnsvn_8703
       Ho:  model has no omitted variables
                 F(3, 145) =      4.25
                  Prob > F =      0.0066
```

C. Exp 8516

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnsvn_8516
       Ho:  model has no omitted variables
                 F(3, 145) =      4.00
                  Prob > F =      0.0090
```

Estonia

A. Exp 8517

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnest_8517
       Ho:  model has no omitted variables
                 F(3, 137) =      5.64
                  Prob > F =      0.0011
```

B. Exp 9406

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnest_9406
       Ho:  model has no omitted variables
                 F(3, 147) =      2.66
                  Prob > F =      0.0504
```

C. Exp 8703

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnest_8703
       Ho:  model has no omitted variables
                 F(3, 152) =      1.76
                  Prob > F =      0.1565
```

Latvia

A. Exp 4407

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnlva_4407
       Ho:  model has no omitted variables
                 F(3, 142) =      2.77
                  Prob > F =      0.0441
```

,

B. Exp 8517

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnlva_8517
       Ho:  model has no omitted variables
                 F(3, 146) =      1.65
                  Prob > F =      0.1811
```

C. Exp 3004

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnlva_3004
       Ho:  model has no omitted variables
                 F(3, 148) =      1.60
                  Prob > F =      0.1915
```

Lithuania

A. Exp 9403

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnltu_9403
       Ho:  model has no omitted variables
                 F(3, 147) =      4.05
                  Prob > F =      0.0084
```

B. Exp 8703

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnltu_8703
       Ho:  model has no omitted variables
                 F(3, 136) =      0.60
                  Prob > F =      0.6184
```

C. Exp 3102

```
. estat ovtest

Ramsey RESET test using powers of the fitted values of D.lnltu_3102
       Ho:  model has no omitted variables
                 F(3, 145) =      2.56
                  Prob > F =      0.0573
```

Lampiran 14 Uji CUSUM NARDL

Republik Ceko

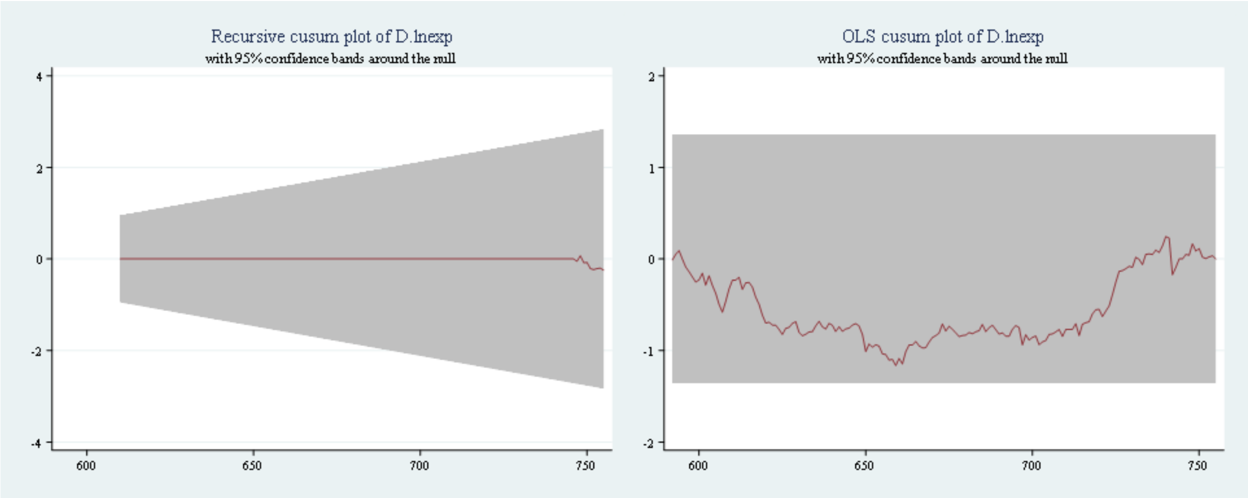
A. Exp 8703

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                              Number of obs =       164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0820	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                              Number of obs =       164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	1.1657	1.6276	1.3581	1.224



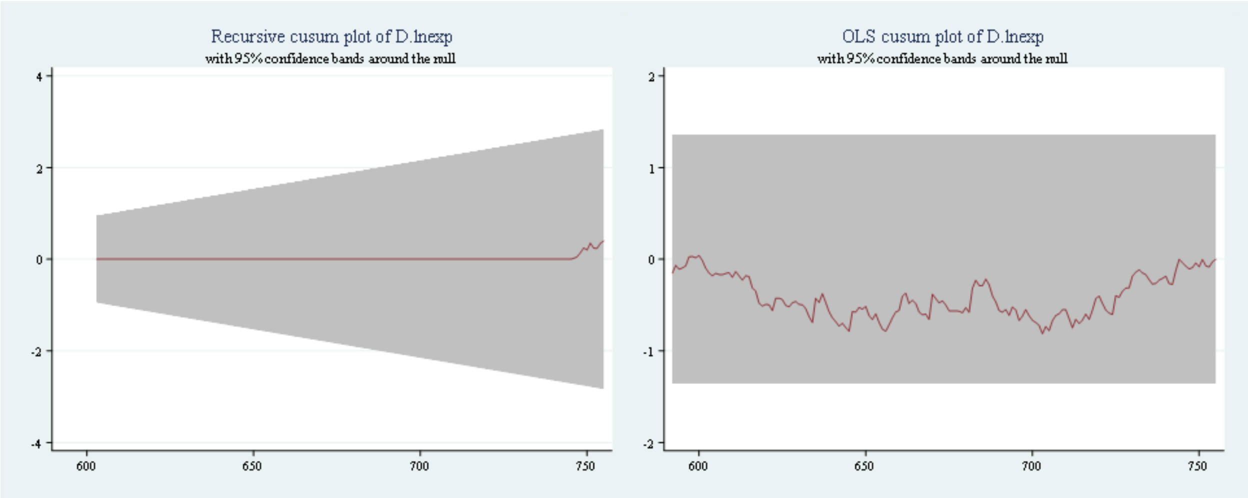
B. Exp 8517

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                              Number of obs =       164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.1329	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                              Number of obs =       164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.8137	1.6276	1.3581	1.224



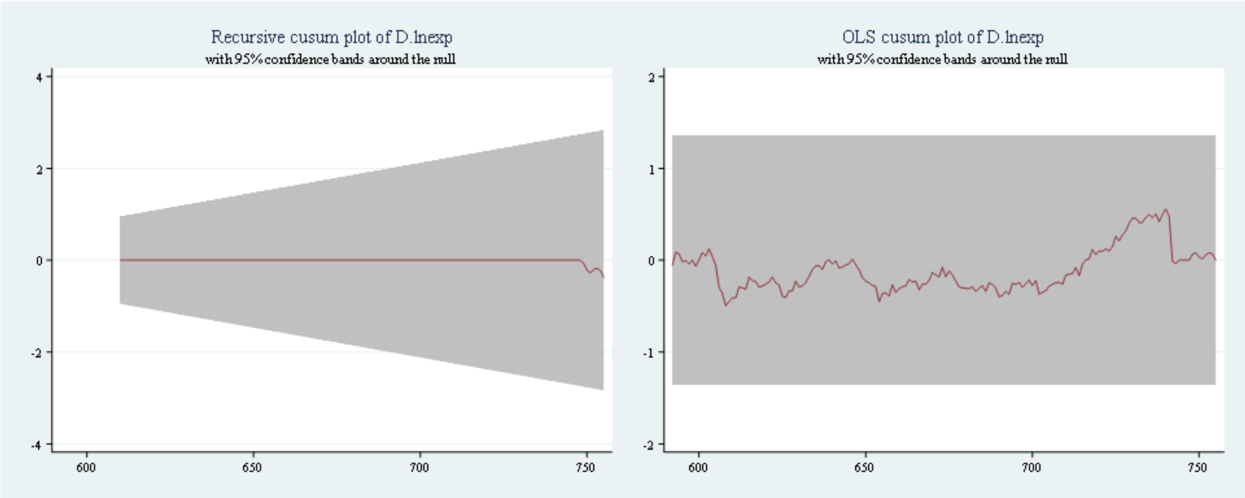
C. Exp 8708

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                   Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.1239	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                   Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.5569	1.6276	1.3581	1.224



Hungaria

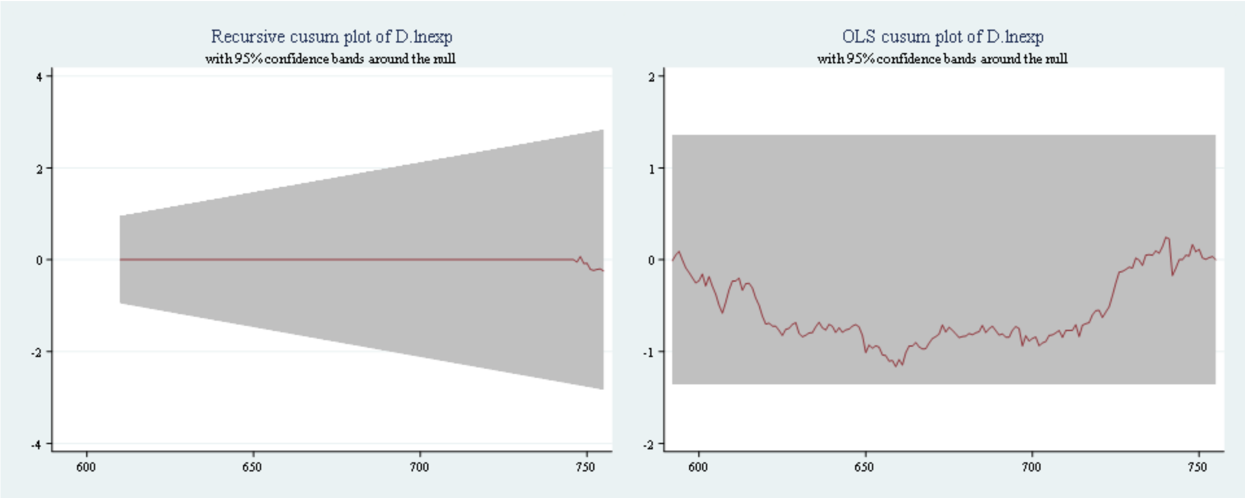
A. Exp 8703

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                   Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.6570	1.6276	1.3581	1.224

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                   Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.1182	1.1430	0.9479	0.850



B. Exp 8708

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

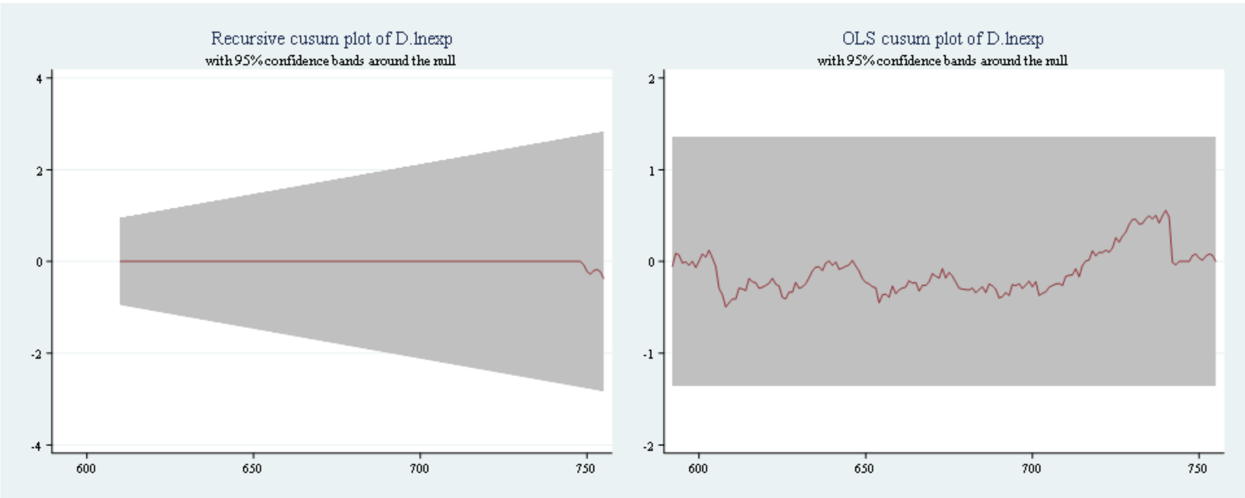
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0732	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

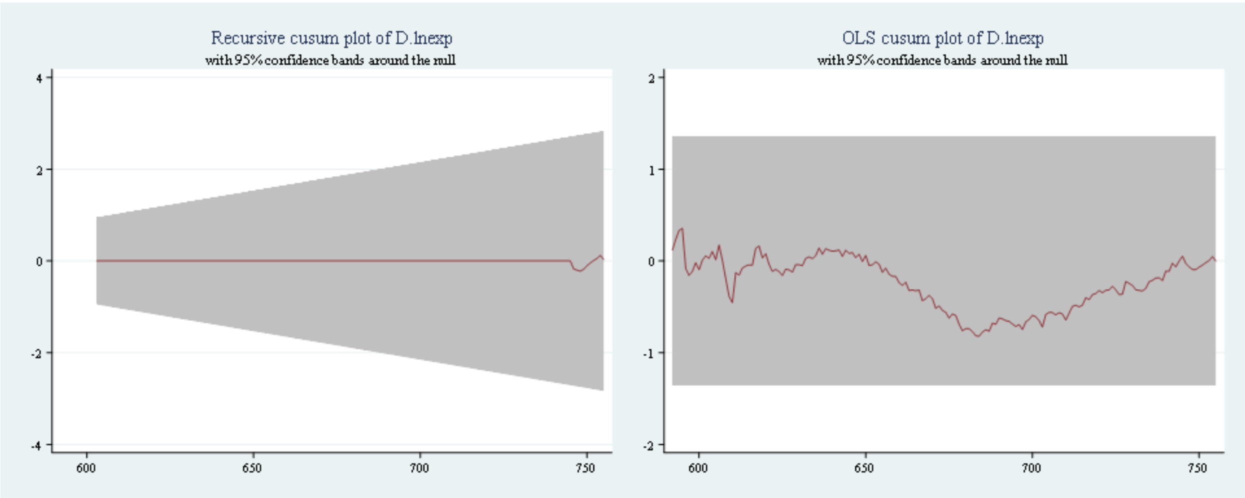
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.5415	1.6276	1.3581	1.224



C. Exp 8507

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =        164
Ho: No structural break

                                   1% Critical       5% Critical      10% Critical
Statistic    Test Statistic          Value             Value             Value
recursive        0.0764              1.1430            0.9479            0.850
```

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =        164
Ho: No structural break

                                   1% Critical       5% Critical      10% Critical
Statistic    Test Statistic          Value             Value             Value
ols              0.8211              1.6276            1.3581            1.224
```

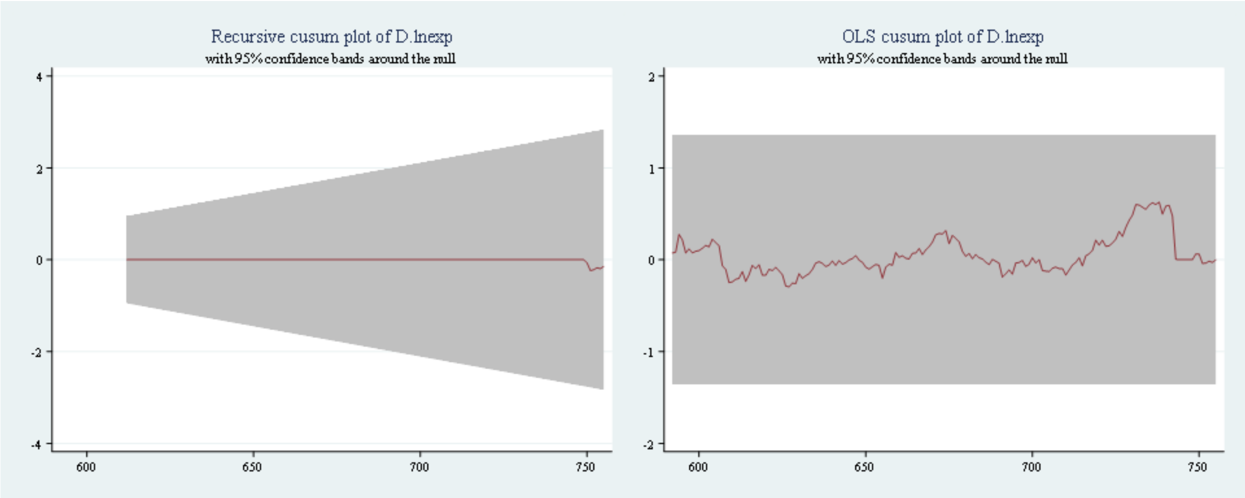


Polandia

A. Exp 8708

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =        164
Ho: No structural break

                                   1% Critical       5% Critical      10% Critical
Statistic    Test Statistic          Value             Value             Value
recursive        0.0809              1.1430            0.9479            0.850
```

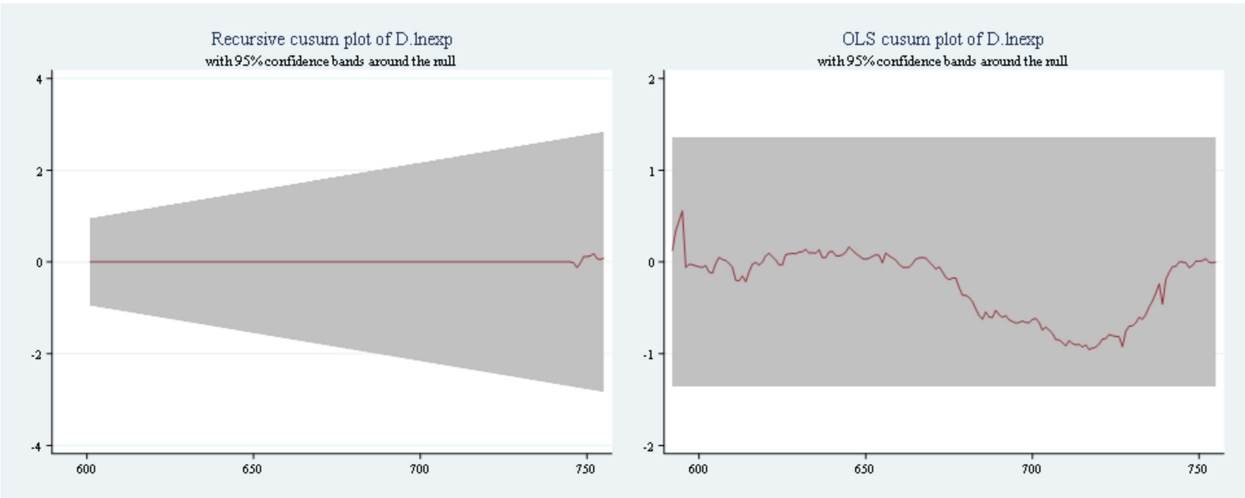
```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =        164
Ho: No structural break

                                   1% Critical       5% Critical      10% Critical
Statistic    Test Statistic          Value             Value             Value
ols              0.6274              1.6276            1.3581            1.224
```



B. Exp 8507

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                   Number of obs =        164
Ho: No structural break

                                   1% Critical      5% Critical     10% Critical
 Statistic    Test Statistic          Value            Value            Value

 recursive        0.0622             1.1430           0.9479            0.850
```

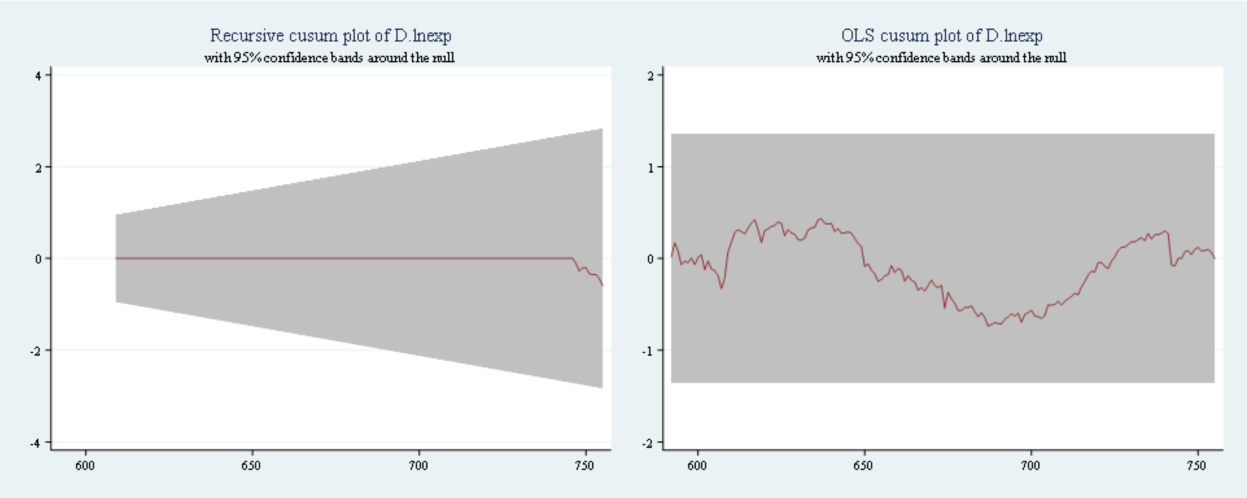
```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                   Number of obs =        164
Ho: No structural break

                                   1% Critical      5% Critical     10% Critical
 Statistic    Test Statistic          Value            Value            Value

 ols              0.9552             1.6276           1.3581            1.224
```



C. Exp 8703

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                 Number of obs =        164
Ho: No structural break

                               1% Critical     5% Critical    10% Critical
 Statistic   Test Statistic       Value           Value           Value
 ---------------------------------------------------------------------------
 recursive       0.1975          1.1430          0.9479           0.850
 ---------------------------------------------------------------------------
```

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                 Number of obs =        164
Ho: No structural break

                               1% Critical     5% Critical    10% Critical
 Statistic   Test Statistic       Value           Value           Value
 ---------------------------------------------------------------------------
 ols             0.7378          1.6276          1.3581           1.224
 ---------------------------------------------------------------------------
```

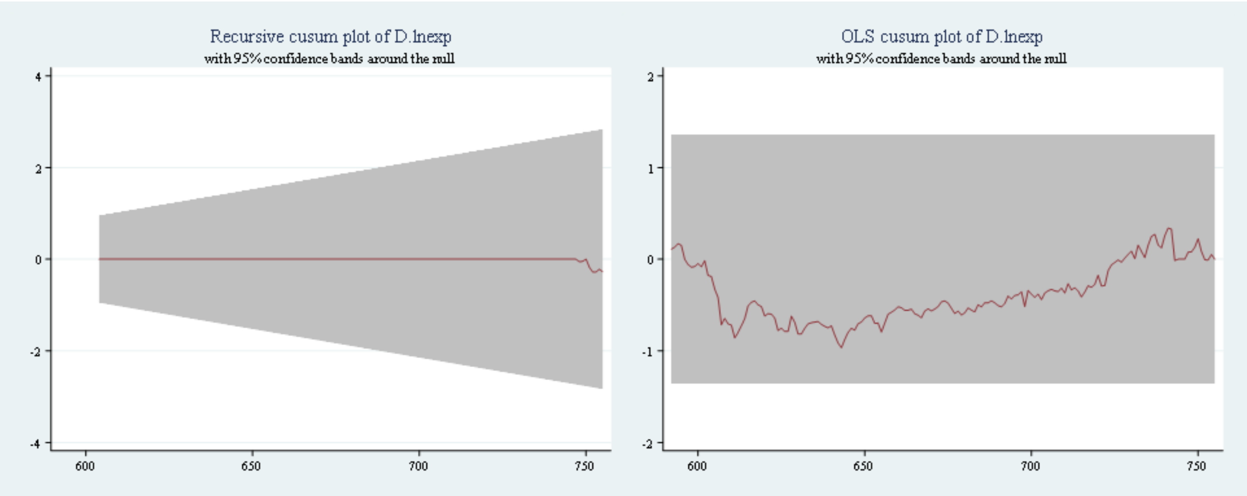


Slovakia

A. Exp 8703

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                 Number of obs =        164
Ho: No structural break

                               1% Critical     5% Critical    10% Critical
 Statistic   Test Statistic       Value           Value           Value
 ---------------------------------------------------------------------------
 recursive       0.0941          1.1430          0.9479           0.850
 ---------------------------------------------------------------------------
```

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                 Number of obs =        164
Ho: No structural break

                               1% Critical     5% Critical    10% Critical
 Statistic   Test Statistic       Value           Value           Value
 ---------------------------------------------------------------------------
 ols             0.9671          1.6276          1.3581           1.224
 ---------------------------------------------------------------------------
```



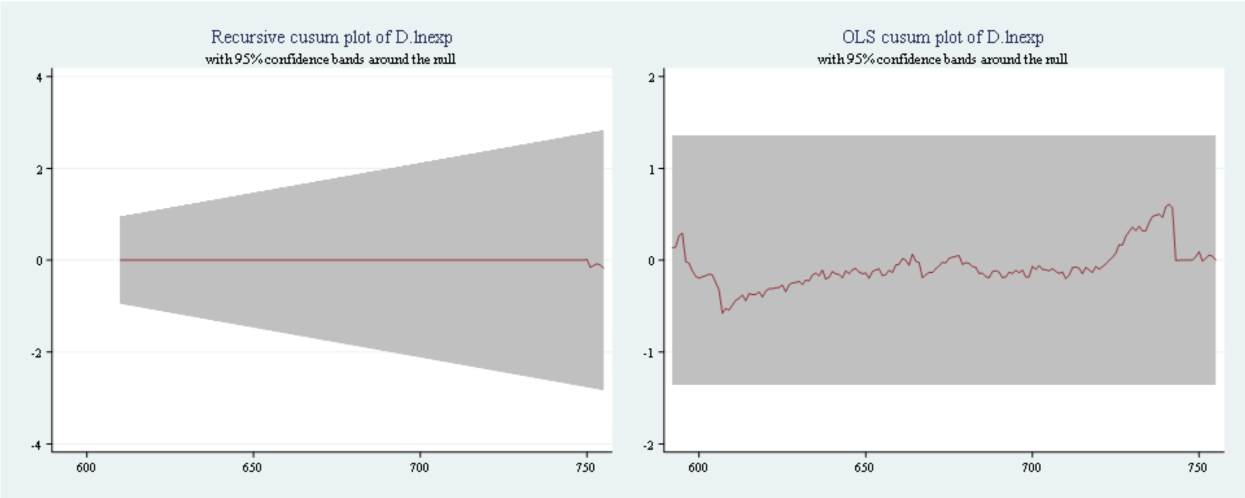
B. Exp 8708

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0578	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.6097	1.6276	1.3581	1.224



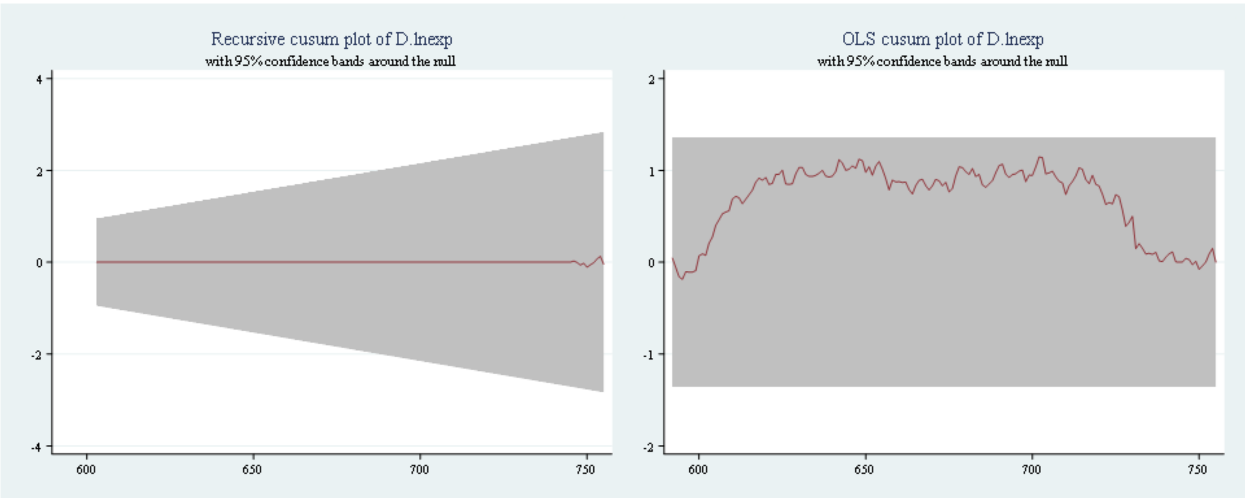
C. Exp 8528

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0434	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	1.1471	1.6276	1.3581	1.224



Slovenia

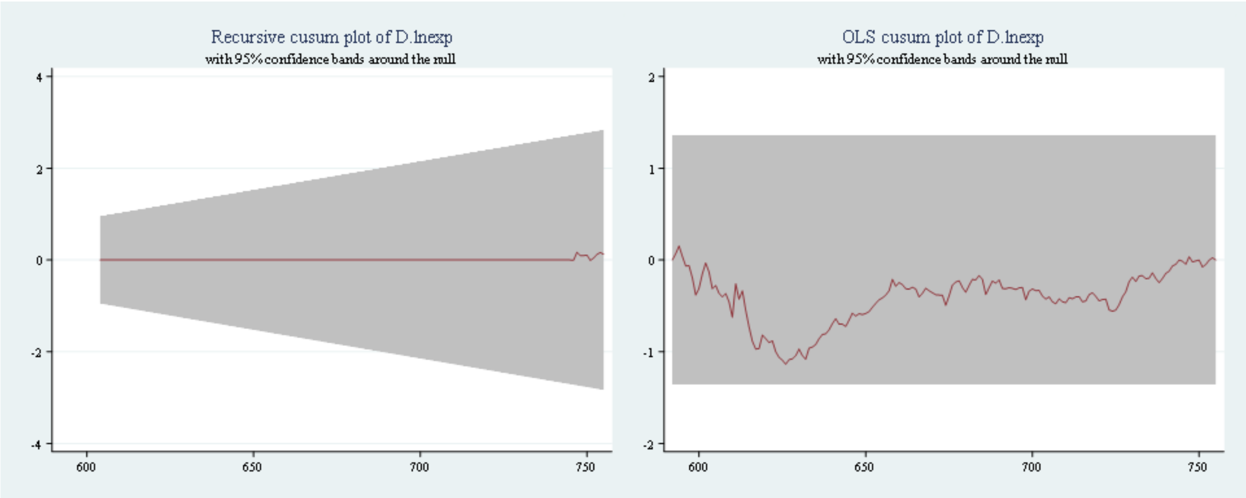
A. Exp 3004

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0586	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	1.1374	1.6276	1.3581	1.224



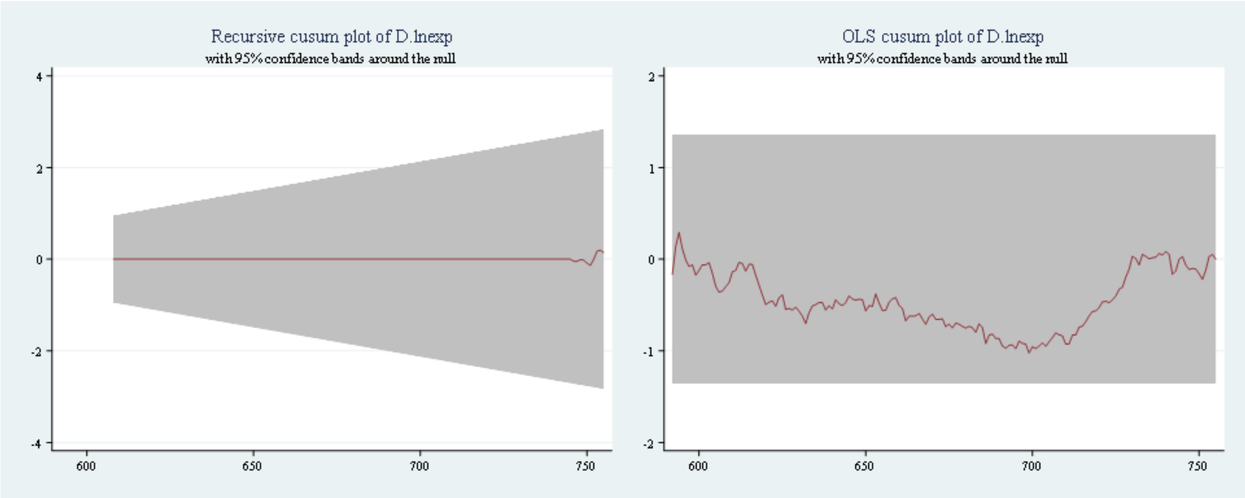
B. Exp 8703

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0659	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	1.0244	1.6276	1.3581	1.224



C. Exp 8516

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

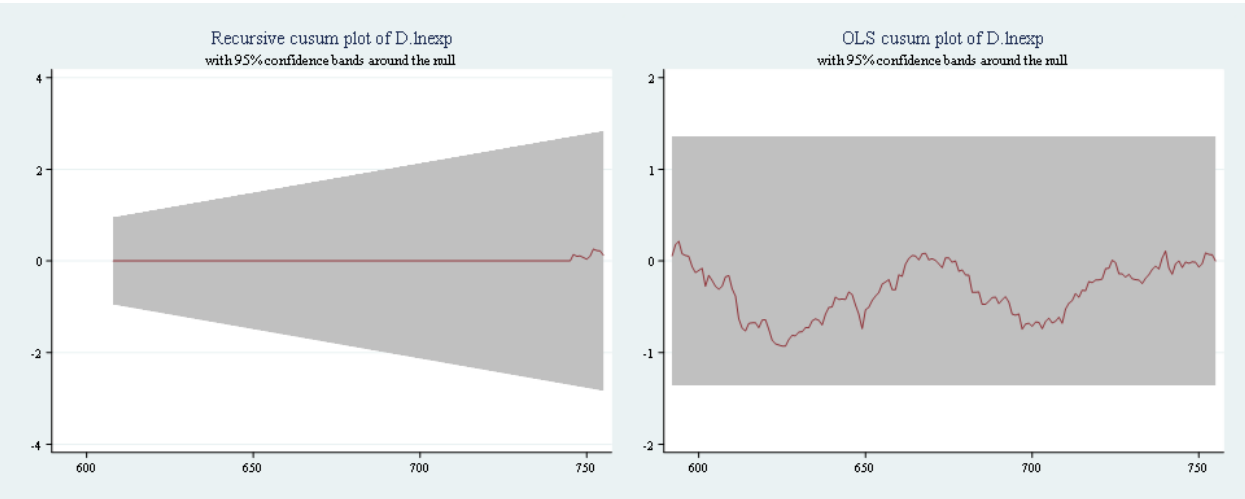
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0879	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.9287	1.6276	1.3581	1.224



Estonia

A. Exp 8517

.
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168   Number of obs = 164
Ho: No structural break

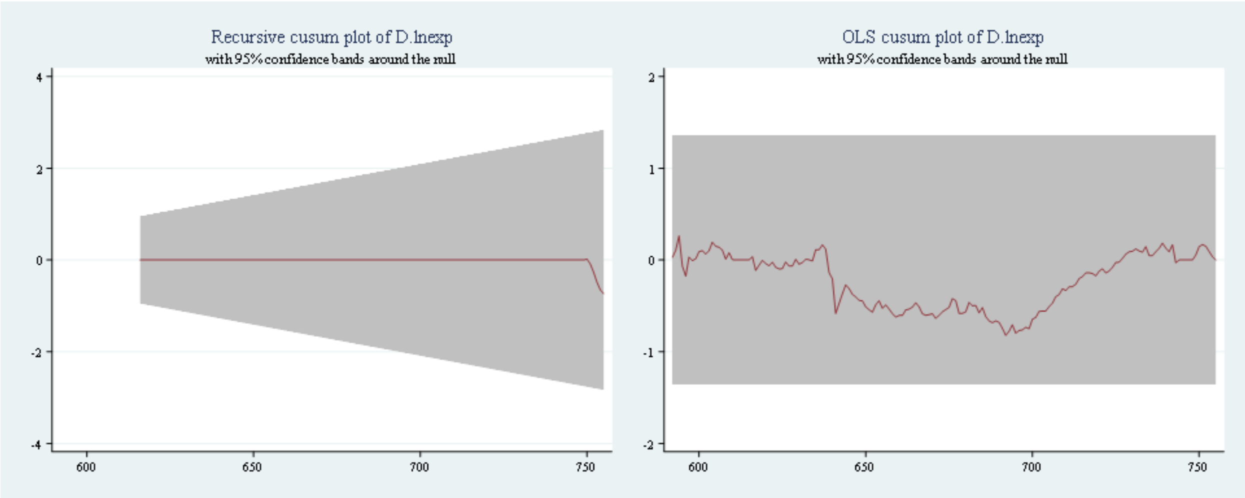
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.2448	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168   Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.8229	1.6276	1.3581	1.224



B. Exp 9406

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168   Number of obs = 164
Ho: No structural break

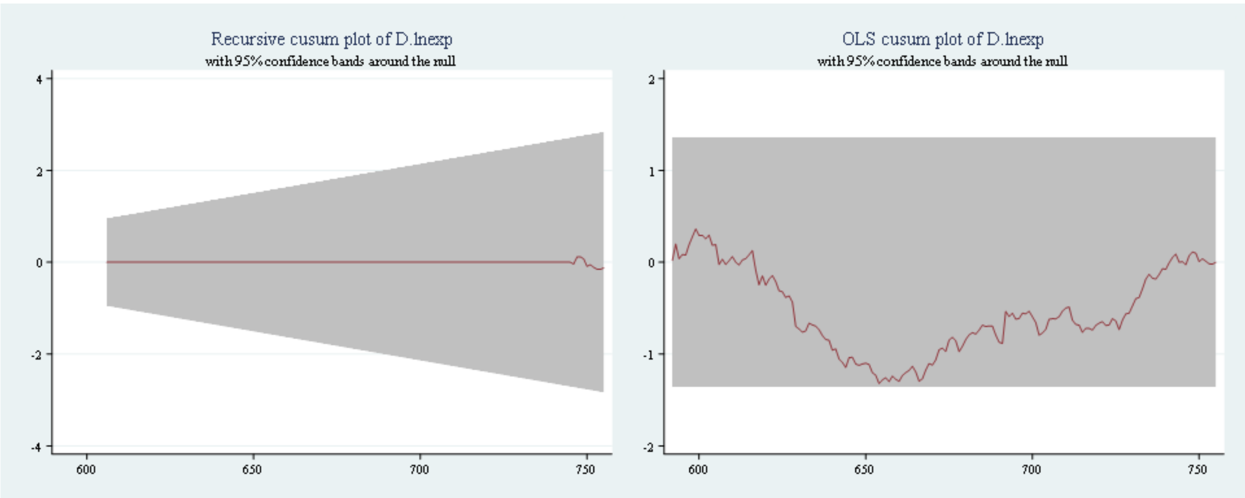
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0525	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168   Number of obs = 164
Ho: No structural break

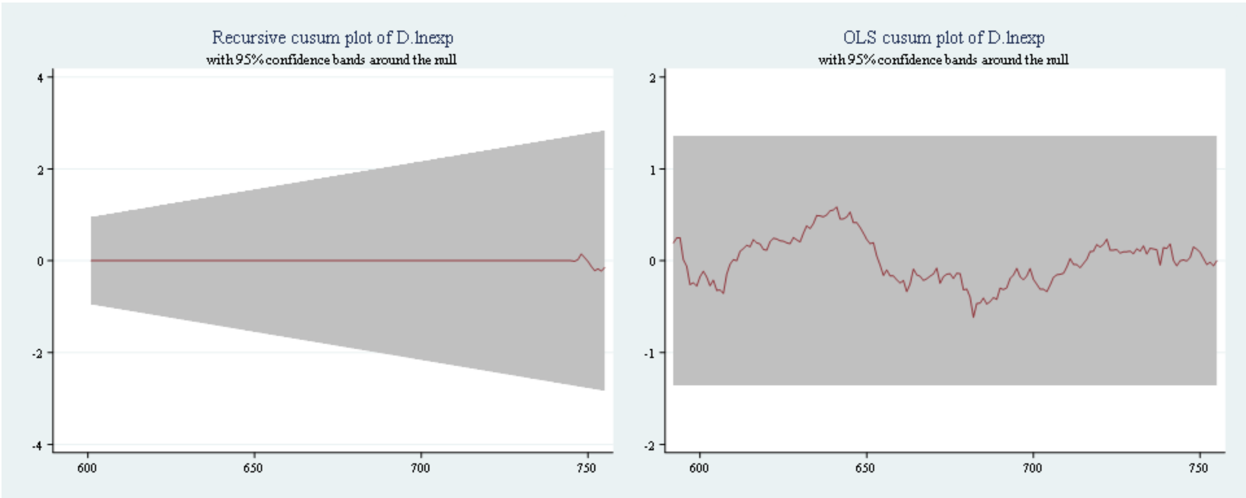
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	1.3214	1.6276	1.3581	1.224



C. Exp 8703

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =       164
Ho: No structural break

                                  1% Critical      5% Critical     10% Critical
  Statistic    Test Statistic        Value            Value            Value
--------------------------------------------------------------------------------
  recursive       0.0759            1.1430           0.9479           0.850
--------------------------------------------------------------------------------
```

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =       164
Ho: No structural break

                                  1% Critical      5% Critical     10% Critical
  Statistic    Test Statistic        Value            Value            Value
--------------------------------------------------------------------------------
  ols             0.6181            1.6276           1.3581           1.224
--------------------------------------------------------------------------------
```

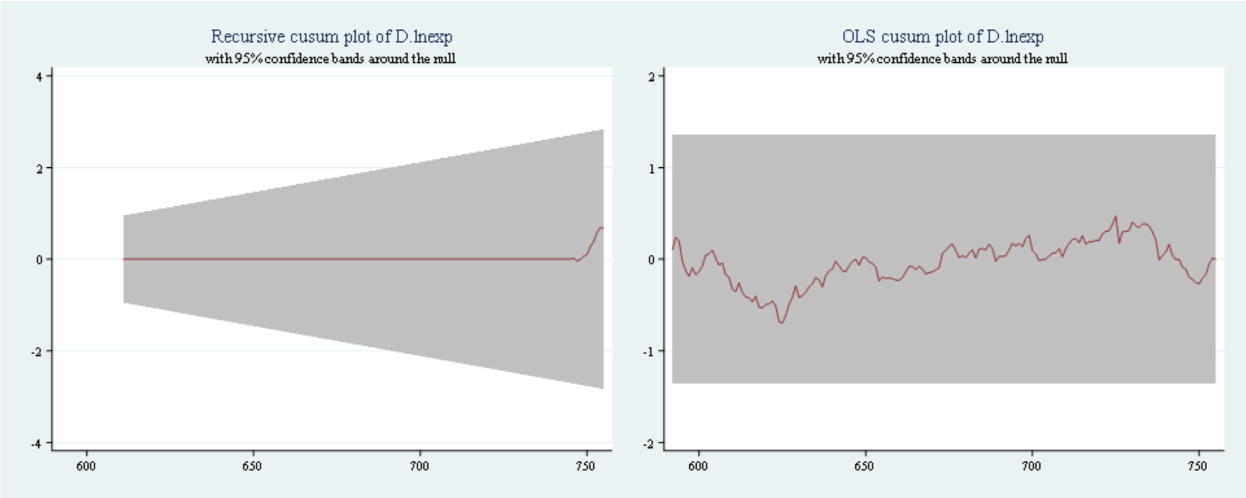


Latvia

A. Exp 4407

```
.
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =       164
Ho: No structural break

                                  1% Critical      5% Critical     10% Critical
  Statistic    Test Statistic        Value            Value            Value
--------------------------------------------------------------------------------
  recursive       0.2316            1.1430           0.9479           0.850
--------------------------------------------------------------------------------
```

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =       164
Ho: No structural break

                                  1% Critical      5% Critical     10% Critical
  Statistic    Test Statistic        Value            Value            Value
--------------------------------------------------------------------------------
  ols             0.6970            1.6276           1.3581           1.224
--------------------------------------------------------------------------------
```



B. Exp 8517

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

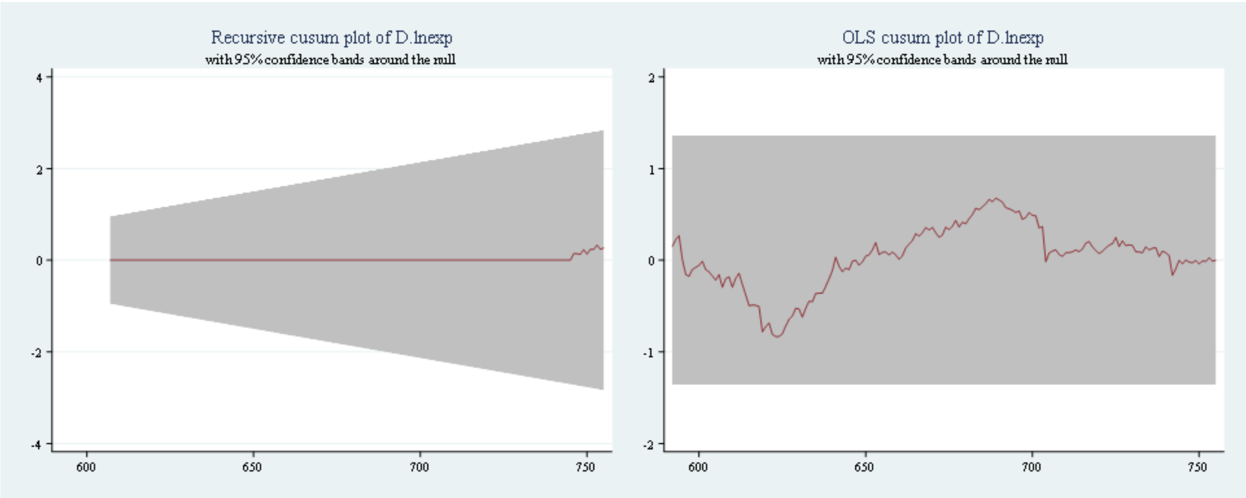
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.1113	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.8337	1.6276	1.3581	1.224



C. Exp 3004

. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

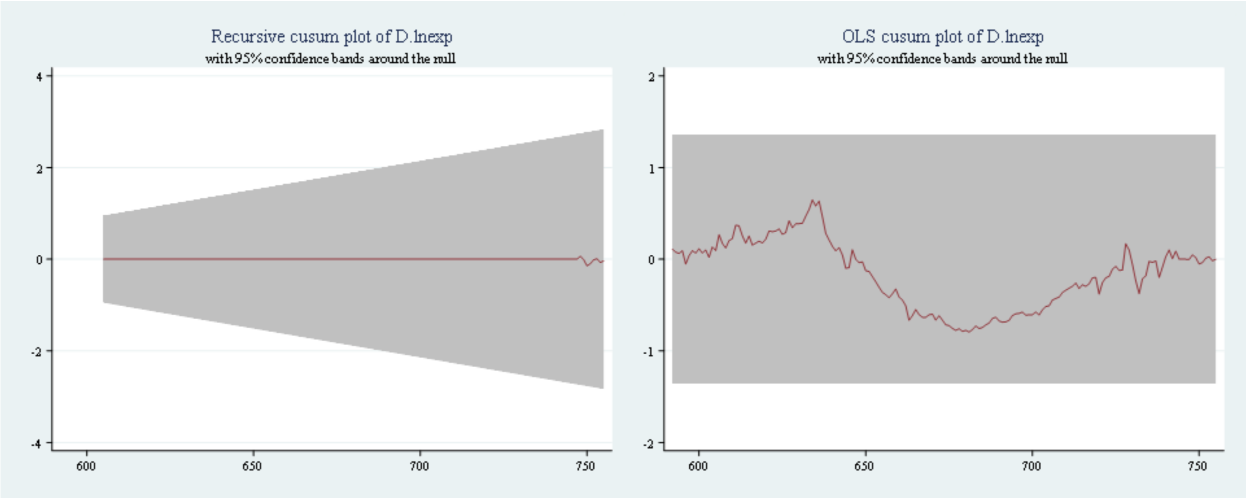
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0499	1.1430	0.9479	0.850

. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168 Number of obs = 164
Ho: No structural break

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	0.7957	1.6276	1.3581	1.224



Lithuania

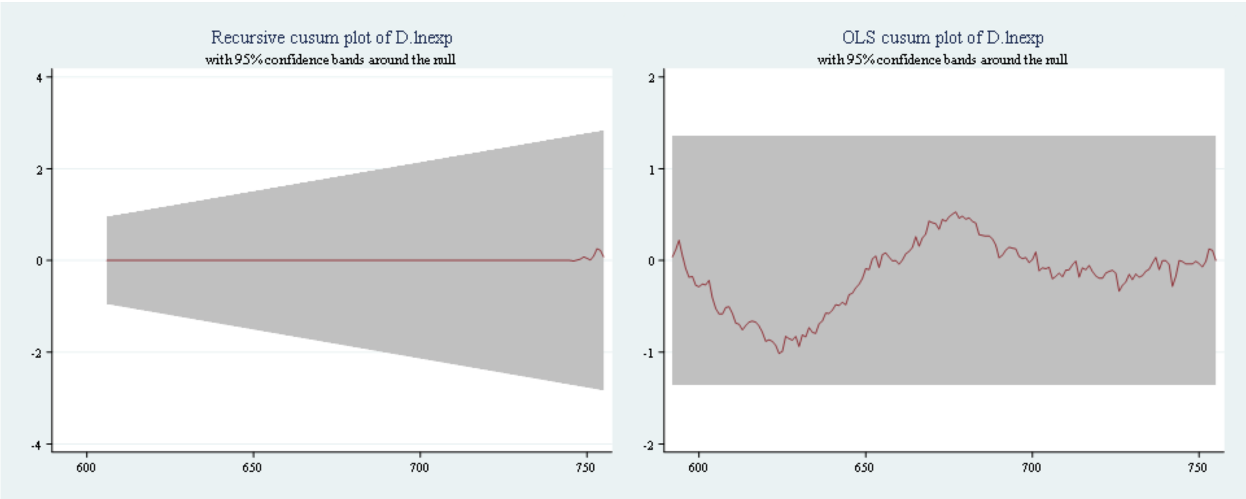
A. Exp 9403

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
recursive	0.0863	1.1430	0.9479	0.850

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                  Number of obs =        164
Ho: No structural break
```

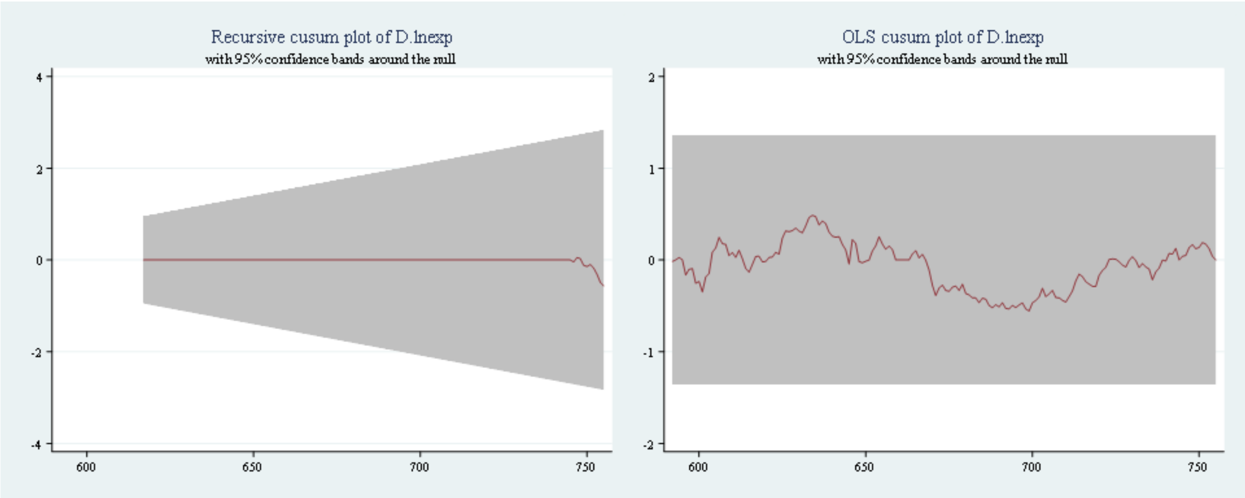
Statistic	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
ols	1.0135	1.6276	1.3581	1.224



B. Exp 8703

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =       164
Ho: No structural break

                              1% Critical      5% Critical     10% Critical
 Statistic   Test Statistic      Value            Value            Value
----------------------------------------------------------------------------
 recursive       0.1885          1.1430           0.9479           0.850
----------------------------------------------------------------------------
```

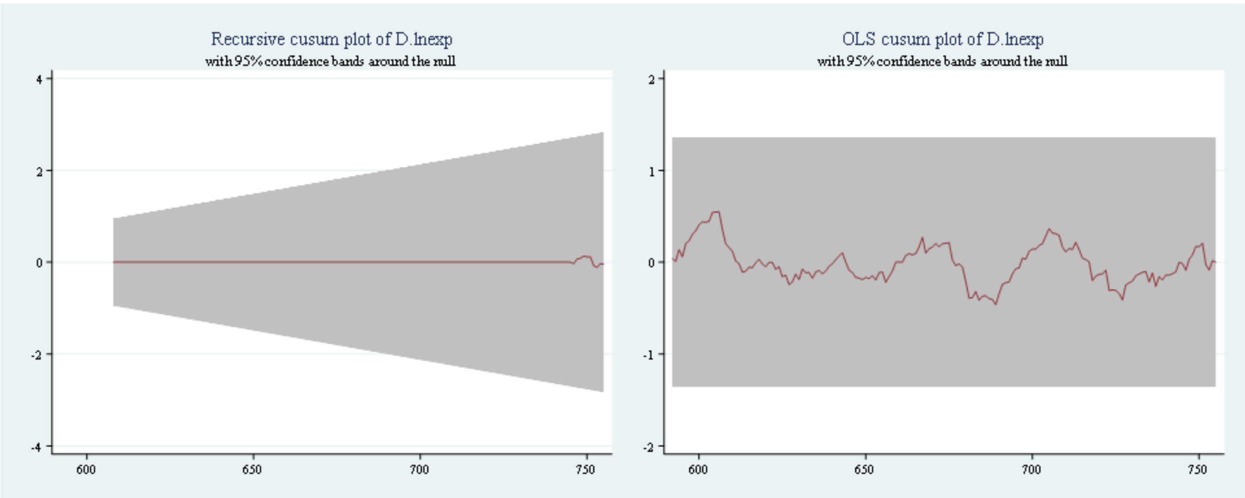
```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =       164
Ho: No structural break

                              1% Critical      5% Critical     10% Critical
 Statistic   Test Statistic      Value            Value            Value
----------------------------------------------------------------------------
 ols             0.5579          1.6276           1.3581           1.224
----------------------------------------------------------------------------
```



C. Exp 3102

```
. estat sbcusum

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =       164
Ho: No structural break

                              1% Critical      5% Critical     10% Critical
 Statistic   Test Statistic      Value            Value            Value
----------------------------------------------------------------------------
 recursive       0.0459          1.1430           0.9479           0.850
----------------------------------------------------------------------------
```

```
. estat sbcusum, ols

Cumulative sum test for parameter stability

Sample: 5 - 168                                    Number of obs =       164
Ho: No structural break

                              1% Critical      5% Critical     10% Critical
 Statistic   Test Statistic      Value            Value            Value
----------------------------------------------------------------------------
 ols             0.5490          1.6276           1.3581           1.224
----------------------------------------------------------------------------
```



Lampiran 15 Uji Wald

Republik Ceko

A. Exp 8703

```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   146) =    1.00
            Prob > F =    0.3190
```

B. Exp 8517

```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   153) =    0.13
            Prob > F =    0.7190
```

C. Exp 8708

```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   146) =    1.55
            Prob > F =    0.2157
```

Long Run

Hungaria

A. Exp 8703

```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   147) =    0.00
            Prob > F =    0.9843
```

B. Exp 8708

```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   148) =    1.08
            Prob > F =    0.2996
```

C. Exp 8507

```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   153) =    0.20
            Prob > F =    0.6543
```

Long Run

```
. test [SR]D.vol_pos = 0

 ( 1)  [SR]D.vol_pos = 0

       F(  1,   148) =    3.60
            Prob > F =    0.0596
```

```
. test [SR]D.vol_neg + [SR]LD.vol_neg + [SR]L2D.vol_neg + [SR]L3D.vol_neg  = 0

 ( 1)  [SR]D.vol_neg + [SR]LD.vol_neg + [SR]L2D.vol_neg + [SR]L3D.vol_neg = 0

       F(  1,   153) =    0.63
            Prob > F =    0.4296
```

Short Run

Polandia

A. Exp 8708

```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   144) =    0.36
            Prob > F =    0.5491
```

B. Exp 8507

```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   155) =    4.11
            Prob > F =    0.0444
```

C. Exp 8703

```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   147) =    0.00
            Prob > F =    0.9724
```

Long Run

```
. test [SR]D.vol_neg = 0

 ( 1)  [SR]D.vol_neg = 0

       F(  1,   144) =    2.32
            Prob > F =    0.1303
```

```
. test [SR]D.vol_neg = 0

 ( 1)  [SR]D.vol_neg = 0

       F(  1,   155) =   14.04
            Prob > F =    0.0003
```

Short Run

Slovakia

A. Exp 8703

```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   152) =    0.29
            Prob > F =    0.5894
```

B. Exp 8708

```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   146) =    0.55
            Prob > F =    0.4576
```

C. Exp 8528

```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   153) =    0.00
            Prob > F =    0.9941
```

Long Run

Slovenia

A. Exp 3004 | B. Exp 8703 | C. Exp 8516

Long Run

A. Exp 3004
```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   152) =    0.03
            Prob > F =    0.8718
```

B. Exp 8703
```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   148) =    1.25
            Prob > F =    0.2661
```

C. Exp 8516
```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   148) =    0.50
            Prob > F =    0.4799
```

Short Run

B. Exp 8703
```
. test [SR]D.vol_pos = 0

 ( 1)  [SR]D.vol_pos = 0

       F(  1,   148) =    2.41
            Prob > F =    0.1226
```

C. Exp 8516
```
. test [SR]D.vol_neg = 0

 ( 1)  [SR]D.vol_neg = 0

       F(  1,   148) =    4.52
            Prob > F =    0.0352
```

Estonia

A. Exp 8517 | B. Exp 9406 | C. Exp 8703

Long Run

A. Exp 8517
```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   140) =    2.65
            Prob > F =    0.1055
```

B. Exp 9406
```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   150) =    0.55
            Prob > F =    0.4580
```

C. Exp 8703
```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   155) =    0.44
            Prob > F =    0.5058
```

Short Run

A. Exp 8517
```
. test [SR]D.vol_neg + [SR]LD.vol_neg + [SR]L2D.vol_neg = 0

 ( 1)  [SR]D.vol_neg + [SR]LD.vol_neg + [SR]L2D.vol_neg = 0

       F(  1,   140) =    6.72
            Prob > F =    0.0105
```

B. Exp 9406
```
. test [SR]D.vol_neg + [SR]LD.vol_neg + [SR]L2D.vol_neg = 0

 ( 1)  [SR]D.vol_neg + [SR]LD.vol_neg + [SR]L2D.vol_neg = 0

       F(  1,   150) =    8.93
            Prob > F =    0.0033
```

C. Exp 8703
```
. test [SR]D.vol_pos = 0

 ( 1)  [SR]D.vol_pos = 0

       F(  1,   155) =    4.83
            Prob > F =    0.0295
```

Latvia

A. Exp 4407 | B. Exp 8517 | C. Exp 3004

Long Run

A. Exp 4407
```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   145) =    0.36
            Prob > F =    0.5514
```

B. Exp 8517
```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   149) =    2.02
            Prob > F =    0.1572
```

C. Exp 3004
```
. test [LR]vol_pos = [LR]vol_neg

 ( 1)  [LR]vol_pos - [LR]vol_neg = 0

       F(  1,   151) =    1.38
            Prob > F =    0.2424
```

Short Run

```
. test [SR]D.vol_pos + [SR]LD.vol_pos + [SR]L2D.vol_pos + [SR]L3D.vol_pos  = 0

 ( 1)  [SR]D.vol_pos + [SR]LD.vol_pos + [SR]L2D.vol_pos + [SR]L3D.vol_pos = 0

       F(  1,   145) =    0.17
            Prob > F =    0.6845
```

Lithuania

A. Exp 9403 | B. Exp 8703 | C. Exp 3102

Long-Run	. test [LR]vol_pos = [LR]vol_neg (1) [LR]vol_pos - [LR]vol_neg = 0 F(1, 150) = 5.97 Prob > F = 0.0157	. test [LR]vol_pos = [LR]vol_neg (1) [LR]vol_pos - [LR]vol_neg = 0 F(1, 139) = 0.12 Prob > F = 0.7265	. test [LR]vol_pos = [LR]vol_neg (1) [LR]vol_pos - [LR]vol_neg = 0 F(1, 148) = 0.13 Prob > F = 0.7229
Short Run	. test [SR]D.vol_pos + [SR]LD.vol_pos + [SR]L2D.vol_pos + [SR]L3D.vol_pos = 0 (1) [SR]D.vol_pos + [SR]LD.vol_pos + [SR]L2D.vol_pos + [SR]L3D.vol_pos = 0 F(1, 150) = 3.14 Prob > F = 0.0784	. test [SR]D.vol_pos + [SR]LD.vol_pos + [SR]L2D.vol_pos + [SR]L3D.vol_pos = [SR]D.vol_neg + [SR]LD.vol_neg + [SR]L2D.vol_neg (1) [SR]D.vol_pos + [SR]LD.vol_pos + [SR]L2D.vol_pos + [SR]L3D.vol_pos - [SR]D.vol_neg - [SR]LD.vol_neg - [SR]L2D.vol_neg = 0 F(1, 139) = 0.08 Prob > F = 0.7843	. test [SR]D.vol_pos + [SR]LD.vol_pos + [SR]L2D.vol_pos + [SR]L3D.vol_pos = 0 (1) [SR]D.vol_pos + [SR]LD.vol_pos + [SR]L2D.vol_pos + [SR]L3D.vol_pos = 0 F(1, 148) = 8.06 Prob > F = 0.0052

Lampiran 16 Hasil Uji Turnitin

UNIVERSITAS AIRLANGGA
FAKULTAS EKONOMI DAN BISNIS
Kampus B Jl. Airlangga 4 Surabaya 60286 Telp. (031) 5033642, 5036584 Faks. (031)5026288
Laman : http://feb.unair.ac.id e-mail : info@feb.unair.ac.id, humas@feb.unair.ac.id

SURAT KETERANGAN
Nomor: 10390/UN3.FEB/PK.05.00/2024

TES KESAMAAN *(SIMILARITY)*

Setelah melakukan tes uji similarity, yang bertanda tangan di bawah ini:

Nama	: Abdul Munir
Nip	: 196701261990031001
Jabatan	: Kepala Urusan Ruang Baca

Menerangkan bawah mahasiswa dibawah ini:

Nama	: Heidir Royyan Firdaus
Nim	: 042111133107
Program Studi	: S1 - Ekonomi Pembangunan
Judul Karya Ilmiah	: Pengaruh Volatilitas Nilai Tukar, Geopolitik, dan Euroization terhadap Ekspor Manufaktur di CEECs EU-8
Paper ID	: 2558851471
Class ID	: 43944422
Date	: December 30, 2024
Hasil menunjukkan	SIMILARITY INDEX : 12%

Demikian surat pernyataan ini kami buat untuk dipergunakan sebagaimana mestinya.

30 Desember 2024
a.n.Kasubag.Akademik,
Kaur Ruang Baca,

ABDUL MUNIR
NIP 196701261990031001

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